

APPENDIX J

MULTILAYER NANOPOROUS SEPARATOR

Transaction History

Date	Transaction Description
04-10-2024	Information Disclosure Statement (IDS) Filed
04-10-2024	Patent Term Adjustment - Ready for Examination
04-10-2024	PTO-SB-69-Authorize EPO Access to Search Results
04-10-2024	Applicants have given acceptable permission for participating foreign
04-10-2024	Information Disclosure Statement (IDS) Filed
04-10-2024	Entity Status Set To Undiscounted (Initial Default Setting or Status Change)
04-10-2024	Initial Exam Team nn
04-26-2024	Sent to Classification Contractor
04-26-2024	FITF set to YES - revise initial setting
04-26-2024	Mail Pre-Exam Notice
04-26-2024	Application Is Now Complete
04-26-2024	Filing Receipt
04-28-2024	Case Docketed to Examiner in GAU
04-28-2024	Application Dispatched from OIPE
05-02-2024	Application ready for PDX access by participating foreign offices
05-02-2024	Application ready for PDX access by participating foreign offices
08-08-2024	PG-Pub Issue Notification
08-09-2024	Information Disclosure Statement (IDS) Filed
08-09-2024	Information Disclosure Statement (IDS) Filed
10-12-2024	Case Docketed to Examiner in GAU
03-27-2025	Information Disclosure Statement (IDS) Filed
03-27-2025	Information Disclosure Statement (IDS) Filed



UNITED STATES
PATENT AND TRADEMARK OFFICE

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION # 18/631,783	RECEIPT DATE / TIME 04/10/2024 04:21:09 PM Z ET	ATTORNEY DOCKET # 00249/62
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Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	1021	FILED BY	Joy Chapman
PATENT CENTER #	65043215	FILING DATE	-
CUSTOMER #	1912	FIRST NAMED INVENTOR	David W. Avison
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Brian Amos

Documents

TOTAL DOCUMENTS: 10

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
00249-62_ADS_fillable-pdf.pdf	9	Application Data Sheet	2174 KB
00249-62_DOCX-Application-APP.TEXT.docx	30	Application body structured text document	68 KB
Warning: Bookmarks were found and have been removed.			
00249-62_Drawings.pdf	6	Drawings-only black and white line drawings	63 KB
00249-62_Preliminary-Amendment.pdf	4	-	133 KB

00249-62_Preliminary-Amendment-A.PE.pdf	(1-1)	1	Preliminary Amendment	96 KB
00249-62_Preliminary-Amendment-SPEC.pdf	(2-2)	1	Specification	131 KB
00249-62_Preliminary-Amendment-REM.pdf	(3-4)	2	Applicant Arguments/Remarks Made in an Amendment	133 KB
00249-62_Auxiliary-PDF-Application.pdf		30	Auxiliary PDF of Application	261 KB
00249-62_Declarations-for-the-3-Inventors.pdf		3	Oath or Declaration filed	2107 KB
00249-62_POA-with-Transmittal.pdf		2	Power of Attorney	399 KB
00249-62_IDS_fillable-pdf.pdf		7	Information Disclosure Statement (IDS) Form (SB08)	1019 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
00249-62_ADS_fillable-pdf.pdf	8AC8C4B38EE5C1B21E9941E020C535BEFE629EC4DE13CD2D11648785590B61791386AFA05DD5E3431ED9CF7CE7FA5F3ACA42156867C41B691B871FA320FCFB22
00249-62_DOCX-Application-APP.TEXT.docx	35613F1653E2E68C11BAAE3E8DC6132F7E77EE6D3C0415AEF66365D40CB05780F28700092821D823260DBFD5278DCDAEF773E5139731B661BD5637A0795E502A
00249-62_Drawings.pdf	CFD156D51F265977E085274C81A6CBA29A9E699C171ACB37FD7CC3395716574F9F8E8515CCC980664E6D59D7D916FD4250F2541BC244E6A20F596D29080D19D9
00249-62_Preliminary-	89076B5262CB892746958F665441C87CF5626A6312EC02DA3E

Amendment.pdf	E115C8157DC207C023CCD3DFC70361A737E228AEFA44EEFD4780D1FC67A33D41FA57F242F5AC14
00249-62__Preliminary-Amendment-A.PE.pdf	470D5F3FA94AC09D36AD0ED22C7046128EF326F3968F7E98B84D4D84D0E1B8AF2BB10F7C49F7C35A2DB64AD6FEC8EE8B15E8F93394557454A319A647EE9D13CA
00249-62__Preliminary-Amendment-SPEC.pdf	C8D1A520F9274546331679B12172FB80EA9CAA13E4CF164E2D198581AFDECA32C34880E23E96CFAB394FD0A2D27C3B47237FB2BDCFAEA02D2FA35D9382331EDE
00249-62__Preliminary-Amendment-REM.pdf	089A657ED85A5E299217B5C040AC08F80ABA3E72AF7619545A160F66BDAF4173040ED5C0382CC7AFA73093B136C276BF09330FEA35DB7A7A9AAF36CFCF72589D
00249-62__Auxiliary-PDF-Application.pdf	4AF2A9B5A676E31C5316FC131B7F7406DD8A7360408617C138FD586E5F6787896B5B72D91B8A4542B4402D1FD88BD88C341AEB41165038AEA9832ABE838A797A
00249-62__Declarations-for-the-3-Inventors.pdf	81E4170A9D3E0A92562735ACF0DC44083F0D3717A0ECB57DDC78E936C99DA9561BA791EA37081DB8AB8CA5BF2E46E0B5EE1C529EB9255A39CA4AD6201E4B0428
00249-62__POA-with-Transmittal.pdf	B80CEE269509174314F170DE8508B34A92841FB56260630BAAF623893D3EEA4D299B7E4634116867CC679320B5DFE91A3E924735A27404D93F6263A4527427E8
00249-62__IDS_fillable-pdf.pdf	1E472A7427DFF9DDA900A77C649330855424EB670DA6096AD373DF0479AF7EAE9990BAF00892C7ACCBF675B08C8BEFD0AF64F7FEF41469D419076DB603D6A21B

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage

submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/62
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Inventor Information:

Inventor 1

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	David	W.	Avison	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Boxborough	State/Province	MA	Country of Residence ⁱ	US
------	------------	----------------	----	-----------------------------------	----

Mailing Address of Inventor:

Address 1	31D Liberty Road				
Address 2					
City	Boxborough	State/Province	MA		
Postal Code	01719	Country ⁱ	US		

Inventor 2

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Steven	A.	Carlson	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Cambridge	State/Province	MA	Country of Residence ⁱ	US
------	-----------	----------------	----	-----------------------------------	----

Mailing Address of Inventor:

Address 1	993 Memorial Drive, Apt. 101				
Address 2					
City	Cambridge	State/Province	MA		
Postal Code	02138	Country ⁱ	US		

Inventor 3

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Benjamin		Sloan	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

Application Data Sheet 37 CFR 1.76		Attorney Docket Number		00249/62		
		Application Number				
Title of Invention		MULTILAYER NANOPOROUS SEPARATOR				
City	Exeter	State/Province	NH	Country of Residence	US	
Mailing Address of Inventor:						
Address 1		11 Gardner Street				
Address 2						
City	Exeter	State/Province	NH			
Postal Code	03833	Country	US			
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					Add	

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	1912		
Email Address	ptodocket@arelaw.com	Add Email	Remove Email

Application Information:

Title of the Invention	MULTILAYER NANOPOROUS SEPARATOR				
Attorney Docket Number	00249/62	Small Entity Status Claimed ☐			
Application Type	Nonprovisional				
Subject Matter	Utility				
Total Number of Drawing Sheets (if any)	6	Suggested Figure for Publication (if any)			

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/62
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	1912		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78. When referring to the current application, please leave the “Application Number” field blank.

Prior Application Status		Pending		Remove	
Application Number		Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
		Continuation of		17/543259	2021-12-06
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
17/543259	Continuation of	17/223845	2021-04-06	11769903	2023-09-26
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
17/223845	Continuation of	16/755294	2020-04-10	10998536	2021-05-04
Prior Application Status		Expired		Remove	
Application Number		Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
16/755294		a 371 of international		PCT/US2018/055862	2018-10-15
Prior Application Status		Expired		Remove	
Application Number		Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
PCT/US2018/055862		Claims benefit of provisional		62/572083	2017-10-13
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.					Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/62
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Remove Access Code ⁱ (if applicable)

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	00249/62
	Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People’s Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO’s Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. ☐ If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. ☐ If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

PTO/AIA/14 (01-22)
Approved for use through 05/31/2024. OMB 0651-0032
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/62
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

Remove

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

Clear

☒ Assignee

☐ Legal Representative under 35 U.S.C. 117

☐ Joint Inventor

☐ Person to whom the inventor is obligated to assign.

☐ Person who shows sufficient proprietary interest

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor:

If the Applicant is an Organization check here.

☒

Organization Name

LG Energy Solution, Ltd.

Mailing Address Information For Applicant:

Address 1

Parc.1, 108 Yeoui-daero, Yeongdeungpo-gu

Address 2

City

Seoul

State/Province

Country

KR

Postal Code

07335

Phone Number

Fax Number

Email Address

Additional Applicant Data may be generated within this form by selecting the Add button.

Add

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/62
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Assignee 1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

Remove

If the Assignee or Non-Applicant Assignee is an Organization check here. ☐

Prefix	Given Name	Middle Name	Family Name	Suffix

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1				
Address 2				
City			State/Province	
Country ⁱ			Postal Code	
Phone Number			Fax Number	
Email Address				

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add

Signature: Remove

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the “Authorization or Opt-Out of Authorization to Permit Access” section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Brian J. Amos/			Date (YYYY-MM-DD)	2024-04-10
First Name	Brian J.	Last Name	Amos	Registration Number	63,679

Additional Signature may be generated within this form by selecting the Add button. Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/62
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

MULTILAYER NANOPOROUS SEPARATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of and priority to U.S. Provisional Patent Application No. 62/572,083, filed on October 13, 2017, the entire contents of which are incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present invention relates generally to porous separators for batteries and other electric current producing cells. In particular, it pertains to multilayer porous separators comprising inorganic oxide/polymer layers on a polymeric layer, and to lithium batteries comprising such separators.

BACKGROUND OF THE INVENTION

[0003] Lithium batteries are widely used in portable electronics, such as smartphones and portable computers. Among the new applications for lithium batteries are high power batteries for hybrid, plug-in hybrid, and fully electric vehicles. Existing lithium batteries, including rechargeable and non-rechargeable lithium batteries, often utilize multilayer separators. Such separators may include an extruded polyethylene or polypropylene porous film that is coated with an inorganic oxide/polymer coating layer. As the demand for lower cost lithium batteries with increased size and energy density has grown, manufacturers have turned to less safe battery active materials in order to meet these goals. This has resulted in a need for battery separators with increased dimensional stability at temperatures such as 150°C, or higher.

[0004] Thus, it would be advantageous if the inorganic oxide/polymer coating layers could provide even greater dimensional stability at elevated temperatures while, at the same time, requiring lower coating weights and coating thicknesses, and providing lower manufacturing costs.

SUMMARY OF THE INVENTION

[0005] It is an object of the present invention to provide lighter weight, lower cost, and thinner inorganic oxide/polymer coating layers for polymer-based separators for lithium batteries and other electric current producing cells.

[0006] The present invention achieves these goals by providing a separator for a lithium battery comprising: (a) a porous polymeric layer, and (b) nanoporous inorganic oxide/polymer composite layers coated on both sides of the polymeric layer. In one embodiment, the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO_2) and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic oxide particles is 5 nm to 90 nm. This combination of inorganic oxide particles (e.g., boehmite particles) having an extremely small crystallite size, and a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator. This combination was further found to enable a significantly lower coating weight and coating thickness of the heat stable layer.

[0007] In another embodiment the nanoporous layer comprises: (a) a blend of inorganic oxide particles (e.g., boehmite particles) and inorganic nitride particles (e.g., boron nitride or aluminum nitride particles), and (b) one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic particles is 5 nm to 90 nm. This combination of: (1) inorganic oxide and nitride particles (e.g., boehmite and boron nitride or aluminum nitride particles) having extremely small crystallite sizes, and (2) a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator.

[0008] Preferably, the heat stable inorganic oxide/polymer layer (or inorganic oxide and nitride/polymer layer) is coated on both sides of the porous polymeric layer, such as a porous polyethylene layer or a porous polypropylene layer, as opposed to being on only one side of the polymeric layer. It has been found that this approach provides improved thermal dimensional stability in a thin and low cost separator. However, having the heat stable layer on only one side of the polymeric layer is also useful.

[0009] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO₂) and a polymer, and wherein the crystallite size of the inorganic oxide is 5 nm to 25 nm. It has been found that this smaller crystallite size results in a thinner coating layer with more cohesive strength and with more adhesion to the polymeric layer at an even lower coating weight than comparable layers where the crystallite size for the inorganic oxide is larger such as, for example, about 80 nm.

[0010] In another embodiment, the present invention includes lithium batteries that comprise the improved separators of this invention.

[0011] A further aspect of the present invention are methods for making separators that comprise coating a solution comprising an inorganic oxide and/or inorganic nitride, and one or more polymers, on one or both surfaces of a porous polymeric layer to form the separators of this invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The features and advantages of the present disclosure will be more fully understood with reference to the following, detailed description when taken in conjunction with the accompanying figures, wherein:

[0013] FIG. 1 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0014] FIG. 2 is a graph of the shrinkage as a function of coating thickness of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0015] FIG. 3 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0016] FIG. 4 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0017] FIG. 5 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0018] FIG. 6 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0019] FIG. 7 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0020] FIG. 8 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0021] FIG. 9 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0022] FIG. 10 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0023] FIG. 11 is a graph of the moisture content as function of the weight fraction of boron nitride in a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 4.

DETAILED DESCRIPTION

[0024] The separators of the present invention provide superior safety, lower cost, lower thickness, and other key performance properties for use in lithium batteries and in other electric current producing cells.

[0025] One aspect of this invention pertains to a separator for a lithium battery comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on both sides of the polymeric layer, wherein the nanoporous layer comprises boehmite and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to 45%, and wherein the crystallite size of the boehmite is 5 nm to 90 nm. Alternatively, the nanoporous layer may further comprise other inorganic oxide (e.g., SiO₂) or nitride (e.g., BN or AlN) particles, or combinations thereof.

[0026] The term “lithium battery” may include, for example, batteries in which the electroactive anode material comprises lithium. This includes, but is not limited to, secondary or rechargeable lithium ion batteries, secondary lithium metal batteries, and primary or non-rechargeable lithium metal batteries.

[0027] The term “separator” may include, for example, porous materials in a battery that are interposed between the cathode and the anode of the battery to provide electrical insulation against short circuiting and to provide pores for filling with the electrolyte of the battery.

[0028] The term “porous polymeric layer” may include, for example, a layer of a polymer material that is porous such that it has permeability to the flow of air through the polymeric layer. The battery industry typically measures this air permeability using a porosimeter that provides a value for the time in seconds for 100 cubic centimeters (cc) of air to pass through the porous layer. This value is commonly called the Gurley number. The battery industry prefers that the Gurley number for a separator be as low as possible without sacrificing mechanical integrity. Typical Gurley numbers for the air permeability of porous polyethylene or porous polypropylene separators, as used in the battery industry, are 100 seconds/100 cc to 300 seconds/100 cc. For this invention, porous polymer layer does not include any of the non-woven polymeric layers, such as with polyester fibers, that are used as separators in the battery industry.

[0029] The term “nanoporous” may include layers having pores up to about 100 nm in diameter, preferably, with at least 90% of the pores being less than 100 nm in diameter. The size and relative number of pores can be measured by mercury porosimetry, or by scanning electron microscope (SEM) or transmission electron microscope (TEM) cross-sections of the layer.

[0030] As discussed herein, the nanoporous layer may be present on both sides of a substrate, support, layer, or coating. For example, the nanoporous layer may be present on both sides of the porous polymeric layer, such as a porous polyethylene layer. The general benefits of the nanoporous layers of this invention are also found when the nanoporous layer is present on only one side of the porous polymeric layer, but the overall performance is typically superior when the nanoporous layer is present on both sides of the porous polymeric layer.

[0031] The superior performance features of two-side coating of the polymeric layer, compared to one-side coating at the same total coating weight, includes lower shrinkage at

high temperatures and improved battery safety. Improved safety is typically the most important performance feature desired in separators for lithium batteries, especially as the batteries become larger and higher in energy density and use less safe battery active materials in order to achieve higher energy densities and lower costs.

[0032] Another advantage of two-side coating of the porous polymeric layer with the nanoporous coating layer is that it provides the opportunity to make the coatings on each side different in composition. This enables the nanoporous layer to be optimized for the best performance when it is facing the anode and when it is facing the cathode. In general, by tailoring the formulations and coating thicknesses of the ceramic coatings on either side of the separator, it is possible to optimize: (a) adhesion between separator and a specific electrode, either anode or cathode; (b) interfacial impedance between separator and a specific electrode, either anode or cathode; (c) complexation of migratory species (e.g., solubilized transition metal ions from cathode).

[0033] For example, when the coating is facing the cathode, it is particularly useful that the coating be resistant to oxidation, especially at higher charge voltages, e.g., above 4.25 volts. This can be done, for example, by selecting the organic polymer of the nanoporous layer to be resistant to oxidation up to the higher voltage desired, e.g., up to 5.0 volts. In addition, the nanoporous layer on one of the sides can be designed with particularly small pore sizes, such as under 10 nm diameter pores, for example, by using boehmite with a primary crystallite size of less than 10 nm in diameter. It should be understood that the present invention is not limited to the examples provided herein. For example, surface-modified alumina particles, including boehmite particles, as described in PCT/IB2015/000272 to Loscutova *et al.* of Sasol, may be utilized in the separator layers described herein. This patent application discloses the dispersion of these surface-modified aluminas into nano-sized single particle crystallites. This reference also describes that the average crystallite size can be measured by X-ray diffraction on a 120 plane. As an example, the crystallite size (or an average crystallite size) of the inorganic particles in a separator comprising a nanoporous layer according to an embodiment of the invention may be measured by: (i) taking a cross-section sample of the nanoporous layer; (ii) using a TEM or very high power SEM on this sample to measure the diameter/width, in their widest dimension, of 100 single primary (inorganic) particles selected at random in the TEM or SEM image; (iii) excluding the 25 highest and 25 lowest diameter numbers in the 100 diameter/width numbers; and (iv) recording the crystallite size as the average of the middle

50 diameter/width numbers, i.e., an average crystallite size of the measured particles. Since the pore size diameters of ceramic particles in a nanoporous layer usually approximate the crystallite size of the ceramic particles, such as boehmite particles, used to make the nanoporous layer, the pore size diameters may indicate the crystallite size of the ceramic particles for nanoporous layers containing ceramic particles of a narrow crystallite size. However, for blends of ceramic particles of different crystallite sizes in a nanoporous coating, the pore size diameters of the coating layer typically are intermediate, or a blend, among the crystallite size diameters of the different ceramic particles and not individual peaks reflecting the different crystallite sizes. The TEM or SEM method described above for measuring the crystallite size does not have this limitation since it is analyzing single particles, and not blends, and thus is a preferred method to use for this invention. These separator configurations using very small pore sizes are useful when placed against a lithium metal anode, where the very small size of the pores assists in preventing the growth of lithium metal dendrites into and through the separator, which reduces cycle life, capacity, and safety for the cell. Where very small boehmite particles are used and some of these boehmite particles penetrate into the pores of the polymeric separator substrate, this is particularly useful in preventing the growth of lithium metal dendrites in and through the separator.

[0034] In another example, boehmite could be modified by reaction with a fluorinated ethylene carbonate for improved cycle life when such nanoporous layer is positioned against an anode comprising silicon or lithium metal. This nanoporous layer with the boehmite covalently reacted with a fluorinated ethylene carbonate could also be used against the cathode to help achieve excellent stability under high voltage operation of the cell, such as at up to 5.0 volts.

[0035] In a further example, the nanoporous layer with pores that are less than 10 nm in diameter could be positioned against the cathode to inhibit the diffusion of transition metals, such as nickel ions, from the cathode to the anode, which degrades the cycle life and other performance properties of the cell. Optionally, the very small boehmite crystallite size of less than 10 nm could be combined with reaction of this boehmite with fluorinated ethylene carbonate or another organic carbonate to make a nanoporous layer positioned against the cathode to lessen the diffusion of transition metal ions from the cathode.

[0036] The term “Boehmite” is understood in the art. It, for example, may include hydrated aluminum oxide of the empirical chemical formula of $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ where x is 1.0 to 2.0. Another term used interchangeably herein for boehmite is AlOOH . Boehmite may be

characterized by its distinctive Fourier-transform infrared spectroscopy (“FTIR”) fingerprint. Boehmite may also be characterized by its distinctive X-ray diffraction pattern and by its distinctive infrared spectrum of double peaks of hydroxyl groups around 3200 cm^{-1} . Boehmite may have values of x for molecules of water that are above 1.5 and even up to about 2. However, the commercial grades of boehmite of all particle sizes typically have a value of x that is in the range of 1.0 to 1.5. The value of x can be measured by first drying the boehmite at about 120°C for 1 hour to remove the surface adsorbed water and then measuring the loss of water that occurs when the boehmite sample is heated to 1000°C for 1 hour and is thereby dehydrated to anhydrous aluminum oxide, Al_2O_3 .

[0037] Examples of polymers that may be utilized in the present invention include, but are not limited to, cellulosic polymers such as hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides such as chitosan, polyethylene oxides, and polyvinyl alcohols. Preferably, the polymers, alone or in combination, provide cohesive and adhesive strength to the nanoporous layer, are insoluble in the electrolyte of the lithium battery, and are electrochemically stable over the voltage range of use of the lithium battery. If needed to achieve these properties, one or more of the polymers may be crosslinked.

[0038] By the term “volume fraction of the one or more polymers in the solid portion of the nanoporous layer” is meant the fraction expressed as a percentage by volume of the polymer content of the solid portion of the inorganic particle/polymer composite layer. As described hereinafter, the nanoporous layer has a % porosity of between about 30% to about 70%. In other words, much of the volume of the nanoporous layer is air, and not a solid material. The air volume (e.g., pores) is not included in the solid portion for purposes of determining the volume fraction of the one or more polymers in the solid portion of the nanoporous layer. The volume fraction of the polymer and inorganic particle content of the nanoporous layer can be calculated based on the relative densities of the materials. For example, when the inorganic particles are only boehmite with a density of 3.03 g/cc , the organic polymer has a density of 1.3 g/cc , and the weight ratio of the boehmite to the organic polymer is 12:1, the volume fraction of boehmite would be 83.7% and the volume fraction of organic polymer would be 16.3%. Changing the weight ratio of the boehmite to the organic polymer to 4:1 results in a volume fraction of boehmite of 63.2% and of organic polymer of 36.8%. If the density of the organic polymer is lower, such as 1.1 g/cc , then its volume fraction at the same weight ratio of inorganic particle/polymer will be higher than at the higher polymer density of 1.3 g/cc . To determine the volume fraction of one or more organic

polymers in the solid portion of the nanoporous layer, for example, the nanoporous layer is removed from the separator, such as by careful scraping of the layer and not removing any of the polymer substrate. Next, a weighed nanoporous layer sample in a crucible is combusted at a sufficient temperature to combust the organic polymer to leave the residue of the ceramic particles. This provides the weight of combusted ceramic particles. When the particles are only boehmite, this weight of the combusted residue is divided by 0.85 in the case of the equivalent of one water molecule per boehmite molecule to obtain the weight of boehmite in the sample prior to the combustion. Although most boehmite particles have a 15% weight loss upon combustion, it should be noted that boehmite particles may have a different weight loss factor, such as 0.84 or 0.82. This same information can be obtained by using a thermogravimetric analyzer to record the weight of the nanoporous layer sample as it is heated to 900 °C or higher, until a constant weight is achieved. As noted above, boehmite has a density of 3.03 g/cc. Most organic polymers have a density in the range of 1.1 to 1.4 g/cc, although some polymers, such as polyvinylidene difluoride (PVdF), have a density of about 1.8 g/cc. After the density of the organic polymers is known by analyzing for the identity and weight ratios of the organic polymers, for boehmite-only nanoporous layers, the volume % of the organic polymers in the solid portion of the nanoporous layer can be calculated. For example, if the weight percent of boehmite is 88.9%, and the weight percent of the organic polymers is 11.1% and its density is 1.3 g/cc--with a pigment:binder ratio of 8:1 by weight--dividing the weight percents of boehmite and the organic polymers by their densities gives relative volume percents of 29.34% for boehmite and 8.54% for the organic polymers. Normalizing these volume percents to a total volume percent of 100% for the solid portion of the nanoporous layer gives a volume percent of 22.54% for the organic polymers. If the nanoporous layer is a blend of boehmite and other ceramic particles or if the density of the organic polymers is not known, further analysis, as known in the art, may be undertaken to measure the volume percent of the organic polymers.

[0039] The term “crystallite size” of the particles refers to the average size of the primary or single crystals of the inorganic particles. This is typically measured using X-ray diffraction, a transmission electron microscope (TEM), or a very high resolution scanning electron microscope (SEM). PCT/IB2015/000272 to Loscutova *et al.* of Sasol cited above provides examples of using X-ray diffraction to determine an average size of the crystals of the inorganic particles, such as alumina and boehmite. As described earlier, a TEM or SEM method may be used to determine the crystallite size. Many inorganic oxide particles, such

as boehmite particles, are nearly cubic in shape, although they can have other shapes such as platelets and needles. In the dry solid state and in solution, boehmite particles are typically agglomerated into larger particles comprised of many primary boehmite particles, such as, for example, 10 to 20 primary particles. The inorganic particles in the nanoporous layers of this invention result in features, such as pore size diameter, that are related to the crystallite size of the particles.

[0040] In one embodiment of the separators of the present invention, the ratio of the total thickness of the nanoporous layer on both sides of the polymeric layer to the total thickness of the separator is 10% to 80%. The thickness of these layers may be measured by SEM analysis of cross-sections of the separator. Because of the nanoporous pore sizes and the high volume fraction of organic polymer in the nanoporous layer, it has been found that a lower limit for the proportional thickness of the nanoporous layer of about 10% of the total thickness of the separator, rather than 15% or more typically 25%, is able to provide an excellent improvement in the dimensional stability of the separator at high temperatures, which is needed for improved safety. It has been found that the combination of nanoporous pore sizes and very high volume fractions of organic polymer in the nanoporous layer (e.g., from about 10% to about 50%) provides acceptable % porosity (e.g., from about 30% to about 70%) with good ionic conductivity of the electrolyte. It has also been found that the separator of this invention functions well with the total thickness of the nanoporous layer as high as 80% of the total thickness of the separator.

[0041] In one embodiment, the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. This heat shrinkage may be measured by the change in the area of the separator after heating in an oven for 1 hour. This measurement of the heat shrinkage may be done by hanging a large sheet of the separator in the oven after marking and measuring the dimensions and area of a square or rectangle such as a 10 cm by 10 cm square, on a smaller section of the large sheet. After the heating in the oven, the marked area is measured for the % change or % shrinkage of the dimensions and the area.

[0042] In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%. The nanoporous separators of the present invention have been found to retain their dimensional stability even when heated well above the melting point of the polymeric layer, such as a porous polyethylene layer.

[0043] In one embodiment of the separators of this invention, the total coating weight of the nanoporous layer on both sides is less than 4.5 grams per square meter (gsm), while

providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. It should be noted that the use of boehmite with its lower density of about 3.03 g/cc, as compared to the higher density anhydrous aluminum oxide or alumina (which has a density of about 4 g/cc), helps to further lower the coating weight of the heat stabilizing layer. The lower density of the inorganic oxide particle proportionately lowers the weight of coating needed for a specific thickness and % porosity.

[0044] For example, a 4 micron thick alumina/polymer layer with a 45% porosity for polyethylene-based separators typically weighs about 8 grams per square meter (gsm). However, switching to boehmite (or a blend of boehmite and other inorganic particles of a similar or lower density) at the same thickness and % porosity lowers the coating weight to about 6 gsm or less. The high volume fraction of polymer in the nanoporous layers of the separators of this invention, combined with the use of boehmite with a crystallite size of 5 nm to 90 nm, provides increased dimensional stability for a specific thickness of the separator layer. Although it is possible to still make the total thickness of the inorganic particle/polymer layer in the separator 4 microns or higher, it is desirable for lower cost and for lower total thickness of the separator to make the thickness of the inorganic particle/polymer layer as thin as possible, such as about 3 microns total, while still achieving an excellent dimensional stability at high temperature. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0045] In one embodiment, other inorganic oxides and/or inorganic nitrides with a density of about 3.0 g/cc or lower, such as silicon dioxide and boron nitride, may be blended with the boehmite pigment. As discussed further below, this provides lower total coating weight of the nanoporous layer and other desirable properties, such as lower water content.

[0046] In another embodiment of the separators of this invention, the total coating weight of the nanoporous layer of both sides is less than 3.5 grams per square meter (gsm), while providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0047] In one embodiment of the separators of the present invention, the crystallite size of the inorganic particles may be about 30 nm to about 50 nm, preferably about 5 nm to about 25 nm, and more preferably about 5 nm to about 10 nm. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of about 30 nm to about 50 nm. In one embodiment, the crystallite size of the boehmite may be about 5 nm to about 10 nm, and preferably about 5 nm to about 8 nm.

[0048] In another embodiment of this invention, the % porosity of the nanoporous layer is between 30% and 55%. The % porosity of the nanoporous layer may be calculated from the density in g/cc of the materials in the dry coating assuming it is a 100% solid coating with 0% porosity and comparing it to the actual density of the coating. The actual density of the nanoporous layer may be calculated by dividing its coat weight in gsm by its coating thickness to provide the density in g/cc of the nanoporous layer. Dividing the density of the nanoporous layer by the calculated density of the same materials in a non-porous layer and multiplying by 100% provides the % solid fraction by volume in the nanoporous layer. Subtracting this value of the % solid fraction from 100% gives the % pore volume or % porosity of the nanoporous layer. The % porosity can also be measured by mercury porosimetry analysis.

[0049] In one embodiment, the pore volume of the nanoporous layer is between 0.8 cc/g and 1.0 cc/g. The pore volume can be calculated by measuring the coat weight of the nanoporous layer in gsm and measuring the thickness of the nanoporous layer to determine the total cc in a square meter. The pore volume in cc/g is then calculated by dividing the total cc in a square meter by the grams of the gsm measurement. In one embodiment, the % porosity of the nanoporous layer is 55% to 70%. Above this level of % porosity, it is difficult to obtain the cohesive and adhesive strength of the nanoporous layer that are preferred for long term, safer lithium battery separator applications.

[0050] In one embodiment of the separators of the present invention, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 15% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 20% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 25% to 45%.

[0051] In one embodiment of the separators of this invention, the nanoporous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of the pores have a diameter less than 80 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of the pores have a diameter less than 50 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 25 nm or less and at least 70% of the pores have a diameter less than 25 nm.

[0052] In one embodiment of the separators of this invention, the ratio of the air permeability of the separator to the air permeability of the polymeric layer without any

nanoporous layer present is 1.0 to 1.4. As described hereinabove, this air permeability is typically measured on an instrument which provides a Gurley number of the number of seconds for 100 cc of air to pass through the separator. A low level of increase of the air permeability is an important indicator of an expected low decrease in the ionic conductivity of electrolyte in the separator due to adding the inorganic oxide/polymer layer. If the ionic conductivity decreases too much, this can interfere with the cycling rate capability of the lithium battery particularly its power rate properties, and also interfere with the cycling lifetime of the lithium battery.

[0053] In one embodiment of the separators of the present invention, the polymeric layer comprises a polyolefin. Suitable polyolefins include, but are not limited to, polyethylenes and polypropylenes. Other examples of the polyolefin polymers include, but are not limited to, homopolymers of 1-butene, 1-pentene, 1-hexene, and 1-octene, as well as copolymers and terpolymers thereof, used alone or in combination with one or more types of polymers. In one embodiment, the polymeric layer comprises polyethylene. The polymeric layer may be of the “wet process” type which is produced by extrusion followed by solvent extraction to provide the porosity or of the “dry process” type which is produced by extrusion followed by stretching to provide the porosity. Various molecular weights, densities, and stereochemistries of polyethylene and the other polymers, with various melt flow and mechanical properties, may be utilized in the porous polymeric layers.

[0054] For the purposes of better uniform wetting of the coating fluid of the nanoporous layer and of better adhesion of the nanoporous layer to the substrate, the porous polymeric layer may have a coating layer on the surface to which the nanoporous layer is applied.

[0055] In one embodiment of the separators comprising polyethylene in the polymeric layer, the heat shrinkage of the separator at 120°C for 1 hour is less than 1.0%. Without the nanoporous layer of the separators of this invention, the heat shrinkage of a polyethylene separator of comparable total thickness is typically greater than 1.0% at 105°C. In one embodiment, the ratio of the air permeability of the separator after the heat shrinkage at 120°C for 1 hour to the air permeability of the separator before the heat shrinkage at 120°C is 0.8 to 1.2. This is an important feature for allowing vacuum drying of the polyethylene-based separators of this invention at a higher temperature of 120°C without any negative effect on the air permeability and ionic conductivity and, in some cases, with a positive effect on ionic conductivity due to the heat treatment. Vacuum drying is beneficial for lowering the moisture

content of separators for increased lithium battery cycling lifetimes and capacities and for higher lithium battery cycling rate capabilities. Vacuum drying is typically done for 4 to 24 hours at about 80°C for polyethylene separators and at about 90°C for polyethylene separators with one or more inorganic oxide/polymer layers. The capability of using a higher temperature of 120°C and a shorter drying time of 1 to 3 hours for the polyethylene-based separators of this invention is useful.

[0056] In another embodiment of the separators of the present invention, the nanoporous layer is a laminated layer on the polymeric layer. By the term “laminated layer” is meant that the nanoporous layer is not initially formed on the polymeric layer, such as by direct coating, but is instead laminated to the polymeric layer in a subsequent step. For example, the nanoporous layer is coated on a release substrate, and then this nanoporous layer is transferred by lamination onto the polymeric layer. This lamination may optionally include the use of heat and/or the use of some organic solvent to adhere the nanoporous layer to the polymeric layer and to aid in the subsequent delamination of the release substrate without interfering with the quality of the laminated multilayer separator.

[0057] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic particle/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite), or a blend of an inorganic oxide and an inorganic nitride (e.g., BN or AlN), and a polymer, and wherein the crystallite size of the inorganic particles is about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of about 25 nm or less and at least about 70% of the pores have a diameter of about 25 nm or less.

[0058] A further aspect of the separators for a lithium battery of the present invention pertains to a separator comprising (a) a porous polyethylene layer, and (b) a nanoporous inorganic particles/polymer composite layer on both sides of the polyethylene layer, wherein the nanoporous layer comprises boehmite, or a blend of boehmite and boron nitride, and one or more polymers, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is about 10% to about 50%, the total coating weight of the nanoporous layer on both sides is 5.5 grams per square meter or less, and the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. In one embodiment, the total coating weight of the

nanoporous layer on both sides is 3.5 grams per square meter or less and the thickness of the polyethylene layer is between 5 microns and 16 microns.

[0059] In another embodiment, this invention is lithium batteries comprising a separator of the present invention.

[0060] Another aspect of the present invention are methods for making the separator of this invention, which method comprises coating a solution comprising inorganic particles and one or more polymers on one or both surfaces of a porous polymeric layer. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of 5 nm to 90 nm. In one embodiment, the boehmite particles have a crystallite size of about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the coating after drying is about 10% to about 50%, preferably about 12% to about 45 %, and more preferably about 15% to about 45%.

[0061] EXAMPLES

[0062] Several embodiments of the present invention are described in the following examples, which are offered by way of illustration and not by way of limitation.

[0063] Example 1

[0064] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPAL 10F4 supplied by Sasol Inc., Houston, Texas) and 1 part of chitosan (deacetylated chitin, poly(D-glucosamine), medium molecular weight, of a polysaccharide supplied by Sigma-Aldrich, St. Louis, MO) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 40 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 1.2 grams per square meter (gsm) to about 5.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. The boehmite coating was then applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0065] The 12:1 by weight ratio of boehmite pigment to chitosan binder is calculated to have 15.3% by volume of chitosan binder in the solid portion of the porous boehmite

coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.4 g/cc for chitosan.

[0066] The coating thickness was measured using a Dorsey gauge on the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the thickness values before and after the coating is the coating thickness in microns. An alternative method for measuring the coating thickness is using the Dorsey gauge on the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0067] The coating weight was measured by weighing a 10 cm x 10 cm sample of the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the weights before and after the coating multiplied by 1,000 is the coat weight in grams per square meter (gsm). An alternative method for measuring the coating weight was weighing a 10 cm x 10 cm sample of the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0068] Figure 1 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.6 gsm to about 2.6 gsm. For two sided coatings, these numbers are doubled to give a total coating weight range of about 1.2 gsm to about 5.2 gsm. The total boehmite coating thickness for both sides varies from about 1.2 microns to about 4.4 microns. If a more accurate measurement of the coating thickness is desired, scanning electron microscope (SEM) analysis of cross-sections of the boehmite-coated plastic separator can be utilized.

[0069] Using a SEM analysis for the total coating thickness showed a density of about 1.06 gsm per micron of coating thickness. From this, the % porosity for this 12:1 pigment to binder ratio is estimated as 62%.

[0070] One objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C. Figure 2 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 1.9 microns or thicker on the 12 micron polyethylene substrate of this example. For coatings on two sides, this number is doubled to give a total coating thickness of about 3.8 microns or thicker. Based on the density of the

boehmite coating determined by SEM analysis, a total coating thickness of about 3.8 microns on both sides equates to about 4.0 gsm of coating weight.

[0071] For better wetting and coating adhesion of the boehmite slurry, a corona treatment of the polyethylene substrate is preferred. Besides wire wound coating rods, a variety of other coating application methods may be used, such as, but not limited to, gravure coating, screen printer coating, reverse roll coating, blade coating, and slot die coating.

[0072] Example 2

[0073] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPERAL D60 supplied by Sasol Inc., Hamburg, Germany) and 1 part of a vinyl pyrrolidone copolymer (Soteras CCS grade supplied by Ashland Inc., Wilmington, DE) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 75 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 2.4 grams per square meter (gsm) to about 9.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the boehmite coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0074] The 12:1 by weight ratio of boehmite pigment to vinyl pyrrolidone copolymer binder is calculated to have 18.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.1 g/cc for the vinyl pyrrolidone copolymer.

[0075] Figure 3 shows a plot of the average boehmite coating thickness and coating weight per side of about 1.2 gsm to about 4.6 gsm. For two sided coating, these numbers are doubled to give a total coating weight range of about 2.4 gsm to about 9.2 gsm. The total boehmite coating thickness for both sides varies from about 2.8 microns to about 9.0 microns.

[0076] As discussed, one objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C for 1 hour. Figure 4 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 2.0 microns or thicker on the 12 micron polyethylene substrate of this example. For the coatings on two sides, this number is doubled to give a total coating thickness of about 4.0 microns or thicker.

Based on Figure 3, a total coating thickness of about 4.0 microns on both sides equates to about 3.9 gsm of coating weight.

[0077] A boehmite slurry of an 8 to 1 ratio by weight of the DISPERAL D60 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was also coated using wire wound rods on the 12 micron polyethylene substrate and dried, as for the 12:1 boehmite D60:polymer coating in this example. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 5 shows a plot of the average boehmite coating weight and coating thickness per side of about 0.8 gsm to about 4.2 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a range of about 1.6 gsm to 8.4 gsm. The corresponding total boehmite coating thicknesses for both sides varies from about 1.6 microns to 8.2 microns. Figure 6 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 2.2 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 4.4 microns or thicker. Based on Figure 5, a total coating thickness of about 4.4 microns equates to about 4.8 gsm of coating weight.

[0078] Example 3

[0079] Using the procedure of Example 2, a boehmite slurry of a 12:1 by weight of DISPAL 25F4 boehmite and the vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was prepared. The DISPAL 25F4 is supplied by Sasol, Lake Charles, LA, and has a primary crystallite particle size of 8 nm. This boehmite slurry was coated on the EP12 12 micron polyethylene substrate from ENTEK and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 18.7%. Figure 7 shows a plot of the average boehmite coating thickness and the coating weight per side of about 0.8 gsm to about 3.0 gsm. For the coatings on two sides for this boehmite coating with a 12:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to about 6.0 gsm. Figure 8 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 1.8 microns or thicker. For two sided coatings, this number is doubled to give a total coating thickness on both sides of about 3.6 microns or thicker. Based on Figure 7, a total coating thickness of about 3.6 microns equates to about 3.6 gsm.

[0080] A boehmite slurry of an 8 to 1 ratio by weight of the DISPAL 25F4 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol was also coated on the 12 micron polyethylene substrate and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 9 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.8 gsm to about 4.7 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to 9.4 gsm. Figure 10 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 0.9 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 1.8 microns or thicker. Based on Figure 9, a total coating thickness of about 1.8 microns equates to about 2.9 gsm of coating weight. For a thickness of about 1.8 microns of the boehmite coating, the coating weight should have been about 2.0 gsm based on the density of the boehmite coating of about 1.1 g/cc shown in Example 1, while factoring in the lower boehmite weight % of the 8:1 boehmite pigment:vinyl pyrrolidone copolymer and the higher vinyl pyrrolidone copolymer weight % and lower density of the vinyl pyrrolidone copolymer compared to chitosan. This shows that about 0.9 gsm of the boehmite coating penetrated into and dried in the pores of the polyethylene substrate to produce a boehmite/polyethylene substrate with a boehmite coating on its surface. At the 8:1 pigment:binder (P:B) ratio of this coating, the about 0.9 gsm of boehmite coating in the pores equates to about 0.8 gsm of boehmite particles in the pores of the boehmite/polyethylene substrate or layer. Since this boehmite coating was applied on both sides of the polyethylene substrate, about one-half, or about 0.4 gsm of boehmite, would have penetrated and dried in the pores on each side of the polyethylene substrate to form the boehmite/polyethylene substrate or layer that resulted in a boehmite-coated boehmite/polyethylene substrate.

[0081] A similar 8:1 boehmite coating with Dispall 25F4, with chitosan substituted for the vinyl pyrrolidone copolymer and a 12 micron thick polyethylene substrate from Shenzhen Senior, Shenzhen, China, used instead of the polyethylene substrate from Entek, was made using the procedure of this Example to give a coat weight of 1.8 gsm. This coating had a heat shrinkage of less than 5% when heated at 150°C for 1 hour. When the procedure of removing the boehmite coating with an adhesive tape was used, even with very careful tape pulls, it was not possible to remove the boehmite coating from the 10 x 10 cm samples without pulling off

some of the polyethylene substrate. This was in contrast to the tape pull results of easy and complete removal of the boehmite coatings from the polyethylene substrate for Examples 1 and 2 and Comparative Examples 1 and 2. This result was consistent with some of the boehmite coating penetrating into the pores of the polyethylene layer and, upon drying, providing increased adhesive and cohesive strength to the boehmite coating such that the point of failure in the tape pull was below the surface of the polyethylene layer at most areas, rather than at the interface of the polyethylene surface and the boehmite coating on the polyethylene surface. After the tape pull, some areas were found where the polyethylene layer had not been disturbed. From these areas, the added thickness of the boehmite layer was found to be about 1.8 microns.

[0082] Since the tape pulls were not able to cleanly remove the boehmite coating from the polyethylene layer, a methanol-saturated soft cloth was used to wash and rub the coating off. The 10 x 10 cm sample after washing had a weight of 0.0757 grams (g). Two more methanol rubs did not change this weight of the washed and rubbed sample. The polyethylene substrate used in this Example contained no inorganic pigment, such as alumina or boehmite. A furnace combustion of the methanol-washed sample in a crucible, at a temperature of 1000 °C for 70 minutes showed 0.0045 g of anhydrous aluminum oxide after dehydrating the boehmite pigment with a weight loss of 15% of the boehmite pigment weight. Using this boehmite to aluminum oxide conversion factor by dividing the aluminum oxide weight by 0.85, this gives a boehmite content of about 0.0053, or at least about 0.53 gsm of boehmite pigment in the pores of the polyethylene layer. Since the wet rubbing to remove the boehmite coating on the surface may have removed a small amount of the boehmite coating in the upper pores of the polyethylene layer, it is possible that the weight of boehmite pigment in the pores was somewhat greater than 0.53 gsm. Since there were two coatings on the polyethylene layer, the amount of boehmite in the pores from a single coating is about one-half of this total, or about 0.27 gsm. As further evidence for penetration of the boehmite coating into the pores of the polyethylene substrate, the dimensional stability of this “washed” polyethylene with about 0.53 gsm of boehmite in the pores after heating at 150°C for 1 hour was 54% and 36.5% in the two directions(--machine direction (MD) and cross direction (CD)). By contrast, the polyethylene substrate before coating showed heat shrinkage of 70% and 75.5%. This increased heat stability with boehmite pigment in the pores of the polyethylene substrate is one explanation for the greater dimensional stability at 150°C with a much lower coat weight and coating thickness added to the surface of the

polyethylene layer. For analyzing the presence and amount of boehmite particles in the pores of the polymeric layer, the removal of the surface coating of the nanoporous layer and subsequent combustion of all of the organic polymer, as described above, in a combustion and/or thermogravimetric method may be used for the case of only boehmite particles in the nanoporous layer and a polymer layer that contained no inorganic or boehmite particles before the application of the boehmite coating. Another method is to use the TEM or SEM method used for measuring the average crystallite size to make images of the presence of boehmite particles in the pores of the polymeric layer and to determine their crystallite size at the same time. Combining this method with the combustion and/or thermogravimetric method above to quantify the weight of the boehmite, or other ceramic particles in the pores, if any, would provide the weight in gsm of boehmite particles in the separator. The known analytical techniques for boehmite particles, such as FTIR, X-ray diffraction, thermogravimetric analysis (TGA), high resolution photoelectron spectroscopy (ESCA), and ²⁷Al nuclear magnetic resonance (nmr) can be utilized to further confirm the presence and quantity of boehmite particles in the pores of the polymeric layer, if needed. An alternative approach for collecting the ceramic particles, such as boehmite particles, in the pores of the polymeric layer, after removing any surface coating from the polymeric layer, would be to do a Soxhlet extraction of the polymeric layer with a hot solvent that dissolves the polymer from the polymeric layer to isolate the insoluble inorganic particles. Besides providing the weight of inorganic particles, this could be combined with the above analytical methods, combustion and/or thermogrametric analysis, and TEM or SEM method, to confirm the presence and quantity of boehmite particles. One benefit of this penetration of nano-sized boehmite particles would be the ability to make and sell boehmite/polyethylene substrates, with no or minimal surface boehmite or other coating, for various applications, including for manufacturing CCS, where the boehmite/polyethylene substrate would have multiple advantages over polyethylene substrates, including, but not limited to, a much greater dimensional stability at high heats, a modified and smaller pore size distribution, and other benefits, such as ease of wetting by subsequent coatings. One example of a particularly useful application for boehmite/polymeric substrates would be filtration applications, where the very small nano-sized boehmite particles, with and without various surface treatments, in the pores of the polymer could provide the desired nanofiltration and ultrafiltration properties not available from the polymeric substrate without boehmite particles in its pores, and could be adapted for desalination of sea water. This benefit would also be available for other

boehmite/polymeric substrates besides polyethylene, such as for boehmite/polypropylene and boehmite/other polyolefin substrates.

[0083] Example 4

[0084] The procedure of Example 1 was used to make a nanoporous separator comprising a 50:50 blend by weight of boehmite and boron nitride pigments, except that a 12 micron thick porous polyethylene separator from Toray Industries, Tokyo, Japan, was substituted for the Entek polyethylene separator, and the vinyl pyrrolidone polymer, Soteras CCS, was substituted for the chitosan. The boron nitride pigment was NX1, supplied by Momentive Performance Materials. The total coating weight on both sides was 2.5 gsm. Figure 11 shows the reduced water content of this separator compared to the control with only boehmite pigment and no boron nitride. Specifically, as shown in Figure 11, at a 50:50 by weight ratio of boehmite and boron nitride, the water content at 20% RH was reduced by one-third from its value with boehmite pigment only.

[0085] Comparative Example 1

[0086] An alumina mix was prepared in water by adding 20 parts of alumina pigment (HJA-0719 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the alumina was greater than 400 nm. The alumina mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 4.4 gsm. The alumina coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0087] The 20:1 by weight ratio of alumina pigment to hydroxypropyl cellulose polymer binder is calculated to have 13.6% by volume of polymer binder in the solid portion of the porous alumina coating layer. This calculation is based on a specific gravity of 4.0 g/cc for alumina and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating

weight of about 4.4 gsm, this alumina coating had a shrinkage of about 60% at 150°C for 1 hour.

[0088] Comparative Example 2

[0089] A boehmite mix was prepared in water by adding 20 parts of boehmite pigment (BG-613 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the boehmite was greater than 400 nm. The boehmite mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 7.0 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0090] The 20:1 by weight ratio of boehmite pigment to hydroxypropyl cellulose polymer binder is calculated to have 10.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating weight of about 7.0 gsm, this boehmite coating had a shrinkage of about 60% at 150°C for 1 hour.

[0091] Table 1 shows a summary of the data from Examples 1 to 3.

Table 1

Example	Boehmite Type	Primary Particle Size (nm)	Pigment:Polymer Binder Weight Ratio	Volume % Polymer	Total Coat Weight for <5% Shrinkage at 150°C for 1 hour
1	10F4	40	12:1	15.3%	≥4.0 gsm
2	D60	75	12:1	18.7%	≥3.9 gsm
2	D60	75	8:1	25.6%	≥4.8 gsm

3	25F4	8	12:1	18.7%	≥ 3.6 gsm
3	25F4	8	8:1	25.6%	≥ 2.9 gsm

[0092] One challenge in using nano-sized ceramic particles, such as nano-sized boehmite particles, is their increased tendency to absorb water because of their much larger surface areas compared to micro-sized ceramic particles, such as micro-sized alumina and micro-sized boehmite particles. For lithium ion cells, lithium metal cells, and other electrochemical cells with sensitivity to water content, the separator needs to have a low water content, such as under 600 ppm of water in the separator, preferably less than 300 ppm of water, and more preferably under 100 ppm of water as typically found in porous polymer only separators with no ceramic materials in them. Although vacuum drying of the separator at up to 80 °C can help to lower this moisture content in separators with nano-sized ceramic particles, and the separators of this invention have increased thermal stability that permits even higher vacuum drying temperatures of up to 90 °C to about 110 °C, some users of separators may prefer not to perform this extra step of vacuum drying. It would be preferable if the greater heat stability, lower cost from the use of a lower amount of ceramic coating materials, and lower ceramic coating thickness, including from the penetration of the ceramic coating and ceramic particles into the porous polymeric separator layer, of the separators of this invention would include a lower moisture content in the separators from the use of more hydrophobic ceramic particles and organic polymers. Having a lower amount of ceramic particles from the lower coating weights of the invention is a significant benefit for lowering the moisture content of the separator, but it would be further useful if the ceramic particles were more hydrophobic and less prone to moisture pickup.

[0093] One approach which provides this lower moisture absorption is to treat the surface of the ceramic particles with a hydrophobic material, such as by using hydrophobically-treated boehmite particles, particularly if the hydrophobic material is chemically reacted with the boehmite particles and thus is permanently bonded to the boehmite particles. Suitable hydrophobic materials for chemical modification of boehmite pigments include, but are not limited to, organic sulfonic acids, organic carbonates, and polymers with hydrophobic groups and with reactive groups, such as carboxylic acid and hydroxyl groups, that can be made to react with hydroxyl groups of the boehmite or other ceramic particles.

[0094] Examples of the organic sulfonic acid-treated boehmite particles include Dispal® 25SR, available from Sasol in Lake Charles, LA, a boehmite pigment chemically reacted with

p-toluene sulfonic acid through the sulfonic acid groups. The reaction leaves the hydrophobic p-toluene group on the outer surface of the boehmite particle to make it less prone to pick up water. An even more hydrophobic example from Sasol is p-dodecylbenzene sulfonic acid surface reacted to boehmite particles through the sulfonic acid group and having the very hydrophobic dodecyl group on the outer surface of the boehmite pigment. This makes the boehmite particles so hydrophobic that this treated boehmite can be dispersed in hydrocarbon solvents. The use of boehmite particles surface treated with organic sulfonic acids in boehmite coatings for separators is disclosed in U.S. Patent Application Publication No. 2013/0171500 by Xu et al.

[0095] Examples of the organic carbonate-treated boehmite particles include, but are not limited to, those with ethylene carbonate and fluoroethylene carbonate as the organic carbonates. Besides the option of choosing an organic carbonate that is very hydrophobic in the part of the molecule that will face away from the surface of the boehmite, the choice can also be made with a view to picking an organic carbonate which will provide increased ionic conductivity to the separator. The use of boehmite particles surface treated with ethylene carbonate to provide separators with increased conductivity is described in U.S. Pat. No. 8,883,354 to Carlson et al.

[0096] The use of boehmite particles surface treated with organic polymers to provide separators is described in U.S. Pat. No. 9,871,239, to Carlson et al. As the primary crystallite size and resulting pore size become smaller in the boehmite separators and the ionic conductivity of the separator becomes lower because of the smaller pore sizes, it is useful to have surface treatments, such as the three types described above, that can be utilized to increase the ionic conductivity of the separator, as well as to decrease the water content of the separator.

[0097] The preparation of a ceramic mix or slurry with the surface-treated hydrophobic materials can be done conveniently by adding a precursor, such as an organic carbonate, to the slurry containing the untreated boehmite particles and heating the slurry to a temperature, such as 80 °C, for a sufficient time, such as 1 hour, to react the surface treatment material, such as an organic carbonate, with the boehmite particles to the desired level of surface treatment. There is no need to isolate the surface-treated boehmite particles by a drying process, but rather the slurry can be used as is, usually after some dilution with water and/or solvents to the desired % solids level for the coating of the porous polymer substrate. If it is desired to further reduce the particle size of the ceramic particles, such as boehmite particles,

and/or to homogenize the slurry for more uniformity and perhaps a lower viscosity at the desired % solids level for coating, the slurry, before and/or after any surface treatment, can be further mixed in an impingement mill or similar milling device. Examples of the use of an impingement mill for reducing and homogenizing coating mixes containing pigments is described in U.S. Pat. Nos. 5,210,114 and 5,292,588, to Katsen.

[0098] While the invention has been described in detail and with reference to specific and general embodiments thereof, it will be apparent to one skilled in the art that various changes and modifications can be made therein without departing from the spirit and scope thereof.

CLAIMS

1. A separator for a lithium battery comprising:
a porous polyethylene layer, and
a second porous layer adjacent said porous polyethylene layer, said second porous layer comprises a first portion and second portion, wherein:
the first portion is solid; and
 - a. the first portion comprises boehmite particles and one or more polymers:
 - b. a volume fraction of said one or more polymers in the first portion is between about 15% and about 50%;
 - c. a total coating weight of said second porous layer on said porous polyethylene layer is about 5.5 grams per square meter or less, and
 - d. a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.
2. The separator of claim 1, wherein the volume fraction of said one or more polymers in the first portion is between 25% and 40%.
3. The separator of claim 1, wherein said boehmite particles have an average crystallite size between 5 nm and 90 nm.
4. The separator of claim 3, wherein the average crystallite size of said boehmite particles is between 30 nm and 50 nm.
5. The separator of claim 1, wherein said second porous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of said pores have a diameter of 80 nm or less.
6. The separator of claim 1, wherein said second porous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of said pores have a diameter of 50 nm or less.
7. The separator of Claim 1, wherein the second porous layer is a nanoporous layer.

8. The separator of Claim 1, wherein the second portion is not a solid.
9. A process for making a separator for a lithium battery comprising:
 - (a) treating boehmite with an organic carbonate so as to make hydrophobic boehmite particles;
 - (b) providing a porous polymeric layer;
 - (c) providing a second porous layer, wherein said second porous layer comprises the hydrophobic boehmite particles and one or more polymers, and wherein the total coating weight of said second porous layer is less than 4.5 grams per square meter, and
 - (d) bringing the second porous layer adjacent the porous polymeric layer, so as to make a separator with no more than 600 parts per million of water.
10. The process of Claim 9, wherein the organic carbonate comprises ethylene carbonate.
11. The process of Claim 9, wherein the organic carbonate comprises fluoroethylene carbonate.
12. The process of Claim 9, wherein the organic carbonate reacts with hydroxyl groups of the boehmite.
13. A device comprising:
 - a porous polyethylene layer, and
 - a second porous layer adjacent said porous polyethylene layer, said second porous layer comprises a first portion and second portion, wherein:
 - the first portion is solid; and

- a. the first portion comprises boehmite particles have an average crystallite size between 5 nm and 90 nm and one or more polymers:
 - b. a volume fraction of said one or more polymers in the first portion is between about 25% and about 40%;
 - c. a total coating weight of said second porous layer on said porous polyethylene layer is about 5.5 grams per square meter or less, and
 - d. a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.
14. The device of claim 13, wherein the average crystallite size of said boehmite particles is between 30 nm and 50 nm.
15. The device of claim 13, wherein said second porous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of said pores have a diameter of 80 nm or less.
16. The separator of claim 13, wherein said second porous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of said pores have a diameter of 50 nm or less.
17. The device of Claim 13, wherein the second porous layer is a nanoporous layer.
18. The device of claim 13, wherein the second portion is not solid.
19. The device of claim 13, wherein the second portion is not solid.

ABSTRACT

A separator for a lithium battery having (a) a porous polymeric layer, such as a polyethylene layer; and (b) a nanoporous inorganic particle/polymer layer on both sides of the polymeric layer, the nanoporous layer having an inorganic oxide and one or more polymers; the volume fraction of the polymers in the nanoporous layer is about 15% to about 50%, and the crystallite size of the inorganic oxide is 5 nm to 90 nm.

Figure 1

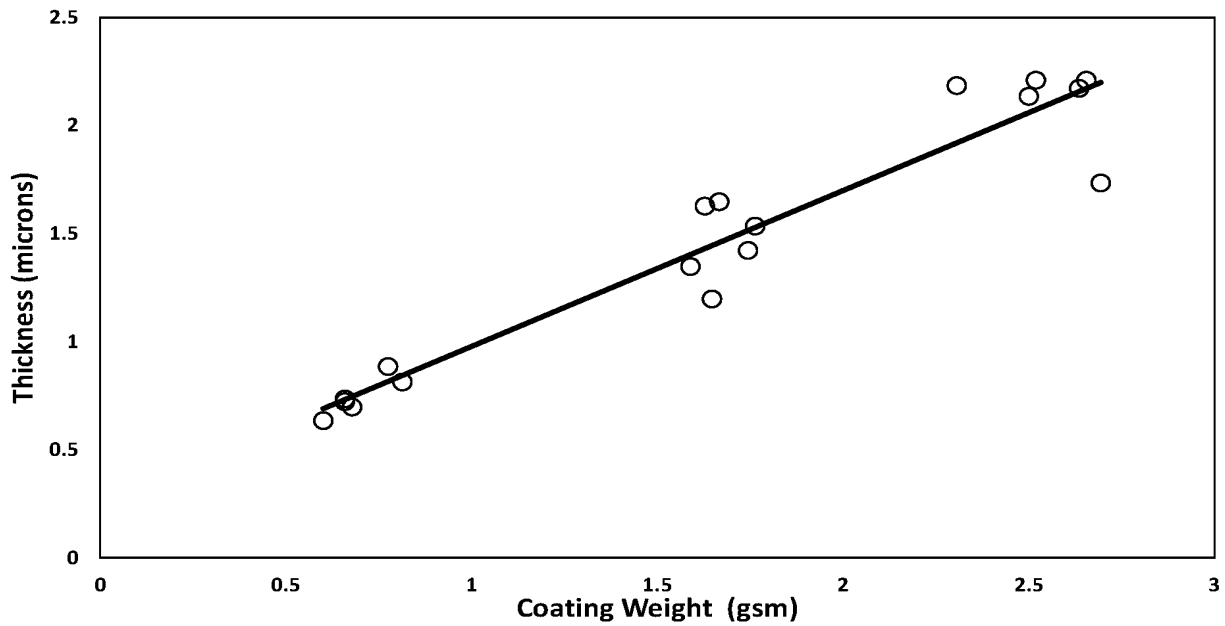


Figure 2

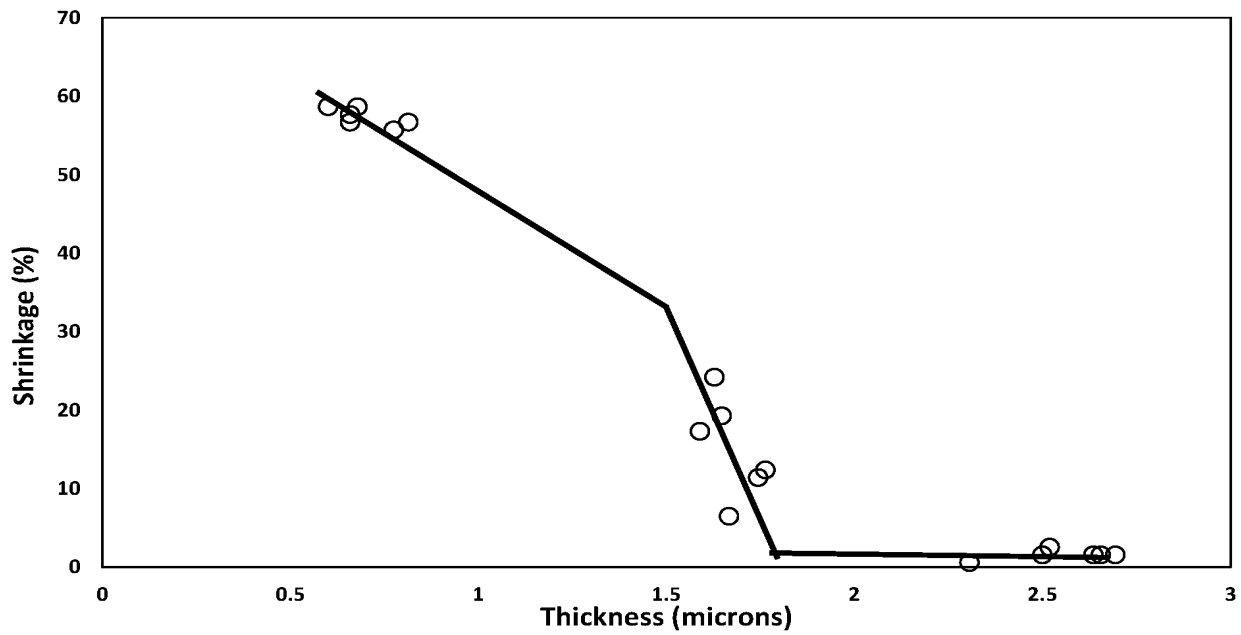


Figure 3

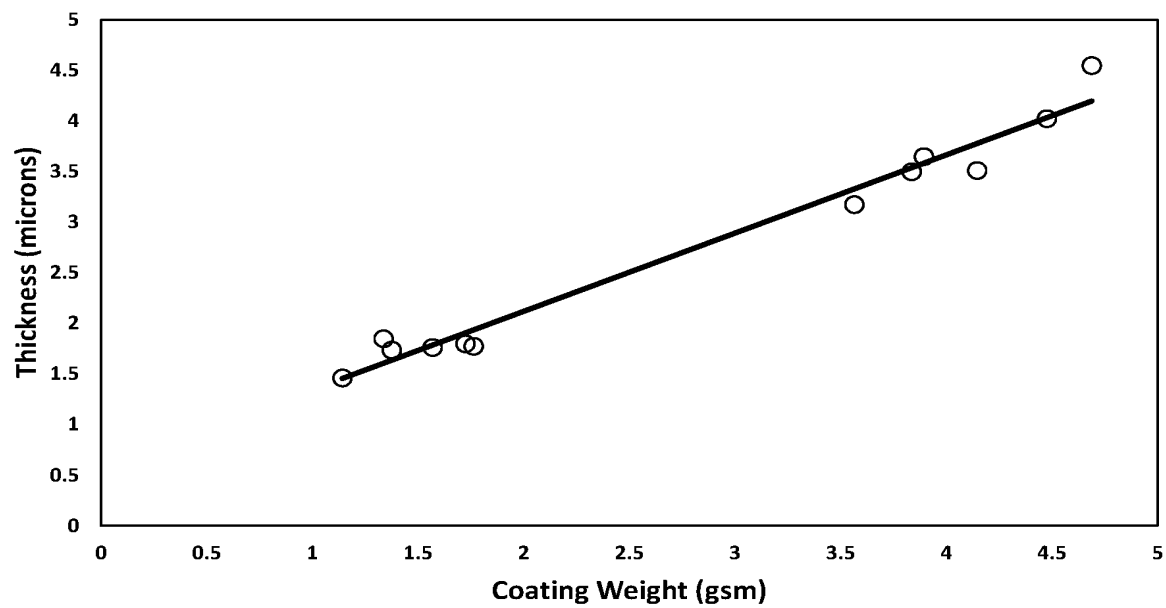


Figure 4

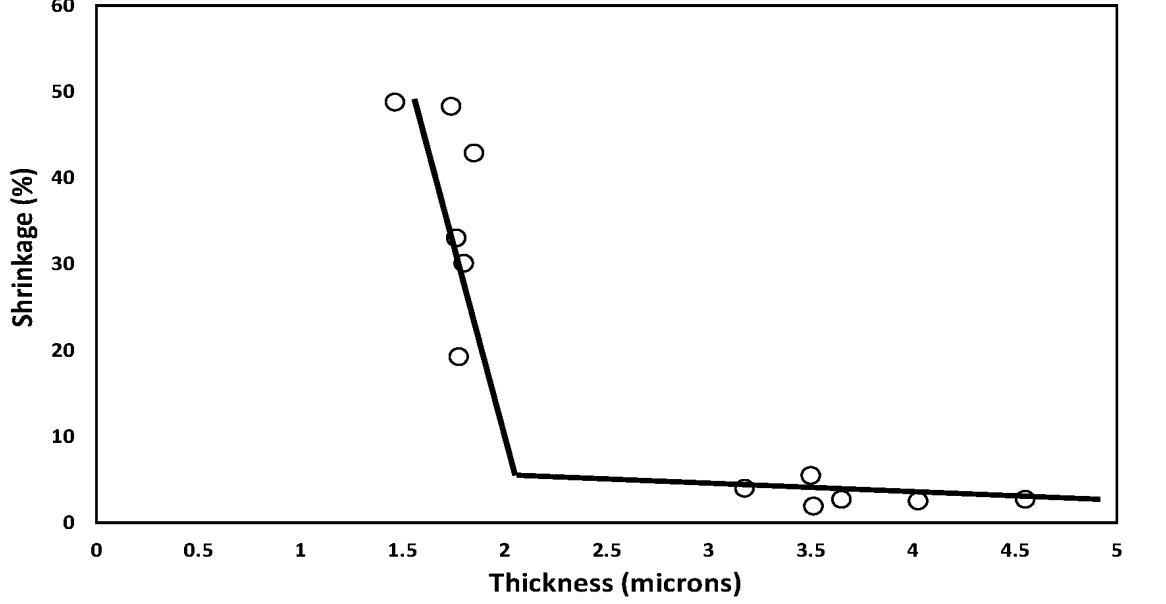


Figure 5

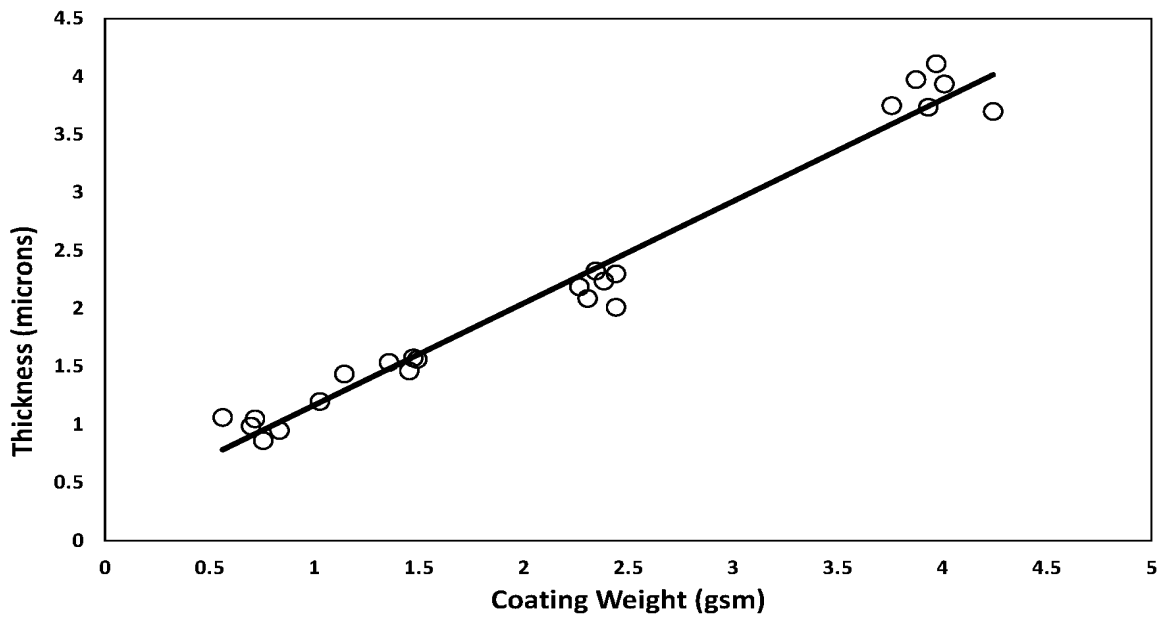


Figure 6

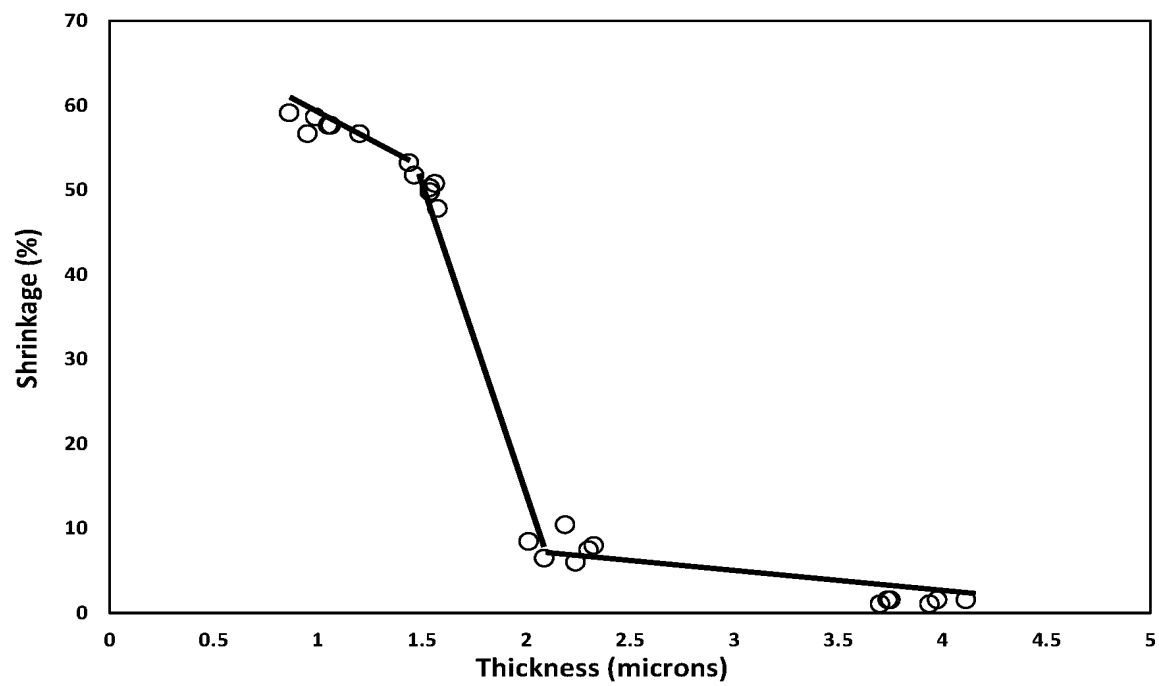


Figure 7

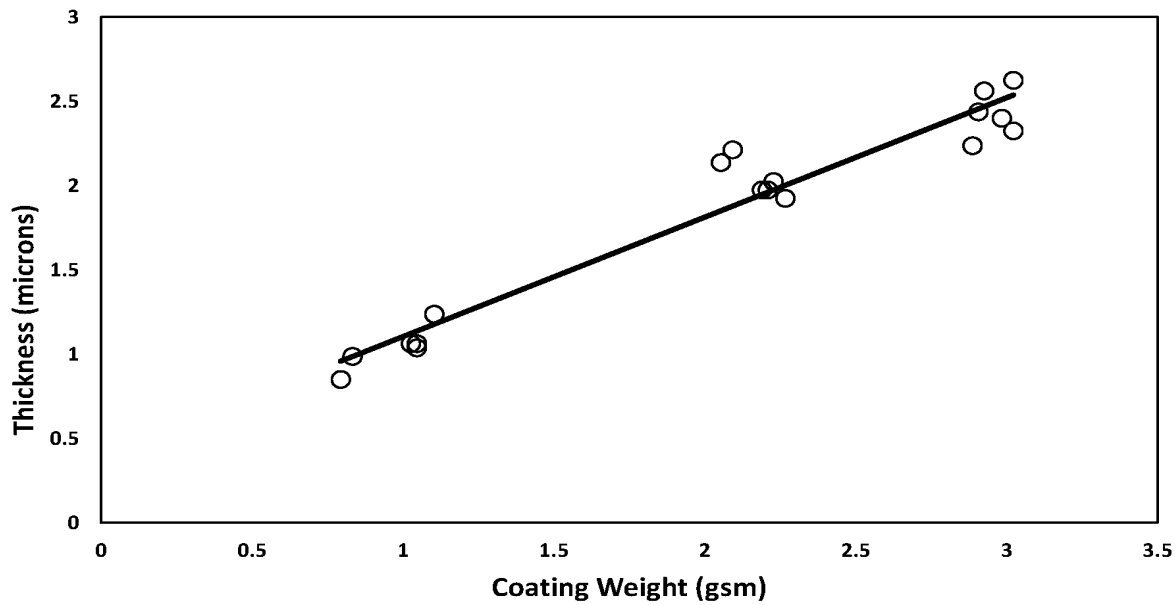


Figure 8

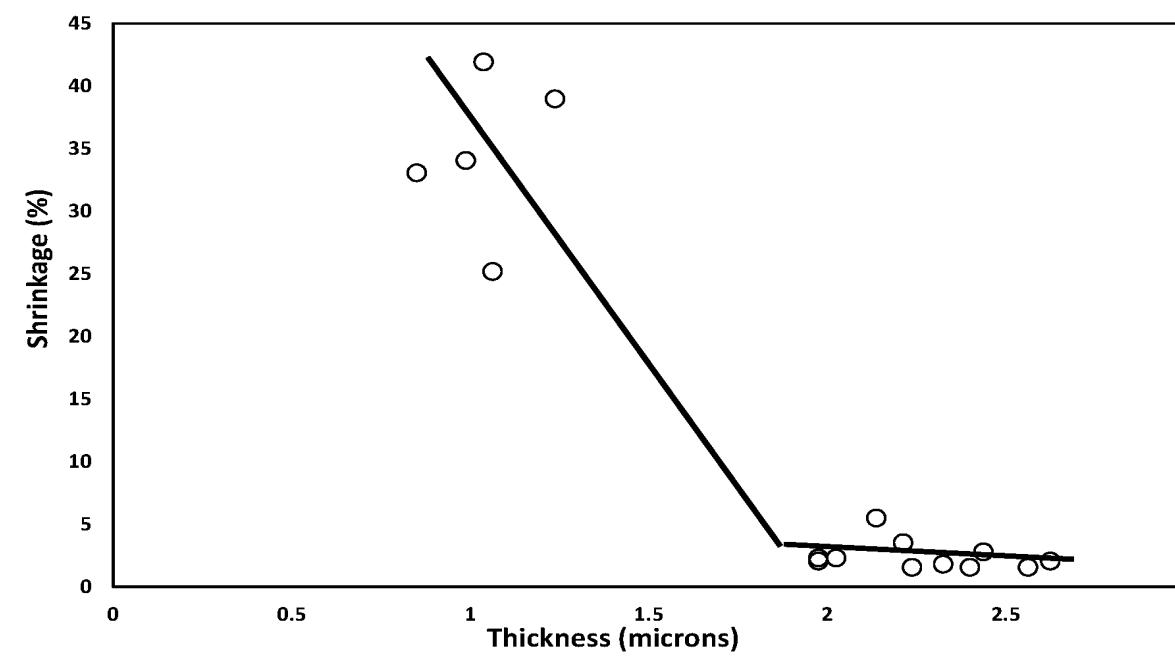


Figure 9

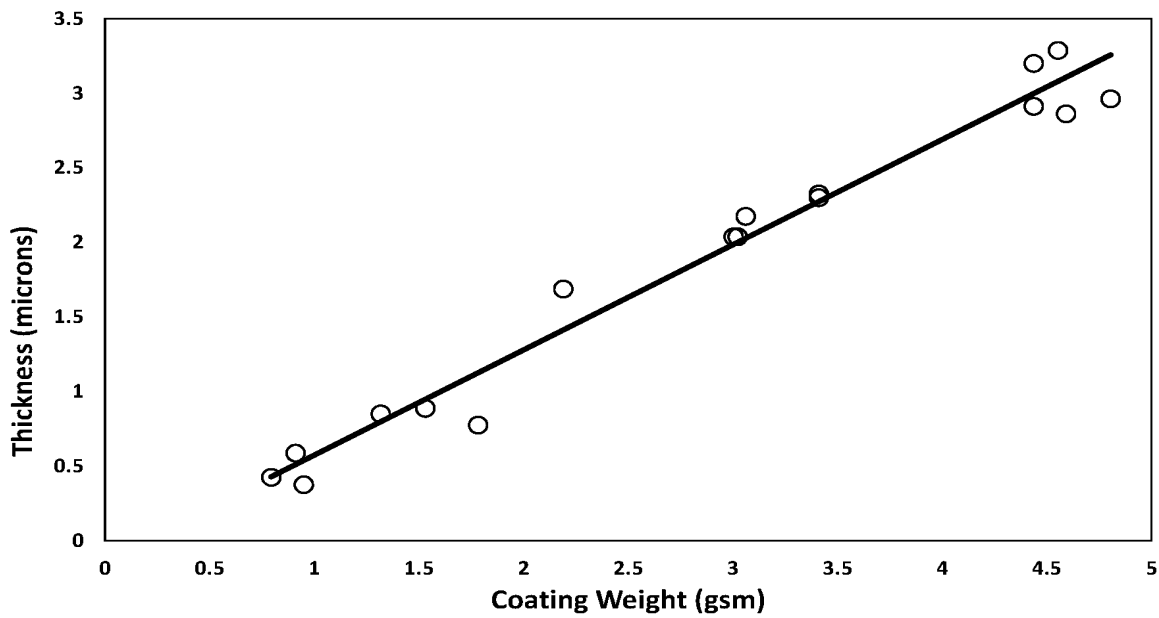
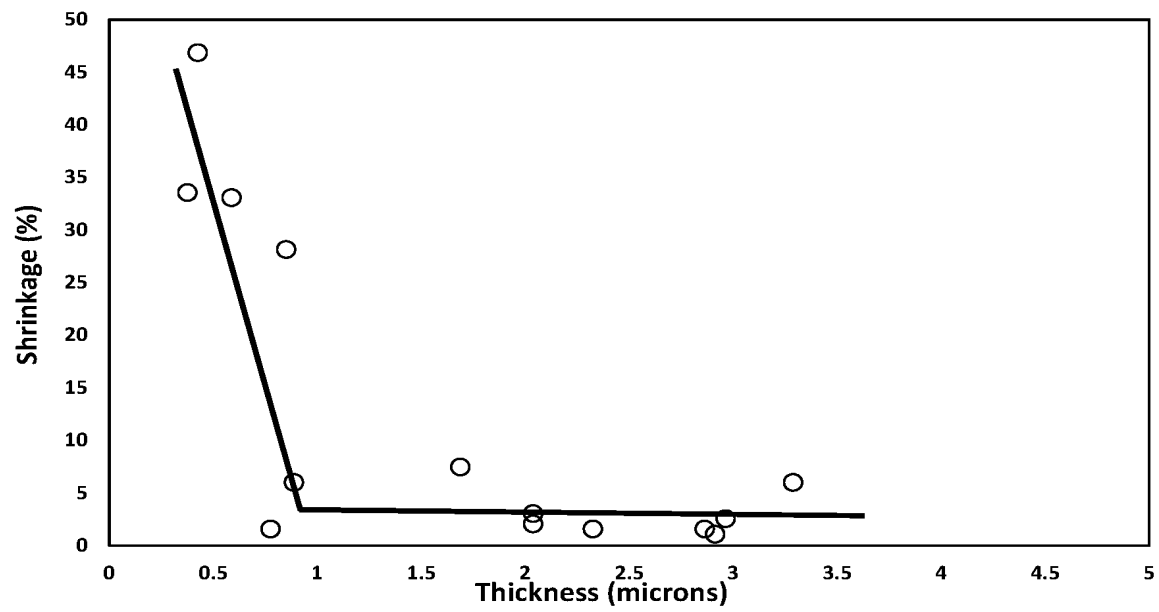


Figure 10



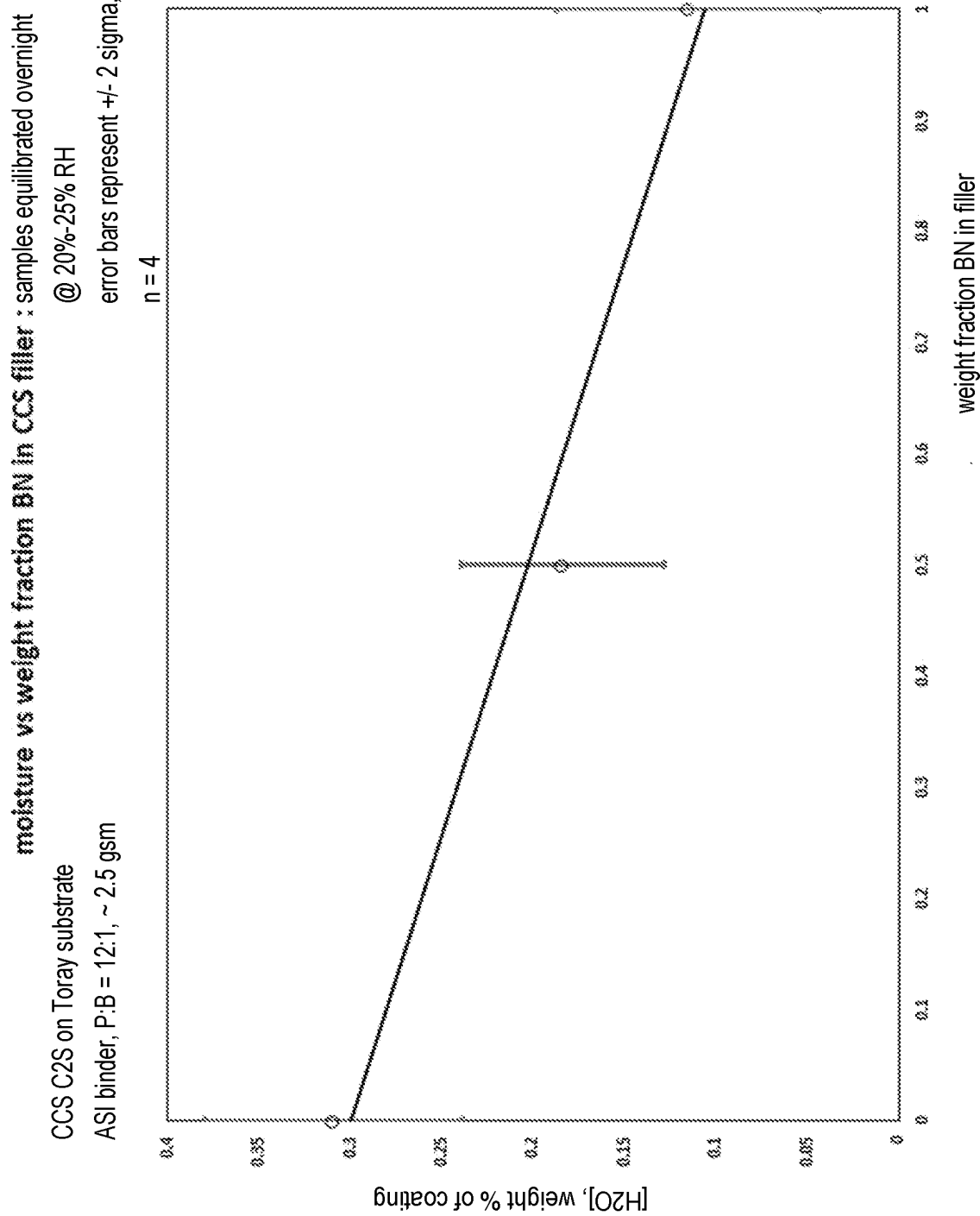


Fig. 11

Docket No. 00249/62

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No. : Not Yet Assigned (Continuation of U.S. Appln. No. 17/543,259)
Filing Date : Herewith
For : MULTILAYER NANOPOROUS SEPARATOR
Customer No. : 1912

**PRELIMINARY AMENDMENT AND
INFORMATION DISCLOSURE STATEMENT**

EFS
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Please amend the above-identified application as set forth below.

EFS

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No. : Not Yet Assigned (Continuation of U.S. Appln. No. 17/543,259)
Filing Date : Herewith
Page 2 of 4

Amendments to the Specification:

On page 1, please amend paragraph [0001] as follows:

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Patent Application No. 17/543,259, filed on December 6, 2021, which is a continuation of U.S. Patent Application No. 17/223,845, filed on April 6, 2021, now U.S. Patent No. 11,769,903, which is a continuation of U.S. Patent Application No. 16/755,294, filed on April 10, 2020, now U.S. Patent No. 10,998,536, which is a U.S. National Stage of PCT International Application No. PCT/US2018/055862, filed on October 15, 2018, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/572,083, filed on October 13, 2017, the entire contents of which are incorporated by reference herein.

EFS

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No. : Not Yet Assigned (Continuation of U.S. Appln. No. 17/543,259)
Filing Date : Herewith
page 3 of 4

REMARKS

Claims 1-19 are pending in the subject application. By this Preliminary Amendment, the specification has been amended to update the priority data for the subject application.

Applicant maintains that the amendments do not raise an issue of new matter. Entry of the amendments is respectfully requested.

Information Disclosure Statement

In accordance with the duty of disclosure under 37 C.F.R. §1.56, applicant would like to direct the Examiner's attention to the references that are listed on the attached Form PTO/SB/08A. These references were previously submitted to, or cited by, the Patent Office during prosecution of U.S. Patent Application No. 17/543,259, from which the subject application claims priority under 35 U.S.C. §120. Accordingly, pursuant to 37 C.F.R. §1.98(d), copies of previously cited or submitted non-U.S. patent references are not included with this Information Disclosure Statement.

EFS

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No. : Not Yet Assigned (Continuation of U.S. Appln. No. 17/543,259)
Filing Date : Herewith
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CONCLUSION

The following fees for a large entity are being paid herewith by credit card: 1) the basic filing fee (\$320.00), 2) examination fee (\$800.00), and 3) search fee (\$700.00). No other fee is deemed necessary in connection with the filing of this application, Preliminary Amendment, and Information Disclosure Statement. However, if any other fee is required to preserve the pendency of the subject application, authorization is hereby given to charge the amount of any such fee to Deposit Account No. 01-1785. Overpayments may also be credited to Deposit Account No. 01-1785.

Respectfully submitted,
AMSTER, ROTHSTEIN & EBENSTEIN LLP
Attorneys for Applicant
405 Lexington Avenue
New York, New York 10174
(212) 336-8000

Dated: April 10, 2024
New York, New York

By /Brian J. Amos/
Brian J. Amos, Reg. No. 63,679

MULTILAYER NANOPOROUS SEPARATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of and priority to U.S. Provisional Patent Application No. 62/572,083, filed on October 13, 2017, the entire contents of which are incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present invention relates generally to porous separators for batteries and other electric current producing cells. In particular, it pertains to multilayer porous separators comprising inorganic oxide/polymer layers on a polymeric layer, and to lithium batteries comprising such separators.

BACKGROUND OF THE INVENTION

[0003] Lithium batteries are widely used in portable electronics, such as smartphones and portable computers. Among the new applications for lithium batteries are high power batteries for hybrid, plug-in hybrid, and fully electric vehicles. Existing lithium batteries, including rechargeable and non-rechargeable lithium batteries, often utilize multilayer separators. Such separators may include an extruded polyethylene or polypropylene porous film that is coated with an inorganic oxide/polymer coating layer. As the demand for lower cost lithium batteries with increased size and energy density has grown, manufacturers have turned to less safe battery active materials in order to meet these goals. This has resulted in a need for battery separators with increased dimensional stability at temperatures such as 150°C, or higher.

[0004] Thus, it would be advantageous if the inorganic oxide/polymer coating layers could provide even greater dimensional stability at elevated temperatures while, at the same time, requiring lower coating weights and coating thicknesses, and providing lower manufacturing costs.

SUMMARY OF THE INVENTION

[0005] It is an object of the present invention to provide lighter weight, lower cost, and thinner inorganic oxide/polymer coating layers for polymer-based separators for lithium batteries and other electric current producing cells.

[0006] The present invention achieves these goals by providing a separator for a lithium battery comprising: (a) a porous polymeric layer, and (b) nanoporous inorganic oxide/polymer composite layers coated on both sides of the polymeric layer. In one embodiment, the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO_2) and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic oxide particles is 5 nm to 90 nm. This combination of inorganic oxide particles (e.g., boehmite particles) having an extremely small crystallite size, and a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator. This combination was further found to enable a significantly lower coating weight and coating thickness of the heat stable layer.

[0007] In another embodiment the nanoporous layer comprises: (a) a blend of inorganic oxide particles (e.g., boehmite particles) and inorganic nitride particles (e.g., boron nitride or aluminum nitride particles), and (b) one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic particles is 5 nm to 90 nm. This combination of: (1) inorganic oxide and nitride particles (e.g., boehmite and boron nitride or aluminum nitride particles) having extremely small crystallite sizes, and (2) a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator.

[0008] Preferably, the heat stable inorganic oxide/polymer layer (or inorganic oxide and nitride/polymer layer) is coated on both sides of the porous polymeric layer, such as a porous polyethylene layer or a porous polypropylene layer, as opposed to being on only one side of the polymeric layer. It has been found that this approach provides improved thermal dimensional stability in a thin and low cost separator. However, having the heat stable layer on only one side of the polymeric layer is also useful.

[0009] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO_2) and a polymer, and wherein the crystallite size of the inorganic oxide is 5 nm to 25 nm. It has been found that this smaller crystallite size results in a thinner coating layer with more cohesive strength and with more adhesion to the polymeric layer at an even lower coating weight than comparable layers where the crystallite size for the inorganic oxide is larger such as, for example, about 80 nm.

[0010] In another embodiment, the present invention includes lithium batteries that comprise the improved separators of this invention.

[0011] A further aspect of the present invention are methods for making separators that comprise coating a solution comprising an inorganic oxide and/or inorganic nitride, and one or more polymers, on one or both surfaces of a porous polymeric layer to form the separators of this invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The features and advantages of the present disclosure will be more fully understood with reference to the following, detailed description when taken in conjunction with the accompanying figures, wherein:

[0013] FIG. 1 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0014] FIG. 2 is a graph of the shrinkage as a function of coating thickness of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0015] FIG. 3 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0016] FIG. 4 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0017] FIG. 5 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0018] FIG. 6 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0019] FIG. 7 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0020] FIG. 8 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0021] FIG. 9 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0022] FIG. 10 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0023] FIG. 11 is a graph of the moisture content as function of the weight fraction of boron nitride in a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 4.

DETAILED DESCRIPTION

[0024] The separators of the present invention provide superior safety, lower cost, lower thickness, and other key performance properties for use in lithium batteries and in other electric current producing cells.

[0025] One aspect of this invention pertains to a separator for a lithium battery comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on both sides of the polymeric layer, wherein the nanoporous layer comprises boehmite and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to 45%, and wherein the crystallite size of the boehmite is 5 nm to 90 nm. Alternatively, the nanoporous layer may further comprise other inorganic oxide (e.g., SiO₂) or nitride (e.g., BN or AlN) particles, or combinations thereof.

[0026] The term “lithium battery” may include, for example, batteries in which the electroactive anode material comprises lithium. This includes, but is not limited to, secondary or rechargeable lithium ion batteries, secondary lithium metal batteries, and primary or non-rechargeable lithium metal batteries.

[0027] The term “separator” may include, for example, porous materials in a battery that are interposed between the cathode and the anode of the battery to provide electrical insulation against short circuiting and to provide pores for filling with the electrolyte of the battery.

[0028] The term “porous polymeric layer” may include, for example, a layer of a polymer material that is porous such that it has permeability to the flow of air through the polymeric layer. The battery industry typically measures this air permeability using a porosimeter that provides a value for the time in seconds for 100 cubic centimeters (cc) of air to pass through the porous layer. This value is commonly called the Gurley number. The battery industry prefers that the Gurley number for a separator be as low as possible without sacrificing mechanical integrity. Typical Gurley numbers for the air permeability of porous polyethylene or porous polypropylene separators, as used in the battery industry, are 100 seconds/100 cc to 300 seconds/100 cc. For this invention, porous polymer layer does not include any of the non-woven polymeric layers, such as with polyester fibers, that are used as separators in the battery industry.

[0029] The term “nanoporous” may include layers having pores up to about 100 nm in diameter, preferably, with at least 90% of the pores being less than 100 nm in diameter. The size and relative number of pores can be measured by mercury porosimetry, or by scanning electron microscope (SEM) or transmission electron microscope (TEM) cross-sections of the layer.

[0030] As discussed herein, the nanoporous layer may be present on both sides of a substrate, support, layer, or coating. For example, the nanoporous layer may be present on both sides of the porous polymeric layer, such as a porous polyethylene layer. The general benefits of the nanoporous layers of this invention are also found when the nanoporous layer is present on only one side of the porous polymeric layer, but the overall performance is typically superior when the nanoporous layer is present on both sides of the porous polymeric layer.

[0031] The superior performance features of two-side coating of the polymeric layer, compared to one-side coating at the same total coating weight, includes lower shrinkage at

high temperatures and improved battery safety. Improved safety is typically the most important performance feature desired in separators for lithium batteries, especially as the batteries become larger and higher in energy density and use less safe battery active materials in order to achieve higher energy densities and lower costs.

[0032] Another advantage of two-side coating of the porous polymeric layer with the nanoporous coating layer is that it provides the opportunity to make the coatings on each side different in composition. This enables the nanoporous layer to be optimized for the best performance when it is facing the anode and when it is facing the cathode. In general, by tailoring the formulations and coating thicknesses of the ceramic coatings on either side of the separator, it is possible to optimize: (a) adhesion between separator and a specific electrode, either anode or cathode; (b) interfacial impedance between separator and a specific electrode, either anode or cathode; (c) complexation of migratory species (e.g., solubilized transition metal ions from cathode).

[0033] For example, when the coating is facing the cathode, it is particularly useful that the coating be resistant to oxidation, especially at higher charge voltages, e.g., above 4.25 volts. This can be done, for example, by selecting the organic polymer of the nanoporous layer to be resistant to oxidation up to the higher voltage desired, e.g., up to 5.0 volts. In addition, the nanoporous layer on one of the sides can be designed with particularly small pore sizes, such as under 10 nm diameter pores, for example, by using boehmite with a primary crystallite size of less than 10 nm in diameter. It should be understood that the present invention is not limited to the examples provided herein. For example, surface-modified alumina particles, including boehmite particles, as described in PCT/IB2015/000272 to Loscutova *et al.* of Sasol, may be utilized in the separator layers described herein. This patent application discloses the dispersion of these surface-modified aluminas into nano-sized single particle crystallites. This reference also describes that the average crystallite size can be measured by X-ray diffraction on a 120 plane. As an example, the crystallite size (or an average crystallite size) of the inorganic particles in a separator comprising a nanoporous layer according to an embodiment of the invention may be measured by: (i) taking a cross-section sample of the nanoporous layer; (ii) using a TEM or very high power SEM on this sample to measure the diameter/width, in their widest dimension, of 100 single primary (inorganic) particles selected at random in the TEM or SEM image; (iii) excluding the 25 highest and 25 lowest diameter numbers in the 100 diameter/width numbers; and (iv) recording the crystallite size as the average of the middle

50 diameter/width numbers, i.e., an average crystallite size of the measured particles. Since the pore size diameters of ceramic particles in a nanoporous layer usually approximate the crystallite size of the ceramic particles, such as boehmite particles, used to make the nanoporous layer, the pore size diameters may indicate the crystallite size of the ceramic particles for nanoporous layers containing ceramic particles of a narrow crystallite size. However, for blends of ceramic particles of different crystallite sizes in a nanoporous coating, the pore size diameters of the coating layer typically are intermediate, or a blend, among the crystallite size diameters of the different ceramic particles and not individual peaks reflecting the different crystallite sizes. The TEM or SEM method described above for measuring the crystallite size does not have this limitation since it is analyzing single particles, and not blends, and thus is a preferred method to use for this invention. These separator configurations using very small pore sizes are useful when placed against a lithium metal anode, where the very small size of the pores assists in preventing the growth of lithium metal dendrites into and through the separator, which reduces cycle life, capacity, and safety for the cell. Where very small boehmite particles are used and some of these boehmite particles penetrate into the pores of the polymeric separator substrate, this is particularly useful in preventing the growth of lithium metal dendrites in and through the separator.

[0034] In another example, boehmite could be modified by reaction with a fluorinated ethylene carbonate for improved cycle life when such nanoporous layer is positioned against an anode comprising silicon or lithium metal. This nanoporous layer with the boehmite covalently reacted with a fluorinated ethylene carbonate could also be used against the cathode to help achieve excellent stability under high voltage operation of the cell, such as at up to 5.0 volts.

[0035] In a further example, the nanoporous layer with pores that are less than 10 nm in diameter could be positioned against the cathode to inhibit the diffusion of transition metals, such as nickel ions, from the cathode to the anode, which degrades the cycle life and other performance properties of the cell. Optionally, the very small boehmite crystallite size of less than 10 nm could be combined with reaction of this boehmite with fluorinated ethylene carbonate or another organic carbonate to make a nanoporous layer positioned against the cathode to lessen the diffusion of transition metal ions from the cathode.

[0036] The term “Boehmite” is understood in the art. It, for example, may include hydrated aluminum oxide of the empirical chemical formula of $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ where x is 1.0 to 2.0. Another term used interchangeably herein for boehmite is AlOOH . Boehmite may be

characterized by its distinctive Fourier-transform infrared spectroscopy ("FTIR") fingerprint. Boehmite may also be characterized by its distinctive X-ray diffraction pattern and by its distinctive infrared spectrum of double peaks of hydroxyl groups around 3200 cm^{-1} . Boehmite may have values of x for molecules of water that are above 1.5 and even up to about 2. However, the commercial grades of boehmite of all particle sizes typically have a value of x that is in the range of 1.0 to 1.5. The value of x can be measured by first drying the boehmite at about 120°C for 1 hour to remove the surface adsorbed water and then measuring the loss of water that occurs when the boehmite sample is heated to 1000°C for 1 hour and is thereby dehydrated to anhydrous aluminum oxide, Al_2O_3 .

[0037] Examples of polymers that may be utilized in the present invention include, but are not limited to, cellulosic polymers such as hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides such as chitosan, polyethylene oxides, and polyvinyl alcohols. Preferably, the polymers, alone or in combination, provide cohesive and adhesive strength to the nanoporous layer, are insoluble in the electrolyte of the lithium battery, and are electrochemically stable over the voltage range of use of the lithium battery. If needed to achieve these properties, one or more of the polymers may be crosslinked.

[0038] By the term "volume fraction of the one or more polymers in the solid portion of the nanoporous layer" is meant the fraction expressed as a percentage by volume of the polymer content of the solid portion of the inorganic particle/polymer composite layer. As described hereinafter, the nanoporous layer has a % porosity of between about 30% to about 70%. In other words, much of the volume of the nanoporous layer is air, and not a solid material. The air volume (e.g., pores) is not included in the solid portion for purposes of determining the volume fraction of the one or more polymers in the solid portion of the nanoporous layer. The volume fraction of the polymer and inorganic particle content of the nanoporous layer can be calculated based on the relative densities of the materials. For example, when the inorganic particles are only boehmite with a density of 3.03 g/cc , the organic polymer has a density of 1.3 g/cc , and the weight ratio of the boehmite to the organic polymer is 12:1, the volume fraction of boehmite would be 83.7% and the volume fraction of organic polymer would be 16.3%. Changing the weight ratio of the boehmite to the organic polymer to 4:1 results in a volume fraction of boehmite of 63.2% and of organic polymer of 36.8%. If the density of the organic polymer is lower, such as 1.1 g/cc , then its volume fraction at the same weight ratio of inorganic particle/polymer will be higher than at the higher polymer density of 1.3 g/cc . To determine the volume fraction of one or more organic

polymers in the solid portion of the nanoporous layer, for example, the nanoporous layer is removed from the separator, such as by careful scraping of the layer and not removing any of the polymer substrate. Next, a weighed nanoporous layer sample in a crucible is combusted at a sufficient temperature to combust the organic polymer to leave the residue of the ceramic particles. This provides the weight of combusted ceramic particles. When the particles are only boehmite, this weight of the combusted residue is divided by 0.85 in the case of the equivalent of one water molecule per boehmite molecule to obtain the weight of boehmite in the sample prior to the combustion. Although most boehmite particles have a 15% weight loss upon combustion, it should be noted that boehmite particles may have a different weight loss factor, such as 0.84 or 0.82. This same information can be obtained by using a thermogravimetric analyzer to record the weight of the nanoporous layer sample as it is heated to 900 °C or higher, until a constant weight is achieved. As noted above, boehmite has a density of 3.03 g/cc. Most organic polymers have a density in the range of 1.1 to 1.4 g/cc, although some polymers, such as polyvinylidene difluoride (PVdF), have a density of about 1.8 g/cc. After the density of the organic polymers is known by analyzing for the identity and weight ratios of the organic polymers, for boehmite-only nanoporous layers, the volume % of the organic polymers in the solid portion of the nanoporous layer can be calculated. For example, if the weight percent of boehmite is 88.9%, and the weight percent of the organic polymers is 11.1% and its density is 1.3 g/cc--with a pigment:binder ratio of 8:1 by weight--dividing the weight percents of boehmite and the organic polymers by their densities gives relative volume percents of 29.34% for boehmite and 8.54% for the organic polymers. Normalizing these volume percents to a total volume percent of 100% for the solid portion of the nanoporous layer gives a volume percent of 22.54% for the organic polymers. If the nanoporous layer is a blend of boehmite and other ceramic particles or if the density of the organic polymers is not known, further analysis, as known in the art, may be undertaken to measure the volume percent of the organic polymers.

[0039] The term “crystallite size” of the particles refers to the average size of the primary or single crystals of the inorganic particles. This is typically measured using X-ray diffraction, a transmission electron microscope (TEM), or a very high resolution scanning electron microscope (SEM). PCT/IB2015/000272 to Loscutova *et al.* of Sasol cited above provides examples of using X-ray diffraction to determine an average size of the crystals of the inorganic particles, such as alumina and boehmite. As described earlier, a TEM or SEM method may be used to determine the crystallite size. Many inorganic oxide particles, such

as boehmite particles, are nearly cubic in shape, although they can have other shapes such as platelets and needles. In the dry solid state and in solution, boehmite particles are typically agglomerated into larger particles comprised of many primary boehmite particles, such as, for example, 10 to 20 primary particles. The inorganic particles in the nanoporous layers of this invention result in features, such as pore size diameter, that are related to the crystallite size of the particles.

[0040] In one embodiment of the separators of the present invention, the ratio of the total thickness of the nanoporous layer on both sides of the polymeric layer to the total thickness of the separator is 10% to 80%. The thickness of these layers may be measured by SEM analysis of cross-sections of the separator. Because of the nanoporous pore sizes and the high volume fraction of organic polymer in the nanoporous layer, it has been found that a lower limit for the proportional thickness of the nanoporous layer of about 10% of the total thickness of the separator, rather than 15% or more typically 25%, is able to provide an excellent improvement in the dimensional stability of the separator at high temperatures, which is needed for improved safety. It has been found that the combination of nanoporous pore sizes and very high volume fractions of organic polymer in the nanoporous layer (e.g., from about 10% to about 50%) provides acceptable % porosity (e.g., from about 30% to about 70%) with good ionic conductivity of the electrolyte. It has also been found that the separator of this invention functions well with the total thickness of the nanoporous layer as high as 80% of the total thickness of the separator.

[0041] In one embodiment, the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. This heat shrinkage may be measured by the change in the area of the separator after heating in an oven for 1 hour. This measurement of the heat shrinkage may be done by hanging a large sheet of the separator in the oven after marking and measuring the dimensions and area of a square or rectangle such as a 10 cm by 10 cm square, on a smaller section of the large sheet. After the heating in the oven, the marked area is measured for the % change or % shrinkage of the dimensions and the area.

[0042] In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%. The nanoporous separators of the present invention have been found to retain their dimensional stability even when heated well above the melting point of the polymeric layer, such as a porous polyethylene layer.

[0043] In one embodiment of the separators of this invention, the total coating weight of the nanoporous layer on both sides is less than 4.5 grams per square meter (gsm), while

providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. It should be noted that the use of boehmite with its lower density of about 3.03 g/cc, as compared to the higher density anhydrous aluminum oxide or alumina (which has a density of about 4 g/cc), helps to further lower the coating weight of the heat stabilizing layer. The lower density of the inorganic oxide particle proportionately lowers the weight of coating needed for a specific thickness and % porosity.

[0044] For example, a 4 micron thick alumina/polymer layer with a 45% porosity for polyethylene-based separators typically weighs about 8 grams per square meter (gsm). However, switching to boehmite (or a blend of boehmite and other inorganic particles of a similar or lower density) at the same thickness and % porosity lowers the coating weight to about 6 gsm or less. The high volume fraction of polymer in the nanoporous layers of the separators of this invention, combined with the use of boehmite with a crystallite size of 5 nm to 90 nm, provides increased dimensional stability for a specific thickness of the separator layer. Although it is possible to still make the total thickness of the inorganic particle/polymer layer in the separator 4 microns or higher, it is desirable for lower cost and for lower total thickness of the separator to make the thickness of the inorganic particle/polymer layer as thin as possible, such as about 3 microns total, while still achieving an excellent dimensional stability at high temperature. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0045] In one embodiment, other inorganic oxides and/or inorganic nitrides with a density of about 3.0 g/cc or lower, such as silicon dioxide and boron nitride, may be blended with the boehmite pigment. As discussed further below, this provides lower total coating weight of the nanoporous layer and other desirable properties, such as lower water content.

[0046] In another embodiment of the separators of this invention, the total coating weight of the nanoporous layer of both sides is less than 3.5 grams per square meter (gsm), while providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0047] In one embodiment of the separators of the present invention, the crystallite size of the inorganic particles may be about 30 nm to about 50 nm, preferably about 5 nm to about 25 nm, and more preferably about 5 nm to about 10 nm. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of about 30 nm to about 50 nm. In one embodiment, the crystallite size of the boehmite may be about 5 nm to about 10 nm, and preferably about 5 nm to about 8 nm.

[0048] In another embodiment of this invention, the % porosity of the nanoporous layer is between 30% and 55%. The % porosity of the nanoporous layer may be calculated from the density in g/cc of the materials in the dry coating assuming it is a 100% solid coating with 0% porosity and comparing it to the actual density of the coating. The actual density of the nanoporous layer may be calculated by dividing its coat weight in gsm by its coating thickness to provide the density in g/cc of the nanoporous layer. Dividing the density of the nanoporous layer by the calculated density of the same materials in a non-porous layer and multiplying by 100% provides the % solid fraction by volume in the nanoporous layer. Subtracting this value of the % solid fraction from 100% gives the % pore volume or % porosity of the nanoporous layer. The % porosity can also be measured by mercury porosimetry analysis.

[0049] In one embodiment, the pore volume of the nanoporous layer is between 0.8 cc/g and 1.0 cc/g. The pore volume can be calculated by measuring the coat weight of the nanoporous layer in gsm and measuring the thickness of the nanoporous layer to determine the total cc in a square meter. The pore volume in cc/g is then calculated by dividing the total cc in a square meter by the grams of the gsm measurement. In one embodiment, the % porosity of the nanoporous layer is 55% to 70%. Above this level of % porosity, it is difficult to obtain the cohesive and adhesive strength of the nanoporous layer that are preferred for long term, safer lithium battery separator applications.

[0050] In one embodiment of the separators of the present invention, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 15% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 20% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 25% to 45%.

[0051] In one embodiment of the separators of this invention, the nanoporous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of the pores have a diameter less than 80 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of the pores have a diameter less than 50 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 25 nm or less and at least 70% of the pores have a diameter less than 25 nm.

[0052] In one embodiment of the separators of this invention, the ratio of the air permeability of the separator to the air permeability of the polymeric layer without any

nanoporous layer present is 1.0 to 1.4. As described hereinabove, this air permeability is typically measured on an instrument which provides a Gurley number of the number of seconds for 100 cc of air to pass through the separator. A low level of increase of the air permeability is an important indicator of an expected low decrease in the ionic conductivity of electrolyte in the separator due to adding the inorganic oxide/polymer layer. If the ionic conductivity decreases too much, this can interfere with the cycling rate capability of the lithium battery particularly its power rate properties, and also interfere with the cycling lifetime of the lithium battery.

[0053] In one embodiment of the separators of the present invention, the polymeric layer comprises a polyolefin. Suitable polyolefins include, but are not limited to, polyethylenes and polypropylenes. Other examples of the polyolefin polymers include, but are not limited to, homopolymers of 1-butene, 1-pentene, 1-hexene, and 1-octene, as well as copolymers and terpolymers thereof, used alone or in combination with one or more types of polymers. In one embodiment, the polymeric layer comprises polyethylene. The polymeric layer may be of the “wet process” type which is produced by extrusion followed by solvent extraction to provide the porosity or of the “dry process” type which is produced by extrusion followed by stretching to provide the porosity. Various molecular weights, densities, and stereochemistries of polyethylene and the other polymers, with various melt flow and mechanical properties, may be utilized in the porous polymeric layers.

[0054] For the purposes of better uniform wetting of the coating fluid of the nanoporous layer and of better adhesion of the nanoporous layer to the substrate, the porous polymeric layer may have a coating layer on the surface to which the nanoporous layer is applied.

[0055] In one embodiment of the separators comprising polyethylene in the polymeric layer, the heat shrinkage of the separator at 120°C for 1 hour is less than 1.0%. Without the nanoporous layer of the separators of this invention, the heat shrinkage of a polyethylene separator of comparable total thickness is typically greater than 1.0% at 105°C. In one embodiment, the ratio of the air permeability of the separator after the heat shrinkage at 120°C for 1 hour to the air permeability of the separator before the heat shrinkage at 120°C is 0.8 to 1.2. This is an important feature for allowing vacuum drying of the polyethylene-based separators of this invention at a higher temperature of 120°C without any negative effect on the air permeability and ionic conductivity and, in some cases, with a positive effect on ionic conductivity due to the heat treatment. Vacuum drying is beneficial for lowering the moisture

content of separators for increased lithium battery cycling lifetimes and capacities and for higher lithium battery cycling rate capabilities. Vacuum drying is typically done for 4 to 24 hours at about 80°C for polyethylene separators and at about 90°C for polyethylene separators with one or more inorganic oxide/polymer layers. The capability of using a higher temperature of 120°C and a shorter drying time of 1 to 3 hours for the polyethylene-based separators of this invention is useful.

[0056] In another embodiment of the separators of the present invention, the nanoporous layer is a laminated layer on the polymeric layer. By the term “laminated layer” is meant that the nanoporous layer is not initially formed on the polymeric layer, such as by direct coating, but is instead laminated to the polymeric layer in a subsequent step. For example, the nanoporous layer is coated on a release substrate, and then this nanoporous layer is transferred by lamination onto the polymeric layer. This lamination may optionally include the use of heat and/or the use of some organic solvent to adhere the nanoporous layer to the polymeric layer and to aid in the subsequent delamination of the release substrate without interfering with the quality of the laminated multilayer separator.

[0057] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic particle/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite), or a blend of an inorganic oxide and an inorganic nitride (e.g., BN or AlN), and a polymer, and wherein the crystallite size of the inorganic particles is about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of about 25 nm or less and at least about 70% of the pores have a diameter of about 25 nm or less.

[0058] A further aspect of the separators for a lithium battery of the present invention pertains to a separator comprising (a) a porous polyethylene layer, and (b) a nanoporous inorganic particles/polymer composite layer on both sides of the polyethylene layer, wherein the nanoporous layer comprises boehmite, or a blend of boehmite and boron nitride, and one or more polymers, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is about 10% to about 50%, the total coating weight of the nanoporous layer on both sides is 5.5 grams per square meter or less, and the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. In one embodiment, the total coating weight of the

nanoporous layer on both sides is 3.5 grams per square meter or less and the thickness of the polyethylene layer is between 5 microns and 16 microns.

[0059] In another embodiment, this invention is lithium batteries comprising a separator of the present invention.

[0060] Another aspect of the present invention are methods for making the separator of this invention, which method comprises coating a solution comprising inorganic particles and one or more polymers on one or both surfaces of a porous polymeric layer. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of 5 nm to 90 nm. In one embodiment, the boehmite particles have a crystallite size of about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the coating after drying is about 10% to about 50%, preferably about 12% to about 45 %, and more preferably about 15% to about 45%.

[0061] EXAMPLES

[0062] Several embodiments of the present invention are described in the following examples, which are offered by way of illustration and not by way of limitation.

[0063] Example 1

[0064] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPAL 10F4 supplied by Sasol Inc., Houston, Texas) and 1 part of chitosan (deacetylated chitin, poly(D-glucosamine), medium molecular weight, of a polysaccharide supplied by Sigma-Aldrich, St. Louis, MO) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 40 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 1.2 grams per square meter (gsm) to about 5.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. The boehmite coating was then applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0065] The 12:1 by weight ratio of boehmite pigment to chitosan binder is calculated to have 15.3% by volume of chitosan binder in the solid portion of the porous boehmite

coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.4 g/cc for chitosan.

[0066] The coating thickness was measured using a Dorsey gauge on the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the thickness values before and after the coating is the coating thickness in microns. An alternative method for measuring the coating thickness is using the Dorsey gauge on the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0067] The coating weight was measured by weighing a 10 cm x 10 cm sample of the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the weights before and after the coating multiplied by 1,000 is the coat weight in grams per square meter (gsm). An alternative method for measuring the coating weight was weighing a 10 cm x 10 cm sample of the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0068] Figure 1 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.6 gsm to about 2.6 gsm. For two sided coatings, these numbers are doubled to give a total coating weight range of about 1.2 gsm to about 5.2 gsm. The total boehmite coating thickness for both sides varies from about 1.2 microns to about 4.4 microns. If a more accurate measurement of the coating thickness is desired, scanning electron microscope (SEM) analysis of cross-sections of the boehmite-coated plastic separator can be utilized.

[0069] Using a SEM analysis for the total coating thickness showed a density of about 1.06 gsm per micron of coating thickness. From this, the % porosity for this 12:1 pigment to binder ratio is estimated as 62%.

[0070] One objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C. Figure 2 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 1.9 microns or thicker on the 12 micron polyethylene substrate of this example. For coatings on two sides, this number is doubled to give a total coating thickness of about 3.8 microns or thicker. Based on the density of the

boehmite coating determined by SEM analysis, a total coating thickness of about 3.8 microns on both sides equates to about 4.0 gsm of coating weight.

[0071] For better wetting and coating adhesion of the boehmite slurry, a corona treatment of the polyethylene substrate is preferred. Besides wire wound coating rods, a variety of other coating application methods may be used, such as, but not limited to, gravure coating, screen printer coating, reverse roll coating, blade coating, and slot die coating.

[0072] **Example 2**

[0073] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPERAL D60 supplied by Sasol Inc., Hamburg, Germany) and 1 part of a vinyl pyrrolidone copolymer (Soteras CCS grade supplied by Ashland Inc., Wilmington, DE) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 75 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 2.4 grams per square meter (gsm) to about 9.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the boehmite coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0074] The 12:1 by weight ratio of boehmite pigment to vinyl pyrrolidone copolymer binder is calculated to have 18.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.1 g/cc for the vinyl pyrrolidone copolymer.

[0075] Figure 3 shows a plot of the average boehmite coating thickness and coating weight per side of about 1.2 gsm to about 4.6 gsm. For two sided coating, these numbers are doubled to give a total coating weight range of about 2.4 gsm to about 9.2 gsm. The total boehmite coating thickness for both sides varies from about 2.8 microns to about 9.0 microns.

[0076] As discussed, one objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C for 1 hour. Figure 4 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 2.0 microns or thicker on the 12 micron polyethylene substrate of this example. For the coatings on two sides, this number is doubled to give a total coating thickness of about 4.0 microns or thicker.

Based on Figure 3, a total coating thickness of about 4.0 microns on both sides equates to about 3.9 gsm of coating weight.

[0077] A boehmite slurry of an 8 to 1 ratio by weight of the DISPERAL D60 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was also coated using wire wound rods on the 12 micron polyethylene substrate and dried, as for the 12:1 boehmite D60:polymer coating in this example. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 5 shows a plot of the average boehmite coating weight and coating thickness per side of about 0.8 gsm to about 4.2 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a range of about 1.6 gsm to 8.4 gsm. The corresponding total boehmite coating thicknesses for both sides varies from about 1.6 microns to 8.2 microns. Figure 6 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 2.2 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 4.4 microns or thicker. Based on Figure 5, a total coating thickness of about 4.4 microns equates to about 4.8 gsm of coating weight.

[0078] Example 3

[0079] Using the procedure of Example 2, a boehmite slurry of a 12:1 by weight of DISPAL 25F4 boehmite and the vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was prepared. The DISPAL 25F4 is supplied by Sasol, Lake Charles, LA, and has a primary crystallite particle size of 8 nm. This boehmite slurry was coated on the EP12 12 micron polyethylene substrate from ENTEK and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 18.7%. Figure 7 shows a plot of the average boehmite coating thickness and the coating weight per side of about 0.8 gsm to about 3.0 gsm. For the coatings on two sides for this boehmite coating with a 12:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to about 6.0 gsm. Figure 8 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 1.8 microns or thicker. For two sided coatings, this number is doubled to give a total coating thickness on both sides of about 3.6 microns or thicker. Based on Figure 7, a total coating thickness of about 3.6 microns equates to about 3.6 gsm.

[0080] A boehmite slurry of an 8 to 1 ratio by weight of the DISPAL 25F4 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol was also coated on the 12 micron polyethylene substrate and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 9 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.8 gsm to about 4.7 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to 9.4 gsm. Figure 10 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 0.9 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 1.8 microns or thicker. Based on Figure 9, a total coating thickness of about 1.8 microns equates to about 2.9 gsm of coating weight. For a thickness of about 1.8 microns of the boehmite coating, the coating weight should have been about 2.0 gsm based on the density of the boehmite coating of about 1.1 g/cc shown in Example 1, while factoring in the lower boehmite weight % of the 8:1 boehmite pigment:vinyl pyrrolidone copolymer and the higher vinyl pyrrolidone copolymer weight % and lower density of the vinyl pyrrolidone copolymer compared to chitosan. This shows that about 0.9 gsm of the boehmite coating penetrated into and dried in the pores of the polyethylene substrate to produce a boehmite/polyethylene substrate with a boehmite coating on its surface. At the 8:1 pigment:binder (P:B) ratio of this coating, the about 0.9 gsm of boehmite coating in the pores equates to about 0.8 gsm of boehmite particles in the pores of the boehmite/polyethylene substrate or layer. Since this boehmite coating was applied on both sides of the polyethylene substrate, about one-half, or about 0.4 gsm of boehmite, would have penetrated and dried in the pores on each side of the polyethylene substrate to form the boehmite/polyethylene substrate or layer that resulted in a boehmite-coated boehmite/polyethylene substrate.

[0081] A similar 8:1 boehmite coating with Dispall 25F4, with chitosan substituted for the vinyl pyrrolidone copolymer and a 12 micron thick polyethylene substrate from Shenzhen Senior, Shenzhen, China, used instead of the polyethylene substrate from Entek, was made using the procedure of this Example to give a coat weight of 1.8 gsm. This coating had a heat shrinkage of less than 5% when heated at 150°C for 1 hour. When the procedure of removing the boehmite coating with an adhesive tape was used, even with very careful tape pulls, it was not possible to remove the boehmite coating from the 10 x 10 cm samples without pulling off

some of the polyethylene substrate. This was in contrast to the tape pull results of easy and complete removal of the boehmite coatings from the polyethylene substrate for Examples 1 and 2 and Comparative Examples 1 and 2. This result was consistent with some of the boehmite coating penetrating into the pores of the polyethylene layer and, upon drying, providing increased adhesive and cohesive strength to the boehmite coating such that the point of failure in the tape pull was below the surface of the polyethylene layer at most areas, rather than at the interface of the polyethylene surface and the boehmite coating on the polyethylene surface. After the tape pull, some areas were found where the polyethylene layer had not been disturbed. From these areas, the added thickness of the boehmite layer was found to be about 1.8 microns.

[0082] Since the tape pulls were not able to cleanly remove the boehmite coating from the polyethylene layer, a methanol-saturated soft cloth was used to wash and rub the coating off. The 10 x 10 cm sample after washing had a weight of 0.0757 grams (g). Two more methanol rubs did not change this weight of the washed and rubbed sample. The polyethylene substrate used in this Example contained no inorganic pigment, such as alumina or boehmite. A furnace combustion of the methanol-washed sample in a crucible, at a temperature of 1000 °C for 70 minutes showed 0.0045 g of anhydrous aluminum oxide after dehydrating the boehmite pigment with a weight loss of 15% of the boehmite pigment weight. Using this boehmite to aluminum oxide conversion factor by dividing the aluminum oxide weight by 0.85, this gives a boehmite content of about 0.0053, or at least about 0.53 gsm of boehmite pigment in the pores of the polyethylene layer. Since the wet rubbing to remove the boehmite coating on the surface may have removed a small amount of the boehmite coating in the upper pores of the polyethylene layer, it is possible that the weight of boehmite pigment in the pores was somewhat greater than 0.53 gsm. Since there were two coatings on the polyethylene layer, the amount of boehmite in the pores from a single coating is about one-half of this total, or about 0.27 gsm. As further evidence for penetration of the boehmite coating into the pores of the polyethylene substrate, the dimensional stability of this “washed” polyethylene with about 0.53 gsm of boehmite in the pores after heating at 150°C for 1 hour was 54% and 36.5% in the two directions(--machine direction (MD) and cross direction (CD)). By contrast, the polyethylene substrate before coating showed heat shrinkage of 70% and 75.5%. This increased heat stability with boehmite pigment in the pores of the polyethylene substrate is one explanation for the greater dimensional stability at 150°C with a much lower coat weight and coating thickness added to the surface of the

polyethylene layer. For analyzing the presence and amount of boehmite particles in the pores of the polymeric layer, the removal of the surface coating of the nanoporous layer and subsequent combustion of all of the organic polymer, as described above, in a combustion and/or thermogravimetric method may be used for the case of only boehmite particles in the nanoporous layer and a polymer layer that contained no inorganic or boehmite particles before the application of the boehmite coating. Another method is to use the TEM or SEM method used for measuring the average crystallite size to make images of the presence of boehmite particles in the pores of the polymeric layer and to determine their crystallite size at the same time. Combining this method with the combustion and/or thermogravimetric method above to quantify the weight of the boehmite, or other ceramic particles in the pores, if any, would provide the weight in gsm of boehmite particles in the separator. The known analytical techniques for boehmite particles, such as FTIR, X-ray diffraction, thermogravimetric analysis (TGA), high resolution photoelectron spectroscopy (ESCA), and ²⁷Al nuclear magnetic resonance (nmr) can be utilized to further confirm the presence and quantity of boehmite particles in the pores of the polymeric layer, if needed. An alternative approach for collecting the ceramic particles, such as boehmite particles, in the pores of the polymeric layer, after removing any surface coating from the polymeric layer, would be to do a Soxhlet extraction of the polymeric layer with a hot solvent that dissolves the polymer from the polymeric layer to isolate the insoluble inorganic particles. Besides providing the weight of inorganic particles, this could be combined with the above analytical methods, combustion and/or thermogravimetric analysis, and TEM or SEM method, to confirm the presence and quantity of boehmite particles. One benefit of this penetration of nano-sized boehmite particles would be the ability to make and sell boehmite/polyethylene substrates, with no or minimal surface boehmite or other coating, for various applications, including for manufacturing CCS, where the boehmite/polyethylene substrate would have multiple advantages over polyethylene substrates, including, but not limited to, a much greater dimensional stability at high heats, a modified and smaller pore size distribution, and other benefits, such as ease of wetting by subsequent coatings. One example of a particularly useful application for boehmite/polymeric substrates would be filtration applications, where the very small nano-sized boehmite particles, with and without various surface treatments, in the pores of the polymer could provide the desired nanofiltration and ultrafiltration properties not available from the polymeric substrate without boehmite particles in its pores, and could be adapted for desalination of sea water. This benefit would also be available for other

boehmite/polymeric substrates besides polyethylene, such as for boehmite/polypropylene and boehmite/other polyolefin substrates.

[0083] Example 4

[0084] The procedure of Example 1 was used to make a nanoporous separator comprising a 50:50 blend by weight of boehmite and boron nitride pigments, except that a 12 micron thick porous polyethylene separator from Toray Industries, Tokyo, Japan, was substituted for the Entek polyethylene separator, and the vinyl pyrrolidone polymer, Soteras CCS, was substituted for the chitosan. The boron nitride pigment was NX1, supplied by Momentive Performance Materials. The total coating weight on both sides was 2.5 gsm. Figure 11 shows the reduced water content of this separator compared to the control with only boehmite pigment and no boron nitride. Specifically, as shown in Figure 11, at a 50:50 by weight ratio of boehmite and boron nitride, the water content at 20% RH was reduced by one-third from its value with boehmite pigment only.

[0085] Comparative Example 1

[0086] An alumina mix was prepared in water by adding 20 parts of alumina pigment (HJA-0719 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the alumina was greater than 400 nm. The alumina mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 4.4 gsm. The alumina coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0087] The 20:1 by weight ratio of alumina pigment to hydroxypropyl cellulose polymer binder is calculated to have 13.6% by volume of polymer binder in the solid portion of the porous alumina coating layer. This calculation is based on a specific gravity of 4.0 g/cc for alumina and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating

weight of about 4.4 gsm, this alumina coating had a shrinkage of about 60% at 150°C for 1 hour.

[0088] Comparative Example 2

[0089] A boehmite mix was prepared in water by adding 20 parts of boehmite pigment (BG-613 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the boehmite was greater than 400 nm. The boehmite mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 7.0 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0090] The 20:1 by weight ratio of boehmite pigment to hydroxypropyl cellulose polymer binder is calculated to have 10.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating weight of about 7.0 gsm, this boehmite coating had a shrinkage of about 60% at 150°C for 1 hour.

[0091] Table 1 shows a summary of the data from Examples 1 to 3.

Table 1

Example	Boehmite Type	Primary Particle Size (nm)	Pigment:Polymer Binder Weight Ratio	Volume % Polymer	Total Coat Weight for <5% Shrinkage at 150°C for 1 hour
1	10F4	40	12:1	15.3%	≥4.0 gsm
2	D60	75	12:1	18.7%	≥3.9 gsm
2	D60	75	8:1	25.6%	≥4.8 gsm

3	25F4	8	12:1	18.7%	≥3.6 gsm
3	25F4	8	8:1	25.6%	≥2.9 gsm

[0092] One challenge in using nano-sized ceramic particles, such as nano-sized boehmite particles, is their increased tendency to absorb water because of their much larger surface areas compared to micro-sized ceramic particles, such as micro-sized alumina and micro-sized boehmite particles. For lithium ion cells, lithium metal cells, and other electrochemical cells with sensitivity to water content, the separator needs to have a low water content, such as under 600 ppm of water in the separator, preferably less than 300 ppm of water, and more preferably under 100 ppm of water as typically found in porous polymer only separators with no ceramic materials in them. Although vacuum drying of the separator at up to 80 °C can help to lower this moisture content in separators with nano-sized ceramic particles, and the separators of this invention have increased thermal stability that permits even higher vacuum drying temperatures of up to 90 °C to about 110 °C, some users of separators may prefer not to perform this extra step of vacuum drying. It would be preferable if the greater heat stability, lower cost from the use of a lower amount of ceramic coating materials, and lower ceramic coating thickness, including from the penetration of the ceramic coating and ceramic particles into the porous polymeric separator layer, of the separators of this invention would include a lower moisture content in the separators from the use of more hydrophobic ceramic particles and organic polymers. Having a lower amount of ceramic particles from the lower coating weights of the invention is a significant benefit for lowering the moisture content of the separator, but it would be further useful if the ceramic particles were more hydrophobic and less prone to moisture pickup.

[0093] One approach which provides this lower moisture absorption is to treat the surface of the ceramic particles with a hydrophobic material, such as by using hydrophobically-treated boehmite particles, particularly if the hydrophobic material is chemically reacted with the boehmite particles and thus is permanently bonded to the boehmite particles. Suitable hydrophobic materials for chemical modification of boehmite pigments include, but are not limited to, organic sulfonic acids, organic carbonates, and polymers with hydrophobic groups and with reactive groups, such as carboxylic acid and hydroxyl groups, that can be made to react with hydroxyl groups of the boehmite or other ceramic particles.

[0094] Examples of the organic sulfonic acid-treated boehmite particles include Dispal® 25SR, available from Sasol in Lake Charles, LA, a boehmite pigment chemically reacted with

p-toluene sulfonic acid through the sulfonic acid groups. The reaction leaves the hydrophobic p-toluene group on the outer surface of the boehmite particle to make it less prone to pick up water. An even more hydrophobic example from Sasol is p-dodecylbenzene sulfonic acid surface reacted to boehmite particles through the sulfonic acid group and having the very hydrophobic dodecyl group on the outer surface of the boehmite pigment. This makes the boehmite particles so hydrophobic that this treated boehmite can be dispersed in hydrocarbon solvents. The use of boehmite particles surface treated with organic sulfonic acids in boehmite coatings for separators is disclosed in U.S. Patent Application Publication No. 2013/0171500 by Xu et al.

[0095] Examples of the organic carbonate-treated boehmite particles include, but are not limited to, those with ethylene carbonate and fluoroethylene carbonate as the organic carbonates. Besides the option of choosing an organic carbonate that is very hydrophobic in the part of the molecule that will face away from the surface of the boehmite, the choice can also be made with a view to picking an organic carbonate which will provide increased ionic conductivity to the separator. The use of boehmite particles surface treated with ethylene carbonate to provide separators with increased conductivity is described in U.S. Pat. No. 8,883,354 to Carlson et al.

[0096] The use of boehmite particles surface treated with organic polymers to provide separators is described in U.S. Pat. No. 9,871,239, to Carlson et al. As the primary crystallite size and resulting pore size become smaller in the boehmite separators and the ionic conductivity of the separator becomes lower because of the smaller pore sizes, it is useful to have surface treatments, such as the three types described above, that can be utilized to increase the ionic conductivity of the separator, as well as to decrease the water content of the separator.

[0097] The preparation of a ceramic mix or slurry with the surface-treated hydrophobic materials can be done conveniently by adding a precursor, such as an organic carbonate, to the slurry containing the untreated boehmite particles and heating the slurry to a temperature, such as 80 °C, for a sufficient time, such as 1 hour, to react the surface treatment material, such as an organic carbonate, with the boehmite particles to the desired level of surface treatment. There is no need to isolate the surface-treated boehmite particles by a drying process, but rather the slurry can be used as is, usually after some dilution with water and/or solvents to the desired % solids level for the coating of the porous polymer substrate. If it is desired to further reduce the particle size of the ceramic particles, such as boehmite particles,

and/or to homogenize the slurry for more uniformity and perhaps a lower viscosity at the desired % solids level for coating, the slurry, before and/or after any surface treatment, can be further mixed in an impingement mill or similar milling device. Examples of the use of an impingement mill for reducing and homogenizing coating mixes containing pigments is described in U.S. Pat. Nos. 5,210,114 and 5,292,588, to Katsen.

[0098] While the invention has been described in detail and with reference to specific and general embodiments thereof, it will be apparent to one skilled in the art that various changes and modifications can be made therein without departing from the spirit and scope thereof.

CLAIMS

1. A separator for a lithium battery comprising:
a porous polyethylene layer, and
a second porous layer adjacent said porous polyethylene layer, said second porous layer comprises a first portion and second portion, wherein:
the first portion is solid; and
 - a. the first portion comprises boehmite particles and one or more polymers:
 - b. a volume fraction of said one or more polymers in the first portion is between about 15% and about 50%;
 - c. a total coating weight of said second porous layer on said porous polyethylene layer is about 5.5 grams per square meter or less, and
 - d. a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.
2. The separator of claim 1, wherein the volume fraction of said one or more polymers in the first portion is between 25% and 40%.
3. The separator of claim 1, wherein said boehmite particles have an average crystallite size between 5 nm and 90 nm.
4. The separator of claim 3, wherein the average crystallite size of said boehmite particles is between 30 nm and 50 nm.
5. The separator of claim 1, wherein said second porous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of said pores have a diameter of 80 nm or less.
6. The separator of claim 1, wherein said second porous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of said pores have a diameter of 50 nm or less.
7. The separator of Claim 1, wherein the second porous layer is a nanoporous layer.

8. The separator of Claim 1, wherein the second portion is not a solid.
9. A process for making a separator for a lithium battery comprising:
 - (a) treating boehmite with an organic carbonate so as to make hydrophobic boehmite particles;
 - (b) providing a porous polymeric layer;
 - (c) providing a second porous layer, wherein said second porous layer comprises the hydrophobic boehmite particles and one or more polymers, and wherein the total coating weight of said second porous layer is less than 4.5 grams per square meter, and
 - (d) bringing the second porous layer adjacent the porous polymeric layer, so as to make a separator with no more than 600 parts per million of water.
10. The process of Claim 9, wherein the organic carbonate comprises ethylene carbonate.
11. The process of Claim 9, wherein the organic carbonate comprises fluoroethylene carbonate.
12. The process of Claim 9, wherein the organic carbonate reacts with hydroxyl groups of the boehmite.
13. A device comprising:
 - a porous polyethylene layer, and
 - a second porous layer adjacent said porous polyethylene layer, said second porous layer comprises a first portion and second portion, wherein:
 - the first portion is solid; and

- a. the first portion comprises boehmite particles have an average crystallite size between 5 nm and 90 nm and one or more polymers:
 - b. a volume fraction of said one or more polymers in the first portion is between about 25% and about 40%;
 - c. a total coating weight of said second porous layer on said porous polyethylene layer is about 5.5 grams per square meter or less, and
 - d. a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.
14. The device of claim 13, wherein the average crystallite size of said boehmite particles is between 30 nm and 50 nm.
15. The device of claim 13, wherein said second porous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of said pores have a diameter of 80 nm or less.
16. The separator of claim 13, wherein said second porous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of said pores have a diameter of 50 nm or less.
17. The device of Claim 13, wherein the second porous layer is a nanoporous layer.
18. The device of claim 13, wherein the second portion is not solid.
19. The device of claim 13, wherein the second portion is not solid.

ABSTRACT

A separator for a lithium battery having (a) a porous polymeric layer, such as a polyethylene layer; and (b) a nanoporous inorganic particle/polymer layer on both sides of the polymeric layer, the nanoporous layer having an inorganic oxide and one or more polymers; the volume fraction of the polymers in the nanoporous layer is about 15% to about 50%, and the crystallite size of the inorganic oxide is 5 nm to 90 nm.

PTO/AIA/01 (06-12)

Approved for use through 11/30/2020. OMB 0651-0032

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of
Invention

MULTILAYER NANOPOROUS SEPARATOR

As the below named inventor, I hereby declare that:

This declaration
is directed to:



The attached application, or



United States application or PCT international application number 16/755,294

filed on April 10, 2020

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

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LEGAL NAME OF INVENTOR

Inventor: David W. Avison

Date (Optional): _____

Signature: David W. Avison

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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LEGAL NAME OF INVENTOR

Inventor: Steven A. Carlson

Date (Optional) _____

Signature: Steven A. Carlson

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Invention

MULTILAYER NANOPOROUS SEPARATOR

As the below named inventor, I hereby declare that:

This declaration
is directed to:☐

The attached application, or

☒United States application or PCT international application number 16/755,294filed on April 10, 2020

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

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LEGAL NAME OF INVENTORInventor: Benjamin Sloan

Date (Optional): _____

Signature: B. Sloan

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Application Number	
Filing Date	Herewith
First Named Inventor	David W. Avison
Title	MULTILAYER NANOPOROUS SEPARATOR
Art Unit	
Examiner Name	
Attorney Docket Number	00249/62

SIGNATURE of Applicant or Patent Practitioner			
Signature	/Brian J. Amos/	Date (Optional)	April 10, 2024
Name	Brian J. Amos	Registration Number	63,679
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.



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LG Energy Solution, Ltd.

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Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
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INFORMATION DISCLOSURE
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(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number	00249/62	

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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

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- ☐ See attached certification statement.
- ☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.
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A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2024-04-10
Name/Print	Brian J. Amos	Registration Number	63,679

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INFORMATION DISCLOSURE
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(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number		00249/62

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	1	9293752	B2	2016-03-22	Murata et al.	
	2	9847519	B2	2017-12-19	Pekalaet et al.	
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Attorney Docket Number	00249/62	

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A new scanning electron microscopy approach to image aerogels at the nanoscale

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Abstract

A new scanning electron microscopy (SEM) technique to image poor electrically conductive aerogels is presented. The process can be performed by non-expert SEM users. We showed that negative charging effects on aerogels can be minimized significantly by inserting dry nitrogen gas close to the region of interest. The process involves the local recombination of accumulated negative charges with positive ions generated from ionization processes. This new technique made possible the acquisition of images of aerogels with pores down to approximately 3 nm in diameter using a positively biased Everhart–Thornley (ET) detector.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Aerogels are highly nanoporous materials that with a thin, three-dimensional (3D) skeletal structure forming the walls of the interconnecting pores. The very high pore volume (often greater than 90%), tortuosity of the pore structure and thinness of the solid struts render the aerogels excellent insulating materials by decreasing both gaseous and solid conduction [1, 2]. Their low optical refractive index properties have been proved to be suitable to detect Cherenkov radiation in high energy physics experiments [3]. Moreover, the highly porous (and low density) structure, exceptional ability to capture hypervelocity dust and optical transparency of aerogels have been exploited to collect cosmic dust in aerospace experiments [4]. Due to their low dielectric constant, aerogels have been used in metal–insulator–semiconductor devices to study current leakage related to high electric fields generated in such devices [5]. It should be noted that important quantities that define the final physical properties of aerogels include: density (or porosity), surface area, dielectric constant, refractive index and thermal conductivity. In some cases, these quantities have a linear dependence with porosity (or density) and thus can be estimated if the porosity content is known [6]. For aerogels to be used as insulators at high temperatures it is important to distinguish thermal exposure

conditions under which the pore structure is maintained as well as to differentiate among aerogel formulations that densify and those that maintain high concentrations of nanopores. Therefore, it is technologically important to find ways to study the porous structure of aerogels.

Both transmission electron microscopy (TEM) and scanning electron microscopy (SEM) techniques are commonly used to estimate the pore size in aerogels [7–9]. One common problem encountered in both techniques during aerogel imaging is the charging effect. These effects are caused by charge accumulation and are due to the dielectric nature and poor electrical properties of the silica and aluminosilicate aerogels of most interest as thermal insulators. Care must be taken during TEM inspection of these materials at the nanoscale, because high electric fields resulting from charge accumulation can cause surface mass migration of SiO_2 species and may form local areas with collapsed porous structures [10]. If a SiO_2 aerogel or aerogel containing SiO_2 is to be investigated by SEM, the most common employed solution to minimize charging effects is to deposit a thin layer of a conducting material (e.g. Au). However, this technique may drastically mask (or even alter) the intrinsic porous structure of the aerogel, which makes the image difficult to interpret [8, 9]. Here we present a new approach to minimize charging effects on aerogels during imaging by *in situ* SEM.

The process involves the local recombination of accumulated charges due to irradiation with charges induced from inert gas injection near the surface of the sample.

2. Experimental details

The aluminosilicate aerogel samples were synthesized by hydrolysis and condensation of a Boehmite powder (Sasol Dispal 25P4, a 7.2 nm crystallite size powder) with tetraethylorthosilicate (TEOS, Sigma Aldrich). The powder was dispersed in acid solution (Jar A). TEOS was added to a 200 proof ethanol solution containing stoichiometric quantities of water (Jar B), and allowed to hydrolyze for 1 h. The contents of the two jars were combined, stirred briefly, and the sol poured into polyethylene molds. The molds were covered with parafilm, and the sol allowed to gel overnight at ambient temperature. The hydrogels were extracted into absolute ethanol, which was exchanged after 24 h. The gels were allowed to age for several days to a few weeks prior to removal of the ethanol solvent by supercritical CO₂ extraction. Surface area and pore size distribution were determined using nitrogen desorption techniques.

For the *in situ* SEM charge compensation experiments we used a Carl Zeiss UltraPlus-Schottky thermal field emission scanning electron microscope, operated at 3 keV and at a working distance of about 6 mm. The instrument was equipped with a charge compensator (CC) device and its movement was driven pneumatically. The CC device used a needle to inject dry nitrogen gas locally, close to the sample surface at about 1 mm. The amount of the injected gas (CC pressure) was regulated between 0% and 100% by a software slider and the corresponding chamber pressure ranged from about 10⁻⁴ to 1 Pa. A positively biased ET detector and line scan at a scan rate of 2 with a line average of 64 and corresponding cycle time of 12.6 s was used for signal collection. It should be noted that the positively biased ET detector is designed to collect both SE and BSE signals. For each collected image, the full three scans were performed after each systematic increase of CC pressure. The aerogel samples were attached to a grounded aluminum disc using conductive silver paste. Finite element (FE) calculations were performed with the Maxwell 2D program from Ansoft Corporation.

3. Results and discussion

When a sample is irradiated with a primary beam of electrons (PE), several signals are generated. A crude estimation for the penetration (or range R) of PE in these aerogels can be obtained using

$$R (\mu\text{m}) = \frac{0.115 E_0^n (\text{keV})}{\rho (\text{g cm}^{-3})} \quad (1)$$

where E_0 is the primary beam energy and is equal to 3 keV and ρ is the density of the aerogels [11]. With $n = 1.35$ (valid for $E_0 < 5$ keV) and average ρ value of 0.06 g cm⁻³, R is approximately 9 μm . Pertinent to this work is the generation of backscattered electrons (BSE) and secondary electron (SE)

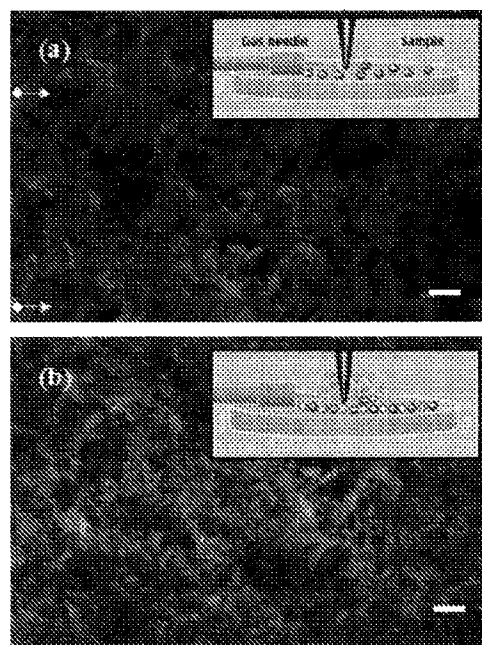


Figure 1. SEM images of aerogels with charging artifacts (gas needle off) (a) and when the charge compensation device is working at 100% (b). Some artifact defocused lines are marked with solid arrows. The scale bar is 100 nm and magnification is $\sim 202\times$. Insets are the corresponding schematic representations of case (a) and (b), respectively.

signals. BSE are produced by multiple elastic scattering events and have relatively high energies, while low energy SE are produced by inelastic scattering events. Associated with the generation of SE and BSE, is the secondary electron (δ) and the backscattering (η) yield coefficients. Generally, during the irradiation process a sample can be charged positively, negatively or be in a neutralized state [12]. The process can be described by using the law of charge conservation, which must hold for all times (t). In general terms

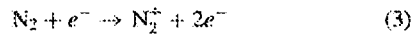
$$I_{PE}[1 - (\delta + \eta)] - I_L = \frac{\partial Q}{\partial t} \quad (2)$$

where I_{PE} is the current injected to the sample by PE, I_L is the leakage current and $\partial Q/\partial t$ is the charge build-up rate [11]. It follows that if the current injected into the sample by PE is greater than the total emission/leakage current, then the sample will have an excess of electrons. This will create a region (of the order of R) of accumulated negative charges on the surface of the sample, and hence the sample will be negatively charged. Typical consequences of this negative charging are the production of image artifacts which include, for instance, image distortion (due to deflection and defocus of PE) and irregular white contrast features.

Figure 1(a) shows an SEM image of an aluminosilicate aerogel sample having a surface area of 256 m² g⁻¹ and 9.45 nm average pore size undergoing charging effects. These artifacts are induced by negative charging. A negative potential

near the surface of the sample defects (and defocuses) the PE to other regions, creating this type of image distortion. For clarity, some distorted lines are marked with white solid arrows in figure 1(a). It is well known that charging is usually not a static problem and may change under scanning events [12]. The defocused lines can be further explained by charging and discharging events during scanning, using capacitor models. Both charging and discharging for a given scanned pixel or set of pixels can be described with a function that has a dominant exponential dependence with the ratio of pixel time to capacitor time constant [12]. The overall trend of such negative surface electric potential function follows a saw-tooth-time dependence with maximum and minimum values that correspond to charging and discharging events, respectively [13]. Thus, the array of pixels in figure 1(a) undergoing charging events with a relatively higher negative surface potential than pixels under discharging will deviate (and defocus) PE more strongly. To minimize these charging problems we used a simple method developed recently by Carl Zeiss NTS, LLC. The process can be performed by non-expert SEM users. The key is to neutralize the negative charges by the recombination of local induced positive charge near the surface of the sample.

For the generation of positive charges (ions), dry nitrogen (N_2) gas was introduced locally (close to the sample surface) in a controlled manner. Details of the process can be found in section 2. Collisions of PE, SE and BSE with the N_2 gas cloud ionizes the nitrogen molecules forming nitrogen positive ions (N_2^+) according to



where the ionization energy is about 15.6 eV [14]. Then N_2^+ are attracted to the surface of the sample by local electric fields and recombine with negative charges, leading to charge neutralization, as shown in schematically in the inset of figure 1(b). In figure 1(b) the CC device was operated at 100%. The porous structure typical of aerogels clearly is visible with porous diameters ranging from ~ 3 to ~ 40 nm. It should be noted that the current leakage term in equation (2) actually has two terms ($I_L = I_G + I_{ion}$), where I_G is the sample leakage current to the ground and I_{ion} is the current leakage due to electron-ion recombination [15]. To a first approximation, it is expected that electron-nitrogen gas molecule collisions with greater cross-sections (σ) will contribute more to the rate of production of ions, because the scattering probability for a given ionization event scales with σ [16]. In figure 2 we present the predicted values for the electron impact ionization σ as a function of the kinetic energy (KE) of the electrons in our experiment, which is for the range of SE above the ionization energy up to PE energy. A close-up of the low KE dependence behavior is plotted in the inset of figure 2. The predicted values are based on the binary-encounter-Bethe (BEB) model, which adds the Mott's cross-section with the high KE factor and Bethe cross-section [17, 18]. Note that for the case of N_2 (black line) the maximum value of σ_{BEB} ($\sim 2.62 \text{ \AA}^2$) is produced with electrons of KE of ~ 100 eV, which corresponds to low energy BSE. For SE with energies of ~ 50 eV, σ_{BEB} is $\sim 2.17 \text{ \AA}^2$.

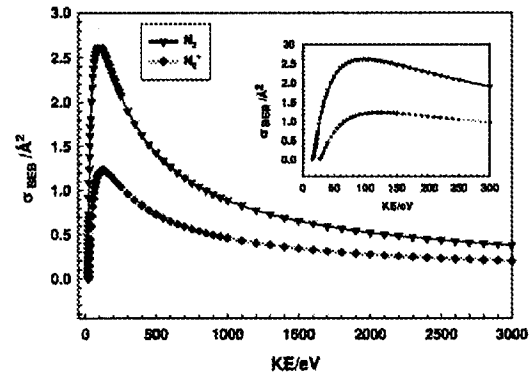


Figure 2. Predicted values for the σ_{BEB} as a function of KE of electrons. The black line indicates results for ionization of N_2 , while the gray line shows the predicted results for N_2^+ .

Moreover, PE and SE (with energy ~ 20.5 eV) produce a σ_{BEB} of $\sim 0.38 \text{ \AA}^2$, and hence should have the same probability to produce N_2^+ during ionization events. However, if we consider the fact that most of the SE ($\sim 90\%$) have KE of 10 eV (or even less), which is below the ionization energy for N_2 , ionization events by SE should be rare. However, there is a possibility that some SE can be accelerated by the positive electric potential at the Faraday cage of the ET detector and may gain sufficient energy to ionize N_2 molecules.

Concerning the production of ions, it is important to consider possible second ionization processes. If N_2^+ produced by the ionization of N_2 molecules does not recombine quickly with the negative charges at the surface of the aerogels, further ionization of N_2^+ may take place and more positive ions will be available for recombination. The σ_{BEB} values for such a situation are presented by the gray line of figure 2, for energies above ~ 27 eV up to the PE energy. The overall trend is to have lower values of σ_{BEB} in contrast with the obtained values for N_2 . Once more, the peak maximum of σ_{BEB} ($\sim 1.23 \text{ \AA}^2$) is produced in the range of BSE (~ 130 eV).

To further explore the effects of N_2 on charging in aerogels we systematically increased the relative amount of nitrogen gas and collected SEM images to see the effects in image contrast. Figures 3(a)–(k) presents a sequence of SEM images showing the effect of changing the percentage of nitrogen, ranging from 0 to 100% in steps of 10%. Due to the highly topographic nature and charging of these aerogels, the overall contrast behavior of the SEM images is expected to be dominated by both topography and voltage contrast. Topography contrast arises due to the relative inclination of the features of the sample with the PE, while voltage contrast is expected to depend on variations of surface potential and hence possible variations in the deflection of the SE due to local electric fields [19]. Although the images of figure 3 are slightly shifted, one can observe the evolution of voltage contrast. Again, for clarity purposes, distorted lines are marked with solid arrows. Note that slight variations in the charging process occur from figure 3(b) (CC operated at 10%) to figure 3(e) (CC operated at 40%), while major effects seem to be observed in figure 3(f) (CC operated at 50%). In figure 3(g) (CC operated at 60%) half

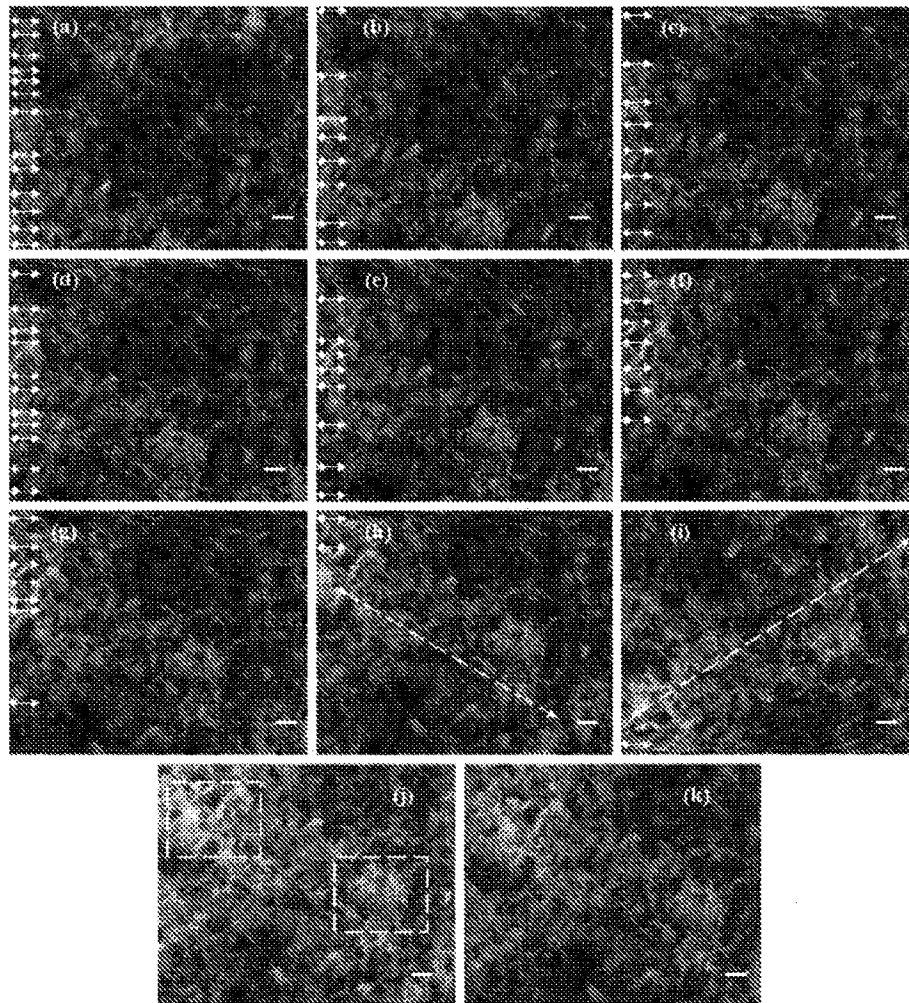


Figure 3. SEM images of aerogels showing the evolution of charging events as a function of relative nitrogen content. CC device off (a), and subsequent increase with steps of 10% ((b)–(k)). Note that positions of artifact lines are marked with solid arrows. For more details refer to the main text. The scale bar is 100 nm and magnification is $\sim 223\times$.

of the image is free of artifact lines due to negative charging. In figure 3(h) (CC operated at 70%) more artifact lines have disappeared but some areas at the top of the image are still not in focus, indicating a relatively strong negative surface potential in that area. In addition, white contrast features (marked with dotted arrows) can be observed distributed along the image from top-left to bottom-right areas, which are also an indication of negative charging. Note that in figure 3(i) (CC operated at 80%) the defocused areas appeared at the lower portion of the image, and the white contrast features (marked with dotted arrows) appear along the image from top-right to bottom-left areas. However, within figure 3(j) (CC operated at 90%) defocused regions were not observed, indicating that strong negative charging has been minimized considerably. Finally, when the CC was operated at 100% (figure 3(k)), the overall bright features observed in figure 3(j) were diminished, indicating that negative charging has been further minimized.

This can be noticed more clearly by comparing the features marked with a dotted box in figure 3(j) and the corresponding areas in figure 3(k). Therefore, to minimize notably negative charging, the CC device needs to be operated at high relative pressures (above 60%). This is expected, since introducing more N_2 gas into the chamber will enhance the generation of positive ions and consequently there will be more positive ions to recombine with negative electrons at the surface of the sample.

At this point we want to indicate that results on imaging of aerogels have been reported previously by Kucheyev *et al* using low vacuum SEM [20]. Although the physics of charge neutralization has similar fundamental principles in both works, we want to briefly clarify the differences and originality of our approach. In regards to the results presented, the SEM images of aerogels in [20] did not show evidence of artifacts and corrections for charging effects, as we clearly

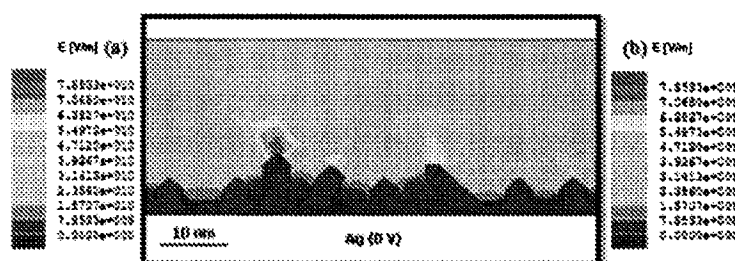


Figure 4. FE model describing the distribution of E for a simplified geometry consisting of an array of silica needles at -1000 V (a) and at -100 V (b), with a grounded silver plate and top electrode, representing the electric potential at the Faraday cage of the ET detector. The direction of the E vector field is from top to bottom.

presented in figures 1 and 3. In terms of the experimental setup, our technique seems to have advantages over the low vacuum SEM technique of Kucheyev *et al* [20]. Firstly, in the low vacuum SEM described in [20] the whole specimen chamber is exposed to a gas (water vapor) and a dedicated gaseous detector (not commonly used) is required for image formation. In addition it has been pointed out that determining the optimum experimental parameters for imaging formation in low vacuum SEM is not trivial, even for ‘experienced’ users [21]. The CC device used in this work is easy to operate and can be installed in conventional SEM equipped with the most common available electron detector (ET).

As pointed out earlier, once the positive ions are generated due to the ionization process, they will be attracted to the surface of the sample by local electric fields (E). This implies that the recombination processes will be affected by such local electric fields. It is important to note that the electric potential at the sample is unknown; therefore the results should be taken as qualitative guidance. Figure 4 depicts the results of the finite element (FE) calculations for a simplified model, which consists of an array of silica needles with different orientations and distances relative to the top electrode (representing the Faraday cage of the ET detector) and a grounded silver electrode. The typical potential at the Faraday cage is ~ 250 V at a distance of ~ 5 cm from the PE impact position. Since in these simulations the distance was decreased to about 30 nm, the potential has been corrected accordingly to be $\sim 1.5 \times 10^{-4}$ V. To solve this electrostatic problem, FE analysis with the Maxwell 2D program was used to solve the Poisson equation using the Neumann boundary conditions and constraining the E field to be continuous across the surface [22]. For a given time, positive ions will preferentially (not exclusively) move to the surface of the aerogels where the magnitude of the local electric field is at its maximum; after that, recombination takes place. In order to explain the differences in defocused lines, and to propose a possible explanation of the evolution process taking into consideration the topography of our aerogels, we have chosen two arbitrary potential values. For instance, consider an area of the sample consisting of an array of silica needles with a local electric potential of -1000 V (which is more than enough to deflect the PE), as described in figure 4(a). Note that in general, the magnitude of E is stronger at the tip-apex area of

the needles, and its maximum value is located at the tip area of the needles closest to the top electrode, and how the E field magnitude diminishes in regions between the needles. This E field behavior is also encountered during the field emission process of rod-like nanostructures [23, 24]. Likewise, if we change the potential of the silica needles to -100 V (which represents an area of the sample with a less severe negative charging) and leave the other parameters unchanged, we notice that the magnitude of the E field is reduced by an order of magnitude but the overall trend is conserved (figure 4(b)). Comparing the results of figures 4(a) and 4(b), we note that the dominant contribution that enhances the local E magnitude is the potential value at the silica needles. Note that needles at -1000 V and further from the top electrode (figure 4(a)) have higher E magnitudes than the needles closest to the top electrode at -100 V potential (figure 4(b)). Based on the FE results one would expect that areas of the aerogels with higher negative charging will attract the positive ions more strongly than areas with relatively lower negative charging. Therefore, for a given time, areas with relatively higher negative electric potentials will present a higher degree of recombination and charging will be minimized faster. Finally, the system will then self-regulate the charging until a relatively uniform electric potential is reached.

4. Conclusions

In summary, we have shown a new SEM approach to image aerogels at relatively high magnifications, where the fine porous structure (down to ~ 3 nm in pore diameter) can be investigated. More importantly, our technique does not use any metal coating to minimize negative charging effects. It uses positive ions which recombine with negative charges at the surface of the sample, revealing the intrinsic porous structure of aerogels. The most common detector (ET) was used for image formation. The results of σ_{FE} implied that the ionization process and hence the generation of positive ions is governed by low energy BSE. Furthermore, taking into account variations in topography of our aerogels, the simplified FE model employed suggested that areas of the aerogels with the highest local electric potentials should attract positive ions more strongly and hence negative charging will be minimized more rapidly in such areas. To substantially minimize negative

charging considerably, it was required to operate the CC above 60%. It is noteworthy that in this work, the dependence of the PE and pixel scan rates on charging was not considered. Finally, it is important to point out that this technique can be used to image other types of non-conductive nanoporous materials.

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Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries



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HIGHLIGHTS

- AlOOH-modified polyethylene separator gains enhanced thermal stability.
- Interlocking interface structure is formed in the modified separator as PE melts.
- AlOOH coating layer shows appropriate electrolyte wettability.
- Cell performance attains improvement as employing AlOOH-modified separator.

GRAPHICAL ABSTRACT



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ABSTRACT

To exploit high-quality separators for lithium ion batteries, current research activities are mainly focused on the modification of microporous polyolefin membranes by coating them with inorganic particles to achieve comprehensive improvements in their thermal stability, electrochemical compatibility, and overcharge protection. Here, we report a separator made by coating boehmite (AlOOH) particles on microporous polyethylene (PE) membranes. Compared to the commercially applied coating materials, e.g., aluminum oxide (Al₂O₃), AlOOH allows for a substantial reduction in the coating thickness, while ensuring excellent thermal stability of the modified PE membrane. Our study shows that this is due to the formation of an interlocking interface structure that interconnects the PE membrane and AlOOH coating layer as soon as PE melts at about 140 °C, preventing the modified PE membrane from shrinking at subsequently elevated temperatures. The modified PE membrane exhibits suitable electrolyte wettability to facilitate ion transport through it. Thus, the lithium ion batteries employing it as a separator could attain substantially improved electrochemical performance. Furthermore, the AlOOH-coated PE separator was also found to provide an excellent overcharge protection.

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1. Introduction

High-power lithium ion batteries show incomparable application potential in electric vehicles and large energy storage equipment owing to their high energy density and long cycle life, but present safety issues [1–6]. In current lithium ion batteries,

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microporous membranes made of polyolefin such as polyethylene (PE) and polypropylene (PP) are commonly used as separators to isolate the cathode and anode and allow the transport of lithium ions and all degradation products, thereby safeguarding the batteries in working. With softening temperatures of ~130–170 °C, polyolefin membranes can shrink to an extent that causes short circuit and even explosion as the batteries are overheated [7,8]. The polyolefin membranes modified by combination with good heat-resistant organic or inorganic materials can exhibit much lower thermal shrinkage [9–12]. For example, with a coating of polyimide nanoballs, the shrinkage of a PE membrane heated at 140 °C for 0.5 h was found to be reduced from 83.3% to 10% [13]. In general, modification with organic polymers often leads to inferior electrolyte wettability, thereby degrading the electrochemical compatibility of the modified membranes [9]. In contrast, hydrophilic inorganic materials such as silicon dioxide (SiO₂) [14–16] and aluminum oxide (Al₂O₃) [17–19] can offer more favorable modification in terms of both thermal stability and electrochemical compatibility of the modified membranes [7].

A common process currently adopted in industries is to modify the polyolefin membranes with a double-faced coating of Al₂O₃ particles [9]. In this case, thermal shrinkage of the membranes is closely related to the Al₂O₃ coating thickness. Zhao et al. [20] studied the Al₂O₃ coating with thicknesses of 3–9 μm and found that, after exposure at 140 °C for 0.5 h, the corresponding shrinkage of the modified PE membranes decreased from 20% to 0%, while the ion conductivity through them dropped from 1.12 to 1.00 mS cm⁻¹. This indicates that increasing the coating thickness may effectively suppress the thermal shrinkage, but would impart an adverse effect on the electrochemical compatibility for the modified membranes for use as separators. To date, the commercially employed polyolefin separator contains an Al₂O₃ coating that is 3 μm or more thick on each face, which accounts for around 25% of the total thickness of the separator. Therefore, there is room for improvement when it comes to increasing the energy density and cut the cost of batteries by reducing the coating thickness of separators while ensuring good thermal stability [21]. To fulfill this goal, the most feasible strategy is to significantly reinforce the interconnection between the coating layer and polyolefin membranes by means of constructing an interlocking interface structure.

In this study, we report a separator made by coating boehmite (AlOOH) particles on a microporous PE membrane. AlOOH was found to enable the formation of an interlocking interface structure when PE melted at about 140 °C, which significantly strengthened their interconnection. Besides, AlOOH is lighter, softer and cheaper and shows better dispersity in an aqueous slurry compared to Al₂O₃. In this article, we present the study on the thermal stability, electrochemical compatibility, and overcharge protection of the modified PE membrane using a 1.15-μm-thick AlOOH coating.

2. Experimental section

2.1. Material preparation

A slurry composed of 35 wt% AlOOH particles of average size 350 nm (Nabaltec, Germany), 1 wt% polyvinyl alcohol (PVA, Lingfeng Chemistry, China), and distilled water was prepared. Before coating, the slurry was well dispersed by grinding in a 1400 r/min grinding machine for 20 min. Using a roller coating instrument, the slurry was coated on both faces of a 16-μm thick PE membrane (Shuang'ao, Shanghai, China), which was subsequently dried at 60 °C for 24 h. The thickness of the AlOOH particle coating was controlled by the rolling speed.

2.2. Characterization and measurement

Field emission scanning electron microscopy (FE-SEM) was used to observe the microscopic structure and morphology of the PE membranes. The thickness of the PE membrane and AlOOH particle coating was measured by a digital display thickness gauge. The shrinkage of the modified PE membrane was determined by measuring the dimensional change with respect to the original one after the thermal treatment at various temperatures for 0.5 h, and was calculated using the following equation:

$$\text{Shrinkage}(\%) = \frac{S_0 - S_T}{S_0} \times 100\% \quad (1)$$

where S_0 and S_T refer to the area of the membrane before and after the thermal treatment, respectively. The permeability (Gurley number) of the separators was tested using a densometer. Mechanical strength, both in the machine direction (MD) and transverse direction (TD) was measured on a universal testing machine. The electrolyte wettability of the separator was examined by two methods: a contact angle test and an electrolyte immersion-height test. The contact angle was examined using a commercial drop shape analysis system (Powereach JC2000C1, Shanghai Zhongchen Digital Technique Equipment Co. Ltd., China). The electrolyte immersion-height test [22] was carried out by immersing the separator vertically in a liquid electrolyte (EC:DEC = 1:1 v/v containing 1 M LiPF₆) and then measuring the soaked height of the liquid electrolyte after 0.5 h.

The ionic conductivities of the pristine and AlOOH modified PE separators were measured by electrochemical impedance spectroscopy (EIS). The membranes were placed between two stainless steel blocking electrodes, and the EIS spectra were obtained by sweeping from 100 kHz to 0.1 Hz with an AC amplitude of 5 mV at room temperature [23]. The ionic conductivity (σ) was calculated using the following equation:

$$\sigma = \frac{d}{R_b \times S} \quad (2)$$

where R_b is the bulk resistance and d and S are the thickness and area of the fibrous membrane, respectively. To evaluate the electrochemical performance of the separators, CR2016 coin-type half cells were assembled in a glove box filled with argon gas, with a Li₄Ti₅O₁₂ working electrode (Li₄Ti₅O₁₂/acetylene black/PVDF = 8/1/1 by weight), lithium metal counter electrode, and liquid electrolyte (1 M LiPF₆ in EC:DEC = 1:1 v/v). Then, a LAND CT2001A battery test system was used to examine the electrochemical performance within the potential window of 1.0 and 2.5 V (vs. Li⁺/Li) at 25 °C.

3. Results and discussion

The separator was developed by coating a microporous PE membrane with AlOOH particles on a roller coating equipment. A slurry composed of AlOOH particles (35 wt%), PVA (1 wt%), and distilled water was used as the starting material. Here, PVA serves as a binder due to its excellent adhesive and film-forming properties. In addition, it also has high tensile strength and good flexibility. Based on the thermogravimetric (TG) analysis undertaken after drying (Fig. S1), the absorbed water content in the AlOOH layer accounted for less than 1% of the total weight, and it can be estimated to be less than five in ten thousands of overall content in a battery. The coating thickness was accurately controlled by adjusting the rolling speed. Here, we provide an example of a double-faced coating with a thickness of 1.15 μm. Scanning electron microscopy (SEM) observation indicates that the pristine PE

membrane consisted of randomly interconnected dendrites and non-uniformly dispersed micropores (Fig. 1a), and the coating on the membrane is formed from the assembly of AlOOH particles with an average size of 350 nm (Fig. 1b). The pristine PE membrane had higher porosity and wider pore size distribution than the AlOOH particle-modified PE membrane.

Fig. 2a shows the effect of AlOOH coating on the thermal shrinking of the modified PE membrane. Clearly, thermal stability of the modified PE membrane is significantly improved. It showed almost no shrinkage below 140 °C and was found to have only lost less than 3% of the original area after exposure to 180 °C for 0.5 h. In contrast, the pristine PE membrane showed obvious shrinkage at 110 °C and was found to have lost more than 85% of the original area with an increase in the temperature to 150 °C. A clear contrast can be observed in the digital photographs shown in Fig. 2b, which shows that the modified PE membrane maintained appreciable integrity and flattening at high temperatures. Additionally, we also observed that the modified PE membrane turned from opaque to transparent at temperatures higher than 140 °C, implying that the pores in the PE membrane were closed shut, owing to its melting. This transformation is essentially beneficial for the safety of lithium ion batteries in the case of overheating.

The excellent thermal stability of the AlOOH-coated PE membrane was strongly related to the coating interface structure rather than the coating thickness. Fig. 3a shows the cross-sectional SEM image of the modified PE membrane preheated at 150 °C for 0.5 h. The inset in Fig. 3a clearly shows that a portion of PE was embedded in the AlOOH particle assembly forming an interlocking interface structure, because the molten PE has good affinity to AlOOH so that it easily flows into the interspace between the AlOOH particles under a capillary force. The formation of the interlocking interface structure can significantly increase the contact area between the AlOOH particles and PE, thereby improving the thermal stability of the modified PE membrane. This phenomenon can also be witnessed from the thickness variation in the modified PE membrane during the thermal treatment. For a pristine membrane, shrinkage is usually accompanied by thickening. As shown in Fig. 3b, this rule was essentially conformed to until 140 °C. However, when the temperature was elevated further to 150 °C, the modified PE membrane showed an obvious thinning although it shrank slightly. The interlocking interface structure can enhance the mechanical strength of the membrane. Consequently, the area and thickness of the modified PE membrane remained unchanged as the temperature was increased from 150 to 180 °C. Through peeling tests, we confirmed that the formation of this interface structure can result in enhanced bonding strength between the pristine PE membrane and AlOOH particles. As shown in Fig. 3c, the AlOOH particle coating layer could be easily stripped off the newly modified PE membrane with an adhesive tape, and its adhesion strength was estimated to be 35 N/m (Fig. S2). After exposure to temperatures beyond 140 °C, the AlOOH particle coating could not be thus

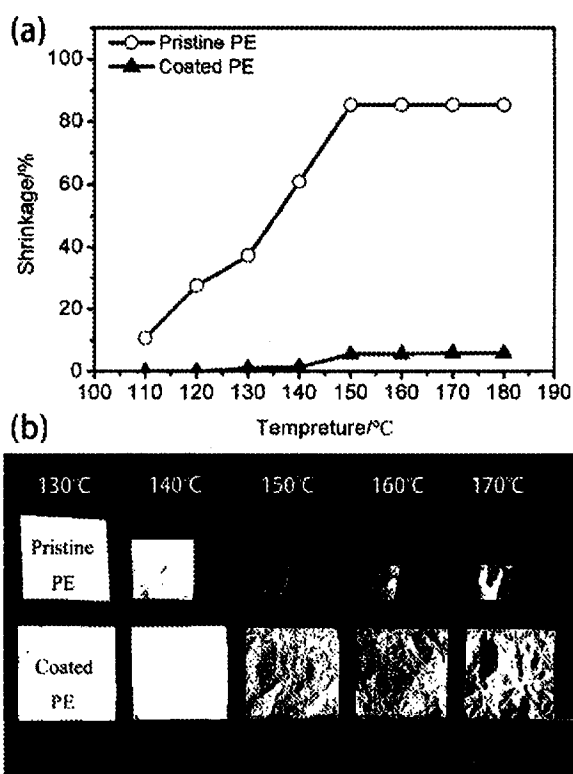


Fig. 2. (a) Temperature dependent shrinkage curves and (b) digital photographs of the pristine PE membrane and AlOOH-coated PE membranes, subjected to thermal treatment between 110 and 180 °C.

stripped off, indicating that the adhesion strength had surpassed 225 N/m.

Besides the thermal stability, some other basic properties of the modified PE membrane were also investigated. As shown in Table 1, the puncture strength and tensile strength of the modified PE membrane were slightly lower than those of the pristine PE membrane, while the permeability was found to decrease considerably by about 20% due to elongation of the gas flow path. Permeability is one of the important factors affecting ion mobility in separators [9]. In general, the thicker the coating, the lower is the permeability and thus the lower is the ion mobility. In this study, the presence of the AlOOH particle coating led to improvement in the wettability to the electrolyte, making up for the decline in the ion mobility resulting from the permeability decrease. As shown in Fig. 4a, the measured contact angle of the electrolyte (1 M LiPF₆ in EC:DEC = 1:1 v/v) on pristine PE membrane was 45.2°, while it was nearly zero on the

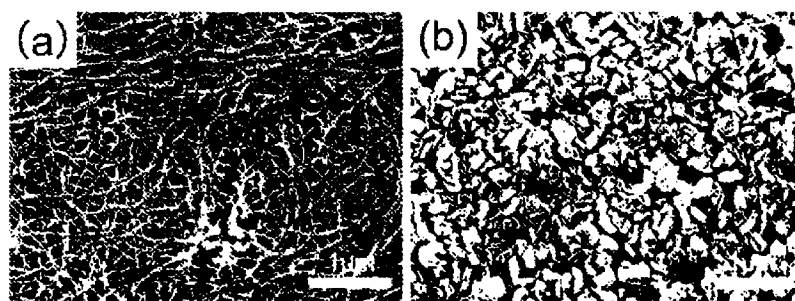


Fig. 3. SEM images showing the surface morphology of the membranes. (a) Pristine PE; (b) AlOOH-coated PE.

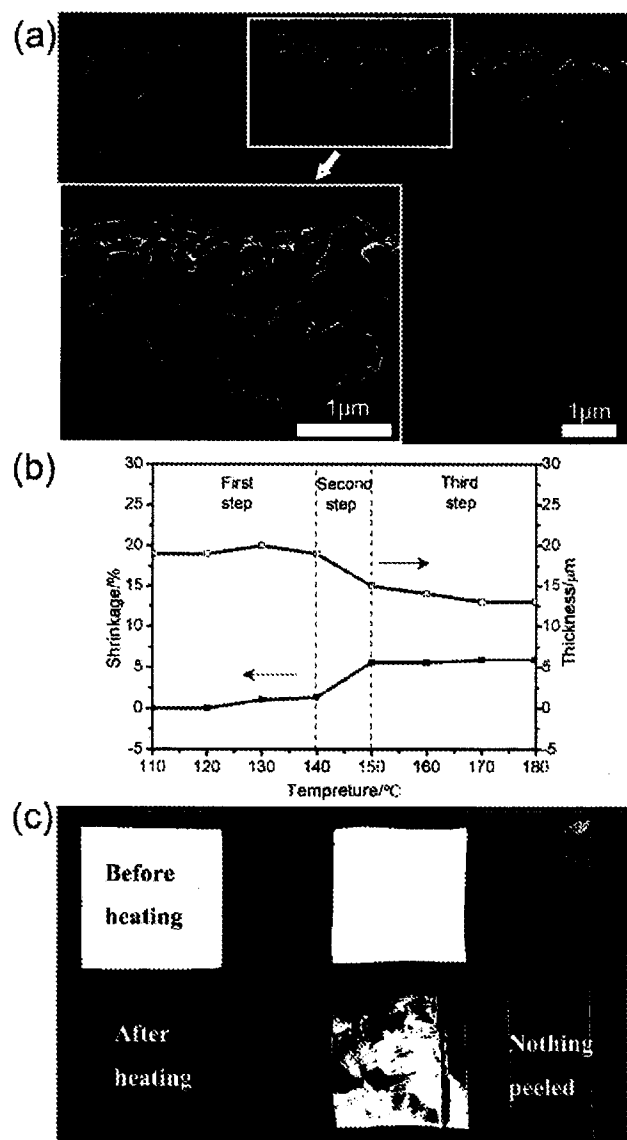


Fig. 3. (a) Cross-sectional SEM image of an AlOOH-coated PE membrane heated at 150 °C for 0.5 h (inset shows an enlarged image of the coating interface); (b) the dependence of the shrinkage and thickness of the AlOOH-coated PE membrane on the heating temperature; (c) digital photographs showing peeling tests on the AlOOH-coated PE membrane before and after heating.

AlOOH-coated PE membrane. Upon simultaneous immersion in a liquid electrolyte, both the pristine and modified PE showed an almost similar soaked height of the electrolyte; however, the modified PE membrane showed stronger capability for electrolyte uptake as shown in Fig. 4b. In summary, the modified PE membrane employed as a separator in lithium ion batteries should facilitate ion transport [22].

The ionic conductivity, an important parameter affecting the

Table 1
Physical properties of the pristine PE separator and the modified PE separator.

Separator	Gurley s/100 ml	Puncture strength/gf	Tensile strength	
			MD/mPa	TD/mPa
Pristine PE	81.0	447.0	105.0	114.5
Modified PE	105.9	430.1	92.7	92.5

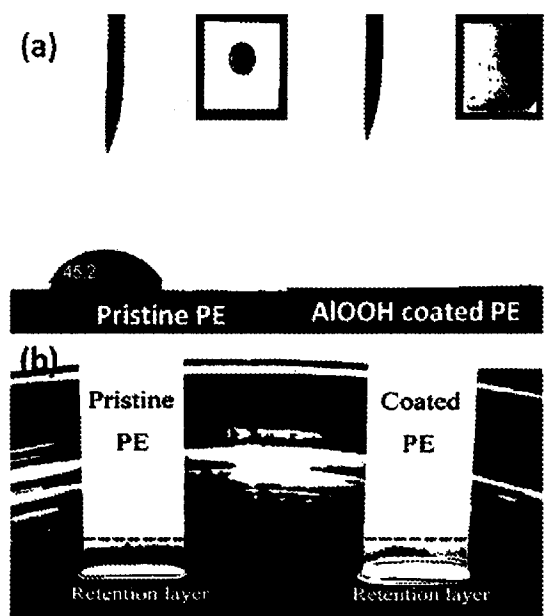


Fig. 4. Digital photographs of (a) contact angle of an electrolyte on the pristine PE membrane (left) and AlOOH coated PE membrane (right); (b) the liquid electrolyte uptake by the pristine PE membrane (left) and the AlOOH-coated PE membrane (right), immersed in a liquid electrolyte, EC: DEC (1:1 v/v containing 1 M LiPF₆) for 0.5 h.

conduction of ionic carriers, was measured. Fig. 5a shows Nyquist plots of the liquid electrolyte-soaked pristine PE membrane and

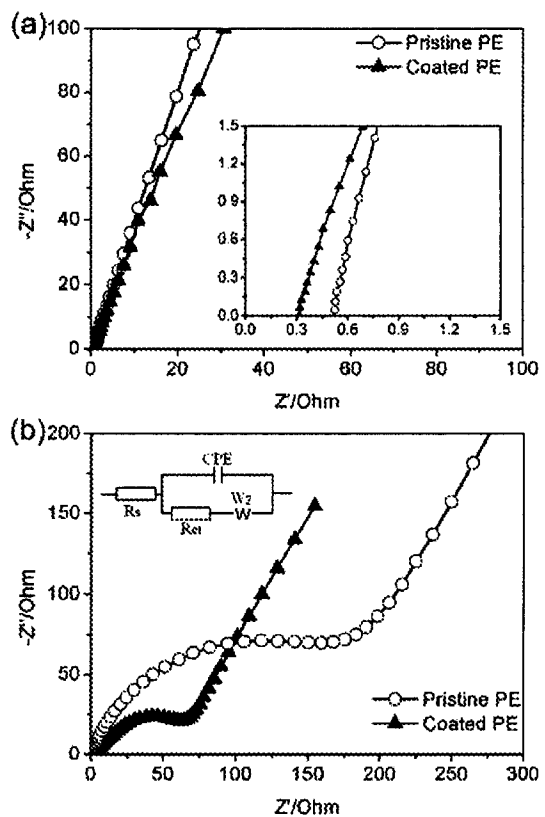


Fig. 5. (a) Ionic conductivities of the liquid electrolyte-soaked separators (inset shows the plots at high-frequency); (b) Nyquist plots for the cells with the equivalent circuit shown in the inset.

modified PE membrane obtained by AC impedance measurements at 25 °C. The intercepts on the real axis of the liquid electrolyte-soaked membranes are 0.51 Ω and 0.31 Ω for the pristine and modified PE membranes (inset in Fig. 5a), respectively. Using Equation (2), ionic conductivity of the modified PE membrane was calculated to be $6.56 \times 10^{-3} \text{ S cm}^{-1}$, while it was $3.48 \times 10^{-3} \text{ S cm}^{-1}$ for the pristine PE membrane. The higher ionic conductivity of the AlOOH-coated membrane may be attributed to its better electrolyte wettability. Furthermore, EIS spectra of the two CR2016 LTO/Li cells that employed the pristine and AlOOH-coated membranes as separators were analyzed to evaluate the interfacial separator resistance in the cells (Fig. 5b). The EIS analysis results were fitted using the equivalent circuit shown in the inset of Fig. 5a [24–26], where R_s denotes the ohmic resistance from the electrode, electrolyte, separator, and current collector, and R_{ct} refers to the charge transfer resistance between the electrolyte and electrode. Based on the fitted data of circuit elements, R_s was calculated to be 2.28 Ω for the AlOOH-coated PE separator, larger than that of the pristine PE separator (0.63 Ω). In contrast, R_{ct} of the AlOOH-coated PE separator was 74.1 Ω , much smaller than that of the pristine PE separator (156.2 Ω), owing to its better electrolyte wettability and membrane-electrode affinity [23,27].

Further evaluation of the modified PE membrane as a separator was done in terms of the C-rate capabilities and cycling performance of two CR2016 coin-type half cells (LTO/Li metal) that employed the pristine and modified PE membranes as separators. Fig. 6 shows the rate performance, where the cells were discharged continuously over current densities ranging from 0.2C to 2C and back to 0.2C for 5 cycles each. For both cells, the discharge capacity

gradually decreased with increase in the discharge current density (Fig. 6a and b). The cell assembled with the modified PE membrane exhibited obviously better rate performance than that of the pristine PE separator at the large current 2C rate (Fig. 6c), which can be attributed to the higher ionic conductivity of the AlOOH-coated PE separator. Cycling performance of the cells, shown in Fig. 6d, was measured with voltages ranging from 1.0 to 2.5 V and a constant charge current density of 1C. In the beginning, the capacity of the cell assembled with the modified PE separator was about 150 mA h g^{-1} , and it retained 96.3% of its initial capacity after 100 cycles. The initial cell capacity of the pristine PE separator was about 142 mA h g^{-1} , and it showed a capacity retention of 94.6% after 100 cycles. The cell employing the modified PE membrane exhibited higher discharge capacity than the one employing a pristine PE membrane for up to 100 cycles.

Moreover, we also found that introduction of the AlOOH particle coating on the PE separator provides excellent overcharge protection. Fig. 7 presents an example of an aluminum-shell full battery consisting of a LiFePO_4 (LFP) cathode, graphite anode, and LiPF_6 electrolyte. The battery that was regularly charged to 4.2 V underwent overcharge up to 10 V under a constant current density of 3 C. It was observed that, with the use of the pristine PE separator, the battery swelled and expelled white smoke at about 15 min after the overcharge (Fig. 7a), and the surface temperature rose to 104.8 °C (Fig. 7b). However, the battery employing an AlOOH-coated PE separator neither deformed nor expelled any smoke (Fig. 7c), thus indicating its superior endurance against overcharge, although the surface temperature rose to 103.4 °C (Fig. 7d). The mechanism of this overcharge protection is proposed to be as

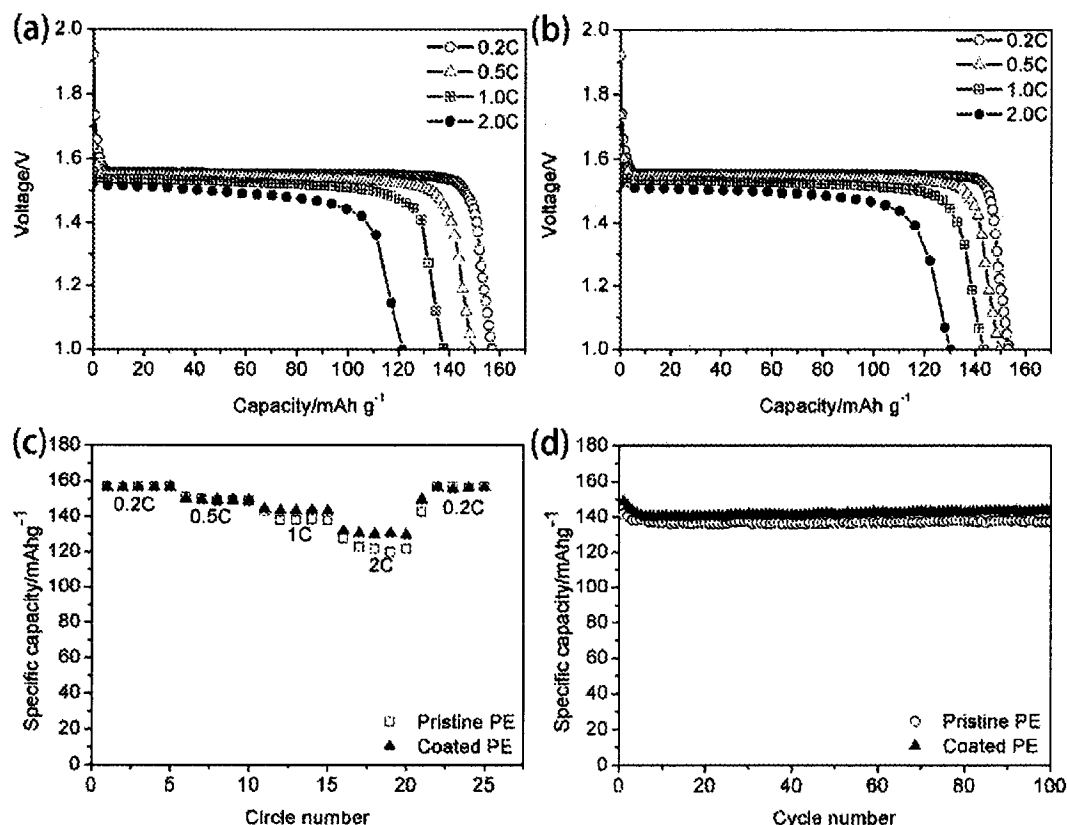


Fig. 6. C-rate charge-discharge profiles of Li/LTO CR2016 coin-type half-cells assembled with (a) a pristine PE separator and (b) an AlOOH-coated PE separator; (c) comparison of the discharge rate capabilities; (d) cycling performances of the two CR2016 coin-type half-cells.

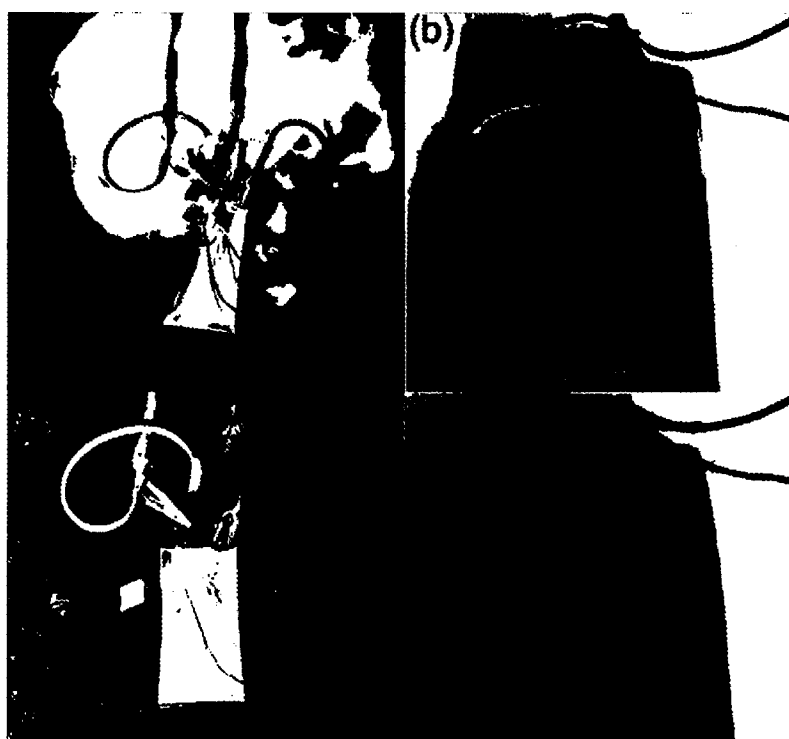


Fig. 7. Demonstration of overcharge in an aluminum-shell full battery employing a pristine PE separator (a, b) and an AlOOH-coated PE separator (c, d).

follows: (i) the strong absorption of LiPF_6 electrolyte on the AlOOH particle coating leads to a reduction in the activity of PF_6^- anions so as to cause an additional overpotential beyond the normal oxidation potential of PF_6^- anions [28]; (ii) the AlOOH particle coating placed in contact with the cathode has a higher oxidation potential and sets the overcharge protection limit to protect the PE from degradation; and (iii) the AlOOH particle coating structurally increases the mechanical strength of the battery. In addition, investigations on the electrochemical performance of LFP/graphite pouch cells (Fig. S3) also show that the AlOOH-coated PE separator can lead to improved C-rate capabilities and cycling performance of cells.

4. Conclusion

In summary, the PE membrane modified by an AlOOH particle coating has been demonstrated to be a promising separator for lithium ion batteries. Due to the formation of an interlocking interface structure with the melting of PE, the modified PE membrane underwent extremely low shrinkage when subjected to thermal treatment up to 180 °C, although the AlOOH coating thickness was only 1.15 μm . Such thin coatings are beneficial in saving space for batteries to gain high energy density. Moreover, the AlOOH-coated PE separator can provide superior electrochemical compatibility and overcharge protection and thus render the batteries more efficient and safer.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jpowsour.2017.02.078>.

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PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference 65213-103	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/US2018/055862	International filing date (<i>day/month/year</i>) 15 October 2018 (15-10-2018)	(Earliest) Priority Date (<i>day/month/year</i>) 13 October 2017 (13-10-2017)
Applicant OPTODOT CORPORATION		

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☒ It is also accompanied by a copy of each prior art document cited in this report.

1. Basis of the report

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☐ a translation of the international application into _____, which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b))

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2. ☐ Certain claims were found unsearchable (See Box No. II)

3. ☒ Unity of invention is lacking (see Box No III)

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This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

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because they relate to subject matter not required to be searched by this Authority, namely:
2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:
3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

see additional sheet

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.
2. ☐ As all searchable claims could be searched without effort justifying an additional fees, this Authority did not invite payment of additional fees.
3. ☒ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

1-10, 13-16
4. ☐ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:

Remark on Protest

- ☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.
- ☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.
- ☒ No protest accompanied the payment of additional search fees.

INTERNATIONAL SEARCH REPORT

International application No
PCT/US2018/055862

A. CLASSIFICATION OF SUBJECT MATTER INV. H01M2/16 H01M10/0525 ADD.		
According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED Minimum documentation searched (classification system followed by classification symbols) H01M		
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched		
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) EPO-Internal, WPI Data		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2016/172705 A1 (HOSHIBA KOJI [JP]) 16 June 2016 (2016-06-16)	1-8,16
Y	par. 83-84, 108, 109 -----	9,10,15
Y	SOLÁ F ET AL: "A new scanning electron microscopy approach to image aerogels at the nanoscale;A new scanning electron microscopy approach to image aerogels at the nanoscale", NANOTECHNOLOGY, IOP, BRISTOL, GB, vol. 22, no. 17, 16 March 2011 (2011-03-16), page 175704, XP020188888, ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704 page 2, second paragraph ----- -/-	9,10,15
☒ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents : "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier application or patent but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone "Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art "&" document member of the same patent family		
Date of the actual completion of the international search 1 April 2019		Date of mailing of the international search report 11/04/2019
Name and mailing address of the ISA/ European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016		Authorized officer Bettio, Andrea

INTERNATIONAL SEARCH REPORT

International application No
PCT/US2018/055862

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	YANG CHONGWEN ET AL: "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries", JOURNAL OF POWER SOURCES, vol. 348, 3 March 2017 (2017-03-03), pages 80-86, XP029949881, ISSN: 0378-7753, DOI: 10.1016/J.JPOWSOUR.2017.02.078	13,14,16
Y	abstract; page 81, left-hand column, 3rd and 4th paragraphs; page 82, figure 2a -----	15
A	US 2017/294638 A1 (JOO DONG JIN [KR] ET AL) 12 October 2017 (2017-10-12) Claim 1 -----	1-10, 13-16

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/US2018/055862

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2016172705 A1	16-06-2016	JP 6399921 B2	03-10-2018
		JP 2016119221 A	30-06-2016
		KR 20160072009 A	22-06-2016
		US 2016172705 A1	16-06-2016

US 2017294638 A1	12-10-2017	CN 107437601 A	05-12-2017
		JP 2017191775 A	19-10-2017
		KR 20170116817 A	20-10-2017
		US 2017294638 A1	12-10-2017

FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

This International Searching Authority found multiple (groups of) inventions in this international application, as follows:

1. claims: 1-10(completely); 16(partially)

A separator for a lithium battery according to present claim
1

2. claims: 11, 12(completely); 16(partially)

A separator for a lithium battery according to present claim
11

3. claims: 13-15(completely); 16(partially)

A separator for a lithium battery according to present claim
13

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

PCT

To:

see form PCT/ISA/220

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY
(PCT Rule 43bis.1)

Date of mailing
(day/month/year) see form PCT/ISA/210 (second sheet)

Applicant's or agent's file reference
see form PCT/ISA/220

FOR FURTHER ACTION
See paragraph 2 below

International application No.
PCT/US2018/055862

International filing date (day/month/year)
15.10.2018

Priority date (day/month/year)
13.10.2017

International Patent Classification (IPC) or both national classification and IPC
INV. H01M2/16 H01M10/0525

Applicant
OPTODOT CORPORATION

1. This opinion contains indications relating to the following items:

- ☒ Box No. I
- Basis of the opinion
- ☐ Box No. II
- Priority
- ☒ Box No. III
- Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☒ Box No. IV
- Lack of unity of invention
- ☒ Box No. V
- Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI
- Certain documents cited
- ☒ Box No. VII
- Certain defects in the international application
- ☐ Box No. VIII
- Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will usually be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA:



European Patent Office
D-80298 Munich
Tel. +49 89 2399 - 0
Fax: +49 89 2399 - 4465

Date of completion of
this opinion

see form
PCT/ISA/210

Authorized Officer

Bettio, Andrea

Telephone No. +49 89 2399-0



**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2018/055862

Box No. I Basis of the opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed.
 - ☐ a translation of the international application into , which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1 (b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43*bis*.1(a))
3. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing:
 - a. ☐ forming part of the international application as filed:
 - ☐ in the form of an Annex C/ST.25 text file.
 - ☐ on paper or in the form of an image file.
 - b. ☐ furnished together with the international application under PCT Rule 13*ter*.1(a) for the purposes of international search only in the form of an Annex C/ST.25 text file.
 - c. ☐ furnished subsequent to the international filing date for the purposes of international search only:
 - ☐ in the form of an Annex C/ST.25 text file (Rule 13*ter*.1(a)).
 - ☐ on paper or in the form of an image file (Rule 13*ter*.1(b) and Administrative Instructions, Section 713).
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that forming part of the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2018/055862

Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability

The questions whether the claimed invention appears to be novel, to involve an inventive step (to be non obvious), or to be industrially applicable have not been examined in respect of

☐ the entire international application

☒ claims Nos. 11, 12

because:

☐ the said international application, or the said claims Nos. relate to the following subject matter which does not require an international search (*specify*):

☐ the description, claims or drawings (*indicate particular elements below*) or said claims Nos. are so unclear that no meaningful opinion could be formed (*specify*):

☐ the claims, or said claims Nos. are so inadequately supported by the description that no meaningful opinion could be formed (*specify*):

☒ no international search report has been established for the whole application or for said claims Nos. 11, 12

☐ a meaningful opinion could not be formed without the sequence listing; the applicant did not, within the prescribed time limit:

☐ furnish a sequence listing in the form of an Annex C/ST.25 text file, and such listing was not available to the International Searching Authority in the form and manner acceptable to it; or the sequence listing furnished did not comply with the standard provided for in Annex C of the Administrative Instructions.

☐ furnish a sequence listing on paper or in the form of an image file complying with the standard provided for in Annex C of the Administrative Instructions, and such listing was not available to the International Searching Authority in the form and manner acceptable to it; or the sequence listing furnished did not comply with the standard provided for in Annex C of the Administrative Instructions.

☐ pay the required late furnishing fee for the furnishing of a sequence listing in response to an invitation under Rule 13ter.1(a) or (b).

☒ See Supplemental Box for further details

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

International application No.
PCT/US2018/055862

Box No. IV Lack of unity of invention

1. ☒ In response to the invitation (Form PCT/ISA/206) to pay additional fees, the applicant has, within the applicable time limit:
- ☒ paid additional fees
 - ☐ paid additional fees under protest and, where applicable, the protest fee
 - ☐ paid additional fees under protest but the applicable protest fee was not paid
 - ☐ not paid additional fees
2. ☐ This Authority found that the requirement of unity of invention is not complied with and chose not to invite the applicant to pay additional fees.
3. This Authority considers that the requirement of unity of invention in accordance with Rule 13.1, 13.2 and 13.3 is
- ☐ complied with
 - ☒ not complied with for the following reasons:
see separate sheet
4. Consequently, this report has been established in respect of the following parts of the international application:
- ☐ all parts.
 - ☒ the parts relating to claims Nos. 1-10, 13-16

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement
- | | | |
|-------------------------------|-------------|------------------------|
| Novelty (N) | Yes: Claims | <u>9, 10, 15</u> |
| | No: Claims | <u>1-8, 13, 14, 16</u> |
| Inventive step (IS) | Yes: Claims | |
| | No: Claims | <u>1-10, 13-16</u> |
| Industrial applicability (IA) | Yes: Claims | <u>1-10, 13-16</u> |
| | No: Claims | |
2. Citations and explanations
- see separate sheet**

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2018/055862

Box No. VII Certain defects in the international application

The following defects in the form or contents of the international application have been noted:

see separate sheet

Re Item III

Non-establishment of opinion with regard to novelty, inventive step and industrial applicability

In reply to the objection to lack of unity, the applicant has paid 1 additional search fee for invention 3. The international search report has been established for inventions 1 and 3.

No opinion will be given in respect of subject-matter which is not covered by the search report (Rule 66.1(e) PCT).

Re Item IV

Lack of unity of invention

Reference is made to the following document:

D1:US 2016/172705 A1 (HOSHIBA KOJI [JP]) 16 June 2016 (2016-06-16)

The present application lacks unity of invention (Rule 13.1 PCT) for the following reasons:

The problem underlying the present invention is the provision of battery separators with increased dimensional stability at temperatures such as 150°C or higher (par. 3 of the present application).

As solution to this problem the applicant proposes:

1)A separator for a lithium battery comprising:

(a) a porous polymeric layer, and

(b) a nanoporous layer adjacent at least one of two opposing sides of said porous polymeric layer, said nanoporous layer comprising boehmite particles and one or more polymers, wherein:

a. a volume fraction of said one or more polymers in the solid portion of said nanoporous layer is between about 15% and about 50%, and

b. said boehmite particles have an average crystallite size between about 5 nm and about 90 nm (present claim 1)

2)A separator for a lithium battery comprising:

(a) a porous polyethylene layer, and

(b) a nanoporous layer adjacent both of two opposing sides of said polyethylene layer, said nanoporous layer comprising boehmite particles and one or more polymers, wherein:

- a. a volume fraction of said one or more polymers in the solid portion of said nanoporous layer is between about 15% and about 50%;
- b. a total coating weight of said nanoporous layer on both of said two opposing sides of said polyethylene layer is about 5.5 grams per square meter or less, and
- c. a heat shrinkage of said separator at 150°C for 1 hour is less than 5% (present claim 11).

3) A separator for a lithium battery comprising:

- a. a porous polymeric layer, and
- b. 0.3 gsm or greater of boehmite particles disposed within a plurality of pores of the porous polymeric layer (present claim 13).

The common features linking the different inventions 1)-2)-3) together are:

a separator for a lithium ion battery comprising

- (a) a porous polymeric layer and,
- (b) boehmite particles.

The common features linking the different inventions (1)-(2) together are:

a separator for a lithium ion battery comprising

- (a) a porous polymeric layer and,
- (b) a nanoporous layer adjacent at least one of two opposing sides of said porous polymeric layer, said nanoporous layer comprising boehmite particles and one or more polymers, said boehmite particles have an average crystallite size between about 5 nm and about 90 nm.

Prior art documents D1 (par. 109) teaches a wound electrode assembly for a non-aqueous electrolyte rechargeable battery in which deformation is suppressed and improving heat resistance and energy density are simultaneously improved by improving a handling property of the separator during the manufacturing process.

According to D1 (par. 83-84) said lithium ion battery is provided with a separator comprising a layer consisting of a fluorine resin-containing particulate (which can be considered as a binder), a polyethylene ionomer (a binder) sodium polyacrylate and

boehmite particles having an average particle diameter of 90 nm mixed in a volume ratio of 40:10:0.3: 50 respectively. Said layer is attached to a 12 microns thick layer consisting of a porous polyethylene film.

The idea of a separator for a lithium ion battery comprising

- (a) a porous polymeric layer and,
- (b) boehmite particles.

is known in the prior art, and cannot serve as single general inventive concept linking the individual subjects in which the application has been divided.

The same applies to the idea of

- (a) a porous polymeric layer and,
- (b) a nanoporous layer adjacent at least one of two opposing sides of said porous polymeric layer, said nanoporous layer comprising boehmite particles and one or more polymers, said boehmite particles have an average crystallite size between about 5 nm and about 90 nm.

Starting from said common concepts, invention 1 comprises the additional features of providing a given volume fraction of polymers (present claim 2), a different average particle size of the boehmite particles, (present claim 3), a different heat shrinkage (present claim 4), a different total coating weight of the nanoporous layer on both of the two opposite sides of the porous polymer layer (present claim 5), a different porosity, or pore volume or average pore diameter, or pore diameter (present claim 6), a different material of the porous polymeric layer (present claim 7), a different structure of the nanoporous layer (present claim 8), providing a separator for a lithium battery wherein the boehmite particles have an average crystallite size between about 5 nm and about 8 nm.

The application as filed does not teach which technical problem is solved by said features.

Invention 2 on the other hand comprises the additional features of

- a) porous polyethylene layer
- b) a nanoporous layer adjacent both of two opposite sides of said polyethylene layers wherein
 - a. a volume fraction of said one or more polymers in the solid portion of said nanoporous layer is between about 15% and about 50%.

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AUTHORITY (SEPARATE SHEET)**

International application No.

PCT/US2018/055862

b. a total coating weight of said nanoporous layer on both of said two opposing sides of said polyethylene layer is about 5.5 grams per square meter or less, and

c. a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.

The application as filed does not teach which technical problem is solved by said features.

Invention 3 comprises the additional feature of

b. 0.3 gsm or greater of boehmite particles disposed within a plurality of pores of the porous polymeric layer.

The application as filed does not teach which technical problem is solved by said features.

Thus there is no same or corresponding special technical feature in the sense of rule 13.1 PCT.

As there is no other technical feature which could fulfil the role of special technical feature in the sense of rule 13 PCT, the present application lacks unity of invention containing the following subjects:

1. claims: 1-10(completely); 16(partially)

A separator for a lithium battery according to present claim 1

2. claims: 11, 12(completely); 16(partially)

A separator for a lithium battery according to present claim 11

3. claims: 13-15(completely); 16(partially)

A separator for a lithium battery according to present claim 13.

In conclusion, the groups of claims are not linked by common or corresponding special technical features and define 3 different inventions not linked by a single general inventive concept.

The application, hence does not meet the requirements of unity of invention as defined in Rules 13.1 and 13.2 PCT.

Re Item V

Reasoned statement with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

Reference is made to the following documents:

- D1 US 2016/172705 A1 (HOSHIBA KOJI [JP]) 16 June 2016 (2016-06-16)
- D2 SOLÁ F ET AL: "A new scanning electron microscopy approach to image aerogels at the nanoscale;A new scanning electron microscopy approach to image aerogels at the nanoscale",
NANOTECHNOLOGY, IOP, BRISTOL, GB,
vol. 22, no. 17, 16 March 2011 (2011-03-16), page 175704,
XP020188888,
ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704
- D3 YANG CHONGWEN ET AL: "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries",
JOURNAL OF POWER SOURCES,
vol. 348, 3 March 2017 (2017-03-03), pages 80-86, XP029949881,
ISSN: 0378-7753, DOI: 10.1016/J.JPOWSOUR.2017.02.078

Invention 1

Art. 33(2) PCT

D1 (paragraphs 83, 84, 108, 109) teaches a wound electrode assembly for a non-aqueous electrolyte rechargeable battery in which deformation is suppressed and improving heat resistance and energy density are simultaneously improved by improving a handling property of the separator during the manufacturing process.

According to D1 (par. 83-84) said lithium ion battery is provided with a separator comprising a layer consisting of a fluorine resin-containing particulate (which can be considered as a binder), a polyethylene ionomer (a binder) sodium polyacrylate and boehmite particles having an average particle diameter of 90 nm mixed in a volume ratio of 40:10:0.3: 50 respectively. Said layer is attached to a 12 microns thick layer consisting of a porous polyethylene film.

As the structure and composition of the separator of D1 fall under the subject matter of present claim 1, the technical features of present claims 4-7, even if not explicitly taught, are necessarily implicitly disclosed in D1.

Consequently the subject matter of present claims 1-8, 16 is not novel in the sense of Art. 33(2) PCT over D1.

Art. 33(3) PCT

The problem underlying the present invention is the provision of battery separators with increased dimensional stability at temperatures such as 150°C or higher (par. 3 of the present application).

D1 may be considered as the closest prior art for the subject matter of present claims 9-10.

The difference between the subject matter of claims 9 and 10 and D1 is that in the latter the average particle diameter is 90 nm instead of between about 5 nm and about 8 nm.

The data disclosed in table 1 on pages 23-24 of the present application show that a separator produced with a boehmite primary particle (Disperal D60) characterized by a primary particle size of 75 nm has a total coat weight for < 5% shrinkage at 150°C for 1 hour of ≥ 3.9 gsm; on the other hand a separator produced with a boehmite particle (Dispall 25f4) characterized by a primary particle size of 8 nm has a total coat weight for < 5% shrinkage at 150°C for 1 hour of ≥ 3.6 gsm.

There is therefore no technical effect deriving from said difference.

The resulting objective technical problem is the provision of alternative materials for the preparation of the separator of D1.

D2 (page 2, second paragraph) discloses Sasol Dispall 25F4 as a boehmite powder characterized by 7.2 nm crystallite size.

It would be therefore obvious to the skilled person to replace the boehmite particles of D1 with Dispall 25F4, thereby coming to the same subject matter of present claim 9.

As the structure and composition of the separator obtained by the combined teachings of D1 and D2 fall under the subject matter of present claim 10, the subject matter of said claim 10 is also obvious.

The subject matter of present claims 9-10 is therefore not inventive in view of the combined teachings of D1 and D2.

Invention 3

Art. 33(2) PCT

D3 (abstract; page 81, left-hand column, 3rd and 4th paragraphs; page 82, figure 2a) discloses a separator for a lithium ion secondary battery obtained by coating both faces of a 16-micron thick polyethylene porous membrane with a slurry composed of 35% boehmite particles having a 350 nm average particle size, 1% wt polyvinyl alcohol as a binder and distilled water; after drying this results in a 1.15 micron thick boehmite layer on both sides of the PE membrane. The separator shrinkage at 150 °C is about 5 % which, according to the teaching of figures 1-10 of the present application, corresponds to a coating weight significantly higher than 0.36 gsm.

Thus the subject matter of present claims 13, 14 and 16 is not novel in the sense of Art. 33(2) PCT over D3.

Art. 33(3) PCT

D3 represents the closest prior art to the subject matter of present claim 15. No technical effect can be derived from the additional features of claim 15. The resulting objective technical problem may be formulated as the provision of an alternative separator for lithium ion secondary batteries.

By combining D3 with D1 the skilled person would obtain all the features of present claim 15 except the particle size of the boehmite particles.

D2 (page 2, second paragraph) discloses a boehmite powder characterized by 7.2 nm crystallite size.

Thus the subject matter of present claim 15 is not inventive in the sense of Art 33(3) PCT in view of the combined teachings of D3, D1 and D2.

Re Item VII

Certain defects in the international application

The wording "about" in present claims 1, 2, 3, 6, 9, 11-13, 15 renders the subject matter of said claims unclear.

Present claims 1, 3, 9 discloses the parameter "average particle size".

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The claims do not disclose the method of experimental determination of said parameter. The subject matter of said claims is therefore not clear.



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华知识产权代理有限公司
王达佐 (010-58732666) 洪欣 (010-58732666)

发文日:

2022 年 04 月 12 日



申请号或专利号: 201880079213.4

发文序号: 2022040702380310

申请人或专利权人: 奥普图多特公司

发明创造名称: 多层纳米多孔分隔物

第一次审查意见通知书

(进入国家阶段的 PCT 申请)

1. ☒ 应申请人提出的实质审查请求, 根据专利法第 35 条第 1 款的规定, 国家知识产权局对上述发明专利申请进行实质审查。
☐ 根据专利法第 35 条第 2 款的规定, 国家知识产权局决定自行对上述发明专利申请进行实质审查。
2. ☒ 申请人要求以其在:
US 专利局的申请日 2017 年 10 月 13 日为优先权日。
3. ☐ 经审查, 申请人于_____提交的修改文件, 不符合专利法实施细则第 51 条第 1 款的规定, 不予接受。
4. ☐ 审查是针对原始提交的国际申请的中文文本或中文译文进行的。
☒ 审查是针对下列申请文件进行的:
依据专利合作条约第 28 条或者第 41 条提交的修改, 权利要求第 1-21 项;
2020 年 6 月 8 日提交的说明书摘要、说明书第 1-102 段、说明书附图。
5. ☒ 本通知书引用下列对比文献 (其编号在今后的审查过程中继续沿用)

编号	文 件 号 或 名 称	公 开 日 期 (或抵触申请的申请日)
1	CN 105247703A	20160113
2	US 2016172705A1	20160616
3	US 2017012264A1	20170112

6. 审查的结论性意见:

关于说明书:

- ☐ 申请的内容属于专利法第 5 条规定的不授予专利权的范围。
☐ 说明书不符合专利法第 26 条第 3 款的规定。
☐ 说明书不符合专利法第 33 条的规定。
☐ 说明书的撰写不符合专利法实施细则第 17 条的规定。
☐ _____

210402
2021. 6

纸件申请, 回函请寄: 100088 北京市海淀区蓟门桥西土城路 6 号 国家知识产权局专利局受理处收
电子申请, 应当通过电子专利申请系统以电子文件形式提交相关文件。除另有规定外, 以纸件等其他形式提交的文件视为未提交。



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关于权利要求书：

- ☐ 权利要求_____不符合专利法第 2 条第 2 款的规定。
- ☐ 权利要求_____不符合专利法第 9 条第 1 款的规定。
- ☐ 权利要求_____不具备专利法第 22 条第 2 款规定的新颖性。
- ☒ 权利要求 1-2, 4-21 不具备专利法第 22 条第 3 款规定的创造性。
- ☐ 权利要求_____不具备专利法第 22 条第 4 款规定的实用性。
- ☐ 权利要求_____属于专利法第 25 条规定的不授予专利权的范围。
- ☐ 权利要求_____不符合专利法第 26 条第 4 款的规定。
- ☐ 权利要求_____不符合专利法第 31 条第 1 款的规定。
- ☒ 权利要求 3 不符合专利法第 33 条的规定。
- ☐ 权利要求_____不符合专利法实施细则第 19 条的规定。
- ☐ 权利要求_____不符合专利法实施细则第 20 条的规定。
- ☐ 权利要求_____不符合专利法实施细则第 21 条的规定。
- ☐ 权利要求_____不符合专利法实施细则第 22 条的规定。

- ☐ 申请不符合专利法第 26 条第 5 款或者实施细则第 26 条的规定。
- ☐ 申请不符合专利法第 19 条第 1 款的规定。
- ☐ 分案申请不符合专利法实施细则第 43 条第 1 款的规定。

上述结论性意见的具体分析见本通知书的正文部分。

7. 基于上述结论性意见，审查员认为：

- ☐ 申请人应当按照通知书正文部分提出的要求，对申请文件进行修改。
- ☐ 申请人应当在意见陈述书中论述其专利申请可以被授予专利权的理由，并对通知书正文部分中指出的不符合规定之处进行修改，否则将不能授予专利权。
- ☒ 专利申请中没有可以被授予专利权的实质性内容，如果申请人没有陈述理由或者陈述理由不充分，其申请将被驳回。
- ☐ _____

8. 申请人应注意下列事项：

- (1) 根据专利法第 37 条的规定，申请人应在收到本通知书之日起的 4 个月内陈述意见，如果申请人无正当理由逾期不答复，其申请被视为撤回。
- (2) 申请人对其申请的修改应当符合专利法第 33 条的规定，不得超出原说明书和权利要求书记载的范围，同时申请人对专利申请文件进行的修改应当符合专利法实施细则第 51 条第 3 款的规定，按照本通知书的要求进行修改。
- (3) 申请人的意见陈述书和 / 或修改文本应当邮寄或递交国家知识产权局专利局受理处，凡未邮寄或递交给受理处的文件不具备法律效力。
- (4) 未经预约，申请人和 / 或代理人不得前来国家知识产权局与审查员举行会晤。
- (5) 对进入实质审查阶段的发明专利申请，在第一次审查意见通知书答复期限届满前（已提交答复意见的除外），主动申请撤回的，可以请求退还 50% 的专利申请实质审查费。

9. 本通知书正文部分共有 3 页，并附有下列附件：

- ☐ 引用的对比文件的复印件共_____份_____页。
- ☐ _____

审查员：司慧玲

联系电话：010-53966533

审查部门：专利审查协作北京中心电学发明审查部

210402
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纸件申请，回函请寄：100088 北京市海淀区蓟门桥西土城路 6 号 国家知识产权局专利局受理处收
电子申请，应当通过电子专利申请系统以电子文件形式提交相关文件。除另有规定外，以纸件等其他形式提交的文件视为未提交。



国家知识产权局

第一次审查意见通知书

(进入国家阶段的 PCT 申请)

申请号:2018800792134

本申请涉及一种多层纳米多孔分隔物, 及包括此类分隔物的锂蓄电池。经审查, 现提出如下审查意见。

一、权利要求 1-2, 4-21 不具备专利法第 22 条第 3 款规定的创造性。

1. 权利要求 1 要求保护一种用于锂蓄电池的分隔物。对比文件 1 (CN105247703A, 参见说明书第 5-10 段, 第 37-41 段, 附图 5-8) 公开了一种电池组用多孔分隔物, 其主要公开了如下技术特征: 电池组用多孔分隔物 (对应本申请中的纳米多孔层) 包含陶瓷颗粒和聚合物黏合剂 (对应本申请中的纳米多孔层中的一种或多种聚合物), 陶瓷颗粒选自无机氧化物颗粒 (对应本申请中的纳米多孔层中的无机氧化物颗粒) 和无机氮化物颗粒, 多孔分隔物在暴露于 200℃ 的温度至少一小时的情况下表现出小于 1% 的收缩率 (对应本申请中的分隔物在 200℃ 下持续 1 小时的任何收缩均小于 5%); 在实施例中的图示 501 的聚合物分隔物是厚度为 18 μm、Gurley 透气性为 300 秒/100cc 的 Tonen® 三层聚烯烃 (对应本申请中的多孔聚合物层), 图示 503 的分隔物在单侧涂覆有包含 5.5:1 比例的勃姆石 (Disperal® 10SR) 和 Arkema Kynar® 761 的 3.5 μm 层; 图示 505 的分隔物在双侧涂覆有包含 5.5:1 比例的勃姆石 (Disperal® 10SR) 和 Arkema Kynar® 761 的 3.5 μm 层, 对应本申请中的在聚合物层的两个相对侧的至少一侧涂覆纳米多孔层; 另外, 对比文件 1 还公开了一种电化学电池, 包括锂盐的无机电解质和此多孔分隔物, 表明此电池组用多孔分隔物可用于锂蓄电池。权利要求 1 所要求保护的技术方案与对比文件 1 公开的技术内容相比, 区别技术特征是: 纳米多孔层的总涂层重量为 4.5 克/平方米或更小。基于上述区别技术特征, 可以确定权利要求 1 相对于对比文件 1 实际解决的技术问题是如何使纳米多孔层的重量较轻且热收缩率较小。

对于上述区别技术特征, 对比文件 2 (US2016172705A1, 参见说明书第 50, 83-109 段) 公开了一种耐热的隔板, 包括含氟树脂和勃姆石的涂层; 同时采用相同的方法制备了具有不同厚度的涂层的隔板并比较了其收缩率, 实验结果表明当涂层的厚度较小时, 隔板的收缩较大, 且当涂层厚度比较小时, 电池的厚度可更薄且具有更高的能量密度。对比文件 2 给出了隔板的收缩率与涂层厚度有关的启示, 因此, 为了寻求电池稳定性与能量密度之间的平衡, 本领域技术人员经过有限次实验即可确定得到的纳米多孔层的总涂层重量的合适范围为 4.5 克/平方米或更小。

由此可见, 本领域技术人员在对比文件 1 的基础上结合对比文件 2 以及公知常识就可以得到权利要求 1 请求保护的技术方案, 即权利要求 1 请求保护的技术方案对本领域技术人员是显而易见的, 不具有突出的实质性特点。因此, 权利要求 1 不具备专利法第 22 条第 3 款规定的创造性。

2. 权利要求 2 对权利要求 1 作了进一步的限定, 对比文件 1 (参见说明书第 5-10 段, 第 37-41 段, 附图 5-8) 中公开了图示 503 的聚合物分隔物是厚度为 18 μm、Gurley 透气性为 300 秒/100cc 的 Tonen® 三层聚烯烃, 因此, 在引用的权利要求 1 不具备创造性时, 权利要求 2 也不具备专利法第 22 条第 3 款规定的创造性。

3. 权利要求 4, 5 对权利要求 1 作了进一步的限定, 对比文件 1 (参见说明书第 5-10 段, 第 37-41 段, 附图 5-8) 中公开了图示 503 的分隔物包含 5.5:1 比例的勃姆石 (Disperal® 10SR) 和 Arkema Kynar® 761。在对比文件 1 给出的勃姆石与聚合物比例的教导下, 为了在提高能量密度时保证分隔物的高温稳定性, 本领域技术人员经过有限次的实验即可确定得到一种或多种聚合物在纳米多孔层的固体部分中的体积分数合适的范围。因此, 在引用的权利要求 1 不具备创造性时, 权利要求 4 和 5 不具备专利法第 22 条第 3 款规定的创造性。

4. 权利要求 6 对权利要求 1 作了进一步的限定, 对比文件 1 (参见说明书第 5-10 段, 第 37-41 段, 附图 5-8) 中已公开无机陶瓷颗粒包括 Al₂O₃、AlO(OH) 或勃姆石、AlN、BN、SiN、ZnO、ZrO₂、SiO₂ 及其组合。因此, 在引用的权利要求 1 不具备创造性时, 权利要求 6 也不具备专利法第 22 条第 3 款规定的创造性。

5. 权利要求 7 对权利要求 6 作了进一步的限定, 对比文件 2 (参见说明书第 50, 83-109 段) 公开了一种耐热的隔板, 包括含氟树脂和耐热颗粒的涂层, 耐热填料颗粒的平均粒径为约 10nm 至约 100nm; 耐热无机填料颗粒可以是例如陶瓷颗粒或例如金属氧化物颗粒, 金属氧化物颗粒可以是例如氧化铝、勃姆石、二氧化钛、氧化锆、氧化镁、氧化锌、氢氧化铝、氢氧化镁等的微粒。另外, 对比文件 2 中还公开了一种隔板中使用的勃姆石平均粒径是 90nm, 落入了权利要求 7 的保护范围。因此, 在引用的权利要求 6 不具备创造性时, 权利



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要求 7 不具备专利法第 22 条第 3 款规定的创造性。

6. 权利要求 8 对权利要求 7 作了进一步的限定，在对比文件 2（参见说明书第 50，83-109 段）公开的耐热填料颗粒的平均粒径为约 10nm 至约 100nm 的教导下，为了提高耐热性，本领域技术人员经过有限次的实验即可确定得到勃姆石颗粒的平均微晶尺寸。因此，在引用的权利要求 7 不具备创造性时，权利要求 8 也不具备专利法第 22 条第 3 款规定的创造性。

7. 权利要求 9 对权利要求 1 作了进一步的限定，对比文件 3（US 2017012264A1，参见说明书 25-34 段）公开了一种隔板层，包括勃姆石、聚合物及其共混物，其中的勃姆石可以是有机磺酸改性处理形成的改性水合氧化铝（对应于本申请中的疏水勃姆石）。因此，在引用的权利要求 1 不具备创造性时，权利要求 9 也不具备专利法第 22 条第 3 款规定的创造性。

8. 权利要求 10 对权利要求 1 作了进一步的限定，对比文件 1（参见说明书第 5-10 段，第 37-41 段，附图 5-8）公开了图示 505 的分隔物在双侧涂覆有包含 5.5:1 比例的勃姆石(Disperal® 10SR)和 Arkema Kynar® 761 的 3.5 μ m 层，对应于本申请中的多孔聚合物层的两个相对侧的两侧上涂覆有纳米多孔层。对比文件 2（参见说明书第 50，83-109 段）公开了分隔物的收缩率与其厚度有关，分隔物的厚度小，收缩就变大，但电池的能量密度也变大。因此，为了寻求电池稳定性和能量密度之间的平衡，本领域技术人员经过有限次的实验即可确定得到相应的总涂层重量的合适范围。因此，在引用的权利要求 1 不具备创造性时，权利要求 10 也不具备专利法第 22 条第 3 款规定的创造性。

9. 权利要求 11，12 对权利要求 1 作了进一步的限定，对比文件 1（参见说明书第 5-10 段，第 37-41 段，附图 5-8）公开了分隔物的平均孔隙尺寸是 10nm 至 90nm；在一些情况下，少于 1% 的孔具有在 10nm 至 90nm 之外的尺寸，落入了权利要求 11 和 12 的保护范围。因此，在引用的权利要求 1 不具备创造性时，权利要求 11 和 12 也不具备专利法第 22 条第 3 款规定的创造性。

10. 权利要求 13 对权利要求 1 作了进一步的限定，对比文件 2（参见说明书第 83-84 段）公开了将粘合剂层混合浆料涂覆到 12 μ m 厚的电晕处理的多孔聚乙烯隔膜（对应本申请中的聚乙烯）的两侧上并干燥，以制造涂层。因此，在引用的权利要求 1 不具备创造性时，权利要求 13 也不具备专利法第 22 条第 3 款规定的创造性。

11. 权利要求 14 要求保护一种锂蓄电池。对比文件 1（参见说明书第 5-10 段，第 37-41 段，附图 5-8）公开了一种电化学电池，其主要公开了以下技术特征：电化学电池，其包括阳极、阴极、包括锂盐的无机电解质、和包括有机聚合物（对应本申请中的纳米多孔层中的聚合物）和陶瓷材料的多孔分隔物层（对应本申请中的纳米多孔层），其中，多孔分隔物层具有 35% 至 50% 的孔隙率和 10nm 至 90nm 的平均孔隙尺寸，并且在暴露于 200℃ 的温度至少一小时的情况下表现出小于 1% 的收缩率（对应本申请中的分隔物在 200℃ 下持续 1 小时的任何收缩均小于 5%）；无机陶瓷颗粒选自无机氧化物颗粒（对应本申请中的纳米多孔层的无机氧化物颗粒）和无机氮化物颗粒；在实施例中的图示 501 的聚合物分隔物是厚度为 18 μ m、Gurley 透气性为 300 秒/100cc 的 Tonen® 三层聚烯烃（对应本申请中的包含聚烯烃的多孔聚合物层），图示 503 的分隔物在单侧涂覆有包含 5.5:1 比例的勃姆石(Disperal® 10SR)和 Arkema Kynar® 761 的 3.5 μ m 层；图示 505 的分隔物在双侧涂覆有包含 5.5:1 比例的勃姆石(Disperal® 10SR)和 Arkema Kynar® 761 的 3.5 μ m 层，对应本申请中的在聚合物层的两个相对侧的至少一侧涂覆纳米多孔层。权利要求 14 所要求保护的技术方案与对比文件 1 公开的技术内容相比，区别技术特征是：纳米多孔层的总涂层重量为 4.5 克/平方米或更小。基于上述区别技术特征，可以确定权利要求 14 相对于对比文件 1 实际解决的技术问题是如何使纳米多孔层的重量较轻且热收缩率较小。

对于上述区别技术特征，参见权利要求 1 的评述。

由此可见，本领域技术人员在对比文件 1 的基础上结合对比文件 2 以及公知常识就可以得到权利要求 14 请求保护的技术方案，即权利要求 14 请求保护的技术方案对本领域技术人员是显而易见的，不具有突出的实质性特点。因此，权利要求 14 不具备专利法第 22 条第 3 款规定的创造性。

12. 权利要求 15，16 对权利要求 14 作了进一步的限定，二者的附加技术特征的评述参见权利要求 4 和 5 的评述。因此，在引用的权利要求 14 不具备创造性时，权利要求 15 和 16 也不具备专利法第 22 条第 3 款规定的创造性。

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纸件申请，回函请寄：100088 北京市海淀区蓟门桥西土城路 6 号 国家知识产权局专利局受理处收
电子申请，应当通过电子专利申请系统以电子文件形式提交相关文件。除另有规定外，以纸件等其他形式提交的文件视为未提交。



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13. 权利要求 17 对权利要求 14 作了进一步的限定，对比文件 2（参见说明书第 50，83-109 段）公开了耐热填料颗粒的平均粒径为约 10nm 至约 100nm，且耐热填料颗粒可以是勃姆石。在对比文件 2 给出的勃姆石颗粒的尺寸的教导下，为了提高耐热性，本领域技术人员经过有限次的实验即可确定得到勃姆石颗粒的平均微晶尺寸。因此，在引用的权利要求 14 不具备创造性时，权利要求 17 也不具备专利法第 22 条第 3 款规定的创造性。

14. 权利要求 18-21 对权利要求 14 作了进一步的限定，对权利要求 18-21 的附加技术特征的评述参见权利要求 10-13 的评述。因此，在引用的权利要求 14 不具备创造性时，权利要求 18-21 也不具备专利法第 22 条第 3 款规定的创造性。

二、权利要求 3 不符合专利法第 33 条的规定。

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审查员姓名:司慧玲
审查员代码:313327



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检索报告

申请号：2018800792134		申请日：20181015		首次检索	
申请人：奥普图多特公司		最早的优先权日：20171013			
权利要求项数：16		说明书段数：102+11			
审查员确定的 IPC 分类号：H01M2/16,H01M10/0525					
检索记录信息：CN105247703A: CNTXT 语义检索,语义基准:2018800792134					
US2017012264A1: 引文查询					
US2016172705A1: 引文查询					
CN101796668A: 引文关联浏览					
相 关 专 利 文 献					
类型	国别以及代码[11] 给出的文献号	代码[43]或[45] 给出的日期	IPC 分类号	相关的段落 和 / 或图号	涉及的权 利要求
Y	CN105247703A	20160113	H01M2/16	说明书第 5-10 段,37-41 段, 附图 5-8	1-2,4-21
Y	US2016172705A1	20160616	H01M4/04	说明书第 50, 83-109 段	1-2,4-21
Y	US2017012264A1	20170112	H01G11/52	说明书第 25-34 段	9
A	CN101796668A	20100804	H01M2/16	全文	1-21



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审 查 员: 司慧玲
2022 年 03 月 24 日

审查部门: 专利审查协作北京中心电学发明审查部



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Primary Examiner
Name: Bettio, Andrea
Tel: +49 89 2399 - 8346

Application No. 18 799 922.2 - 1108	Ref. OPTO-003-EP	Date 08.09.2023
Applicant LG Energy Solution, Ltd.		

Communication pursuant to Article 94(3) EPC

The examination of the above-identified application has revealed that it does not meet the requirements of the European Patent Convention for the reasons enclosed herewith. If the deficiencies indicated are not rectified the application may be refused pursuant to Article 97(2) EPC.

You are invited to file your observations and insofar as the deficiencies are such as to be rectifiable, to correct the indicated deficiencies within a period

of 4 months

from the notification of this communication, this period being computed in accordance with Rules 126(2) and 131(2) and (4) EPC. One set of amendments to the description, claims and drawings is to be filed within the said period on separate sheets (R. 50(1) EPC).

If filing amendments, you must identify them and indicate the basis for them in the application as filed. Failure to meet either requirement may lead to a communication from the Examining Division requesting that you correct this deficiency (R. 137(4) EPC).

Failure to comply with this invitation in due time will result in the application being deemed to be withdrawn (Art. 94(4) EPC).

Registered Letter

EPO Form 2001 11.16AUWF (05.09.2023)



Bettio, Andrea
Primary Examiner
For the Examining Division

Enclosure(s): 2 page/s reasons (EPO Form 2906)

The examination is being carried out on the **following application documents**

Description, Pages

1-26	filed in electronic form on	01-06-2023
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Claims, Numbers

1-9	filed in electronic form on	01-06-2023
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Drawings, Sheets

1/6-6/6	as published
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Reference is made to the following documents:

- | | |
|----|---|
| D1 | US 2016/172705 A1 (HOSHIBA KOJI [JP]) 16 June 2016 (2016-06-16) |
| D2 | SOLÁ F ET AL: "A new scanning electron microscopy approach to image aerogels at the nanoscale;A new scanning electron microscopy approach to image aerogels at the nanoscale",
NANOTECHNOLOGY, IOP, BRISTOL, GB,
vol. 22, no. 17, 16 March 2011 (2011-03-16), page 175704, XP020188888,
ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704 |
| D3 | YANG CHONGWEN ET AL: "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries",
JOURNAL OF POWER SOURCES,
vol. 348, 3 March 2017 (2017-03-03), pages 80-86, XP029949881,
ISSN: 0378-7753, DOI: 10.1016/J.JPOWSOUR.2017.02.078 |

Art. 84 EPC

Claim 1 is not supported by the description as required by Article 84 EPC, as its scope is broader than justified by the description. Claim 1 discloses the wording "one or more polymer binders that provide adhesive and cohesive strength to the nanoporous layer". Whilst the chemical nature of said polymers is not disclosed by the claim's wording, the

teaching of par. 37 of the present application is that only some specific classes of polymers may be used for said purpose, namely "cellulosic polymers...vinyl pyrrolidone polymers and copolymers, polysaccharides, polyethylene oxides and polyvinyl alcohol.". The subject matter of present Claim 1 therefore does not comply with the requirements of ART. 84 EPC.

30/30/24



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PATENT AND TRADEMARK OFFICE

ELECTRONIC PAYMENT RECEIPT

APPLICATION # 18/631,783	RECEIPT DATE / TIME 04/10/2024 04:21:09 PM Z ET	ATTORNEY DOCKET # 00249/62
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Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	1021	FILED BY	Joy Chapman
PATENT CENTER #	65043215	AUTHORIZED BY	Brian Amos
CUSTOMER #	1912	FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	David W. Avison

Payment Information

PAYMENT METHOD CARD / 1010	PAYMENT TRANSACTION ID E202440G22228508	PAYMENT AUTHORIZED BY Joy Chapman
PRE-AUTHORIZED ACCOUNT 011785	PRE-AUTHORIZED CATEGORY 37 CFR 1.16 (National application filing, search, and examination fees); 37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.19 (Document supply fees); 37 CFR 1.20 (Post issuance fees); 37 CFR 1.21 (Miscellaneous fees and charges)	

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1111	UTILITY PATENT APPL. SEARCH FEE	700.00	1	700.00
1011	BASIC FILING FEE - UTILITY (PAPER FILING ALSO REQUIRES NON-ELECTRONIC FILING FEE UNDER 1.16(T))	320.00	1	320.00
1311	PATENT APPL. EXAMINATION FEE	800.00	1	800.00
			TOTAL	\$1,820.00

AMOUNT:

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Document Code:WFEE
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18631783	04/10/2024

Document Number	Fee Code	Fee Code Description	Amount Paid	Payment Method
I20244MJ52277508	1054	Non-DOC-X Filing Surcharge	\$400.00	Deposit Account

File History Content Report

The following content is missing from the original file history record obtained from the United States Patent and Trademark Office. No additional information is available.

Document Date - 2024-04-10

Document Title - Applicant Communication Re: Entity Status Set to Undiscounted (Initial Default Setting or Status Change)



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APPLICATION NUMBER	FILING of 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/631,783	04/10/2024		2220	00249/62	19	3

CONFIRMATION NO. 1021

1912
AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

FILING RECEIPT



0000000071535724

Date Mailed: 04/26/2024

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

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Inventor(s)

David W. Avison, Boxborough, MA;
Steven A. Carlson, Cambridge, MA;
Benjamin Sloan, Exeter, NH;

Applicant(s)

LG Energy Solution, Ltd., Seoul, KOREA, REPUBLIC OF;

Power of Attorney: The patent practitioners associated with Customer Number 1912

Domestic Priority data as claimed by applicant

This application is a CON of 17/543,259 12/06/2021 PAT 11,973,181
which is a CON of 17/223,845 04/06/2021 PAT 11,769,903
which is a CON of 16/755,294 04/10/2020 PAT 10,998,536
which is a 371 of PCT/US2018/055862 10/15/2018
which claims benefit of 62/572,083 10/13/2017

Foreign Applications for which priority is claimed (You may be eligible to benefit from the Patent Prosecution Highway program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.
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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/631,783**

Projected Publication Date: 08/08/2024

Non-Publication Request: No

Early Publication Request: No

Title

MULTILAYER NANOPOROUS SEPARATOR

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/631,783	04/10/2024	David W. Avison	00249/62

CONFIRMATION NO. 1021

PUBLICATION NOTICE



OC000000076256257

1912
AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

Date Mailed: 08/08/2024

Title:MULTILAYER NANOPOROUS SEPARATOR

Publication No.US-2024-0266593-A1
Publication Date:08/08/2024

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/631,783	08/09/2024 03:18:31 PM Z ET	00249/62

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	1021	FILED BY	Joy Chapman
PATENT CENTER #	66728690	FILING DATE	04/10/2024
CUSTOMER #	1912	FIRST NAMED INVENTOR	David W. Avison
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Brian Amos

Documents

TOTAL DOCUMENTS: 3

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
00249-62_IDS_non-fillable-pdf.pdf	4	Information Disclosure Statement (IDS) Form (SB08)	1382 KB
Warning: This is not a USPTO supplied IDS fillable form. Data in the form cannot be automatically loaded to other USPTO systems.			
KR_10-2016-0072009_with-Eng-Abstract.pdf	18	Foreign Reference	12304 KB
KR_10-2011-0116489_with-Eng-Abstract.pdf	23	Foreign Reference	17456 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
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KR_10-2016-0072009_with-Eng-Abstract.pdf	E06F614365F85B38E9F915C94CF2E7CC57225BD4980A568D22629EF2E88C76213374DF5FB01A6E921521C00C6F8EBA0504787CF8FEABD97BE6270ACFD FDC6E38
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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		18631783
Filing Date		2024-04-10
First Named Inventor	David W. Avison	
Art Unit		1729
Examiner Name		
Attorney Docket Number		00249/62

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

- ☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).
- OR
- ☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).
- ☐ See attached certification statement.
- ☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.
- ☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2024-08-09
Name/Print	Brian J. Amos	Registration Number	63,679

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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		18631783	
	Filing Date		2024-04-10	
	First Named Inventor	David W. Avison		
	Art Unit		1729	
	Examiner Name			
	Attorney Docket Number		00249/62	

U.S.PATENTS								
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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	10-2016-0072009	KR	A	2016-06-22	Samsung SDI Co., Ltd.		☒
	2	10-2011-0116489	KR	A	2011-10-26	Korea Institute of Science and Technology		☒
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

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Filing Date		2024-04-10
First Named Inventor	David W. Avison	
Art Unit		1729
Examiner Name		
Attorney Docket Number		00249/62

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1		☐

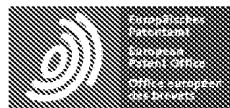
If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.



Espacenet

Bibliographic data: KR20160072009 (A) — 2016-06-22

WOUND ELECTRODE ASSEMBLY FOR NONAQUEOUS ELECTROLYTE RECHARGEABLE BATTERY, NONAQUEOUS ELECTROLYTE RECHARGEABLE BATTERY INCLUDING THE SAME, AND METHODE FOR MANUFACTURING THE SAME

Inventor(s): HOSHIBA KOJI [JP] ± (호시바 코지)

Applicant(s): SAMSUNG SDI CO LTD [KR] ± (삼성에스디아이 주식회사)

Classification: - international: H01M10/0525; H01M10/0587; H01M4/62
- cooperative: H01M10/0431 (EP, US); H01M4/623 (EP, US); H01M50/461 (EP, US); H01M10/0525 (EP, US); H01M10/0587 (EP, US); Y02E60/10 (EP); Y02P70/50 (EP)

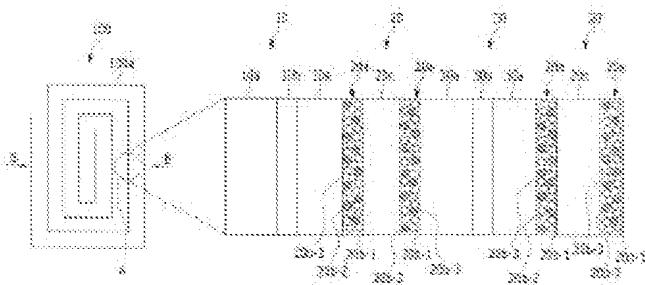
Application number: KR20150142804 20151013 [Global Dossier](#)

Priority number(s): JP20140258190 20141222

Also published as: JP2016119221 (A) JP6399921 (B2) KR102434888 (B1) US11239487 (B2) US2016172705 (A1)

Abstract of KR20160072009 (A)

The present invention provides a wound electrode assembly for a nonaqueous electrolyte rechargeable battery that is provided with a band-shaped anode, a band-shaped cathode, a band-shaped porous membrane placed between the band-shaped anode and the band-shaped cathode, and an adhesion layer formed on a surface of the band-shaped porous membrane. The adhesion layer includes fluororesin-containing microparticles, binder particles supporting the fluororesin-containing microparticles and having a total volume smaller than a total volume of the fluororesin-containing microparticles, and heat-resistant filler particles. The binder particles have an average particle diameter of 100 to 500 nm and the heat-resistant filler particles have an average particle diameter of 10 to 100 nm.



	(19) 대한민국특허청(KR)	(11) 공개번호	10-2016-0072009
	(12) 공개특허공보(A)	(43) 공개일자	2016년06월22일
(51) 국제특허분류(Int. Cl.)	(71) 출원인		
HO1M 10/0587 (2010.01) HO1M 10/0525 (2010.01)	삼성에스디아이 주식회사		
HO1M 4/62 (2006.01)	경기도 용인시 기흥구 공세로 150-20 (공세동)		
(52) CPC특허분류	(72) 발명자		
HO1M 10/0587 (2013.01)	호시바 코지		
HO1M 10/0525 (2013.01)	일본국 가나가와켄 요코하마시 쓰루미쿠 스가사와 초 2-7 가부시키키가이샤 삼성요코하마 겐큐쇼 내		
(21) 출원번호	10-2015-0142804	(74) 대리인	
(22) 출원일자	2015년10월13일	팬코리아특허법인	
심사청구일자	없음		
(30) 우선권주장			
JP-P-2014-258190	2014년12월12일	일본(JP)	

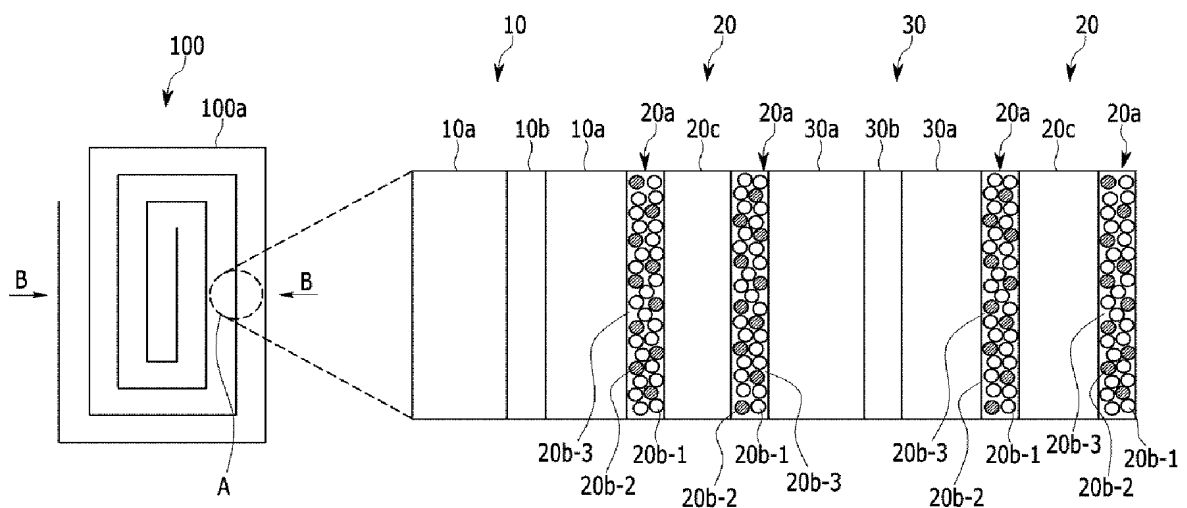
전체 청구항 수 : 총 8 항

(54) 발명의 명칭 비수전해질 이차전지용 전극권회 소자, 이를 이용한 비수전해질 이차전지, 및 비수전해질 이차전지용 전극권회 소자의 제조 방법

(57) 요약

본 발명에 의하면, 띠형 양극과, 띠형 음극과, 띠형 양극 및 띠형 음극의 사이에 배치된 띠형 다공질막과, 띠형 다공질막의 표면에 형성되는 접착층을 구비하고, 상기 접착층은 플루오르 수지 함유 미립자와, 플루오르 수지 함유 미립자를 담지하고, 동시에 충부피가 플루오르 수지 함유 미립자의 충부피보다도 작은 결합제 입자와, 내열성 필러 입자를 포함하고, 상기 결합제 입자의 평균 입경은 100~500nm이며, 상기 내열성 필러 입자의 평균 입경은 10~100nm인 것을 특징으로 하는 비수전해질 이차전지용 전극권회 소자가 제공된다.

도 1 내지 도 3



(52) CPC특허분류
H01M 4/623 (2013.01)
Y02E 60/122 (2013.01)

명세서

청구항

청구항 1

띠형 양극과,

띠형 음극과,

상기 띠형 양극 및 띠형 음극의 사이에 배치된 띠형 다공질막과,

상기 띠형 다공질막의 표면에 형성되는 접착층을 구비하고,

상기 접착층은 플루오르 수지 함유 미립자와, 상기 플루오르 수지 함유 미립자를 담지하고, 동시에 총부피가 상기 플루오르 수지 함유 미립자의 총부피보다도 작은 결합제 입자와, 내열성 필러 입자를 포함하고,

상기 결합제 입자의 평균 입경은 100~500nm이며,

상기 내열성 필러 입자의 평균 입경은 10~100nm인 것을 특징으로 하는 비수전해질 이차전지용 전극권회 소자.

청구항 2

제 1 항에 있어서,

상기 결합제 입자의 평균 입경 및 상기 내열성 필러 입자의 평균 입경은 이하의 수식(1) 및 (2) 중, 적어도 어느 하나 만족시키는 것을 특징으로 하는 청구항 1 기재의 비수전해질 이차전지용 전극권회 소자.

$$y < 50/17 \cdot (x+70) \cdots (1)$$

$$y > 10/7 \cdot (120-x) \cdots (2)$$

상기 수식(1), (2) 중, y는 상기 결합제 입자의 평균 입경이며, x는 상기 내열성 필러 입자의 평균 입경이다.

청구항 3

제 2 항에 있어서,

상기 결합제 입자의 평균 입경 및 상기 내열성 필러 입자의 평균 입경은 상기 수식(1) 및 (2)의 모두를 충족하는 것을 특징으로 하는 비수전해질 이차전지용 전극권회 소자.

청구항 4

제 1 항 내지 제 3 항 중 어느 하나에 있어서,

상기 띠형 음극은 음극 활물질과, 상기 플루오르 수지 함유 미립자를 포함하는 음극 활물질층을 구비하고,

상기 접착층은 상기 음극 활물질층에 결합 하고 있는 것을 특징으로 하는 비수전해질 이차전지용 전극권회 소자.

청구항 5

제 1 항 내지 제 3 항 중 어느 하나에 있어서,

상기 플루오르 수지 함유 미립자는 구형입자인 것을 특징으로 하는 비수전해질 이차전지용 전극권회 소자.

청구항 6

제 1 항 내지 제 3 항 중 어느 하나에 있어서,

상기 플루오르 수지는 폴리 불화 비닐리덴을 포함하는 것을 특징으로 하는 비수전해질 이차전지용 전극권회 소자.

청구항 7

제 1 항 내지 제 3 항 중 어느 하나에 기재된 비수전해질 이차전지용 전극권회 소자를 구비하는 것을 특징으로 하는 비수전해질 이차전지.

청구항 8

플루오르 수지 함유 미립자와, 상기 플루오르 수지 함유 미립자를 담지하고, 동시에 총부피가 상기 플루오르 수지 함유 미립자의 총부피보다도 작은 결합제 입자와, 내열성 필러 입자를 포함하는 수계 슬러리를 띠형 다공질 막의 표면에 도포하고, 건조하는 공정을 포함하고,

상기 결합제 입자의 평균 입경은 100~500nm이며,

상기 내열성 필러 입자의 평균 입경은 10~100nm인 것을 특징으로 하는 비수전해질 이차전지용 전극권회 소자의 제조 방법.

발명의 설명

기술 분야

[0001] 본 발명은 비수전해질 이차전지용 전극권회 소자, 이를 이용한 비수전해질 이차전지, 및 비수전해질 이차전지용 전극권회 소자의 제조 방법에 관한 것이다.

배경 기술

[0002] 폴리 불화 비닐리덴(PVDF)계의 플루오르 수지를 리튬이온(lithiumion) 이차전지의 겔(gel)전해질의 매트릭스 폴리머(matrix polymer)로서 사용하는 예는 많이 존재한다. 예를 들면, PVDF계의 플루오르(fluorine)수지로 이루어지는 다공질막을 세퍼레이터(separator)의 표면에 형성하는 기술이 알려져 있다. 이 기술에서는 예를 들면 이하에 나타내는 방법에 의해 다공질막을 세퍼레이터의 표면에 형성한다. 제1 방법으로는 NMP(N-메틸 피롤리돈), 디메틸 아세트아미도, 아세톤(acetone) 등의 유기용제 중에 플루오르 수지를 용해시켜 슬러리(slurry)을 제작한다. 그리고, 이 슬러리를 세퍼레이터나 전극에 도포 후, 물, 메탄올, 트리프로필렌글리콜 등의 반응매, 또는 이들의 증기를 이용해서 플루오르 수지를 상분리시키는 것으로 플루오르 수지를 다공질화시킨 도포층을 형성한다. 제2 방법으로는 플루오르 수지를 디메틸 카보네이트, 프로필렌 카보네이트, 에틸렌 카보네이트 등을 용매로 하는 가열 전해액 중에 용해시키는 것으로 가열 슬러리를 제작한다. 그리고, 이 가열 슬러리를 세퍼레이터나 전극에 도포하는 것으로 도포층을 형성한다. 그리고, 도포층을 냉각 하는 것으로, 플루오르 수지를 겔(전해액으로 팽윤된 다공질막)에 전이시킨다.

[0003] 그런데, 상기의 방법에 의해 PVDF의 다공질막이 표면에 형성된 세퍼레이터는 미형성된 세퍼레이터에 비해 미끄러짐성이 나쁘고, 정전기도 발생하기 쉬우므로, 제조 공정상에서 취급하기 어렵다는 문제가 있었다. 구체적으로는 상기 세퍼레이터를 권회 소자의 세퍼레이터로 했을 경우에, 띠형의 양극, 음극, 세퍼레이터를 겹쳤을 때의 상호의 미끄러짐성이 나쁘기 때문에, 권회 소자가 일그러져버린다는 문제가 있었다. 권회 소자가 일그러졌을 경우, 권회 소자를 케이스에 수납하기 어려워지는 문제가 생길 수 있다. 또, 이 왜곡의 영향에 의해 이 권회 소자를 사용한 비수전해질 이차전지는 사이클(cycle) 수명이 충분하지 않다는 문제도 있었다. .

[0004] 또한, 근래에는 세퍼레이터 가열시의 열수축을 억제하기 위해, 내열성 필러를 다공질막에 첨가하는 것이 제안되어 있다. 그러나, 상기의 다공질막(겔 전해질막)에는 내열성 필러 입자를 고밀도로 포함시키는 것이 어려웠다. 이 때문에, 다공질막에 내열성을 발현시키고자 하면, 다공질막이 두꺼워져 버린다는 문제가 있었다. 다공질막이 두꺼워지면, 이차전지의 에너지 밀도가 작아진다.

발명의 내용

해결하려는 과제

[0005] 본 발명은 이러한 문제를 해결하기 위한 것으로, 세퍼레이터의 제조 공정상에서의 취급성을 개선하고, 권회 소자의 왜곡을 억제하고, 동시에, 내열성 및 에너지 밀도를 향상시키는 것이 가능한 비수전해질 이차전지용 전극권회 소자, 이를 이용한 비수전해질 이차전지, 및 비수전해질 이차전지용 전극권회 소자의 제조 방법을 제공함에 있다.

과제의 해결 수단

- [0006] 상술한 기술적 과제를 달성하기 위한 본 발명의 일 측면은 띠형 양극과, 띠형 음극과, 띠형 양극 및 띠형 음극의 사이에 배치된 띠형 다공질막과, 띠형 다공질막의 표면에 형성되는 집착층을 구비하며, 집착층은 플루오르수지 함유 미립자와, 플루오르수지 함유 미립자를 담지하고, 동시에, 총부피가 플루오르수지 함유 미립자의 총부피보다도 작은 결착제 입자와, 내열성 필러 입자를 포함하고, 결착제 입자의 평균 입경은 100~500nm이며, 내열성 필러 입자의 평균 입경은 10~100nm인 것을 특징으로 하는 비수전해질 이차전지용 전극권회 소자가 제공된다.
- [0007] 상기 결착제 입자의 평균 입경 및 내열성 필러 입자의 평균 입경은 이하의 수식(1) 및 (2)중, 적어도 어느 하나를 만족시킬 수 있다.
- [0008] $y < 50/17 \cdot (x+70) \cdots (1)$
- [0009] $y > 10/7 \cdot (120-x) \cdots (2)$
- [0010] 수식(1), (2) 중, y는 결착제 입자의 평균 입경이며, x는 내열성 필러 입자의 평균 입경이다.
- [0011] 상기 결착제 입자의 평균 입경 및 내열성 필러 입자의 평균 입경은 상기 수식(1) 및 (2) 모두를 충족할 수 있다.
- [0012] 상기 띠형 음극은 음극 활물질과, 플루오르수지 함유 미립자를 포함하는 음극 활물질층을 구비하고, 집착층은 음극 활물질층에 결착하고 있을 수 있다.
- [0013] 상기 플루오르수지 함유 미립자는 구형입자일 수 있다.
- [0014] 상기 플루오르수지는 폴리 불화 비닐리텐을 포함할 수 있다.
- [0015] 또한, 본 발명의 다른 측면은 상기 비수전해질 이차전지용 전극권회 소자를 구비하는 비수전해질 이차전지를 제공한다.
- [0016] 또한, 본 발명의 다른 측면은 플루오르수지 함유 미립자와, 플루오르수지 함유 미립자를 담지하고, 동시에, 총부피가 플루오르수지 함유 미립자의 총부피보다도 작은 결착제 입자와, 내열성 필러 입자를 포함하는 수계 슬러리를 띠형 다공질막의 표면에 도포하고 건조하는 공정을 포함하고, 결착제 입자의 평균 입경은 100~500nm이며, 내열성 필러 입자의 평균 입경은 10~100nm인 비수전해질 이차전지용 전극권회 소자의 제조 방법을 제공한다.

발명의 효과

- [0017] 상술한 본 발명의 과제 해결 수단의 일부 실시예 중 하나에 의하면, 세퍼레이터의 취급성 및 권회 소자의 왜곡을 개선시킬 수 있다.
- [0018] 또한, 집착층이 박막화되어, 에너지 밀도를 향상시킬 수 있다.
- [0019] 또한, 권회 소자의 내열성, 즉 권회 소자를 사용한 비수전해질 이차전지의 내열성을 향상시킬 수 있다.

도면의 간단한 설명

- [0020] 도 1은 본 발명의 실시형태에 따른 리튬이온 이차전지의 개략적인 구성을 나타내는 단면도이다.
- 도 2는 본 발명의 실시형태에 따른 전극 적층체의 개략적인 구성을 나타내는 단면도이다.
- 도 3은 결착제 입자의 평균 입경과, 내열성 필러 입자의 평균 입경과, 이차전지의 특성(종합 평가)과의 대응 관계를 나타내는 그래프이다.

발명을 실시하기 위한 구체적인 내용

- [0021] 이하에 첨부 도면을 참조하면서, 본 발명의 바람직한 실시형태에 대해서 상세하게 설명한다. 한편, 본 명세서 및 도면에 있어서, 실질적으로 동일한 기능 구성을 갖는 구성 요소에 대해서는 동일한 부호가 부여되는 것에 의해 중복 설명을 생략한다.
- [0022] <1. 리튬이온 이차전지의 구성>

- [0023] (리튬이온 이차전지의 전체구성)
- [0024] 먼저, 도 1 및 도 2를 참조하여, 본 발명의 실시형태에 따른 리튬이온 이차전지의 구성에 대하여 설명한다. 도 1은 권회 소자(100)의 평면도와, 권회 소자(100)의 영역 A를 확대한 확대도를 나타낸다. 도 2는 양극, 음극 및 2장의 세퍼레이터가 적층된 전극 적층체(100a)의 평면도와, 전극 적층체(100a)의 영역 A를 확대한 확대도를 나타낸다.
- [0025] 리튬이온 이차전지는 권회 소자(100)와, 비수전해질용액과, 외장재를 구비한다. 권회 소자(100)는 띠형 음극(10), 띠형 세퍼레이터(20), 띠형 양극(30), 및 띠형 세퍼레이터(20)를 이 순서대로 적층한 전극 적층체(100a)를 길이 방향으로 감고, 화살표 B 방향으로 압축한 것이다.
- [0026] (음극의 구성)
- [0027] 띠형 음극(10)은 음극 집전체(10b)와, 음극 집전체(10b)의 양면에 형성된 음극 활물질층(10a)을 포함한다. 띠형 음극(10)은 소위 수계음극이다. 따라서, 본 실시형태에 따른 권회 소자(100) 및 리튬이온 이차전지는 수계음극을 구비한 것이다.
- [0028] 구체적으로는 음극 활물질층(10a)은 음극 활물질과, 증점제와, 결합제를 포함한다. 음극 활물질층(10a)을 구성하는 음극 활물질로는 리튬과의 합금화, 또는 리튬(Li)의 가역적인 흡장 및 방출이 가능한 물질이라면 특별히 한정되지 않고, 예를 들면, 리튬, 인듐(In), 주석(Sn), 알루미늄(Al), 규소(Si) 등의 금속 및 이들 합금이나 산화물, $\text{Li}_{4/3}\text{Ti}_{5/3}\text{O}_4$, SnO 등의 전이금속 산화물이나, 인조흑연, 천연흑연, 인조흑연과 천연흑연의 혼합물, 인조흑연을 피복한 천연흑연, 난흑연화성 탄소, 흑연탄소섬유, 수지소성 탄소, 열분해 기상성장 탄소, 코크스(coke), 메소카본 마이크로비즈(MCMB), 푸르푸릴 알코올(furfuryl alcohol)수지 소성 탄소, 폴리아센(polyacene), 피치(pitch)계 탄소섬유 등의 탄소 재료 등을 들 수 있다. 이들 음극 활물질은 단독으로 이용될 수도 있고, 2종 이상이 병용되어도 된다. 중에서도, 흑연계 재료를 주재료로 이용하는 것이 바람직하다.
- [0029] 증점제는 음극합제 슬러리를 도포에 알맞은 점도로 조정함과 동시에, 음극 활물질층(10a) 내에서 결합체로서 기능하는 것이다. 증점제로는 수용성 고분자가 바람직하게 이용할 수 있으며, 예를 들면 셀룰로오스계 고분자, 폴리 아크릴산(Polyacrylic acid)계 고분자, 폴리비닐알코올(polyvinyl alcohol), 폴리에틸렌 옥사이드(polyethylene oxide) 등을 들 수 있다. 셀룰로오스계 고분자로는 예를 들면 카르복시메틸셀룰로오스(CMC)의 금속염 또는 암모늄염, 메틸 셀룰로오스(methylcellulose), 에틸 셀룰로오스(ethyl cellulose), 히드록시 알킬 셀룰로오스(hydroxy alkyl cellulose)등의 셀룰로오스(Cellulose) 유도체 등을 들 수 있다. 다른 예로는 폴리비닐피롤리돈(PVP), 전분(starch), 인산 전분, 카제인(casein), 각종 변성 전분(starch), 키틴(chitin), 키토산(chitosan) 유도체 등을 들 수 있다. 이 증점제들은 각각 단독으로 또는 2종 이상을 조합해서 사용할 수 있다. 이 중에서도, 셀룰로오스계 폴리머가 바람직하고, 카르복시 메틸셀룰로오스의 알칼리 금속염이 특히 바람직하다.
- [0030] 결합제는 음극 활물질들을 결합시키는 것이다. 결합제는 수계음극의 결합체로서 사용가능한 것이라면 특별히 제한되지 않는다. 본 실시형태에 사용가능한 결합제로는 예를 들면, 엘라스토머계 고분자의 미립자를 들 수 있다. 엘라스토머계 고분자로는 SBR(스티렌 부타디엔 고무), BR(부타디엔 고무), NBR(니트릴부타디엔 고무), NR(천연 고무), IR(이소프렌 고무), EPDM(에틸렌-프로필렌-디엔 삼원공중합체), CR(클로로프렌 고무), CSM(클로로술폰화 폴리에틸렌), 아크릴산 에스테르, 메타아크릴산 에스테르의 공중합체 및 이들의 부분 수소화물, 혹은 완전 수소화물, 아크릴산 에스테르계 공중합체 등을 들 수 있다. 이들은 결합성 향상을 위해, 카르본산이나 술폰산, 인산, 수산기 등의 극성 작용기를 갖는 단량체에 의해 변성 되어 있어도 좋다. 또, 음극 활물질층(10a)은 결합체로서, 후술하는 플루오르 수지 함유 미립자를 포함할 수 있다. 플루오르 수지 함유 미립자는 본체를 나중 슬러리 중에서 분산시켜도 되며, 수분산체 상태인 것을 슬러리 중에 첨가할 수도 있다. 따라서, 음극 활물질층(10a)을 형성하기 위한 슬러리의 용매로 물을 사용 할 수 있다.
- [0031] 증점제 및 결합제의 음극 활물질층 내의 함유비는 리튬이온 이차전지의 음극 활물질층에 적용 가능한 함유비라면 특별히 제한되지 않는다.
- [0032] 음극집전체(10b)는 도전체라면 어떤 것이라도 좋고, 예를 들면, 구리, 스테인리스강, 및 니켈 도금 강철 등으로 구성된다. 음극집전체(10b)에는 음극단자가 접속된다.
- [0033] 띠형 음극(10)은 예를 들면, 이하의 방법에 의해 제작된다. 즉, 음극 활물질층의 재료를 물에 분산시켜서 음극합제 슬러리(수계 슬러리)을 형성하고, 이 음극합제 슬러리를 집전체 위에 도포한다. 이에 의해, 도포층을 형성

한다. 이어서, 도포층을 건조한다. 음극합제 슬러리 중에는 플루오르 수지 미립자 및 엘라스토머계 고분자의 미립자가 음극 활물질층(10a) 내에 분산되어 있다. 이어서, 건조한 도포층을 음극집전체(10b)와 함께 압연한다. 이에 따라, 띠형 음극(10)이 제작된다.

[0034] 띠형 세퍼레이터(20)은 띠형 다공질막(20c)와, 띠형 다공질막(20c)의 양면에 형성된 접착층(20a)을 포함한다.

[0035] 띠형 다공질막(20c)은 특별히 제한 되지 않고, 리튬이온 이차전지의 세퍼레이터로서 사용되는 것이라면, 어떠한 것이어도 좋다. 띠형 다공질막(20c)로는 우수한 고율방전 성능을 나타내는 다공질막이나 부직포 등을, 단독 혹은 병용하는 것이 바람직하다. 띠형 다공질막(20a)을 구성하는 수지로는 예를 들면 폴리에틸렌(polyethylene), 폴리프로필렌(polypropylene) 등으로 대표되는 폴리올레핀(polyolefin)계 수지, 폴리에틸렌 테레프탈레이트(Polyethylene terephthalate), 폴리부틸렌 테레프탈레이트(polybutylene terephthalate) 등으로 대표되는 폴리에스테르(Polyester)계 수지, PVDF, 불화 비닐리덴(VDF)-헥사플루오로 프로필렌(HFP) 공중합체, 불화 비닐리덴-페플루오로 비닐에테르(par fluorovinyl ether)공중합체, 불화 비닐리덴-테트라플루오로에틸렌(tetrafluoroethylene) 공중합체, 불화 비닐리덴-트리플루오로에틸렌(trifluoroethylene) 공중합체, 불화 비닐리덴-플루오로에틸렌(fluoroethylene) 공중합체, 불화 비닐리덴-헥사플루오로 아세톤(hexafluoroacetone) 공중합체, 불화 비닐리덴-에틸렌(ethylene) 공중합체, 불화 비닐리덴-프로필렌(propylene) 공중합체, 불화 비닐리덴-트리플루오로 프로필렌(trifluoro propylene) 공중합체, 불화 비닐리덴-테트라플루오로에틸렌(tetrafluoroethylene)-헥사플루오로 프로필렌(hexafluoropropylene) 공중합체, 불화 비닐리덴-에틸렌(ethylene)-테트라플루오로에틸렌(tetrafluoroethylene) 공중합체 등을 들 수 있다.

[0036] 접착층(20a)은 상술한 플루오르 수지 함유 미립자(20b-1)와, 결합제 입자(20b-2)와, 내열성 필러 입자(20b-3)를 포함하고, 띠형 세퍼레이터(20)와 띠형 음극(10) 및 띠형 양극(30)을 결합 한다. 도 1에서는 접착층(20a)은 띠형 세퍼레이터(20)의 양면에 형성되어 있지만, 적어도 한 쪽 표면에 형성되어 있을 수도 있다.

[0037] 플루오르 수지 함유 미립자(20b-1)는 플루오르 수지를 포함하는 미립자이다. 플루오르 수지 함유 미립자(20b-1)를 구성하는 플루오르 수지의 바람직한 예로는 PVDF 및 PVDF를 포함하는 공중합체 등을 들 수 있다. PVDF를 포함하는 공중합체로는 불화 비닐리덴(VDF)과 헥사플루오로 프로필렌(HFP)과의 공중합체, 불화 비닐리덴(VDF)과 테트라플루오로에틸렌(TFE)과의 공중합체 등을 들 수 있다. 이 플루오르 수지들은 카르복산 등의 극성기에 의해 변성 되어 있어도 좋다.

[0038] 플루오르 수지 함유 미립자(20b-1)의 입경(플루오르 수지 함유 미립자(20b-1)를 구형으로 간주했을 때의 직경)은 특별히 제한 되지 않고, 음극 활물질층(10a) 내에 분산할 수 있는 입경이라면 어떤 값이어도 좋다. 예를 들면, 플루오르 수지 함유 미립자(20b-1)의 평균 입경(입경의 산술평균값)은 80~500nm정도일 수 있다. 플루오르 수지 함유 미립자(20b-1)의 평균 입경은 예를 들면 레이저 회절법(laser diffractometry)에 의해 측정된다. 구체적으로는 레이저 회절법에 의해 플루오르 수지 함유 미립자(20b-1)의 입도 분포를 측정하고, 이 입도 분포에 기초하여 입경의 산술평균값을 산출하면 된다. 다른 입자의 평균 입경도 동일한 방법에 의해 측정 가능하다.

[0039] 한편, 플루오르 수지 함유 미립자(20b-1)는 본 실시형태의 효과를 해치지 않은 범위 내에서 각종 가공, 예를 들면 다른 수지에 의해 복합화되어 있어도 된다. 예를 들면, 플루오르 수지 함유 미립자(20b-1)는 아크릴 수지와 복합화되어 있어도 된다. 이 플루오르 수지 함유 미립자(20b-1)는 IPN(Inter-penetrating network polymer)형의 구조를 하고 있다.

[0040] 플루오르 수지 함유 미립자(20b-1)는 예를 들면 플루오르 수지를 구성하는 모노머(monomer)(예를 들면 VDF)를 유화 중합 함으로써 제작(합성)된다. 플루오르 수지 함유 미립자(20b-1)는 플루오르 수지를 구성하는 모노머를 현탁 중합 시키고, 이에 따라 얻어진 조립자(粗粒子)를 분쇄하는 것으로 제작될 수 있다.

[0041] 본 실시형태의 플루오르 수지 함유 미립자(20b-1)는 구형입자인 것이 특히 바람직하다. 구형의 플루오르 수지 함유 미립자(20b-1)는 예를 들면 상술한 유화 중합법에 의해 제작 가능하다. 또, 플루오르 수지 함유 미립자(20b-1)의 형상은 예를 들면 SEM(주사전자 현미경)에 의해 확인할 수 있다.

[0042] 결합제 입자(20b-2)는 접착층(20a) 내에서 플루오르 수지 함유 미립자(20b-1)를 담지하는 것이다. 접착층(20a) 내에서 차지하는 결합제 입자(20b-2)의 총부피는 접착층(20a) 내에서 차지하는 플루오르 수지 함유 미립자(20b-1)의 총부피보다도 작다. 구체적으로는 (플루오르 수지 함유 미립자(20b-1)의 총부피)/(결합제 입자(20b-2)의 총부피)는 2~20정도인 것이 바람직하다.

[0043] 접착층(20a)은 플루오르 수지 함유 미립자(20b-1) 및 결합제 입자(20b-2)를 상기의 부피비로 포함하므로, 후술하는 실시예에서 나타난 바와 같이, 띠형 세퍼레이터(20)의 제조 공정상에서의 취급성을 개선 할 수 있다. 구체

적으로는 띠형 세퍼레이터(20)의 미끄러짐성을 개선하고, 권회 소자(100)의 왜곡을 억제 할 수 있다. 그 결과, 사이클 수명도 향상된다.

- [0044] 결착제 입자(20b-2)의 평균 입경은 100~500nm이다. 또, 결착제 입자(20b-2)의 종류는 특별히 제한 되지 않지만, 예를 들면, 이온성 비수용성 결착제 입자, 및 비이온성 비수용성 결착제 입자로부터 이루어지는 군 중, 적어도 1종 이상으로 구성되는 것이 바람직하다. 한편, 결착제 입자(20b-2)는 상기의 결착제 입자에 더하여, 비이온성 수용성 결착제 및 이온성 수용성 결착제 중 적어도 1종을 추가로 포함하고 있어도 된다. 단, 이온성 수용성 결착제의 함유율 (플루오르 수지 함유 미립자의 질량에 대한 함유율)은 2질량% 이하가 된다. 이온성 수용성 결착제의 함유율은 1.0질량% 이하가 되는 것이 바람직하다.
- [0045] 이온성 결착제의 함유율이 2질량%를 넘을 경우, 접착층(20a)의 접착성이 저하된다. 본 발명자는 이 이유를 아래와 같이 추측하고 있다. 즉, 접착층(20a)의 접착성은 접착층(20a)을 구성하는 플루오르 수지 함유 미립자(20b-1), 및 이온성 비수용성 결착제 입자 등이 갖는 극성기가 전극 (특히 음극)과의 계면에서 특정 방향으로 배향함으로써 발현된다. 그리고, 이온성 수용성 결착제의 함유율이 2질량%보다 크면, 도포 건조 공정에서 균일하게 분포된 이온성 수용성 결착제가 전극 표면의 극성기의 배향에 약영향을 줄 가능성이 있다. 그 결과, 접착층(20a)의 접착성이 저하된다.
- [0046] 본 실시형태에 적용 가능한 이온성 비수용성 결착제 입자는 특별히 제한 되지 않는다. 이온성 비수용성 결착제 입자를 구성하는 결착제의 예로는 예를 들면, 카르복산 변성 아크릴산 에스테르, 폴리올레핀 아이오노머, 및 카르복산 변성 스티렌-부타디엔 공중합체 등을 들 수 있다. 이온성 비수용성 결착제 입자는 이들 중에서 1종으로 구성되어도 되며, 2종 이상으로 구성되어도 된다.
- [0047] 본 실시형태에 적용 가능한 비이온성 비수용성 결착제 입자도 특별히 제한 되지 않는다. 비이온성 비수용성 결착제 입자를 구성하는 결착제의 예로는 비이온성 비수용성 결착제의 예로는 아크릴산 부틸 등의 라디칼 중합성 모노머, 라우릴 황산 나트륨 등의 음이온계 계면활성제, 과황산칼륨과 같은 수용성 개시제로부터 유화 중합으로 얻어지는 폴리 아크릴산 부틸 수분산체 등을 들 수 있다. 아크릴산-2-히드록시에틸과 같은 수산기를 함유한 모노머를 적당히 공중합시켜서, 수중의 분산 안정성을 향상시킨 것이라도 된다. 비이온성 비수용성 결착제 입자는 이들 중에서 1종으로 구성되어도 되며, 2종 이상으로 구성되어도 된다.
- [0048] 본 실시형태에 적용 가능한 비이온성 수용성 결착제도 특별히 제한 되지 않는다. 비이온성 수용성결착제를 구성하는 결착제의 예로는 폴리-N-비닐 아세트아미드(PNVA), 폴리비닐알코올(PVA), 폴리비닐피롤리돈(PVP), 히드록시에틸 셀룰로오스, 메틸 셀룰로오스, 히드록시프로필 셀룰로오스, 히드록시프로필 구아검, 로카스트빈검, 및 폴리옥시에틸렌 등을 들 수 있다. 비이온성 수용성 결착제는 이들 중에서 1종으로 구성될 수 있으며, 2종 이상으로 구성되어도 좋다.
- [0049] 본 실시형태에 적용 가능한 이온성 수용성 결착제도 특별히 제한 되지 않는다. 이온성 수용성 결착제를 구성하는 결착제의 예로는 폴리 아크릴산, 카르복시메틸셀룰로오스(CMC), 스티렌-말레인산 공중합체, 이소부틸렌-말레인산 공중합체, N-비닐 아크릴아미드-아크릴산 공중합체, 이들 알칼리 금속 염, 및 이들 암모늄 염 등을 들 수 있다. 이온성 수용성 결착제는 이들 중에서 1종으로 구성되어도 되며, 2종 이상으로 구성되어도 된다.
- [0050] 접착층(20a)은 상술한 도포에 적합한 점도를 부여하고, 내열성 필러의 분산 안정성을 확보하기 위해, 증점제를 더욱 포함하고 있어도 좋다. 증점제로는 상기의 비이온성 수용성 결착제가 바람직하다. 또, 접착층(20a)은 다공도(porosity) 조정이나 열안정성을 위해 내열성 필러 입자(20b-3)를 더욱 포함한다.
- [0051] 여기에서, 내열성 필러 입자(20b-3)의 평균 입경은 10~100nm이다. 이와 같이, 내열성 필러 입자(20b-3)의 평균 입경은 결착제 입자(20b-2)의 평균 입경 이하이다. 이는 이하의 이유에 따른다. 즉, 리튬이온 이차전지의 에너지 밀도를 향상시키기 위해서는 접착층(20a)을 박막화 할 필요가 있다. 접착층(20a)을 박막화하는 방법의 하나로, 접착층(20a)을 구성하는 입자의 평균 입경을 작게 하는 것을 생각할 수 있다. 따라서, 본 실시형태에서는 내열성 필러 입자(20b-3)의 평균 입경을 작게, 즉 10~100nm으로 했다.
- [0052] 단, 결착제 입자(20b-2)의 평균 입경까지 작게 해버리면, 결착제 입자(20b-2) 및 내열성 필러 입자(20b-3)가 띠형 다공질막(20c)의 기공 내에 침입해서 막힘을 일으킨다. 이 경우, 리튬이온 이차전지의 특성이 오히려 저하되어버린다. 따라서, 본 실시형태에서는 결착제 입자(20b-2)의 평균 입경을 내열성 필러 입자(20b-3)의 평균 입경 이상으로 하는 것으로, 상기와 같은 클로깅(clogging)을 억제하고 있다. 구체적으로는 결착제 입자(20b-2)의 평균 입경을 100~500nm으로 한다.
- [0053] 이와 같이, 본 실시형태에서는 내열성 필러 입자(20b-3)의 평균 입경을 작게 하고, 결착제 입자(20b-2)의 평균

입경을 크게 하는 것에 의해, 접착층(20a)을 박막화하면서, 에너지 밀도를 향상 할 수 있다. 또, 내열성 필러 입자(20b-3)가 세퍼레이터(20)의 표층(즉, 접착층(20a))에 포함되어 있으므로, 리튬이온 이차전지의 내열성이 향상된다.

[0054] 또한, 본 발명자는 내열성 필러 입자(20b-3)의 평균 입경과 결합제 입자(20b-2)의 평균 입경과의 대응 관계에 대해서 더욱 상세하게 조사했다. 그 결과, 내열성 필러 입자(20b-3)의 평균 입경과 결합제 입자(20b-2)의 평균 입경이 이하의 수식(1), (2) 중 적어도 한 쪽을 만족시키는 것이 바람직하고, 양쪽을 만족시키는 것이 더욱 바람직하다는 결과를 도출할 수 있었다. 이 경우, 리튬이온 이차전지의 특성이 더욱 향상된다.

[0055] $y < 50/17 \cdot (x+70) \cdots (1)$

[0056] $y > 10/7 \cdot (120-x) \cdots (2)$

[0057] 수식(1), (2) 중, y는 결합제 입자(20b-2)의 평균 입경이며, x는 내열성 필러 입자(20b-3)의 평균 입경이다.

[0058] 본 실시형태에 적용 가능한 내열성 필러 입자는 특별히 제한 되지 않는다. 예를 들면, 내열성 필러 입자는 내열성 유기 필러 입자이어도 좋고, 내열성 무기 필러 입자(무기입자)이어도 좋으며, 이들 혼합물이어도 좋다. 단, 내열성 무기 필러 입자는 내열성 유기 필러 입자보다도 내열성이 우수한 것이 많으므로, 보다 바람직하다. 내열성 유기 필러 입자 및 무기 필러 입자의 혼합비는 특별히 제한 되지 않는다.

[0059] 내열성 유기 필러 입자로는 예를 들면, 가교 폴리스티렌(가교PS), 가교 폴리 메타아크릴산 메틸(가교PMMA), 실리콘 수지, 에폭시 경화물, 폴리에테르 술폰, 폴리아미드이미드, 폴리아미드, 펠라민 수지, 폴리페닐렌설파이드 수지의 미립자 등을 들 수 있다. 내열성 유기 필러 입자는 이들 중에서 1종으로 구성되어도 좋으며, 2종 이상으로 구성되어도 좋다. 내열성 무기 필러 입자는 구체적으로는 세라믹 입자이며, 보다 구체적으로는 금속산화물 입자이다. 금속산화물 입자로는 예를 들면 알루미늄, 베마이트, 티타니아, 지르코니아, 마그네시아, 산화 아연, 수산화 알루미늄, 수산화 마그네슘 등의 미립자를 들 수 있다. 내열성 필러 입자의 함유율은 본 실시형태의 효과가 얻어지는 범위 내라면 특별히 제한 되지 않지만, 예를 들면 접착층(20a)의 총질량에 대하여 70질량% 이하라면 좋다.

[0060] 접착층(20a)은 이하의 방법에 의해 제작된다. 즉, 접착층(20a)의 재료를 물에 분산, 용해시키는 것으로 접착층 합제 슬러리(수계 슬러리)를 작성한다. 이어서, 이 접착층 합제 슬러리를 띠형 다공질막(20c)의 양면 중, 적어도 한 쪽 표면에 도포하는 것으로 도포층을 형성한다. 이어서, 이 도포층을 건조한다. 이에 의해, 접착층(20a)이 형성된다.

[0061] 띠형 양극(30)은 양극집전체(30b)와, 양극집전체(30b)의 양면에 형성된 양극 활물질층(30a)을 가진다. 양극 활물질층(30a)은 적어도 양극 활물질을 포함하고, 도전제와, 결합제를 더욱 포함하고 있어도 된다. 양극 활물질은 리튬이온을 가역적으로 흡장 및 방출하는 것이 가능한 물질이라면 특별히 한정되지 않고, 예를 들면, 코발트 산 리튬(LCO), 니켈산 리튬, 니켈 코발트산 리튬, 니켈코발트 알루미늄산 리튬 (이하, 「NCA」라고 하는 경우도 있음), 니켈코발트 망간산 리튬 (이하, 「NCM」이라고 하는 경우도 있음), 망간산 리튬, 인산철 리튬, 황화 니켈, 황화구리, 황, 산화철, 산화 바나듐 등을 들 수 있다. 이들 양극 활물질은 단독으로 이용될 수도 있고, 2종 이상이 병용되어도 좋다.

[0062] 양극 활물질은 상기에 든 양극 활물질의 예 중, 특히, 층상암염형 구조를 갖는 전이금속산화물의 리튬 염인 것이 바람직하다. 이러한 층상암염형 구조를 갖는 전이금속산화물의 리튬 염으로는 예를 들면, $\text{Li}_{1-x-y-z}\text{Ni}_x\text{Co}_y\text{Al}_z\text{O}_2$ (NCA) 또는 $\text{Li}_{1-x-y-z}\text{Ni}_x\text{Co}_y\text{Mn}_z\text{O}_2$ (NCM) ($0 < x < 1$, $0 < y < 1$, $0 < z < 1$, 동시에 $x+y+z < 1$)으로 표시되는 3원계의 전이금속산화물의 리튬 염을 들 수 있다.

[0063] 도전제로는 예를 들면 케첸 블랙(Ketjenblack), 아세틸렌 블랙(acetylene black) 등의 카본블랙, 천연흑연, 인조흑연 등을 들 수 있지만, 양극의 도전성을 높이기 위한 것이라면 특별히 제한 되지 않는다.

[0064] 결합제는 양극 활물질들을 결합 하는 동시에, 양극 활물질과 양극집전체(30b)를 결합한다. 결합제의 종류는 특별히 한정되지 않고, 종래의 리튬이온 이차전지의 양극 활물질층에 사용된 결합제라면 어떤 것이라도 사용할 수 있다. 예를 들면 폴리 불화 비닐리덴(polyvinylidene fluoride), 불화 비닐리덴(VDF)-헥사플루오로 프로필렌(HFP) 공중합체, 불화 비닐리덴-퍼플루오로 비닐에테르(par fluorovinyl ether) 공중합체, 불화 비닐리덴-테트라플루오로에틸렌(tetrafluoroethylene) 공중합체, 불화 비닐리덴-트리플루오로에틸렌(trifluoroethylene) 공중합체, 에틸렌프로필렌 디엔(ethylene-propylene-diene) 삼원공중합체, 스티렌 부타디엔 고무(Styrene-butadiene rubber), 아크릴로니트릴 부타디엔 고무(acrylonitrile-butadiene rubber), 플루오르 고무

(fluororubber), 폴리 아세트산 비닐(polyvinyl acetate), 폴리메틸 메타크릴레이트(polymethylmethacrylate), 폴리에틸렌(polyethylene), 니트로셀룰로오스(cellulose nitrate) 등을 들 수 있지만, 양극 활물질 및 도전체를 집전체 위로 결착 시킬 수 있는 것이라면, 특별히 제한 되지 않는다.

[0065] 양극집전체(30b)는 도전체라면 어떤 것이라도 좋고, 예를 들면, 알루미늄, 스테인리스강, 및 니켈 도금 강철 등으로 구성될 수 있다. 양극집전체(30b)에는 양극단자가 접속된다.

[0066] 락형 양극(30)은 예를 들면, 이하의 방법에 의해 제작된다. 즉, 양극 활물질층의 재료를 유기용제 또는 물에 분산시키는 것으로 양극합제 슬러리를 형성하고, 이 양극합제 슬러리를 집전체위로 도포한다. 이에 의해, 도포층이 형성된다. 이어서, 도포층을 건조한다. 이어서, 건조한 도포층을 양극집전체(30b)와 함께 압연한다. 이에 의해, 락형 양극(30)이 제작된다.

[0067] 전극 적층체(100a)는 락형 음극(10), 락형 세퍼레이터(20), 락형 양극(30), 및 락형 세퍼레이터(20)을 이 순으로 적층 하는 것에 의해 제조된다. 따라서, 전극 적층체(100a)의 한쪽 면(표면)에는 락형 세퍼레이터(20)가 배치되고, 이면에는 락형 음극(10)이 배치되므로, 전극 적층체(100a)를 감으면, 전극 적층체(100a)의 한 부분의 표면 (즉 락형 세퍼레이터(20))에 전극 적층체(100a)의 다른 부분의 이면 (즉 락형 음극(10))이 접촉한다.

[0068] 비수전해질용액은 전해질을 유기 용매에 용해시킨 용액이다. 전해질은 특별히 한정되지 않고, 예를 들면, 본 실시형태에서는 리튬 염을 사용할 수 있다. 리튬 염으로는 예를 들면, LiClO_4 , LiBF_4 , LiAsF_6 , LiPF_6 , $\text{LiPF}_6 \cdot x(\text{C}_n\text{F}_{2n+1})_x$ (단, $1 < x < 6$, $n=1$ 또는 2), LiSCN , LiBr , LiI , Li_2SO_4 , $\text{Li}_2\text{B}_{10}\text{Cl}_{10}$, NaClO_4 , NaI , NaSCN , NaBr , KClO_4 , KSCN 등의 리튬(Li), 나트륨(Na) 또는 칼륨(K)의 1종을 포함하는 무기 이온 염, LiCF_3SO_3 , $\text{LiN}(\text{CF}_3\text{SO}_2)_2$, $\text{LiN}(\text{C}_2\text{F}_5\text{SO}_2)_2$, $\text{LiN}(\text{CF}_3\text{SO}_2)(\text{C}_4\text{F}_9\text{SO}_2)$, $\text{LiC}(\text{CF}_3\text{SO}_2)_3$, $\text{LiC}(\text{C}_2\text{F}_5\text{SO}_2)_3$, $(\text{CH}_3)_4\text{NBF}_4$, $(\text{CH}_3)_4\text{NBr}$, $(\text{C}_2\text{H}_5)_4\text{NClO}_4$, $(\text{C}_2\text{H}_5)_4\text{NI}$, $(\text{C}_3\text{H}_7)_4\text{NBr}$, $(n\text{-C}_4\text{H}_9)_4\text{NClO}_4$, $(n\text{-C}_4\text{H}_9)_4\text{NI}$, $(\text{C}_2\text{H}_5)_4\text{N-maleate}$, $(\text{C}_2\text{H}_5)_4\text{N-benzoate}$, $(\text{C}_2\text{H}_5)_4\text{N-phthalate}$, 스테아릴 술폰산 리튬(stearyl sulfonic acid lithium), 옥틸 술폰산 리튬(octyl sulfonic acid), 도데킬벤젠술폰산 리튬(dodecyl benzene sulphonic acid) 등의 유기 이온 염 등을 들 수 있고, 이들 이온성 화합물을 단독, 혹은 2종류 이상 혼합해서 이용하는 것이 가능하다. 한편, 전해질염의 농도는 종래의 리튬 이차전지에서 사용되는 비수 전해액과 동일하면 되며, 특별히 제한은 없다. 본 실시형태에서는 적당한 리튬 화합물(전해질염)을 0.8~1.5mol/L정도의 농도로 함유시킨 비수전해액을 사용 할 수 있다.

[0069] 또, 유기 용매로는 예를 들면, 프로필렌 카보네이트(propylene carbonate), 에틸렌 카보네이트(ethylene carbonate), 부틸렌 카보네이트(ethylene carbonate), 클로로에틸렌 카보네이트(chloroethylene carbonate), 비닐렌 카보네이트(vinylene carbonate) 등의 환형 탄산에스테르(ester)류; γ -부티로락톤(butyrolactone), γ -발레로 락톤(valerolactone) 등의 환형에스테르류; 디메틸 카보네이트(dimethyl carbonate), 디에틸카보네이트(diethyl carbonate), 에틸 메틸 카보네이트(ethyl methyl carbonate) 등의 쇠상 카보네이트류; 포름산 메틸(methyl formate), 아세트산 메틸(methyl acetate), 부티르산 메틸(butyric acid methyl), 아세트산에틸(ethyl acetate), 피로피온산에틸(ethyl propionate) 등의 쇠상 에스테르류; 테트라하이드로푸란(Tetrahydrofuran) 또는 그 유도체; 1,3-디옥산(dioxane), 1,4-디옥산(dioxane), 1,2-디메톡시에탄(dimethoxyethane), 1,4-디부톡시에탄(dibutoxyethane), 메틸 디글라임(methyl diglyme) 등의 에테르(ether)류; 아세토니트릴(acetonitrile), 벤조니트릴(benzonitrile) 등의 니트릴(nitrile)류; 디옥솔란(Dioxolane) 또는 그 유도체; 에틸렌 설파이드(ethylene sulfide), 설폴란(sulfolane), 술폰(sultone) 또는 그 유도체 등의 단독 또는 2종 이상의 혼합물 등을 들 수 있지만, 이들에 한정되지 않는다. 비수전해질용액은 락형 세퍼레이터(20)에 함침된다. 한편, 상기의 각 전극에는 공지된 도전 보조제, 첨가제 등을 적당히 더해도 된다. 외장재는 예를 들면 알루미늄 라미네이트일 수 있다.

[0070] <2. 비수전해질 리튬이온 이차전지의 제조 방법>

[0071] 이어서, 비수전해질 리튬이온 이차전지의 제조 방법에 대하여 설명한다.

[0072] (락형 양극의 제조 방법)

[0073] 락형 양극(30)은 예를 들면, 이하의 방법에 의해 제작된다. 즉, 양극 활물질층의 재료를 유기 용매나 물에 분산시키는 것으로 양극합제 슬러리를 형성하고, 이 양극합제 슬러리를 집전체위로 도포한다. 이에 의해, 도포층이 형성된다. 이어서, 도포층을 건조한다. 이어서, 건조한 도포층을 양극집전체(30b)와 함께 압연한다. 이에 의해, 락형 양극(30)이 제작된다.

- [0074]

(띠형 음극의 제조 방법)
- [0075]

띠형 음극(10)은 예를 들면, 이하의 방법에 의해 제작된다. 즉, 음극 활물질층의 재료를 물에 분산시키는 것으로 음극합제 슬러리를 형성하고, 이 음극합제 슬러리를 집전체위로 도포한다. 이에 의해, 도포층이 형성된다. 이어서, 도포층을 건조한다. 음극합제 슬러리 중에는 플루오르 수지미립자 및 엘라스토머계 고분자의 미립자가 음극 활물질층(10a) 내에 분산되어 있다. 이어서, 건조한 도포층을 음극집전체(10b)와 함께 압연한다. 이에 의해, 띠형 음극(10)이 제작된다.
- [0076]

(띠형 세퍼레이터의 제조 방법)
- [0077]

띠형 세퍼레이터(20)은 이하의 방법에 의해 제작된다. 즉, 접착층(20a)의 재료를 물에 분산, 용해시키는 것으로 접착층 합제 슬러리를 형성한다. 이어서, 이 접착층 합제 슬러리를 띠형 다공질막(20c)의 양면 중, 적어도 한 쪽 표면에 도포하는 것으로 도포층을 형성한다. 이어서, 이 도포층을 건조한다. 이에 의해, 접착층(20a)이 형성된다. 즉, 띠형 세퍼레이터(20)가 제작된다.
- [0078]

(권회 소자 및 전지의 제조 방법)
- [0079]

이어서, 띠형 음극(10), 띠형 세퍼레이터(20), 띠형 양극(30), 및 띠형 세퍼레이터(20)를 이 순서대로 적층 하는 것에 의해 전극 적층체(100a)를 제작한다. 이어서, 전극 적층체(100a)를 감는다. 이에 의해, 전극 적층체(100a)의 한쪽 표면(즉, 띠형 세퍼레이터(20))에 전극 적층체(100a)의 다른쪽 이면(즉, 띠형 음극(10))이 접촉 한다. 이에 의해, 권회 소자(100)가 제작된다. 이어, 권회 소자(100)를 압축하는 것으로 편평한 형태의 권회 소자(100)를 제작한다. 이어, 편평한 형태의 권회 소자(100)를 비수전해액과 함께 외장체(예를 들면 라미네이트 필름)에 삽입하고, 외장체를 밀봉 하는 것에 의해, 리튬이온 이차전지를 제작한다. 한편, 외장체를 밀봉할 때는 각 집전체에 도통하는 단자를 외장체의 외부에 돌출시킨다.
- [0080]

[실시예]
- [0081]

(실시예1)
- [0082]

(양극의 제작)
- [0083]

코발트 산 리튬, 카본블랙, 폴리 불화 비닐리덴(PVDF)을 고형분의 질량비 96:2:2로 N-메틸 피롤리돈 중에 용해 분산시키는 것으로 양극합제 슬러리를 제작했다. 이어서, 양극합제 슬러리를 두께 12 μ m의 알루미늄 박 집전체의 양면에 도포 후, 건조했다. 건조후의 도포층을 압연하는 것으로 양극 활물질층을 제작했다. 집전체 및 양극 활 물질층의 총두께는 120 μ m이었다. 이어서, 알루미늄 리드 선을 전극단부에 용접하는 것에 의해 띠형 양극을 얻었다.
- [0084]

(음극의 제작)
- [0085]

흑연, 변성SBR미립자의 수분산체, PVDF 수분산체 내로 아크릴 수지를 중합시켜서 복합화한 플루오르 수지 함유 미립자의 수분산체, 카르복시메틸셀룰로오스의 나트륨 염을 고형분의 질량비 97:1:1:1로 물 용매 중에 용해 분 산시키는 것으로, 음극합제 슬러리를 제작했다. 이어서, 이 음극합제 슬러리를 두께 10 μ m의 구리박 집전체의 양 면에 도포후, 건조했다. 건조후의 도포층을 압연하는 것으로 음극 활물질층을 얻었다. 집전체 및 음극 활물질층 의 총두께는 120 μ m이었다. 그 후, 니켈 리드 선을 단부에 용접하는 것에 의해 띠형 음극을 얻었다. 한편, 본 실 시예에서 사용한 플루오르 수지 함유 미립자의 평균 입경을 레이저 회절법에서 측정한 바, 300nm정도이었다. 한 편, 다른 입자의 평균 입경도 동일한 방법으로 측정했다. 또, 플루오르 수지 함유 미립자를 SEM에서 관찰한 바, 구형입자이었다.
- [0086]

(세퍼레이터의 제작)
- [0087]

상기의 플루오르 수지 함유 미립자의 수분산체와, 평균 입경 480nm의 폴리에틸렌 아이오노머 수분산체(결착제 입자)와, 폴리 아크릴산 나트륨과, 평균 입경 90nm의 베마이트 입자(내열성 필러 입자)를 물 용매 중에 용해 분 산시키는 것에 의해 접착층 합제 슬러리를 제작했다. 여기에서, 플루오르 수지 함유 미립자와, 결착제 입자와, 폴리 아크릴산 나트륨과, 내열성 필러 입자와의 혼합비(부피비)는 40:10:0.3:50로 했다.
- [0088]

이어서, 이 접착층 합제 슬러리를 코로나 처리 완료된 두께 12 μ m의 다공질 폴리에틸렌 세퍼레이터 필름의 양면 에 도포하는 것으로 도포층을 형성했다. 그리고, 이들 도포층을 건조하는 것에 의해, 양면에 두께 3 μ m(세퍼레이 터 한쪽면당의 두께)의 접착층이 형성된 세퍼레이터를 얻었다.

- [0089]

(권회 소자의 제작)
- [0090]

음극, 세퍼레이터, 양극, 세퍼레이터를 이 순서대로 적층하고, 직경 3cm의 심지를 이용하여, 이 적층체를 길이 방향으로 감쌌다. 단부를 테이프로 고정한 후, 심지를 제거하고, 두께 3cm의 2장의 금속 플레이트 사이에 원통상 전극권회 소자를 끼우고, 3초간 유지하는 것으로, 편평한 형태의 전극권회 소자를 얻었다. 여기에서, 양극, 음극, 및 세퍼레이터의 TD방향(권회 방향(MD방향)과 수직인 방향)의 길이는 세퍼레이터(32mm)>음극(30mm)>양극(28mm)로 했다. 이것들을 중앙맞춤하여 배치하는 것으로, 세퍼레이터의 TD방향의 단부로부터 양극의 TD방향의 단부까지의 거리, 소위 마진은 상하 각각 2mm가 된다.
- [0091]

(두께 증가율의 평가)
- [0092]

이 전극권회 소자를 48시간 방치 전후 소자의 두께 증가율을 계측하여, 이로부터 형상 안정성을 평가했다. 두께 증가율이 작을수록 형상 안정성이 양호하므로(즉, 권회 소자의 왜곡이 작으므로) 바람직하다. 두께 증가율은 48시간 방치 전후 소자의 두께 증가량을 방치 전 소자의 두께로 나누는 것으로 얻어진다.
- [0093]

(전지의 제작)
- [0094]

상기 전극권회 소자를 폴리프로필렌/알루미늄/나일론의 3층으로 이루어지는 라미네이트 필름에, 2개의 리드 선이 밖으로 나오게 전해액과 함께 감압밀봉 하는 것으로, 전지를 제작했다. 전해액으로는 에틸렌 카보네이트/에틸 메틸 카보네이트를 3대7(부피비)로 혼합한 용매에 1M의 LiPF₆을 용해시킨 것을 사용했다. 이 전지를 90℃로 가열한 두께 3cm의 2장의 금속 플레이트 사이에 끼우고, 5분간 유지했다. 이 전지를 설계 용량 1/10CA (1CA는 1시간 방전율)로 4.4V까지 정전류 충전을 행하고, 계속해서 4.4V로 1/20CA가 될때까지 정전압 충전을 행했다. 그 후 1/2CA로 3.0V까지 정전류 방전을 행하는 것으로, 이차전지를 제작했다. 또, 이 때의 용량을 초기 방전 용량으로 했다.
- [0095]

(부하 특성)
- [0096]

그 다음에, 상기와 같은 조건으로 충전을 행한 후, 3/2CA로 3.0V까지 저전류 방전을 행하고, 그 때의 방전 용량을 계측했다. 이 방전 용량을 초기 방전 용량으로 나누는 것으로, 방전 용량 유지율 (부하 특성)을 산출했다. 방전 용량유지율이 높을수록 에너지를 효율적으로 이용 할 수 있으므로 바람직하다.
- [0097]

(수명시험)
- [0098]

제작한 전지를 0.5CA, 4.4V의 정전류 충전, 0.05CA까지의 정전압 충전의 충전 공정과, 0.5CA, 3.0V의 정전류 방전의 방전 공정을 반복하는 사이클 시험을 행하고, 100사이클 후의 초기 방전 용량에 대한 방전 용량의 감소율 (유지율)을 계측하여, 수명성능평가로 했다. 방전 용량의 감소율이 작을수록 수명특성이 우수하며, 바람직하다. 유지율은 100사이클 후의 방전 용량을 초기 방전 용량으로 나누는 것으로 얻어진다.
- [0099]

(도포층 필요두께)
- [0100]

상기 (세퍼레이터의 제작)과 동일한 처리에 의해, 필름 양면에 두께 1, 1.5, 2, 2.5 μ m (세퍼레이터 한쪽면당의 두께)의 도포층을 형성하고, 각 도포층을 건조하는 것에 의해, 필름 양면에 접착층을 형성했다. 즉, 접착층의 두께가 서로 다른 여러 종류의 세퍼레이터를 준비했다. 그리고, 이들 세퍼레이터를 이용해서 상술한 처리에 의해 이차전지를 제작했다. 이들 이차전지를 150℃의 항온조에서 1시간 정치 했다. 이차전지를 항온조로부터 꺼내서 방냉한 후, 이차전지를 해체하고, 이차전지 소자 중앙부의 세퍼레이터의 TD방향의 수축을 관찰했다. 도포층 두께(접착층 두께)가 작을수록 수축이 크지만, 시험 후, 양극과 음극이 세퍼레이터로 격리되는데 필요한 최소한의 두께를 도포층 필요두께(접착층 필요두께)로 판정했다. 필요두께가 작을수록 이차전지를 박막화 할 수 있고, 전지의 고에너지 밀도화에 기여하므로 바람직하다.
- [0101]

(실시예2~8)
- [0102]

실시예 2~8에서는 결착제 입자 및 내열성 필러 입자를 표1에 나타내는 구성으로 변경한 것 외에는 실시예 1과 동일한 처리를 행했다.
- [0103]

(비교예1)
- [0104]

세퍼레이터 제조에 있어서, PVDF를 N-메틸 피롤리돈에 용해시킨 용액을, 두께 12 μ m의 다공질 폴리에틸렌 필름의 양면에 도포하고, 용액이 도포된 필름을 수중에 침지시킨 후 건조하는 것으로, 필름의 양면에 그물코형으로 다공질화한 접착층을 형성했다. 접착층의 두께는 3 μ m이었다. 상기 이외는 실시예 1과 동일한 처리를 행했다.

[0105] (비교예 2, 3, 4)

[0106] 비교예 2, 3에서는 절착제 입자 및 내열성 필러 입자를 표1에 나타내는 구성으로 변경한 것 외에는 실시예 1과 동일한 처리를 행했다.

표 1

	세퍼레이터층 중의 플루오르수지함유 미립자의 유무	내열성 필러 입자		절착제 입자	
		조성	평균 입경 (nm)	조성	평균 입경 (nm)
실시예1	있음	베마이트	95	폴리에틸렌 아이오노머	480
실시예2	있음	TiO ₂	15	변성SBR	210
실시예3	있음	TiO ₂	15	변성SBR	180
실시예4	있음	알루미나	60	변성SBR	180
실시예5	있음	알루미나	60	폴리에틸렌 아이오노머	120
실시예6	있음	베마이트	95	폴리에틸렌 아이오노머	120
실시예7	있음	TiO ₂	15	변성SBR	480
실시예8	있음	TiO ₂	15	폴리에틸렌 아이오노머	120
비교예1	없음	-	-	-	-
비교예2	있음	TiO ₂	15	폴리아크릴산 에스테르	70
비교예3	있음	베마이트	95	폴리아크릴산 에스테르	610
비교예4	있음	알루미나	150	변성SBR	180

[0108] 또, 평가 결과를 표2 및 도 3에 나타낸다. 도 3의 가로축x는 내열성 필러 입자의 평균 입경을 나타내고, 세로축 y는 절착제 입자의 평균 입경을 나타낸다. 도 3의 「□」은 「절착제 입자의 평균 입경이 100~500nm이며, 내열성 필러 입자의 평균 입경이 10~100nm이다」라는 조건 외에 상술한 수식(1), (2)의 조건을 모두 충족하는 점이다. 「■」은 「절착제 입자의 평균 입경이 100~500nm이며, 내열성 필러 입자의 평균 입경이 10~100nm이다」라는 조건을 충족시키나, 수식(1), (2)의 조건을 충족하지 않는 점이다. 「◆」은 「절착제 입자의 평균 입경이 100~500nm이며, 내열성 필러 입자의 평균 입경이 10~100nm이다」라는 조건을 충족하지 않는다 (즉 본 발명의 범위 외의) 점이다. 그래프L1은 수식 $y=50/17*(x+70)$ 을 나타내고, 그래프L2은 $y=10/7*(120-x)$ 을 나타낸다.

표 2

	두께 증가율(%)	사이클 수명(%)	접착층 필요두께(μm)	부하특성(%)
실시예1	6	93	2	85
실시예2	6	93	1	84
실시예3	5	93	1	82
실시예4	5	92	1	83
실시예5	6	91	1	81
실시예6	7	91	2	82
실시예7	5	92	2	84
실시예8	6	90	1	81
비교예1	10	85	>3	83
비교예2	6	88	2	77
비교예3	5	89	3	84
비교예4	5	89	3	84

[0110] 실시예 1~8에서는 모두 두께 증가율이 작고, 사이클 수명도 양호했다. 또한, 실시예 1~8에서는 접착층 필요두께도 작고, 부하 특성도 양호했다. 따라서, 실시예 1~8에서는 접착층이 박막화되어, 에너지 밀도도 양호했다. 한편, 비교예 1은 플루오르수지 함유 고분자가 입자상이 아니고 그물코형구조를 하고 있기 때문에 편평한 형상의

권회 소자의 제조후의 형상 안정성이 뒤떨어지고 있었다. 다시 말해, 비교예 1의 권회 소자는 실시예의 권회 소자보다도 크게 일그러지고 있었다. 또한, 비교예 1에서는 사이클 수명도 열화되어 있는데, 이 이유는 편평형 권회 소자의 왜곡에 있다고 생각된다. 다시 말해, 비교예 1의 띠형 세퍼레이터는 실시예 1~8의 띠형 세퍼레이터보다도 미끄러짐성이 나쁘고, 권회 소자를 편평형으로 할 때에 전극 적층체끼리의 접촉 부분이 잘 미끄러지지 않았다. 그 결과, 권회 소자가 일그러졌다. 그리고, 비교예 1에서는 일그러진 권회 소자를 이용해서 전지를 제작하고 있다. 이 때문에, 전지 내에서 전극간 거리가 안정되지 않고, 사이클 수명이 저하되었다고 생각된다.

또, 비교예 2~4에서는 접착층 필요두께 및 부하 특성 중 적어도 한쪽이 실시예 1~8보다도 나빴다. 따라서, 실시예 1~8과 비교예 2~3을 비교하면, 결합제 입자의 평균 입경은 100~500nm이며, 내열성 필터 입자의 평균 입경은 10~100nm인 것이 필요한 것을 알 수 있다.

또, 실시예 1~6과 실시예 7~8을 비교하면, 실시예 1~6의 특성 쪽이 양호했다. 실시예 1~6은 수식(1), (2)의 조건을 모두 충족한다. 따라서, 수식(1), (2)의 적어도 한쪽이 충족되는 것이 바람직하고, 양쪽 채워지는 것이 더욱 바람직한 것도 알 수 있다.

이상에 의해, 본 실시형태에 따른 권회 소자는 제조 공정상에서 취급하기 쉬운 세퍼레이터를 이용해서 제작된다. 또, 권회 소자는 그 왜곡을 억제하고, 동시에, 비수전해질 이차전지의 사이클 수명을 향상시킬 수 있다. 또한, 결합제 입자의 평균 입경 및 내열성 필터 입자의 평균 입경이 상술한 요건을 충족하므로, 접착층이 박막화되어, 나아가서는 에너지 밀도가 향상된다. 또한, 권회 소자의 내열성, 즉 권회 소자를 사용한 비수전해질 이차전지의 내열성이 향상된다.

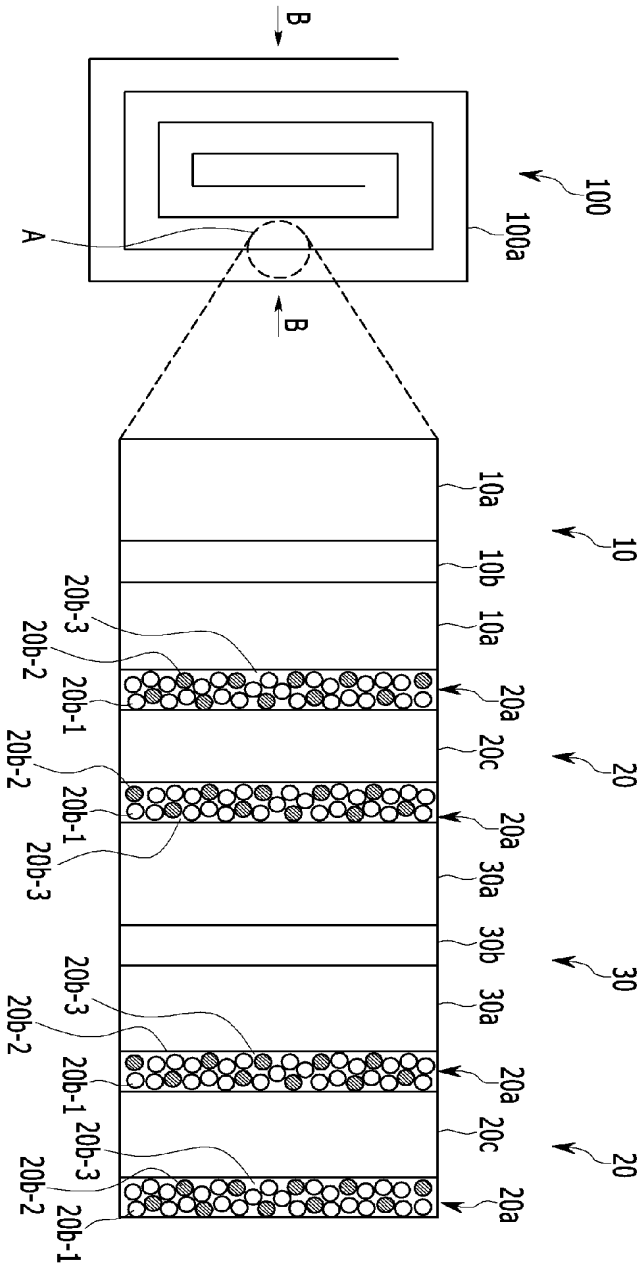
이상, 첨부 도면을 참조하면서 본 발명이 바람직한 실시형태에 대해서 상세하게 설명했지만, 본 발명은 이러한 예에 한정되지 않는다. 본 발명이 속하는 기술의 분야에 있어서의 통상의 지식을 갖는 자라면, 특허청구의 범위에 기재된 기술적 사상의 범주내에 있어서, 각종 변경예 또는 수정예에 이를 수 있는 것은 명확해서, 이것들에 대해서도, 당연히 본 발명의 기술적 범위에 속하는 것이라고 이해된다.

부호의 설명

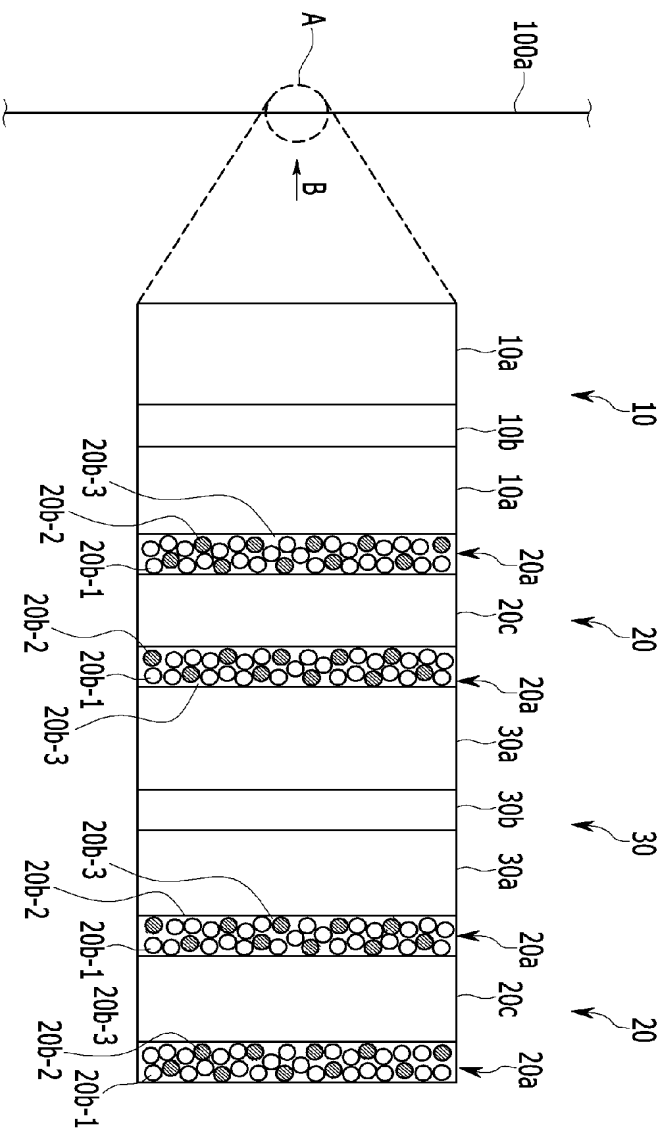
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- 10b 음극집전체
- 20 띠형 세퍼레이터
- 20a 접착층
- 20b-1 플루오르 수지 함유 미립자
- 20b-2 결합제 입자
- 20b-3 내열성 필터 입자
- 20c 띠형 다공질막
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도 11

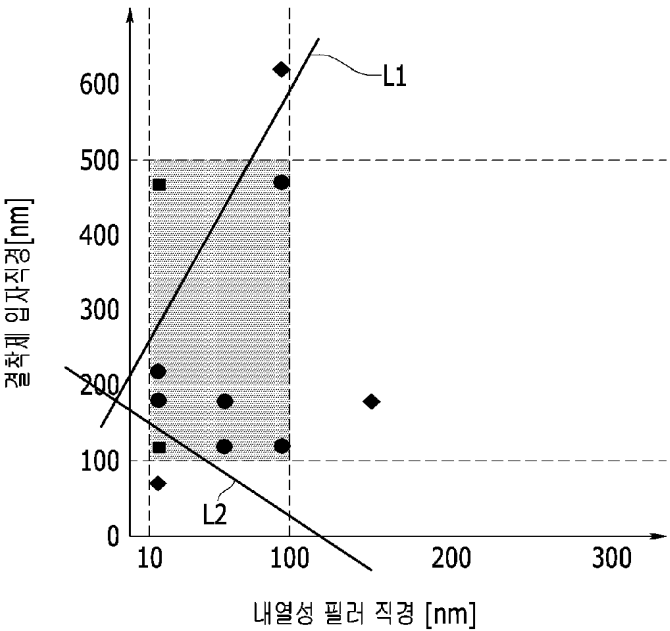
도 12



도 11



도 2





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METAL OXIDE ULTRAFINE FIBER-BASED COMPOSITE SEPARATOR WITH HEAT RESISTANCE AND SECONDARY BATTERY USING SAME

Inventor(s): JO SEONG MU [KR]; KIM DONG YOUNG [KR]; JANG SUNG YEON [KR] ± (JO, SEONG MU, ; KIM, DONG YOUNG, ; JANG, SUNG YEON)

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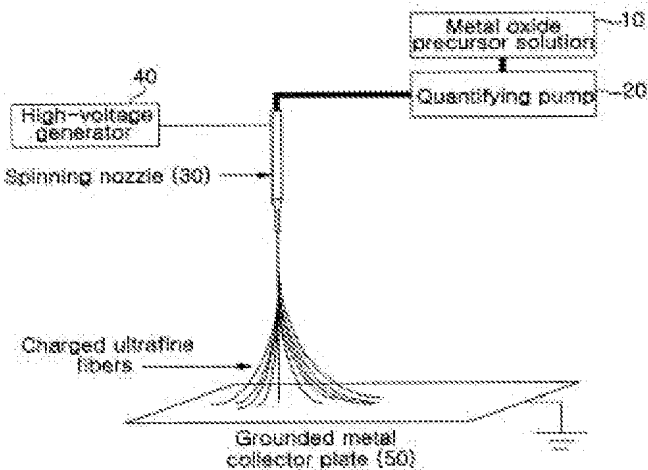
Application number: KR20100035935 20100419 Global Dossier

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Abstract not available for KR20110116489 (A)
Abstract of corresponding document: US2012003524 (A1)

An ultrafine fiber-based composite separator comprising a fibrous porous body which comprises ultrafine metal oxide/polymer composite fibers, or ultrafine metal oxide fibers and a polymer resin coating layer formed on the surface thereof, the ultrafine fibers being continuously randomly arranged and layered, and obtained by electrospinning a metal oxide precursor sol-gel solution or a mixture of a metal oxide precursor sol-gel solution and a polymer resin solution, wherein the surface of the metal oxide/polymer composite fibers has a uniform mixing composition of the metal oxide and the polymer resin, in which the separator has a heat shrinkage rate at 150~250° C. of 10% or less and does not break down due to melting at a temperature of 200° C. or lower, has low heat shrinkage rate, and superior heat resistance and ionic conductivity, being capable of providing improved cycle and power properties when used in manufacturing a battery.



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	심사청구일자		김동영
	2010년04월19일		서울특별시 동대문구 회기동 65번지 신현대아파트
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			대구광역시 중구 남산4동 2466-1 보성황실타운
			103동 1202호
		(74)	대리인
			장성구, 제일특허법인

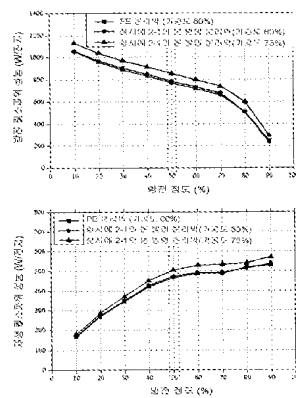
전체 청구항 수 : 총 18 항

(54) 금속산화물 초극세 섬유-기반 내열성 복합 분리막 및 이를 이용한 이차전지

(57) 요약

본 발명은 금속산화물 초극세 섬유-기반 내열성 복합 분리막 및 이를 이용한 이차전지에 관한 것으로, 금속산화물 전구체의 졸-겔 용액 또는 이와 고분자 수지 용액의 혼합물을 전기방사하여 배출된 초극세 금속산화물/고분자 복합섬유 또는 초극세 금속산화물 섬유가 연속적으로 랜덤하게 배열 및 축적되어 이루어진 다공체를 포함하고 (이때 상기 초극세 금속산화물 섬유로 이루어진 다공체는 그 표면에 고분자 수지 코팅층을 가짐), 150 내지 250℃에서의 열수축률이 10% 이하이고 200℃ 이하의 온도에서는 용융에 의한 붕괴를 일으키지 않는 본 발명의 복합 분리막은 열수축률이 낮고 우수한 내열성 및 이온전도도를 가져 전지 구성시 사이클 특성 및 출력 특성을 향상시킬 수 있어, 고에너지 밀도와 고용량의 이차전지의 제조에 유용하게 사용될 수 있다.

배 도 도 - 도8



특허청구의 범위

청구항 1

금속산화물 전구체의 졸-겔 용액 또는 금속산화물 전구체의 졸-겔 용액과 고분자 수지 용액의 혼합물을 전기방사(electrospinning)하여 배출된 초극세 금속산화물/고분자 복합섬유 또는 초극세 금속산화물 섬유가 연속적으로 랜덤하게 배열 및 축적되어 이루어진 다공체를 포함하고, 이때 상기 초극세 금속산화물 섬유로 이루어진 다공체는 그 표면에 고분자 수지 코팅층을 가지며,

150 내지 250℃에서의 열수축률이 10% 이하이고 200℃ 이하의 온도에서는 용융에 의한 붕괴를 일으키지 않는 것을 특징으로 하는,

초극세 섬유상 복합 분리막.

청구항 2

제 1 항에 있어서,

상기 고분자 수지 코팅층을 갖는 섬유 다공체가, 금속산화물 전구체의 졸-겔 용액 또는 금속산화물 전구체의 졸-겔 용액과 고분자 수지 용액(제1 고분자 수지 용액)의 혼합물을 전기방사하여 얻어지는 섬유 다공체를 300 내지 1000℃에서 소결한 후 제2 고분자 수지 용액으로 함침 또는 코팅하여 얻어진 것임을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 3

제 1 항에 있어서,

상기 고분자 수지 용액이 전해액에 팽윤성을 갖는 고분자 수지, 180℃ 이상의 용점을 갖거나 용융되지 않는 고분자 수지, 또는 이들의 혼합물을 포함하는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 4

제 3 항에 있어서,

상기 전해액에 팽윤성을 갖는 고분자 수지가 폴리비닐리덴플루오라이드, 폴리(비닐리덴플루오라이드-코-헥사플루오로프로필렌), 퍼플루오로폴리머, 퍼플루오로폴리머 공중합체, 폴리에틸렌글리콜 디알킬에테르, 폴리에틸렌글리콜 디알킬에스터, 폴리에틸렌옥사이드, 폴리프로필렌옥사이드, 폴리아크릴로니트릴, 폴리아크릴로니트릴 메틸메타크릴레이트 공중합체, 폴리우레탄, 폴리메틸메타크릴레이트, 폴리메틸메타크릴레이트 공중합체 및 이들의 혼합물로 이루어진 군으로부터 선택되는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 5

제 3 항에 있어서,

상기 180℃ 이상의 용점을 갖거나 용융되지 않는 고분자 수지가 폴리아마이드, 폴리이미드, 폴리아마이드이미드, 폴리(메타-페닐렌 이소프탈아마이드), 폴리설폰, 폴리에테르설폰, 폴리페닐렌설폰, 폴리에테르케톤, 폴리에테르이미드, 폴리에틸렌테레프탈레이트, 폴리트리메틸렌테레프탈레이트, 폴리에틸렌나프탈레이트, 폴리테트라플루오로에틸렌, 폴리우레탄, 폴리우레탄 공중합체 및 이들의 혼합물로 이루어진 군으로부터 선택되는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 6

제 2 항에 있어서,

상기 제1 고분자 수지 용액이 폴리비닐피롤리돈, 폴리비닐알콜, 폴리비닐아세테이트, 폴리에틸렌 옥사이드 및 이들의 혼합물로 이루어진 군으로부터 선택되는 고분자 수지를 포함하는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 7

제 2 항에 있어서,

상기 제2 고분자 수지 용액이 전해액에 팽윤성을 갖는 고분자 수지, 180℃ 이상의 용점을 갖거나 용융되지 않는 고분자 수지, 또는 이들의 혼합물을 포함하는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 8

제 1 항에 있어서,

상기 섬유의 다공체를 구성하는 금속산화물이 실리카(SiO_2), 알루미나(Al_2O_3), 보헤마이트(bohemite), ZnO_2 , LiAlO_2 , TiO_2 , BaTiO_3 , LiTiO_3 및 이들의 혼합물로 이루어진 군으로부터 선택되는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 9

제 1 항에 있어서,

상기 초극세 금속산화물/고분자 복합섬유가 5~80 중량%의 금속산화물을 함유하는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 10

제 1 항에 있어서,

상기 금속산화물 전구체의 졸-겔 용액이 실리카(SiO_2), 알루미나(Al_2O_3), 보헤마이트, ZnO_2 , LiAlO_2 , TiO_2 , BaTiO_3 , LiTiO_3 및 이들의 혼합물로 이루어진 군으로부터 선택된 금속산화물의 나노입자를 금속산화물 전구체의 졸-겔 용액으로부터 얻어지는 금속산화물의 중량에 대해 0.1~200 중량%의 양으로 추가로 포함하는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 11

제 1 항에 있어서

상기 섬유의 다공체가 초극세 섬유상 사이에 분산된 바인더를 추가로 포함하는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 12

제 1 항에 있어서,

상기 섬유의 다공체가 고분자 수지의 유리전이온도(T_g) 이상이면서 용융온도(T_m) 이하의 온도에서 열압착되거나, 열압착 후 열연신되거나, 열압착 후 열연신에 이어 추가로 열압착된 것임을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 13

제 12 항에 있어서,

상기 섬유의 다공체가 열압착 또는 열연신 후 100~250℃의 온도에서 열처리된 것임을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 14

제 1 항에 있어서,

상기 전기방사가 용융취입(melt-blowing), 플래쉬 방사(flash spinning) 또는 전기취입(electro-blowing)인 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 15

제 1 항에 있어서,
상기 전기방사가 실온 내지 150℃의 온도에서 수행되는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 16

제 1 항에 있어서,
상기 초극세 섬유가 1 내지 3000nm 범위의 직경을 갖는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 17

제 12 항에 있어서,
상기 분리막이 40~75%의 기공도를 갖는 것을 특징으로 하는, 초극세 섬유상 복합 분리막.

청구항 18

양극, 음극, 및 상기 양극과 음극 사이에 위치한 제 1 항 내지 제 17 항 중 어느 한 항의 초극세 섬유상 복합 분리막과 전해질을 포함하는 이차전지.

명세서

기술분야

본 발명은 열수축률이 낮고 우수한 내열성 및 이온전도도를 갖는, 금속산화물 초극세 섬유-기반 복합 분리막, 및 이를 포함함으로써 우수한 사이클 특성 및 출력 특성을 갖는 고에너지 밀도 및 고용량의 이차전지에 관한 것이다.

배경기술

리튬이온 이차전지, 리튬이온 고분자전지 및 슈퍼 캐패시터(전기이중층캐패시터 및 유사캐패시터)와 같은 이차전지는 고성능화, 경량화, 및 자동차 전원용과 같은 대형화 추세에 따라 고에너지 밀도 및 고용량의 것이 요구되고 있다.

그러나, 폴리올레핀 분리막과 액체 전해질을 사용하는 기존의 리튬이온 이차전지, 및 겔 고분자전해질막 또는 폴리올레핀 분리막에 겔 코팅한 고분자 전해질을 사용하는 기존의 리튬이온 고분자전지는 내열성 측면에서 고에너지 밀도 및 고용량 전지로서 이용하기에 크게 부족한 실정이다.

분리막은 전지의 양극과 음극 사이에 위치하여 절연을 시키며, 전해액을 유지시켜 이온전도의 통로를 제공하며, 전지의 온도가 지나치게 높아지면 전류를 차단하기 위하여 분리막의 일부가 용융되어 기공을 막는 폐쇄(shutdown)기능을 제공한다. 온도가 더 올라가 분리막이 용융되면 큰 홀이 생겨 양극과 음극 사이에 단락이 발생된다. 이 온도를 단락온도(short circuit temperature)라 하는데, 일반적으로 분리막은 낮은 폐쇄온도와 보다 높은 단락온도를 가져야 한다. 폴리에틸렌 분리막의 경우 전지의 이상 발열시 150℃ 이상에서 수축하여 전극 부위가 드러나게 되어 단락이 유발될 가능성이 있다.

그러므로, 고에너지 밀도 및 고용량의 이차전지를 제조하기 위해서는 내열성이 우수하여 열수축률이 작고, 높은 이온전도도에 따라 우수한 사이클 성능을 갖는 분리막이 필요하다.

이러한 분리막을 얻기 위하여, 미국 공개특허 제 2006/0019154 A1 호는 폴리올레핀계 분리막(통기도(air permeability) 200초/분 이하)을 용점이 180℃ 이상인 폴리아마이드, 폴리이미드 또는 폴리아마이드이미드 용액에 함침시킨 후, 용고액에 침지하여 용매를 추출함으로써 다공성 내열성 수지가 코팅된 폴리올레핀 분리막을 제조하는 것을 개시하고 있다.

일본 공개특허 제 2005-209570 호는 고에너지 밀도화 및 대형화시 충분한 안전성을 확보하기 위하여, 200℃ 이상의 용융점을 갖는 방향족 폴리아마이드, 폴리이미드, 폴리에테르 설폰, 폴리에테르 케톤, 폴리에테르이미드 등의 내열성 수지 용액을 폴리올레핀 분리막의 양면에 코팅하고 이를 용고액에 침지, 수세, 건조하여 내열성 수

지가 코팅된 폴리올레핀 분리막을 제조하는 것을 개시하고 있다. 이때, 이온전도도의 저하를 줄이기 위하여 상기 내열성 수지 용액에 다공성 부여를 위한 상분리제를 첨가하고, 내열성 수지의 코팅량도 0.5-6.0g/m²로 제한하고 있다.

[0008] 그러나, 상술한 내열성 수지에 침지 또는 내열성 수지로의 코팅은 폴리올레핀 분리막의 기공을 막아 리튬이온의 이동을 제한하므로 충방전 특성의 저하를 일으킨다. 또한, 폴리올레핀 분리막의 기공구조가 막히지 않는다고 하더라도, 보편적으로 사용되는 폴리올레핀 분리막은 40% 정도의 기공도 및 수십 나노미터의 기공크기를 가지므로 대용량 전지에는 불충분한 이온전도도를 제공한다.

[0009] 미국 특허 제 6,447,958 B1 호는 세라믹 분말과 내열성 절소함유 방향족 고분자를 포함하는 액상 슬러리를 폴리올레핀, 레이온, 비닐론, 폴리에스터, 아크릴, 폴리스티렌, 나일론 등의 다공성 직포, 부직포, 종이, 다공성 필름 등에 코팅한 후 응고액에 침지하여 용매를 추출함으로써 내열성 분리막을 제조하는 것을 개시하고 있다. 그러나, 이 방법은 내열성 고분자를 코팅한 후 응고액에 침지하고 수세 및 건조하는 일련의 공정이 너무 복잡하고 비용을 증가시킨다는 단점을 갖는다.

[0010] 일본 공개특허 제 2001-222988 호 및 제 2006-59717 호는 용점이 150℃ 이상인 폴리아라미드, 폴리이미드의 직포, 부직포, 천, 다공성 필름 등의 지지체를 폴리에틸렌옥사이드, 폴리프로필렌옥사이드, 폴리에테르 및 폴리비닐리덴과 같은 고분자의 겔 전해질에 함침하거나, 상기 지지체에 상기 고분자의 겔 전해질을 코팅하여 내열성 전해질막을 제조하는 것을 개시하고 있다. 그러나, 이 경우도 내열성은 만족스러울 수 있으나, 이온전도 측면에서는 지지체나 내열성 방향족 고분자층에서의 이온이동이 기존의 리튬이온 전지의 분리막이나 겔 전해질의 경우와 비슷하게 여전히 제한을 받는다.

[0011] 국제공개 제 WO 05/057700 A1 호는 미세 무기입자를 포함하는 고체입자를 함유한 고분자 용액을 전기방사하여 다공성 복합막을 제조하는 것을 개시하고 있다. 그러나, 고농도의 고체입자를 함유한 고분자 용액이 불균일한 분산용액임에 기인하여 노즐이 막히는 문제가 발생하고, 제조된 초극세 섬유에 표면의 고체입자가 응집된 상태로 불균일하게 분포된다. 따라서, 균질 혼합된 경우와는 달리 응집된 고체입자와 고분자 사이에 결합에 생겨 균질 혼합의 경우보다 기계적 물성이 저하된다. 또한, 상기 다공성 복합막을 전지의 분리막으로 사용할 경우 노출된 고체입자들의 전기화학적 불안정성으로 인하여 분리막의 전기화학적 안정성이 크게 저하되고, 미세 무기입자들의 혼성으로부터 기대되는 내열성의 증대가 미미하다.

[0012] 이와 같이, 상기 종래 기술에 개시된 분리막 및 전해질막은 여전히 내열성과 이온전도도를 동시에 충족하지 못하며, 내열성 코팅은 출력 특성의 저하 또한 가져온다. 따라서, 내열성과 더불어 급속 충방전과 같은 가혹조건 하에서 우수한 성능이 요구되는 자동차 전원용과 같은 고에너지 밀도 및 대용량의 전지에 사용되기는 어려운 실정이다.

발명의 내용

해결하려는 과제

[0013] 따라서, 본 발명의 목적은 상기와 같은 문제점을 해결하기 위한 것으로서 열수축률이 낮고 우수한 내열성 및 이온전도도를 가짐으로써 전지 구성시 사이클 특성 및 출력 특성을 향상시킬 수 있는 분리막, 및 이를 이용한 이차전지를 제공하는 것이다.

과제의 해결 수단

[0014] 상기 목적을 달성하기 위하여 본 발명은,

[0015] 금속산화물 전구체의 졸-겔 용액 또는 금속산화물 전구체의 졸-겔 용액과 고분자 수지 용액의 혼합물을 전기방사(electrospinning)하여 배출된 초극세 금속산화물/고분자 복합섬유 또는 초극세 금속산화물 섬유가 연속적으로 랜덤하게 배열 및 축적되어 이루어진 다공체를 포함하고, 이때 상기 초극세 금속산화물 섬유로 이루어진 다공체는 그 표면에 고분자 수지 코팅층을 가지며,

[0016] 150 내지 250℃에서의 열수축률이 10% 이하이고 200℃ 이하의 온도에서는 용융에 의한 붕괴를 일으키지 않는 것을 특징으로 하는,

[0017] 초극세 섬유상 복합 분리막을 제공한다.

[0018] 본 발명은 또한, 양극, 음극, 및 이들 두 전극 사이에 위치한 상기 초극세 섬유상 복합 분리막과 전해질을 포함하는 이차전지를 제공한다.

[0019]

발명의 효과

[0020] 이상에서 설명한 바와 같이, 본 발명에 따른 내열성 초극세 섬유상 복합 분리막은 열수축률이 낮고 우수한 내열성 및 이온전도도를 가지며 전극과의 접착성이 우수하여 전지 구성시 싸이클 특성 및 출력 특성을 향상시킬 수 있어, 고에너지 밀도와 고용량의 이차전지의 제조에 유용하게 사용될 수 있다.

[0021] 즉, 본 발명의 내열성 초극세 섬유상 복합 분리막은 전기자동차, 하이브리드 전기자동차 및 연료전지 자동차 등과 같이 높은 내열성과 열안정성을 요구하는 전기화학소자(리튬이온 이차전지, 리튬이온 고분자전지, 슈퍼 캐패시터를 포함하는 이차전지)의 제조에 특히 유용하다.

도면의 간단한 설명

[0022] 도 1은 본 발명에서 사용하는 전기방사 장치의 일례의 모식도이고,

도 2는 비교예 1에서 제조된 금속산화물 나노입자/고분자 복합 초극세 섬유의 전자투과현미경 사진으로서, (a) SiO_2/PAN (1/1.8 중량비), (b) TiO_2/PVdF (1:2 중량비), (c) TiO_2/PVdF (1:1 중량비)이고,

도 3은 비교예 1에서 제조된 금속산화물 나노입자 복합 고분자 분리막의 전기화학적 안정성 측정결과로서, (a) SiO_2/PVdF (1:2 중량비), (b) TiO_2/PVdF (1:2 중량비)이고,

도 4는 실시예 1-1, 1-2 및 1-3에서 각각 제조된 초극세 섬유상 금속산화물 다공체의 전자투과현미경 사진으로서, (a) SiO_2 , (b) Al_2O_3 , (c) 고분자 수지 함침된 SiO_2 초극세 섬유상 다공체이고,

도 5는 실시예 2-1에서 제조된 초극세 섬유상 SiO_2/PVdF (1:1 중량비) 내열성 복합 분리막의 전자투과현미경 사진으로서, (a) 열압착 전, (b) 열압착 후, (c) 용매를 사용하여 PVdF 성분을 추출 제거한 후이고,

도 6은 실시예 2-1에서 제조된 초극세 섬유상 내열성 복합 분리막의 전기화학적 안정성 측정결과로서, (a) SiO_2/PVdF (1:1 중량비), (b) TiO_2/PVdF (1:1 중량비)이고,

도 7은 실시예 2-1에서 제조된 내열성 복합 분리막의 TMA 분석결과(고온 용융 보존성)로서, (a) PE 분리막 및 PVdF 초극세 섬유상 분리막, (b) 초극세 섬유상 SiO_2/PVdF (1:1 중량비), (c) 250℃에서 약 1시간 방치한 후의 초극세 섬유상 SiO_2/PVdF (1:1 중량비) 분리막이고,

도 8은 실시예 2-1에서 제조된 SiO_2/PVdF (1:1 중량비) 내열성 복합 분리막의 출력 특성 그래프이고,

도 9는 실시예 2-1에서 제조된 SiO_2/PVdF (1:1 중량비) 내열성 복합 분리막의 충방전 특성 그래프이다.

발명을 실시하기 위한 구체적인 내용

[0023] 본 발명에 따른 내열성 초극세 섬유상 복합 분리막은 금속산화물 전구체의 졸-겔 용액 또는 금속산화물 전구체의 졸-겔 용액과 고분자 수지 용액의 혼합물의 전기방사(electrospinning)에 의해 제조된 "초극세 금속산화물/고분자 복합섬유의 다공체" 또는 전기방사에 의해 제조된 "표면에 고분자 수지 코팅층을 갖는 초극세 금속산화물 섬유의 다공체"를 포함하되, 150 내지 250℃에서 10% 이하의 열수축률을 가지며 200℃까지는 용융에 의한 붕괴를 일으키지 않는 것을 특징으로 한다.

[0024] 본 발명에 따른 복합 분리막을 구성하는 초극세 섬유의 다공체는 "금속산화물 전구체의 졸-겔 용액" 또는 "금속산화물 전구체의 졸-겔 용액과 고분자 수지 용액의 혼합물"을 전기방사하여 제조된다. 이 초극세 섬유는, 일반적으로 화학적 합성방법에 의해 제조되는 육안으로 볼 때 분말상인 금속산화물계 나노섬유, 나노로드,

나노튜브, 나노입자 등과는 달리, 상술한 해당 용액을 고전압 전기장하에서 노즐을 통해 물리적으로 압출하여 형성되는 섬유이므로 수 nm 내지 수 μm 의 직경 및 수십 cm 내지 수백 m의 길이를 갖는 초극세 구조의 연속 섬유상이다. 본 발명의 금속산화물 초극세 섬유의 다공체는 육안으로 볼 때 전기방사에 의해 제조되는 초극세 섬유가 연속적으로 랜덤하게 배열 및 축적되어 다공성 분리막 형태를 이루고 있다.

[0025] 본 발명에 따라 금속산화물 초극세 섬유상을 형성시키는 전기방사의 원리는 여러 문헌에 잘 나타나 있는데[G. Taylor. Proc. Roy. Soc. London A, 313, 453(1969); J. Doshi and D. H. Reneker, J. Electrostatics, 35 151(1995)], 도 1에 나타난 바와 같이, 임계전압 이상의 고전압 전기장하에서 저점도의 액체가 극미세 방울로 스프레이되는 현상인 정전 스프레이(electrostatic spray)와는 달리, 충분한 점도를 지닌 고분자 수지 용액이나 고분자 수지 용융체가 고전압 정전기력을 부여받을 때 초극세 섬유가 형성되며, 이러한 공정을 전기방사라고 지칭한다. 전기방사 장치는 고분자 수지 용액을 저장하는 배럴, 일정 속도로 고분자 수지 용액을 토출하는 정량 펌프, 및 고전압 발생기가 연결된 방사노즐을 포함한다. 정량 펌프를 통하여 토출되는 금속산화물 전구체 용액은 고전압 발생기에 의하여 하전된 방사 노즐을 통과하면서 초극세 섬유로 방출되고, 이는 일정 속도로 이동하는 컨베이어 형태의 접지된 집전판 위에 다공성 초극세 섬유로서 축적된다. 이와 같은 금속산화물 전구체 용액의 전기방사에 의해 수 내지 수천 나노미터 크기를 갖는 초극세 섬유의 제조가 가능하고, 섬유의 생성과 동시에 3차원의 네트워크 구조로 융착 및 적층되어 목적하는 금속산화물 초극세 섬유상의 다공성 웹이 형성될 수 있다. 이 초극세 섬유상의 다공성 웹은 초박막, 초경량이며, 기존 섬유에 비해 부피 대비 표면적 비가 지극히 높고, 높은 기공도를 갖는다 (도 1 참조).

[0026] 본 발명에 따르면, 금속산화물 초극세 섬유상 다공체를 구성하는 섬유의 평균 직경은 분리막의 기공도 및 기공 크기 분포에 매우 큰 영향을 미친다. 섬유 직경이 작을수록 기공 크기가 작아지며, 기공 크기 분포도 작아진다. 또한, 섬유의 직경이 작을수록 섬유의 비표면적이 증대되어 전해액 보액능력이 커지게 되므로 전해액 누액의 가능성이 줄어들게 된다. 따라서, 금속산화물 초극세 섬유상 다공체의 섬유 직경은 1-3000nm 범위이며, 바람직하게는 1-1000nm 범위이고, 더 바람직하게는 50-800nm 범위이다. 초극세 섬유상 분리막의 기공도는 바람직하게는 40-75%일 수 있다. 기공도가 40% 미만이면 이온전도도가 저하되고, 75%를 초과하면 분리막의 기계적 특성이 열악해진다.

[0027] 본 발명에서 금속산화물 초극세 섬유상의 형성은 상기 전기방사의 개념을 확장한, 용융취입(melt-blowing), 플래쉬 방사(flash spinning), 또는 이들 공정의 변형으로서 고전압 전기장과 에어분사에 의해 초극세 섬유를 제조하는 전기취입(electro-blowing)에 의해서 수행될 수 있다. 이들 방법 모두 전기장하에서 노즐을 통해 압출하는 전기방사법과 동일한 개념을 지니므로 본 발명에서의 전기방사는 이러한 방법을 모두 포함한다.

[0028] 보다 구체적으로, 본 발명에 따른 초극세 섬유상 복합 분리막은, (i) 금속산화물 전구체의 졸-겔 용액과 고분자 수지 용액의 혼합물을 전기방사하여 얻어지는 초극세 금속산화물/고분자 복합섬유의 다공체로부터 제조되거나 (소결공정을 거치지 않는 방법), 또는 (ii) 금속산화물 전구체의 졸-겔 용액, 또는 이 졸-겔 용액과 고분자 수지 용액(제1 고분자 수지 용액)의 혼합물을 전기방사하여 얻어지는 섬유의 다공체를 300 내지 1000℃에서 소결한 후 (소결에 의해 고분자 수지 성분 제거됨) 제2 고분자 수지 용액으로 함침 또는 코팅하여 얻어지는, 표면에 고분자 수지 코팅층을 갖는 초극세 금속산화물 섬유의 다공체로부터 제조될 수 있다 (소결공정을 거치는 방법).

[0029] 소결공정을 거치지 않는 금속산화물/고분자 복합섬유의 다공체의 제조에 사용되는 고분자 수지로는 이차전지의 전해액에 친화력이 있어 전해액에 팽윤성을 갖는 고분자 수지, 180℃ 이상의 용점을 갖거나 용융되지 않는 내열성 고분자 수지, 또는 이들의 혼합물이 적합하다. 전해액에 팽윤성을 갖는 고분자 수지는 전기화학적으로 안정한 것이면 특별히 제한되지는 않으나, 전해액에 팽윤성을 갖는 고분자 수지의 구체적인 예로는 폴리비닐리덴플루오라이드(polyvinylidene fluoride), 폴리(비닐리덴플루오라이드-코-헥사플루오로프로필렌) poly(vinylidenefluoride-co-hexafluoropropylene), 퍼플루오로폴리머(perfluoropolymer), 퍼플루오로폴리머 공중합체, 폴리에틸렌글리콜 디알킬에테르(polyethyleneglycol dialkylether) 및 폴리에틸렌글리콜 디알킬에스터(polyethyleneglycol dialkylester)와 같은 폴리에틸렌글리콜 유도체, 폴리에틸렌옥사이드(polyethylene oxide) 및 폴리프로필렌옥사이드(polypropylene oxide)와 같은 폴리옥사이드, 폴리아크릴로니트릴(polyacrylonitrile), 폴리아크릴로니트릴 메틸메타크릴레이트 공중합체와 같은 폴리아크릴로니트릴 공중합체, 폴리우레탄(polyurethane), 폴리메틸메타크릴레이트(polymethylmethacrylate), 폴리메틸메타크릴레이트 공중합체 및 이들의 혼합물을 들 수 있다.

[0030] 나아가, 180℃ 이상의 용점을 갖거나 용융되지 않는 내열성 고분자 수지의 구체적인 예로는 폴리아마이드(polyamide), 폴리이미드(polyimide), 폴리아마이드이미드(polyamideimide), 폴리(메타-페닐렌 이소프탈아마이드)

드)(poly(meta-phenylene isophthalamide), 폴리설향(polysulfone), 폴리에테르설향(polyethersulfone), 폴리페닐렌설향(polyphenylenesulfone), 폴리에테르케톤(polyetherketone), 폴리에테르이미드(polyetherimide), 폴리에틸렌테레프탈레이트(polyethyleneterephthalate), 폴리트림에틸렌테레프탈레이트(polytrimethyleneterephthalate), 폴리에틸렌나프탈레이트(polyethylenenaphthalate)와 같은 방향족 폴리에스터, 폴리테트라플루오로에틸렌(polytetrafluoroethylene), 폴리우레탄(polyurethane), 폴리에테르우레탄과 같은 폴리우레탄 공중합체 및 이들의 혼합물을 들 수 있다.

[0031] 소결공정을 거치는, 고분자 수지 코팅층을 갖는 초극세 금속산화물 섬유의 다공체의 제조에 사용되는 고분자 수지(제1 고분자 수지)로는 300 내지 1000℃ 범위에서 소결 후 탄소성분이 잔류되지 않는 것이 적합하다. 이러한 고분자 수지의 구체적인 예로는 폴리비닐피롤리돈(Polyvinylpyrrolidone, PVP), 폴리비닐알콜(Polyvinylalcohol, PVA), 폴리비닐아세테이트(polyvinylacetate, PVAc), 폴리에틸렌 옥사이드(Polyethylene oxide, PEO) 및 이들의 혼합물을 들 수 있으며, 소결과정에서 탄소성분이 잔류되지 않는 고분자이면 모두 가능하다.

[0032] 소결공정 이후, 초극세 섬유의 다공체를 함침하거나 코팅하는 데에 사용되는 고분자 수지(제2 고분자 수지)로는 전해액에 팽윤성을 갖는 고분자 수지, 180℃ 이상의 용점을 갖거나 용융되지 않는 내열성 고분자 수지, 또는 이들의 혼합물을 사용할 수 있으며, 이들의 구체적인 예는 상기에서 나열한 바와 같다. 상기 제2 고분자 수지는 초극세 섬유상 금속산화물 다공성 지지체의 섬유간 결합력을 증대시켜 복합분리막의 기계적 특성을 향상시키고, 전해액에 친화력을 지닌 고분자성분으로 인해 전해액 보지력을 증대 및 전기화학적 안정성을 증대시키는 역할을 한다. 종래의 기술에서는 폴리올레핀 다공막을 고분자 수지에 함침 또는 고분자 수지로 코팅한 후 물 또는 유기 용제 수용액의 응고액에 침지 응고시키고 수세 건조하여 내열성 복합 분리막을 형성시키므로 기공구조가 폐쇄되는 문제가 있다. 그러나, 본 발명에 따른 전기방사법에 의해 얻어진 금속산화물 초극세 섬유 다공체는 통상 70~95%의 매우 높은 기공도를 가지고 기공구조도 개방기공으로 이루어져 있으므로 고분자 수지와 함침 또는 코팅에 의한 결과과정에서 기공이 폐쇄되지 않아 분리막 성능이 저하되지 않는다.

[0033] 본 발명에 사용되는 금속산화물 전구체의 졸-겔 용액은 용매로서 물, 알콜 또는 이들의 혼합물을 포함하므로 친수성을 지니고 있으며, 알콜로는 메탄올, 에탄올, 프로판올, 부탄올 등 탄소수가 비교적 적은 저급 알콜을 사용할 수 있다. 반면에, 본 발명에서 사용하는 상기 고분자 수지는 물이나 알콜에 용해되지 않으며, 고분자 수지 용액은 용매로서 디메틸포름아마이드(dimethylformamide, DMF), 디메틸아세트아마이드(dimethylacetamide, DMAc), 디메틸설폭사이드(dimethylsulfoxide, DMSO), 아세톤(acetone), 테트라하이드로퓨란(tetrahydrofuran, THF), 이들의 혼합물 등과 같은 유기용매를 포함한다.

[0034] 금속산화물 전구체의 졸-겔 용액과 고분자 수지 용액의 혼합물은 고형분(금속산화물과 고분자 수지의 함량)의 농도가 낮은 경우에는 균질 용액을 형성하나, 고형분 농도가 높으면 금속산화물 졸-겔 용액과 고분자 수지 용액 간에 상분리가 일어나 불균질 용액을 형성한다. 결과적으로, 고형분 농도가 너무 낮거나, 또는 고형분 농도가 높으면 심각한 상분리가 일어나 두 경우 모두 전기방사시 섬유 형성이 어렵다. 따라서, 혼합용액을 실온에서 전기방사할 경우 섬유 형성이 가능한 고형분 농도를 갖는 용액을 사용할 수도 있지만, 실온에서 상분리가 일어나는 높은 고형분 농도를 갖는 용액이라도 용액의 온도를 높이면 고형분의 용해도가 증대되고 금속산화물 졸-겔 용액과 고분자 수지 용액 간에 친화력이 증대하여 상분리를 줄일 수 있다.

[0035] 따라서, 본 발명에서 금속산화물 전구체의 졸-겔 용액과 고분자 수지 용액의 혼합물의 전기방사는 실온 내지 150℃의 온도에서 실시하는 것이 바람직하며, 이보다 높은 온도에서는 금속산화물 졸-겔 반응이 과도하게 촉진되어 금속산화물 입자가 석출되어 용액이 고화되므로 전기방사에 이용할 수 없게 된다.

[0036] 또한, 금속산화물 전구체의 졸-겔 용액은 고분자 수지 용액의 유기용매에 잘 혼합되나, 고분자 수지는 금속산화물 전구체의 졸-겔 용액에 용해되지 않기 때문에, 금속산화물 전구체의 졸-겔 용액과 고분자 수지 용액의 혼합물에서 금속산화물 전구체의 졸-겔 용액의 함량이 높아지면 고분자가 석출된다. 따라서, 초극세 금속산화물/고분자 복합섬유의 중량에 대해 금속산화물의 함량이 5~80 중량%인 것이 바람직하며, 더욱 바람직하게는 5~50 중량%일 수 있다.

[0037] 또한, 초극세 섬유상 분리막의 강도 증대를 위하여, 초극세 섬유의 다공체에 초극세 섬유간 결합을 강화시킬 수 있는 바인더를 추가로 분산시킬 수 있다. 이러한 바인더는 도입은 전기방사하는 과정에서 금속산화물 전구체의 졸-겔 용액의 방사노즐 또는 이와 고분자 수지의 혼합물의 방사노즐이 아닌, 별도의 방사노즐을 사용하여 바인더 분산액을 함께 분무할 수 있으며, 이로써 초극세 섬유상 사이에 바인더가 추가로 분산되어 분포된 다공체를 제조할 수 있다. 이때 사용되는 바인더는 화학적 가교반응이 가능한 우레탄, 열가소성 수지, 또는 폴리테트라플

루오로에틸렌 및 그의 유도체, 폴리에틸렌, 폴리프로필렌, 저융점 폴리에스터와 같은 용융 또는 화학적 가교반응에 의해 초극세 섬유간 결합을 시킬 수 있는 것이면 모두 가능하나, 전기화학적으로 안정하여야 하고 가능한 용점이 높은 수지로 이루어진 바인더가 바람직하다.

[0038] 본 발명에 사용되는 금속산화물 전구체는 $M(OR)_x$, $MR_x(OR)_y$, MX_y , $M(NO_3)_y$ ($M = Si, Al, Zn, Li, Ti, Ba$ 등을 포함하는 금속; $R =$ 알킬그룹 ; $X = F, Cl, Br, I$; x 및 $y = 1\sim 4$) 또는 이들의 혼합물일 수 있으며, 초극세 섬유의 다공체를 구성하는 금속산화물의 구체적인 예로는 실리카(SiO_2), 알루미나(Al_2O_3), 보헤마이트(bohemite), ZnO_2 , $LiAlO_2$, TiO_2 , $BaTiO_3$, $LiTiO_3$ 및 이들의 혼합물을 들 수 있으나, 이들에 한정된 것은 아니다.

[0039] 유연하면서도 내열성이 우수한 분리막을 제조하기 위해, 본 발명에 사용되는 금속산화물 전구체의 졸-겔 용액은 실리카(SiO_2), 알루미나(Al_2O_3), 보헤마이트, ZnO_2 , $LiAlO_2$, TiO_2 , $BaTiO_3$, $LiTiO_3$ 및 이들의 혼합물로 이루어진 군으로부터 선택된 금속산화물의 나노입자를 추가로 포함할 수 있다. 금속산화물 나노입자는 나노로드, 나노튜브, 나노섬유 등의 형상을 갖는 나노 크기의 분말상이다. 이때 금속산화물 나노입자는 금속산화물 전구체의 졸-겔 용액으로부터 얻어지는 금속산화물의 중량에 대해 0.1~200 중량%, 바람직하게는 0.1~100 중량%, 더욱 바람직하게는 0.1~50 중량%의 양으로 사용할 수 있다. 이와 같은 나노 크기 분말상인 금속산화물 나노입자만을 사용한 종래 기술에서는 나노입자가 고분자 수지 용액에 균일하게 분산되지 않아 고농도의 금속산화물 나노입자가 함유된 고분자 초극세 섬유를 제조하기가 어렵고 불균질 분산으로 인하여 제조되는 섬유의 물리화학적 특성이 만족스럽지 않았다.

[0040] 이러한 본 발명의 전기방사에 의해 얻어진 초극세 섬유의 다공체는 통상적으로 70~95% 정도의 높은 기공도를 가지기 때문에, 적절한 기공도를 부여하고 기계적 특성을 향상시키기 위하여, 고분자 수지의 유리전이온도(T_g) 이상이면서 용융온도(T_m) 이하의 온도에서 열압착하거나, 열압착 후 열연신하거나, 열압착 후 열연신에 이어 추가로 열압착하는 것이 바람직하며, 또한 필요시 상기 열압착 또는 열연신 후 소정시간 동안 100~250℃의 온도에서 열처리할 수 있다. 열압착과 열연신 공정에 의해, 함유된 고분자 성분은 초극세 섬유상 용액과 고분자쇄 배향을 일으켜 초극세 섬유상 다공체의 기계적 특성을 향상시켜 주며, 이 공정을 거친 초극세 섬유상 다공체는 40~75%의 기공도를 갖는다. 바인더가 함유된 경우 상기의 열압착 공정에 의해 섬유간 결합이 일어나므로 보다 기계적 특성이 향상된 다공체를 제조할 수 있다.

[0041] 이와 같이 제조된 초극세 금속산화물/고분자 복합섬유의 다공체 또는 표면에 고분자 수지 코팅층을 갖는 초극세 금속산화물 섬유는 다공체를 포함하는 본 발명의 초극세 섬유상 복합 분리막은 150 내지 250℃에서 1시간 이상 방치할 경우 10% 이하의 낮은 열수축률을 나타내며 막의 형태 안정성이 우수하여 200℃ 이하의 온도에서는 용융에 의한 붕괴를 일으키지 않는다. 이러한 복합 분리막의 내열특성은 열적 기계적 분석(thermo mechanical analysis, TMA) 기기를 사용하여 특정 길이의 복합 분리막에 일정한 하중을 가한 상태에서 온도를 점차 승온시키면서 복합 분리막이 용융에 의해 신장되면서 끊어지는 온도로 정의되며, 이를 고온 용융 보존성(high temperature melt integrity, HTMI)이라 칭한다.

[0042] 또한, 본 발명의 초극세 섬유상 복합 분리막은 높은 통기도와 높은 기공도를 가지므로 이온전도도가 높고 전지 구성시 우수한 내열성을 지니면서도 출력 특성이 우수한 전지를 제공한다. 즉, 기공도가 60% 정도가 되는 본 발명의 초극세 섬유상 복합 분리막은 비슷한 기공도를 지닌 폴리에틸렌 분리막 보다도 우수한 출력 특성을 가지므로, 폴리에틸렌 분리막에 세라믹 입자를 코팅한 내열성 분리막에 비하여 훨씬 우수한 출력 특성을 나타낸다. 본 발명의 초극세 섬유상 다공체와는 달리, 금속산화물 미세 입자와 고분자와의 혼합용액을 전기방사하여 제조한 섬유상 분리막의 경우, 고분자의 용점 이상의 온도에서 고분자가 용융되어 흘러 섬유상 구조를 유지하지 못하고 붕괴되므로 높은 내열성을 지닌 분리막을 제조할 수가 없다.

[0043] 한편, 본 발명의 목적을 달성하기 위한 다른 측면에 따른 이차전지는 양극, 음극, 및 이들 두 전극 사이에 위치한 상기 초극세 섬유상 복합 분리막과 전해질을 포함하는 것을 특징으로 한다.

[0044] 본 발명에 따른 이차전지는, 양극활물질을 함유한 양극과 음극활물질을 함유한 음극 사이에 본 발명의 내열성 초극세 섬유상 분리막을 삽입하고 라미네이션한 후 유기전해액, 고분자 전해질 또는 고체 전해질을 주입함으로써 제조된다. 양극활물질로는 리튬 코발트 복합산화물, 리튬 니켈 복합산화물, 니켈 망간 복합산화물, 포스페이트 화합물 또는 이들의 혼합물이 사용될 수 있으며, 음극활물질은 리튬 이차전지와 같은 비수계 전해질 전지로 사용될 수 있는 한 특별히 제한되지는 않으나 그 구체적인 예로는 흑연과 코우크스 같은 탄소재료, 주석산화물,

금속리튬, 이산화규소, 산화티탄 및 이들의 혼합물을 들 수 있다.

유기전해액, 고분자 전해질 또는 고체 전해질 중에 함유되는 리튬염의 종류는 특별히 제한되지 않고, 리튬 이차 전지 분야에서 통상적으로 사용되는 리튬염이면 어떤 것이라도 가능하며, 그 구체적인 예로는 LiPF_6 , LiClO_4 , LiAsF_6 , LiBF_4 , LiCF_3SO_3 , $\text{LiN}(\text{SO}_2\text{CF}_3)_2$, $\text{LiN}(\text{SO}_2\text{C}_2\text{F}_5)_2$, $\text{LiPF}_{6-x}(\text{C}_n\text{F}_{2n+1})_x$ ($1 < x < 6$, $n=1$ 또는 2) 및 이들의 혼합물을 들 수 있으며, 그 중 LiPF_6 가 보다 바람직하다. 리튬염의 농도는 0.5~3.0 M 일 수 있으며, 주로 1M 유기전해액 이 사용된다.

이하, 본 발명을 하기 실시예에 의거하여 좀더 상세하게 설명하고자 한다. 단, 하기 실시예는 본 발명을 예시하기 위한 것일 뿐, 본 발명의 범위가 이들만으로 제한되는 것은 아니다.

[실시예]

하기 실시예 및 비교예에서 제조된 분리막 각각의 물성 및 이를 포함하는 전지의 물성을 하기와 같은 방법으로 측정하였다.

분리막의 기공도

하기 수학식 1의 겉보기 기공도 및 하기 수학식 2의 부탄올 함침법에 의해 분리막의 기공도를 평가하였다:

수학식 1

겉보기 기공도(%) = {1-(ρ_m/ ρ_p)} X 100

상기 식에서, ρ_m은 섬유층의 밀도이고, ρ_p는 고분자의 밀도이다.

수학식 2

기공도(%) = {(M_{BuOH}/ ρ_{BuOH})/(M_{BuOH}/ ρ_{BuOH}+M_m/ ρ_p)} X 100

상기 식에서, M_{BuOH}은 흡수된 부탄올의 무게이고, M_m는 분리막의 무게이고, ρ_{BuOH}는 부탄올의 밀도이고, ρ_p는 고 분자의 밀도이다.

분리막의 전해액 흡수율

3cm× 3cm 분리막 시편을 1M LiPF_6 EC/DMC/DEC(1/1/1) 전해질 용액에 실온에서 약 2시간 동안 침지한 후, 표면에 묻은 과량의 전해액을 종이 여과지로 제거한 다음, 무게를 측정하여 전해액 흡수율을 결정하였다.

분리막의 열수축률

3cm× 3cm 분리막 시편을 150~250℃에서 1시간 동안 방치한 후 열수축률을 계산하였다.

분리막의 내열특성(고온 용융 보존성)

분리막의 내열특성으로서 TMA(Perkin-Elmer, TMA-7)를 사용하여 50~400℃의 온도범위에서 온도변화에 따른 1 cm X 0.5 cm 크기 시료의 길이변화를 측정하여 고온 용융 보존성(high temperature melt integrity, HTMI)을 조사

하였다. 이때 가열 속도 및 하중은 분당 5℃ 및 200g (정전기력 : 50 mN) 이었다.

[0061] 전지의 내열특성

[0062] LiCoO₂//전해액 함침 분리막//MCMB 구조로 단전지를 구성하여 전지를 50% SOC까지 충전한 후 분당 5℃의 속도로 40~250℃까지 승온하였다. 각각 10℃ 구간마다 5분씩 안정화 시간을 부여한 후 임피던스를 100mHz에서 1MHz까지 150초 동안 측정하고, 이어 10초 동안 OCV, 그리고 1kHz에서 10초 동안 임피던스를 연달아 측정하였다.

[0063] 분리막의 전기화학적 안정성

[0064] 스테인레스 스틸 플레이트(stainless steel plate)//전해액 함침 분리막//스테인레스 스틸 플레이트로 셀을 구성한 다음에, 분리막의 전기화학적 안정성 평가는 소정량의 전해액에 함침한 분리막에 대하여 선형 스위프 전압 전류법(linear sweep voltammetry)으로 결정하였으며, 니켈 작업(working) 전극, 리튬 참조(reference) 전극 및 상대전극으로 이루어진 3-전극 전기화학 셀을 사용하여 전기화학 분석기(electrochemical analyzer, CHI model 600)로부터 실온에서 2~6V(Li⁺/Li) 영역에서 1 mV/분의 스캔 속도로 측정하였다.

[0065] 전지의 충방전 특성

[0066] 전지의 충방전 특성 평가는 전지 싸이클러(WBCS 3000, WonA Tech Co.)를 사용하여 전류 밀도 0.68 mA/cm² (0.2C), 4.2V의 고정전류와 고정전압의 조건으로 충전하고, 2.75V까지 3.4 mA/cm²(1C)의 조건에서 방전시켰다. 충방전 싸이클 시험은 실온에서 500 싸이클까지 유지된 용량%를 평가하였다.

[0067] 전지의 출력 특성

[0068] 전지의 출력 특성은 하이브리드 펄스 파워특성(hybrid pulse power characterization, HPPC)을 평가하여 조사하였다. 단전지를 구성하여 정전류/정전압으로 4.2V까지 충전한 후, 10분 동안 안정화시키고, 1C로 10% SOC까지 방전시키고 20분 동안 안정화시켰다. 이어 방전 펄스파워(discharge pulse power)를 측정하기 위하여 10초 동안 5C로 방전시키고 40초 동안 안정화시켰다. 그리고, 재생 펄스 파워(regeneration pulse power)를 측정하기 위하여 10초 동안 3C로 충전하였다. 1 단계 측정 후 후속 단계 측정을 위하여 다시 전지를 1C로 20% SOC, 30% SOC, 40% SOC, 50% SOC, 60% SOC, 70% SOC, 80% SOC, 및 90% SOC까지 방전시키면서 비슷한 측정을 반복하였다.

[0069] 전극의 제조

[0070] 양극으로서 PVdF 바인더, 수퍼-P 카본, LiCoO₂ (Japan Chemical사 제품)로 구성된 슬러리를 알루미늄 호일에 캐스팅한 것(SKC 공급품)을 사용하였으며, 음극으로서 MCMB (Osaka Gas사 제품), PVdF 바인더, 수퍼-P 카본으로 구성된 슬러리를 구리 호일에 캐스팅한 것(SKC 공급품)을 사용하였다. 이 전극의 이론 용량은 145 mAh/g 이었다.

[0071] 상기 양극과 음극 제조시 입자간 및 금속 호일과의 접착력을 증대시키기 위하여, 해당 슬러리를 각각 캐스팅한 후에 롤 프레싱을 하여 전극의 두께가 약 50 μm가 되도록 하였다.

[0072] 비교예 1

[0073] 38 g의 디메틸포름아미드(DMF)에 실리카 2 g(Aldrich 사 제품)을 넣고, 초음파 발생장치를 이용하여 약 1 시간 동안 분산시킨 후, 폴리아크릴로니트릴(PAN, Mw 150,000, polyscience 사) 3.6 g을 넣고 75℃에서 1시간 동안 교반하여 PAN을 용해시켜 실리카/PAN 혼합용액을 제조하였다. 이 혼합용액을 도 1과 같은 전기방사 장치(17 kV의 고전압 전기장, 27G의 방사노즐과 금속집전판 사이의 거리 14 cm, 30 μl/분의 토출량)를 이용하여 실리카 나노입자가 분산된 PAN (SiO₂/PAN) 초극세 섬유를 제조하였다. 또한, 동일한 방법으로, 38 g의 DMF에 실리카 2 g

과 폴리비닐리덴플루오라이드(PVdF, Kynar 761) 4 g을 사용하여 실리카가 분산된 PVdF (SiO₂/PVdF) 초극세 섬유를 제조하였다.

[0074] 또한, 동일한 방법으로, 이산화티탄 나노입자(P25, 대구사 제품)와 폴리비닐리덴플루오라이드(PVdF, Kynar 761)를 사용하여 TiO₂/PVdF 중량비가 1:2 및 1:1인 TiO₂/PVdF 초극세 섬유를 제조하였다.

[0075] 제조된 SiO₂/PAN 및 TiO₂/PVdF 초극세 섬유의 전자투과현미경 사진을 도 2에 나타내었다 ((a) SiO₂/PAN (1/1.8 중량비), (b) TiO₂/PVdF (1:2 중량비), (c) TiO₂/PVdF (1:1 중량비)). 제조된 섬유의 표면에선 금속산화물 나노입자가 많이 관찰되었으며, 매우 거친 표면을 보여 주었다.

[0076] 또한, 제조된 SiO₂/PVdF 및 TiO₂/PVdF 초극세 섬유상 분리막의 전기화학적 안정성을 도 3에 나타내었다 ((a) SiO₂/PVdF (1:2 중량비), (b) TiO₂/PVdF (1:2 중량비)) 이 SiO₂/PVdF 및 TiO₂/PVdF 초극세 섬유상 분리막은 3.7V 근처에서 전해액 분해가 일어나 매우 좋지 않은 전기화학적 안정성을 보여 주었으며, 3cm X 3cm 크기의 이들 시편을 180℃에서 약 1시간 동안 방치하였을 때 각각 13.7% 및 14.5%의 수축이 일어났다.

[0077] 실시예 1-1

[0078] 테트라에톡시오르토실리케이트(TEOS) 20.8 g, 에틸알콜 9.2 g, 물 3.5g, 염산 수용액 0.1 g의 혼합 용액을 약 70℃에서 약 3시간동안 교반하여 TEOS 졸-겔 용액을 제조한 후, 도 1의 전기방사 장치를 이용하여 20 kV 고전압 전기장하에서 20 μ l/분의 토출속도로 30G의 방사노즐을 사용하여 실리카 초극세 섬유를 제조하였다. 제조된 실리카 섬유를 약 400℃에서 소결하여 평균 직경 680 nm의 실리카 초극세 섬유상 다공체를 제조하였다. 제조된 실리카 초극세 섬유의 전자투과현미경 사진을 도 4의 (a)에 나타내었다.

[0079] 실시예 1-2

[0080] 알루미늄 이소프로폭사이드 7 g, 에틸알콜 40 ml, 물 10 ml, 염산 수용액 25 ml의 혼합하고 교반하여 알루미늄 이소프로폭사이드 졸-겔 용액을 제조하고, 이 용액에 에틸알콜 5 ml에 폴리비닐피롤리돈(PVP) 1.5 g이 용해된 용액을 첨가하고 약 70℃에서 2시간 동안 교반하여 혼합용액을 제조하였다. 이 용액을 도 1의 전기방사 장치를 사용하여 15.5 kV 고전압 전기장하에서 24G 방사노즐을 사용하여 30 μ l/분의 토출속도로 알루미나/PVP 초극세 섬유상 다공체를 제조하였다. 이를 약 500℃에서 소결하면서 PVP를 제거하여 평균 직경 600 nm의 알루미나 초극세 섬유상 다공체를 제조하였다. 제조된 알루미나 초극세 섬유의 전자투과현미경 사진을 도 4의 (b)에 나타내었다.

[0081] 실시예 1-3

[0082] 실시예 1-1에서 제조한 초극세 섬유상 실리카 다공체를 PVdF와 폴리에테르이미드(1:1 중량비)가 5% 중량비로 용해된 DMF 용액에 함침시키고 건조한 후 약 150℃에서 열압착하여 기공도가 65%인 복합 분리막을 제조하였다. 이 분리막의 전자투과현미경 사진을 도 4의 (c)에 나타내었다. 이 분리막을 200℃ 오븐에서 약 1시간 동안 보관한 뒤 열수축률을 측정한 결과 약 4.3%의 수축을 나타내었으며, 내열특성을 TMA로 평가한 결과 약 300℃까지도 용융에 의해 분리막이 붕괴되지 않는 등 우수한 열안정성을 나타내었다.

[0083] 실시예 2-1

[0084] TEOS 113 g, 에틸알콜 50 g, 물 19 g, 염산 수용액 0.56 g의 혼합용액을 반응시켜 TEOS 졸-겔 용액을 제조하고, 이 혼합용액의 일부를 취하여 PVdF(Kynar 761) 3.5 g이 21.5 g의 DMF에 용해된 용액과 혼합하고 약 50℃에서 교반하여 14 중량%의 SiO₂/PVdF (1:1 중량비)의 혼합용액을 제조하였다. 이 혼합용액을 도 1의 전기방사 장치를 이용하여 (용액온도 50℃, 13.8 kV의 고전압 전기장, 27G의 노즐, 용액토출속도 27 μ l/분) 262 nm의 평균 섬유직경 및 90%의 기공도를 갖는 실리카/PVdF 초극세 섬유상 다공체를 제조하였다. 제조된 초극세 섬유상 다공체의 전자투과현미경 사진을 도 5의 (a)(열압착 전)에 나타내었다.

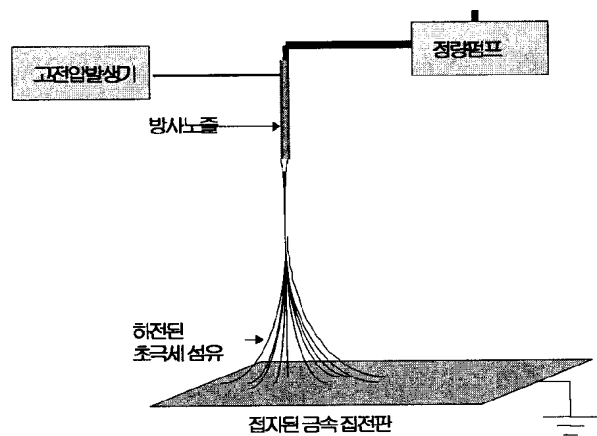
- [0085] 또한, 이들의 기계적 특성을 향상시킴과 동시에 60% 및 75%의 기공도를 가지도록 약 140℃에서 열압착 및 150℃에서 열연신하고 150℃에서 약 30분 동안 열처리하여 실리카/PVdF 내열성 복합 분리막을 제조하였으며, 이의 전자투과현미경 사진을 도 5의 (b)(열압착 후)에 나타내었다.
- [0086] 또한, 이 내열성 복합 분리막을 아세톤을 사용하여 PVdF 성분을 추출 제거한 다음의 실리카 초극세 섬유층의 전자투과현미경 사진을 도 5의 (c)에 나타내었다. 비교예 1에서의 금속산화물 나노입자를 사용한 경우와는 달리 초극세 실리카가 섬유층 전체에 균일하게 분포되어 있음을 알 수 있다. 이는 실리카 성분과 PVdF 성분이 실리카/PVdF 복합 초극세 섬유층 내에 매우 균질하게 분포되었음을 보여 주는 것으로 이로부터 우수한 내열성과 기계적 특성을 가짐을 예상할 수 있다. 또한, 상기와 유사한 방법으로, $Ti(OBu)_4$ 의 졸-겔 용액과 PVdF 혼합용액을 전기방사한 후 얻어진 초극세 섬유층 분리막을 180℃에서 약 10분간 열처리하여 TiO_2 /PVdF 내열성 초극세 복합 분리막을 제조하였다. 이 내열성 분리막을 150~250℃ 오븐에서 약 1시간 동안 보관한 뒤 열수축률을 측정한 결과 모두 약 1.5% 이내의 수축을 나타내었다.
- [0087] 또한, 도 6에서 보듯이, 이 내열성 복합 분리막은 4.5 V 이상의 우수한 전기화학적 안정성을 나타내었다 ((a) SiO_2 /PVdF (1:1 중량비), (b) TiO_2 /PVdF (1:1 중량비)) 또한, TMA를 사용한 내열성 평가(고온 용융 보존성)에서 상용 폴리에틸렌(PE) 분리막 및 PVdF 단독 용액을 전기방사하여 제조한 초극세 섬유층 PVdF 분리막은 도 7의 (a)에서 보듯이 PE 및 PVdF의 용점 부근에서부터 고분자의 용융으로 인하여 급격히 신장되면서 절단이 되는 막 붕괴가 일어났다. 그러나, 상기 실리카/PVdF 초극세 섬유층 분리막은 도 7의 (b)에서 보듯이 200℃ 이상의 온도에서도 PVdF의 용융에 의한 막 붕괴가 일어나지 않는 등 우수한 내열성을 보여 주었다. 또한 실리카/PVdF 초극세 섬유층 분리막은 도 7의 (c)에서 250℃에서 약 1시간 방치한 경우에도 PVdF가 부분적으로 용융되어 용출된 모습을 보여주고 있으나, 용융에 의해 분리막이 붕괴되지는 않는 등 우수한 형태 안정성을 나타내었다.
- [0088] 또한, 도 8에서 보듯이, 60%의 기공도를 갖는 상용 PE 분리막보다 60%의 기공도를 갖는 실리카/PVdF 복합 분리막이 약 2% 정도 더 우수한 출력 특성을 나타내었으며, 75% 기공도를 갖는 실리카/PVdF 복합 분리막은 약 8% 이상의 더 우수한 출력 특성을 나타내었다.
- [0089] 또한, 60%의 기공도를 갖는 상용 PE 분리막과 상기 60% 기공도를 갖는 실리카/PVdF 복합 분리막으로 단전지를 구성하여 500사이클 충방전 시험을 수행한 결과, 도 9에서 보듯이, PE 분리막은 초기 용량의 83%를 유지한 반면에, 실리카/PVdF 복합 분리막은 86%의 더 높은 초기 용량의 유지를 나타내었다.
- [0090] 실시예 2-2
- [0091] TEOS 113 g, 에틸알콜 50 g, 물 19 g, 염산 수용액 0.56 g의 혼합용액을 반응시켜 TEOS 졸-겔 용액을 제조하고, 이 혼합용액을 일부 취하여 PVdF(Kynar 761) 7.62 g이 21 g의 DMF에 용해된 용액과 혼합하고 약 50℃에서 교반하여 16 중량%의 SiO_2 /PVdF (1:1 중량비)의 혼합용액을 제조하였다. 이 혼합용액을 70℃로 유지하면서 18 kV의 고전압 전기장하에서 27G의 노즐을 사용하여 20 μ l/분의 토출속도로 토출하여 88%의 기공도를 갖는 실리카/PVdF 복합 초극세 섬유층을 제조하였다. 이를 열압착 및 150℃에서 열연신하고 약 30분 동안 열처리하여 60%의 기공도를 갖는 내열성 분리막을 제조하였다. 이 내열성 분리막을 200℃ 오븐에서 약 1시간 동안 보관한 뒤 열수축률을 측정한 결과 약 3.0% 이내의 수축을 나타내었으며, 이 분리막은 60% 기공도를 갖는 PE 분리막보다 약 2% 정도 더 우수한 출력 특성을 나타내었다.
- [0092] 실시예 2-3
- [0093] TEOS 113 g, 에틸알콜 50 g, 물 19 g, 염산수용액 0.56 g의 혼합용액을 반응시켜 TEOS 졸-겔 용액을 제조하고, 이 혼합용액을 일부 취하여 PVdF(Kynar 761) 4 g이 21 g의 DMF에 용해된 용액과 혼합하고 약 60℃에서 교반하여 16 중량%의 SiO_2 /PVdF (=6:4 중량비)의 혼합용액을 제조하였다. 이 혼합용액을 70℃로 유지하면서 16 kV의 고전압 전기장하에서 27G의 노즐을 사용하여 30 μ l/분의 토출속도로 토출하여 90%의 기공도를 갖는 실리카/PVdF 복합 초극세 섬유층을 제조하였다. 이를 열압착 및 150℃에서 열연신하고 약 30분 동안 열처리하여 65%의 기공도를 갖는 내열성 분리막을 제조하였다. 이 내열성 분리막을 200℃ 오븐에서 약 1시간 동안 보관한 뒤 열수축률을 측정한 결과 약 2.5% 이내의 수축을 나타내었으며, 이 분리막은 60% 기공도를 갖는 PE 분리막보다 약 2.1% 정도 더 우수한 출력 특성을 나타내었다.

[0094] 실시예 2-4

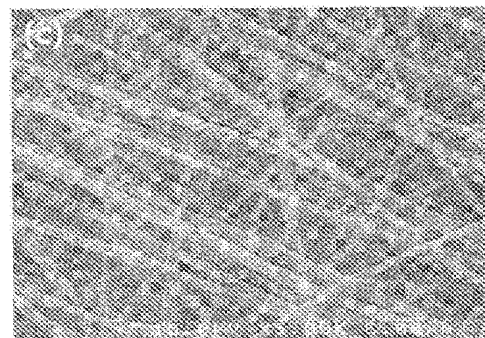
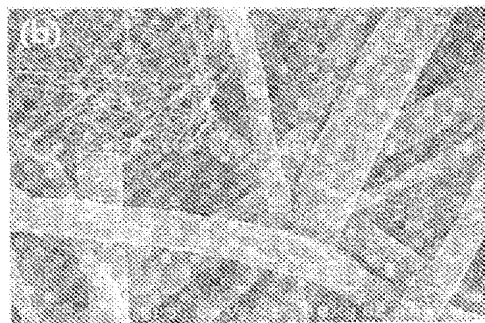
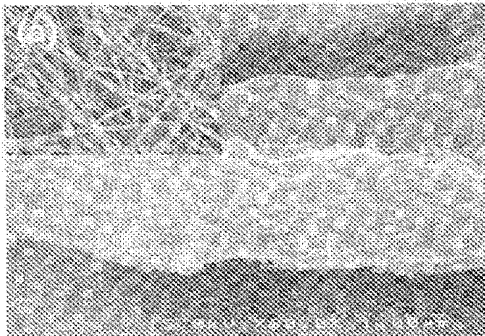
[0095] TEOS 113 g, 에틸알콜 50 g, 물 19 g, 염산수용액 0.56 g의 혼합용액을 반응시켜 TEOS 졸-겔 용액을 제조하고, 이 혼합용액을 일부 취하여 PVdF(Kynar 761) 3.81 g 및 실리카 나노입자 3.81 g이 21 g의 DMF에 용해된 용액과 혼합하고 약 50℃에서 교반하여 16 중량%의 SiO₂/PVdF (1:1 중량비)의 혼합용액을 제조하였다. 이 혼합용액을 70℃로 유지하면서 18 kV의 고전압 전기장하에서 27G의 노즐을 사용하여 20 μ l/분의 토출속도로 토출하여 88%의 기공도를 갖는 실리카/PVdF 복합 초극세 섬유를 제조하였다. 이를 열압착 및 150℃에서 열연신하고 약 30분 동안 열처리하여 60%의 기공도를 갖는 내열성 분리막을 제조하였다. 이 분리막은 실시예 2-2에서 제조한 분리막에 비하여 약간 유연성이 좋았으나, 이 내열성 분리막을 200℃ 오븐에서 약 1 시간 동안 보관한 뒤 열수축률을 측정한 결과 약 5.5% 이내의 수축을 나타내었으며, 이 분리막은 60% 기공도를 갖는 PE 분리막보다 약 2.0% 정도 더 우수한 출력 특성을 나타내었다.

도면

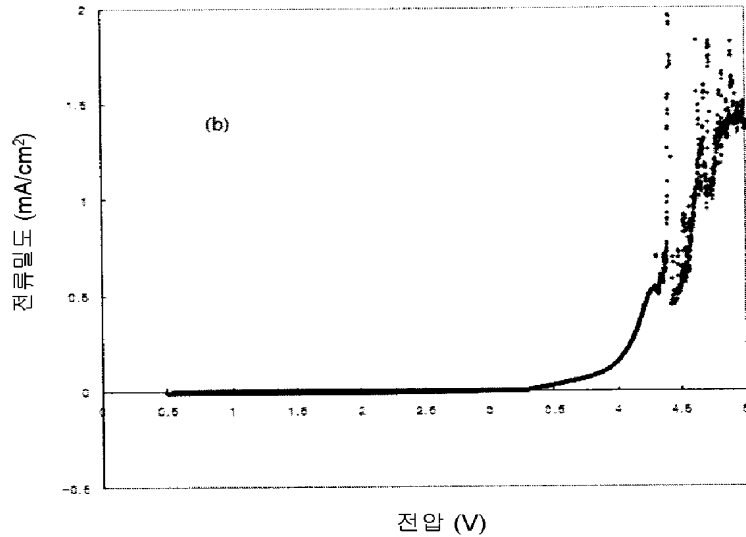
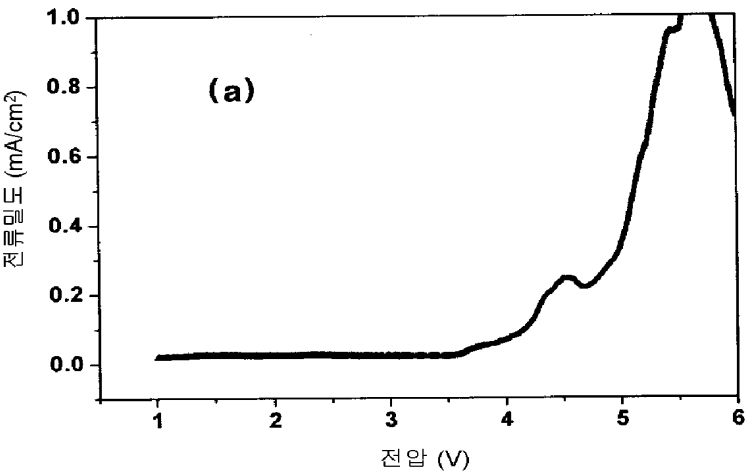
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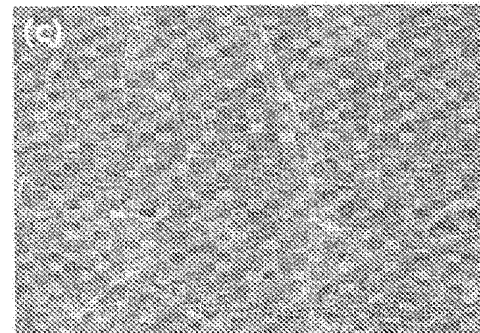
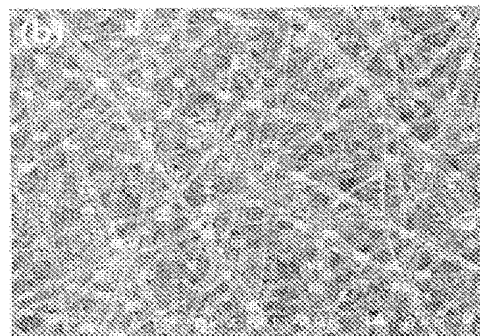
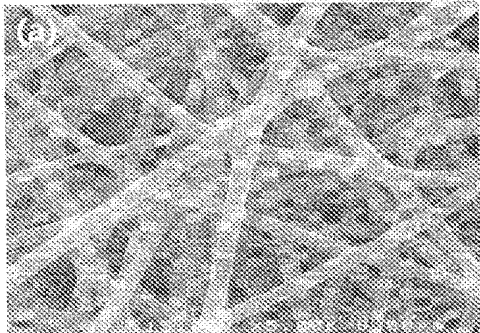
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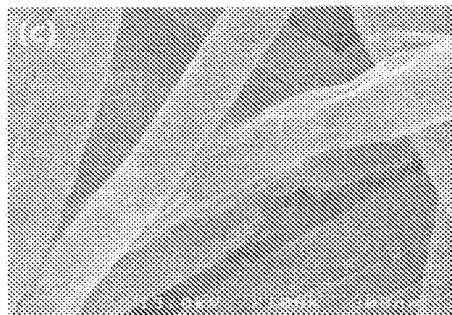
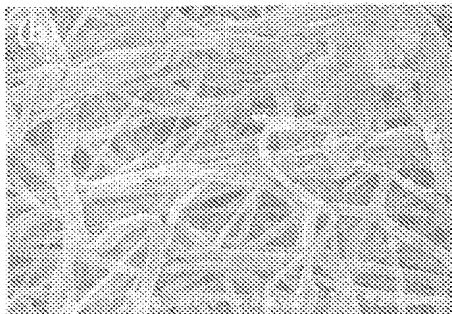
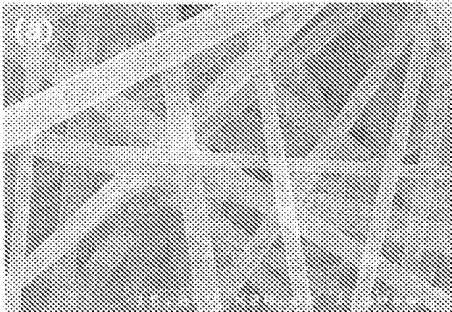
도 10



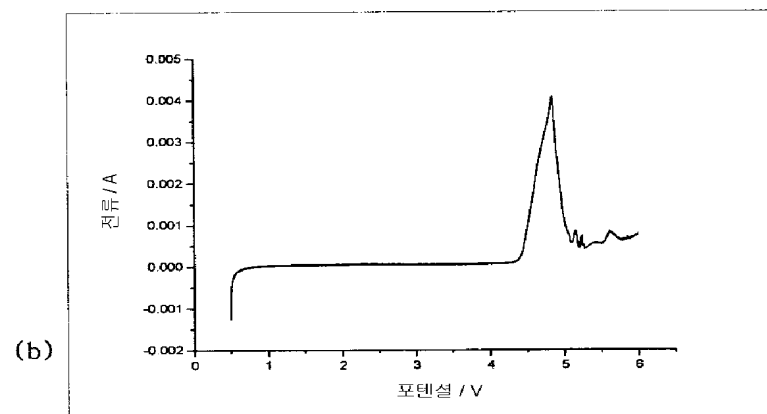
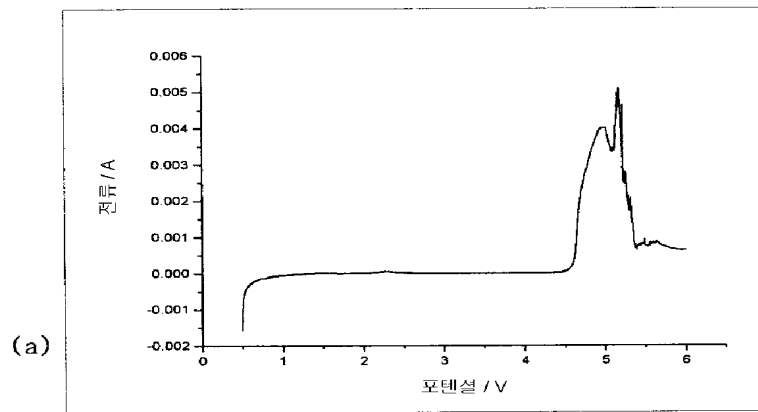
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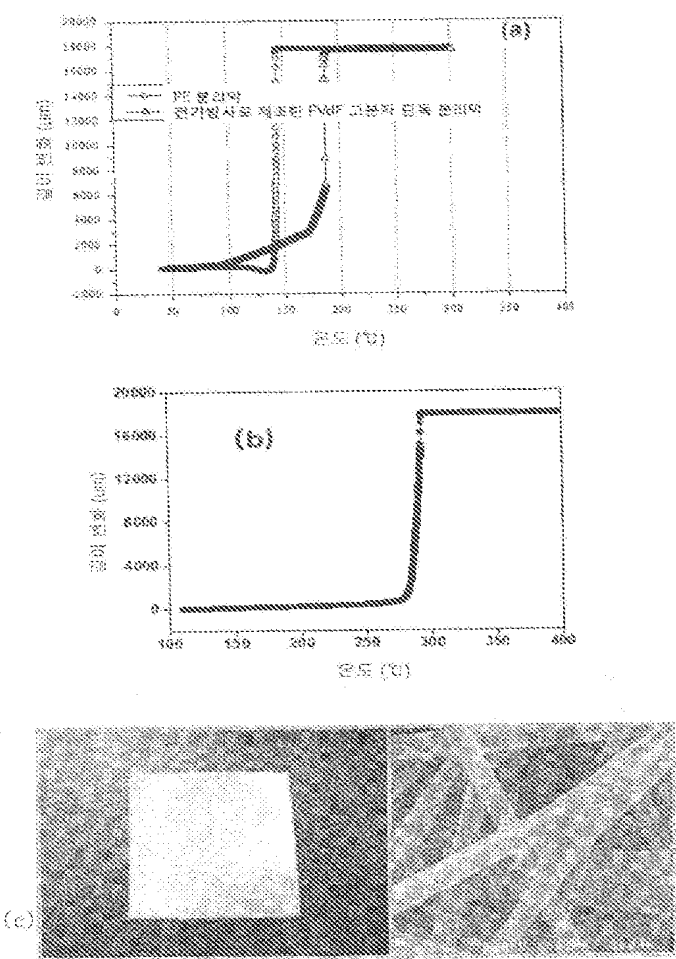
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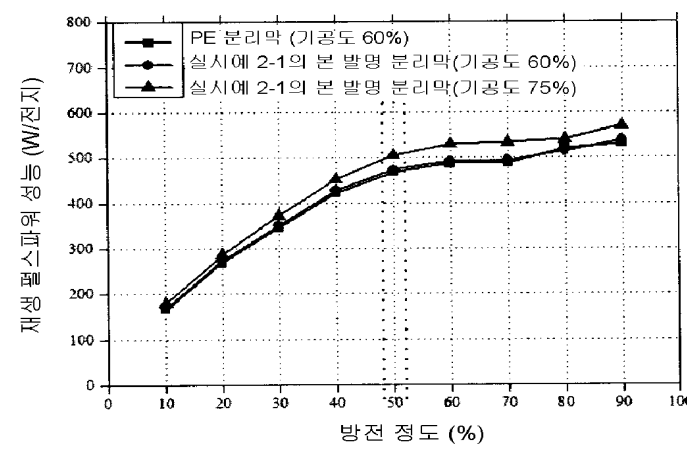
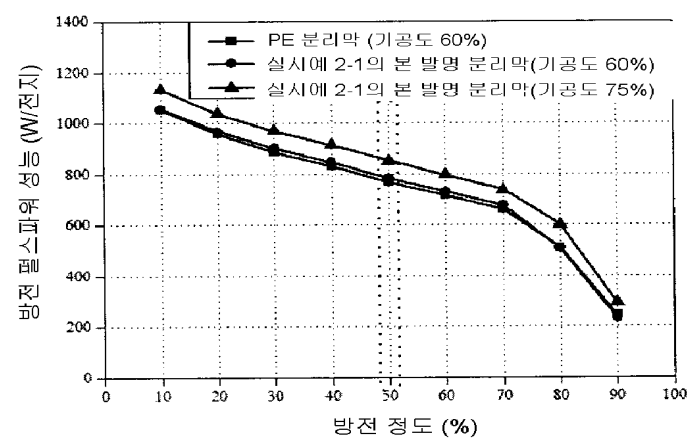
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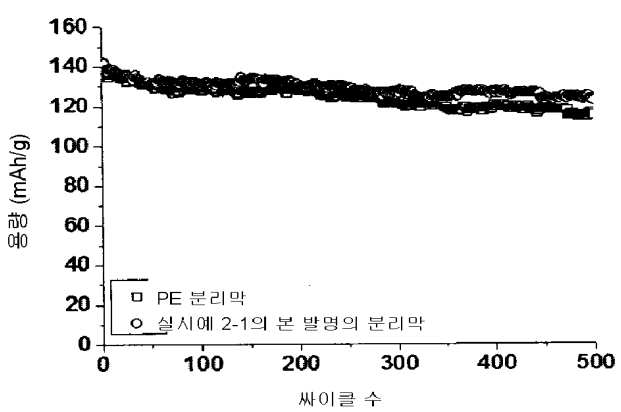
도면7



도 10



도 11





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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/631,783	03/27/2025 12:53:37 PM Z ET	00249/62

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	1021	FILED BY	Joy Chapman
PATENT CENTER #	69762040	FILING DATE	04/10/2024
CUSTOMER #	1912	FIRST NAMED INVENTOR	David W. Avison
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Brian Amos

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
00249-62_IDS_fillable-pdf.pdf	4	Information Disclosure Statement (IDS) Form (SB08)	1234 KB
WO2017010780_with-Eng-Abstract.pdf	22	Foreign Reference	1465 KB

Digest

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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		18631783
Filing Date		2024-04-10
First Named Inventor	David W. Avison	
Art Unit		1729
Examiner Name	Lisa S. Park	
Attorney Docket Number		00249/62

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

- ☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).
- OR
- ☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).
- ☐ See attached certification statement.
- ☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.
- ☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2025-03-27
Name/Print	Brian J. Amos	Registration Number	63,679

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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		18631783	
	Filing Date		2024-04-10	
	First Named Inventor	David W. Avison		
	Art Unit		1729	
	Examiner Name	Lisa S. Park		
	Attorney Docket Number		00249/62	

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Examiner Name	Lisa S. Park	
Attorney Docket Number		00249/62

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Bibliographic data: WO2017010780 (A1) — 2017-01-19

SEPARATOR AND ELECTROCHEMICAL DEVICE COMPRISING SAME

Inventor(s): SUNG DONG-WOOK [KR]; KIM MIN-HYUNG [KR]; SUNG IN-HYOUK [KR]; SHIN HYUN-KYUNG [KR]; LEE JI-EUN [KR]; LEE JOO-SUNG [KR] ± (성동욱, ; 김민형, ; 성인혁, ; 신현경, ; 이지은, ; 이주성)

Applicant(s): LG CHEMICAL LTD [KR] ± (주식회사 엘지화학)

Classification: - international: H01M10/052; H01M50/414; H01M50/434; H01M50/443; H01M50/449; H01M50/417; H01M50/489; H01M50/491
- cooperative: C08K3/22 (KR); H01M10/052 (KR, US); H01M10/0525 (US); H01M50/414 (EP, KR, US); H01M50/417 (EP, KR); H01M50/434 (EP, KR, US); H01M50/443 (EP, KR, US); H01M50/446 (EP, KR, US); H01M50/449 (EP, US); H01M50/451 (KR); H01M50/489 (KR); H01M50/491 (KR); C08K2003/2227 (KR); H01M50/417 (US); H01M50/489 (EP, US); H01M50/491 (EP, US); Y02E60/10 (EP, KR, US); Y02T10/70 (KR, US)

Application number: WO2016KR07532 20160711 Global Dossier

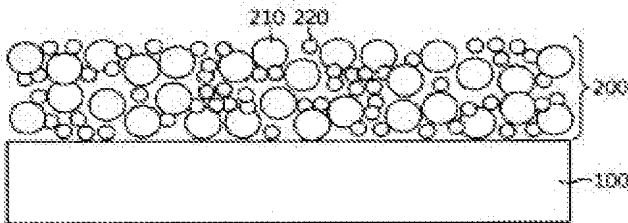
Priority number(s): KR20150098672 20150710

Also published as: CN107851762 (A) CN107851762 (B) EP3316346 (A1) EP3316346 (A4) EP3316346 (B1) more

Abstract of WO2017010780 (A1)

Disclosed are a separator and an electrochemical device comprising the same, the separator comprising: a porous substrate having a plurality of pores; and a porous coating layer formed on at least one surface of the porous substrate or at least one surface of the porous substrate and a portion of the pore, the porous coating layer containing a plurality of inorganic particles and a binder polymer disposed on a part or the entirety of a surface of the inorganic particle to connect and fix the inorganic particles, wherein the inorganic particle comprises an alumina particle and an

[E1]



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aluminum hydroxide particle having a smaller average particle diameter than the alumina particle.

(12) 특허협력조약에 의하여 공개된 국제출원

(19) 세계지식재산권기구
국제사무국

(43) 국제공개일
2017년 1월 19일 (19.01.2017)



(10) 국제공개번호
WO 2017/010780 A1

- (51) 국제특허분류: *H01M 2/16* (2006.01) *H01M 10/52* (2010.01)
- (21) 국제출원번호: PCT/KR2016/007532
- (22) 국제출원일: 2016년 7월 11일 (11.07.2016)
- (25) 출원언어: 한국어
- (26) 공개언어: 한국어
- (30) 우선권정보: 10-2015-0098672 2015년 7월 10일 (10.07.2015) KR
- (71) 출원인: 주식회사 엘지화학 (LG CHEM, LTD.) [KR/KR]; 07336 서울시 영등포구 여의대로 128, Seoul (KR).
- (72) 발명자: 성동욱 (SUNG, Dong-Wook); 34122 대전시 유성구 문지로 188 LG 화학기술연구원, Daejeon (KR). 김민형 (KIM, Min-Hyung); 34122 대전시 유성구 문지로 188 LG 화학기술연구원, Daejeon (KR). 성인혁 (SUNG, In-Hyook); 34122 대전시 유성구 문지로 188 LG 화학기술연구원, Daejeon (KR). 신현경 (SHIN, Hyun-Ky-ung); 34122 대전시 유성구 문지로 188 LG 화학기술연구원, Daejeon (KR). 이지은 (LEE, Ji-Eun); 34122 대전시 유성구 문지로 188 LG 화학기술연구원, Daejeon (KR). 이주성 (LEE, Joo-Sung); 34122 대전시 유성구 문지로 188 LG 화학기술연구원, Daejeon (KR).

(74) 대리인: 특허법인 필앤온지 (PHIL & ONZI INT'L PATENT & LAW FIRM); 06670 서울시 서초구 반포대로 63, 8층, Seoul (KR).

(81) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 국내 권리의 보호를 위하여): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JP, KE, KG, KN, KP, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.

(84) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 역내 권리의 보호를 위하여): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), 유라시아 (AM, AZ, BY, KG, KZ, RU, TJ, TM), 유럽 (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

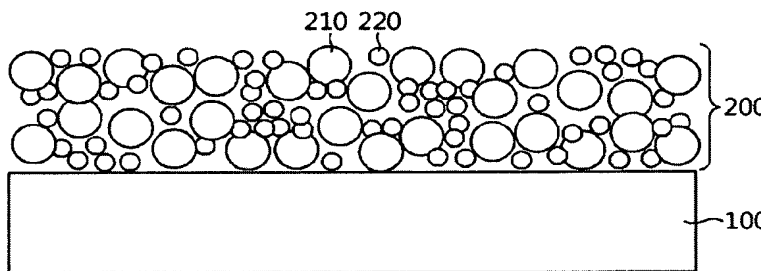
공개:

— 국제조사보고서와 함께 (조약 제 21 조(3))

(54) Title: SEPARATOR AND ELECTROCHEMICAL DEVICE COMPRISING SAME

(54) 발명의 명칭 : 세퍼레이터 및 이를 포함하는 전기화학소자

[도1]



(57) Abstract: Disclosed are a separator and an electrochemical device comprising the same, the separator comprising: a porous substrate having a plurality of pores; and a porous coating layer formed on at least one surface of the porous substrate or at least one surface of the porous substrate and a portion of the pore, the porous coating layer containing a plurality of inorganic particles and a binder polymer disposed on a part or the entirety of a surface of the inorganic particle to connect and fix the inorganic particles, wherein the inorganic particle comprises an alumina particle and an aluminum hydroxide particle having a smaller average particle diameter than the alumina particle.

(57) 요약서: 다수의 기공을 갖는 다공성 기재; 및 상기 다공성 기재의 적어도 일면에 또는 상기 다공성 기재의 적어도 일면 및 기공의 일부에 형성되어 있으며, 다수의 무기물 입자 및 상기 무기물 입자의 표면의 일부 또는 전부에 위치하여 상기 무기물 입자 사이를 연결 및 고정시키는 바인더 고분자를 포함하는 다공성 코팅층을 포함하고, 상기 무기물 입자가 알루미늄 입자 및 상기 알루미늄 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함하는 세퍼레이터 및 이를 포함하는 전기화학 소자가 제시된다.



WO 2017/010780 A1

명세서

발명의 명칭: 세퍼레이터 및 이를 포함하는 전기화학소자 기술분야

- [1] 본 발명은 세퍼레이터 및 이를 포함하는 전기화학소자에 관한 것으로서, 열수축 특성이 세퍼레이터 및 이를 포함하는 전기화학소자에 관한 것이다.
- [2] 본 출원은 2015년 7월 10일에 출원된 한국출원 제10-2015-0098672호에 기초한 우선권을 주장하며, 해당 출원의 명세서 및 도면에 개시된 모든 내용은 본 출원에 원용된다.

배경기술

- [3] 최근 에너지 저장 기술에 대한 관심이 갈수록 높아지고 있다. 휴대폰, 캠코더 및 노트북 PC, 나아가서는 전기 자동차의 에너지까지 적용분야가 확대되면서 전기화학소자의 연구와 개발에 대한 노력이 점점 구체화되고 있다. 전기화학소자는 이러한 측면에서 가장 주목 받고 있는 분야이고 그 중에서도 충방전이 가능한 이차전지의 개발은 관심의 초점이 되고 있으며, 최근에는 이러한 전지를 개발함에 있어서 용량 밀도 및 비에너지를 향상시키기 위하여 새로운 전극과 전지의 설계에 대한 연구개발로 진행되고 있다.
- [4] 현재 적용되고 있는 이차전지 중에서 1990년대 초에 개발된 리튬 이차전지는 수용액 전해액을 사용하는 Ni-MH, Ni-Cd, 황산-납 전지 등의 재래식 전지에 비해서 작동 전압이 높고 에너지 밀도가 월등히 크다는 장점으로 각광을 받고 있다. 그러나 이러한 리튬 이온 전지는 유기 전해액을 사용하는 데 따르는 발화 및 폭발 등의 안전 문제가 존재하고, 제조가 까다로운 단점이 있다.
- [5] 최근의 리튬 이온 고분자 전지는 이러한 리튬 이온 전지의 약점을 개선하여 차세대 전지의 하나로 꼽히고 있으나 아직까지 전지의 용량이 리튬 이온 전지와 비교하여 상대적으로 낮고, 특히 저온에서의 방전 용량이 불충분하여 이에 대한 개선이 시급히 요구되고 있다.
- [6] 상기와 같은 전기화학소자는 많은 회사에서 생산되고 있으나 그들의 안전성 특성은 각각 다른 양상을 보인다. 이러한 전기화학소자의 안전성 평가 및 안전성 확보는 매우 중요하다. 가장 중요한 고려사항은 전기화학소자가 오작동시 사용자에게 상해를 입혀서는 안 된다는 것이며, 이러한 목적으로 안전규격은 전기화학소자 내의 발화 및 발연 등을 엄격히 규제하고 있다. 전기화학소자의 안전성 특성에 있어서, 전기화학소자가 과열되어 열폭주가 일어나거나 분리막이 관통될 경우에는 폭발을 일으키게 될 우려가 크다. 특히, 전기화학소자의 분리막으로서 통상적으로 사용되는 폴리올레핀계 다공성 기재는 재료적 특성과 연신을 포함하는 제조공정상의 특성으로 인하여 100℃ 이상의 온도에서 극심한 열 수축 거동을 보임으로서, 양극과 음극 사이의 단락을 일으켰다.

- [7] 이와 같은 전기화학소자의 안전성 문제를 해결하기 위하여, 다수의 기공을 갖는 다공성 기재의 적어도 일면에, 과량의 무기물 입자와 바인더 고분자의 혼합물을 코팅하여 다공성 유기-무기 코팅층을 형성한 세퍼레이터가 제안되었다.
- [8] 이러한 다공성 유기-무기 코팅층을 형성한 세퍼레이터에서 입경이 지나치게 큰 큰 입자를 사용하는 경우 열수축 제어가 어려웠고, 입경이 지나치게 작은 입자를 사용하는 경우, 코팅층 형성용 조성물의 안정성이나 공정성이 떨어져 사용이 곤란하다는 문제가 있다.

발명의 상세한 설명

기술적 과제

- [9] 따라서 본 발명이 해결하고자 하는 과제는, 열수축을 최소화하여 안정성을 확보한 세퍼레이터를 제공하는 것이다.
- [10] 본 발명이 해결하고자 하는 또 다른 과제는 상기 세퍼레이터를 구비하는 전기화학소자를 제공하는 것이다.

과제 해결 수단

- [11] 상기 과제를 해결하기 위하여 본 발명의 일 측면에 따르면, 하기 구현예들의 세퍼레이터가 제공된다.
- [12] 제1 구현예는,
- [13] 다수의 기공을 갖는 다공성 기재; 및
- [14] 상기 다공성 기재의 적어도 일면에 또는 상기 다공성 기재의 적어도 일면 및 기공의 일부에 형성되어 있으며, 다수의 무기물 입자 및 상기 무기물 입자의 표면의 일부 또는 전부에 위치하여 상기 무기물 입자 사이를 연결 및 고정시키는 바인더 고분자를 포함하는 다공성 코팅층을 포함하고,
- [15] 상기 무기물 입자가 알루미나 입자 및 상기 알루미나 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함하는 세퍼레이터에 관한 것이다.
- [16] 제2 구현예는, 제1 구현예에 있어서,
- [17] 상기 수산화알루미늄 입자의 평균입경이 상기 알루미나 입자의 평균입경의 이 0.01 내지 0.9 배인 세퍼레이터에 관한 것이다.
- [18] 제3 구현예는, 제1 구현예 또는 제2 구현예에 있어서,
- [19] 상기 알루미나 입자의 평균입경이 0.5 내지 3 μm 인 세퍼레이터에 관한 것이다.
- [20] 제4 구현예는, 제1 구현예 내지 제3 구현예 중 어느 한 구현예에 있어서,
- [21] 상기 수산화알루미늄 입자의 평균입경이 0.05 내지 0.4 μm 인 세퍼레이터에 관한 것이다.
- [22] 상기 알루미나 입자 대 상기 수산화알루미늄 입자의 함량비가 50:50 내지 97:3인 세퍼레이터에 관한 것이다.
- [23] 제5 구현예는, 제1 구현예 내지 제4 구현예 중 어느 한 구현예에 있어서,
- [24] 상기 수산화알루미늄 입자가 보헤마이트($\gamma\text{-AlO}(\text{OH})$), 유사 보헤마이트(Al_2O_3

· H₂O), 다이어스포(diaspore, α -AlO(OH)), 바이어리트(bayerlite, α -AlO(OH)₃), 깁사이트(gibbsite, γ -AlO(OH)₃), 노드스트란디트(nordstrandite, AlO(OH)₃) 로 이루어진 군으로부터 선택된 1종 이상인 세퍼레이터에 관한 것이다.

- [25] 제6 구현예는, 제1 구현예 내지 제5 구현예 중 어느 한 구현예에 있어서,
- [26] 상기 다공성 기재가 폴리올레핀계 다공성 기재인 세퍼레이터에 관한 것이다.
- [27] 제7 구현예는, 제6 구현예에 있어서,
- [28] 상기 폴리올레핀계 다공성 기재가 폴리에틸렌, 폴리프로필렌, 폴리부틸렌 및 폴리펜텐으로 이루어진 군으로부터 선택된 어느 하나의 고분자 또는 이들의 혼합물로 형성된 세퍼레이터에 관한 것이다.
- [29] 제8 구현예는, 제1 구현예 내지 제7 구현예 중 어느 한 구현예에 있어서,
- [30] 상기 다공성 기재의 두께가 5 내지 50 μ m이고, 기공 크기 및 기공도는 각각 0.01 내지 50 μ m 및 10 내지 95%인 세퍼레이터에 관한 것이다.
- [31] 제9 구현예는, 제1 구현예 내지 제8 구현예 중 어느 한 구현예에 있어서,
- [32] 상기 무기물 입자와 바인더 고분자의 중량비가 50:50 내지 99:1인 세퍼레이터에 관한 것이다.
- [33] 제10 구현예는, 제1 구현예 내지 제9 구현예 중 어느 한 구현예에 있어서,
- [34] 상기 바인더 고분자가 폴리비닐리덴 플루오라이드-헥사플루오로프로필렌 (polyvinylidene fluoride-co-hexafluoropropylene), 폴리비닐리덴 플루오라이드-트리클로로에틸렌 (polyvinylidene fluoride-co-trichloroethylene), 폴리메틸메타크릴레이트 (polymethylmethacrylate), 폴리부틸아크릴레이트 (polybutylacrylate), 폴리아크릴로니트릴 (polyacrylonitrile), 폴리비닐피롤리돈 (polyvinylpyrrolidone), 폴리비닐아세테이트 (polyvinylacetate), 에틸렌 비닐 아세테이트 공중합체 (polyethylene-co-vinyl acetate), 폴리에틸렌옥사이드 (polyethylene oxide), 폴리아릴레이트(polyarylate), 셀룰로오스 아세테이트 (cellulose acetate), 셀룰로오스 아세테이트 부틸레이트 (cellulose acetate butyrate), 셀룰로오스 아세테이트 프로피오네이트 (cellulose acetate propionate), 시아노에틸플루란 (cyanoethylpullulan), 시아노에틸폴리비닐알콜 (cyanoethylpolyvinylalcohol), 시아노에틸셀룰로오스 (cyanoethylcellulose), 시아노에틸수크로오스 (cyanoethylsucrose), 플루란 (pullulan) 및 카르복실 메틸 셀룰로오스 (carboxyl methyl cellulose)로 이루어진 군으로부터 선택된 어느 하나 또는 이들 중 2종 이상의 혼합물인 세퍼레이터에 관한 것이다.
- [35] 본 발명의 다른 측면에 따르면, 하기 구현예들의 전기화학소자가 제공된다.
- [36] 제11 구현예는, 캐소드, 애노드, 상기 캐소드 및 애노드 사이에 개재된 세퍼레이터를 포함하는 전기화학소자에 있어서, 상기 세퍼레이터가 제1 구현예 내지 제10 구현예 중 어느 한 구현예의 세퍼레이터인 전기화학소자에 관한 것이다.
- [37] 제12 구현예는, 제11 구현예에 있어서,
- [38] 상기 전기화학소자가 리튬 이차전지인 전기화학소자에 관한 것이다.

발명의 효과

- [39] 본 발명의 일 실시예에 따르면, 무기물 입자로서 크기와 재질이 상이한 알루미늄 입자 및 상기 알루미늄 입자 보다 입경이 작은 수산화알루미늄 입자를 사용한 다공성 코팅층을 구비함으로써, 무기물 입자의 다공성 코팅층에 의해 다공성 기재의 열수축율 특성이 크게 개선되어, 전기화학소자가 과열되는 경우에도 캐소드와 애노드 사이의 단락을 억제할 수 있으므로 안정성이 향상된 세퍼레이터와 이를 구비한 전기화학소자를 제공할 수 있다.
- [40] 또한, 밀도가 상대적으로 작은 수산화알루미늄 입자를 사용함으로써, 세퍼레이터 및 전기화학소자의 경량화를 도모할 수 있고, 수산화알루미늄 입자의 우수한 흡열 특성에 의해, 이상 작동에 따른 전기화학소자의 내부 온도 급속한 온도 상승을 제어할 수 있다.

도면의 간단한 설명

- [41] 도 1은 본 발명의 일 실시예에 따른 세퍼레이터를 도시한 개략도이다.
도 2 및 도 3은 실시예 1 및 비교예 1에서 얻어진 세퍼레이터의 열수축율 측정시 외관 사진을 각각 나타낸 것이다.

발명의 실시를 위한 형태

- [42] 이하, 본 발명을 상세히 설명하기로 한다. 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야만 한다.
- [43] 본 발명의 일 측면에 따른 세퍼레이터는, 다수의 기공을 갖는 다공성 기재; 및 상기 다공성 기재의 적어도 일면 및 상기 다공성 기재의 기공 중 1종 이상의 영역에 형성되어 있으며, 다수의 무기물 입자 및 상기 무기물 입자의 표면의 일부 또는 전부에 위치하여 상기 무기물 입자 사이를 연결 및 고정시키는 바인더 고분자를 포함하는 다공성 코팅층;을 포함하고, 상기 무기물 입자가 알루미늄 입자 및 상기 알루미늄 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함한다.
- [44] 본 발명에서는 무기물 입자로서 평균입경 크기와 그 물질이 상이한 2종의 무기물 입자, 알루미늄 입자 및 상기 알루미늄 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함한다.
- [45] 상기 수산화알루미늄 입자의 평균입경은 상기 알루미늄 입자의 평균입경의 0.01 내지 0.9 배, 바람직하게는 0.015 내지 0.85 배, 더 바람직하게는 0.02 내지 0.08 배일 수 있다.
- [46] 본 발명의 일 실시예에 따르면, 알루미늄 입자의 평균입경은 0.5 내지 3 μm , 바람직하게는 0.7 내지 2.7 μm , 더 바람직하게는 0.9 내지 2.5 μm 일 수 있고, 수산화알루미늄 입자의 평균입경은 0.05 내지 0.4 μm , 바람직하게는 0.07 내지 0.37 μm , 더 바람직하게는 0.09 내지 0.35 μm 일 수 있다.
- [47] 본 발명의 무기물 입자는 상기와 같이 서로 상이한 평균입경을 갖는 2종을

포함하고 있으므로, 다공성 코팅층 형성용 조성물 내의 무기물 입자의 분산성 및 코팅 공정성이 향상되고, 코팅층의 두께의 조절이 용이하며, 기계적 물성 및 전기적 특성이 저하되는 것을 개선할 수 있다. 더불어, 대입경 입자들 사이의 큰 공극 내에, 소입경 입자가 위치할 수 있어, 생성되는 다공성 코팅층의 기공의 크기를 제어할 수 있으며, 다공성 코팅층의 치밀성이 개선되어 다공성 기체의 열수축 현상을 억제함으로써 전지의 충전 및 방전시 내부 단락 현상을 방지할 수도 있다.

[48] 특히, 본 발명에서는, 알루미나 입자 및 상기 알루미나 입자 보다 평균입경이 작은 수산화알루미늄 입자를 함께 사용함으로써, 단지 크기가 상이한 동종 무기물 입자 2종을 사용하는 것에 비하여, 보다 유리한 효과를 질 수 있는 바, 이하, 살펴보겠다.

[49] 상기 알루미나 입자는 Al_2O_3 의 화학식을 갖는 알루미늄과 산소의 화합물로서, 상대적으로 높은 열전도성을 갖는 전기 절연체로서 알려져 있고, 그 밀도는 3.95 내지 4.1g/cm^3 이다.

[50] 한편, 상기 수산화알루미늄 입자로는 수산화기의 종류에 따라서, 수산화기가 1개인 보헤마이트($\gamma\text{-AlO}(\text{OH})$), 다이어스포(diaspore, $\alpha\text{-AlO}(\text{OH})$), 수산화기가 2개인 바이어리트(bayerlite, $\alpha\text{-AlO}(\text{OH})_3$), 깁사이트(gibbsite, $\gamma\text{-AlO}(\text{OH})_3$), 노드스트란디트(nordstrandite, $\text{AlO}(\text{OH})_3$)가 있고, 또한 알루미나의 수화물인 유사 보헤마이트($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$)를 들 수 있다. 이들 수산화알루미늄 입자로는 이들을 단독 또는 2종 이상의 혼합물로도 사용할 수 있다. 상기 수산화알루미늄 입자는 통상 밀도가 약 2.4 내지 3.1g/cm^3 의 범위를 가진다.

[51] 또한, 수산화알루미늄 입자는 흡열 특성이 뛰어나고, 원형 이외에, 판 형상(박편 형상), 침 형상(피브릴 형상), 소정의 결정축 방향으로 길게 연장된 육각판 형상, 사각 판 형상을 비롯한 다각 판 형상, 방추 형상 등을 가질 수 있다.

[52] 따라서, 본 발명의 다공성 코팅층에, 대입경 무기물 입자로서 알루미나 입자와 함께, 알루미나 입자 보다 평균입경이 작고 알루미나에 대비하여 상대적으로 밀도가 작은 수산화알루미늄 입자를 사용하는 경우, 세퍼레이터의 전체 중량을 낮출 수 있어, 전기화학소자의 경량화를 도모할 수 있다. 또한, 수산화알루미늄 입자는 구형 외에 판 형상 등의 다양한 형상으로 구현할 수 있어, 구형의 알루미나 입자가 서로 이웃하여 배열되거나 상당히 이격되어 배열되어 있는 경우에도 알루미나 입자들 사이의 다양한 형태의 공극내에 용이하게 배치될 수 있다. 그 결과, 다공성 코팅층의 치밀성과 기계적 물성을 크게 향상시키고, 다공성 기체의 열수축 현상을 억제하여 전기화학 소자 내부의 단락을 방지하는 역할을 할 수 있다.

[53] 또한, 평균입경이 상이한 알루미나 입자, 즉 대입경의 알루미나 입자 및 소입경의 알루미나 입자를 사용하는 경우, 대입경의 알루미나 입자에 대비하여 소입경의 알루미나 입자는 그 가격이 현저히 고가이고, 취급이 용이하지 않은 등의 문제가 있으나, 본 발명에서는 알루미나 입자 및 알루미나 입자 보다

평균입경이 작은 수산화알루미늄 입자를 함께 사용함으로써, 이러한 문제가 해소되는 이점도 있다.

[54] 또한, 본 발명의 일 실시예에 따르면, 상기 알루미나 입자 대 수산화알루미늄 입자의 함량비는 50:50 내지 97:3, 바람직하게는 55:45 내지 95:5, 더 바람직하게는 60:40 내지 93:7일 수 있다.

[55] 상기 함량비가 이러한 범위를 만족하는 경우, 다공성 코팅층 형성용 코팅 조성물의 분산 안정성이나 공정성이 크게 개선되고, 균일한 두께의 다공성 코팅층의 형성이 가능하고, 다공성 코팅층의 공극율의 제어가 용이할 수 있다.

[56] 상기 다공성 기재는 다공성 고분자 기재일 수 있고, 구체적으로 다공성 고분자 필름 기재 또는 다공성 고분자 부직포 기재를 들 수 있다.

[57] 상기 다공성 고분자 필름 기재로는 폴리에틸렌, 폴리프로필렌과 같은 폴리올레핀으로 이루어진 다공성 고분자 필름일 수 있으며, 이러한 폴리올레핀 다공성 고분자 필름 기재는 예를 들어 80 내지 130 °C의 온도에서 섰다운 기능을 발현한다.

[58] 이때, 폴리올레핀 다공성 고분자 필름은 고밀도 폴리에틸렌, 선형 저밀도 폴리에틸렌, 저밀도 폴리에틸렌, 초고분자량 폴리에틸렌과 같은 폴리에틸렌, 폴리프로필렌, 폴리부틸렌, 폴리펜텐 등의 폴리올레핀계 고분자를 각각 단독 또는 이들의 2종 이상 혼합하여 고분자로 형성할 수 있다.

[59] 또한, 상기 다공성 고분자 필름 기재는 폴리올레핀 외에 폴리에스테르 등의 다양한 고분자들을 이용하여 필름 형상으로 성형하여 제조될 수도 있다. 또한, 상기 다공성 고분자 필름 기재는 2층 이상의 필름층이 적층된 구조로 형성될 수 있으며, 각 필름층은 전술한 폴리올레핀, 폴리에스테르 등의 고분자 단독으로 또는 이들을 2종 이상 혼합한 고분자로 형성될 수도 있다.

[60] 또한, 상기 다공성 고분자 필름 기재 및 다공성 부직포 기재는 상기와 같은 폴리올레핀계 외에 폴리에틸렌테레프탈레이트(polyethyleneterephthalate), 폴리부틸렌테레프탈레이트(polybutyleneterephthalate), 폴리에스테르(polyester), 폴리아세탈(polyacetal), 폴리아미드(polyamide), 폴리카보네이트(polycarbonate), 폴리이미드(polyimide), 폴리에테르에테르케톤(polyetheretherketone), 폴리에테르설폰(polyethersulfone), 폴리페닐렌옥사이드(polyphenyleneoxide), 폴리페닐렌설파이드(polyphenylenesulfide), 폴리에틸렌나프탈렌(polyethylenenaphthalene) 등을 각각 단독으로 또는 이들을 혼합한 고분자로 형성될 수 있다. 상기 다공성 기재의 두께는 특별히 제한되지 않으나, 바람직하게는 1 내지 100 μm , 더욱 바람직하게는 5 내지 50 μm 이고, 다공성 기재에 존재하는 기공 크기 및 기공도 역시 특별히 제한되지 않으나 각각 0.01 내지 50 μm 및 10 내지 95%인 것이 바람직하다.

[61] 본 발명의 일 측면에 따른 세퍼레이터에 있어서, 다공성 코팅층 형성에 사용되는 바인더 고분자로는 당 업계에서 다공성 코팅층 형성에 통상적으로 사용되는 고분자를 사용할 수 있다. 특히, 유리 전이 온도(glass transition

temperature, T_g)가 -200 내지 200°C인 고분자를 사용할 수 있는데, 이는 최종적으로 형성되는 다공성 코팅층의 유연성 및 탄성 등과 같은 기계적 물성을 향상시킬 수 있기 때문이다. 이러한 바인더 고분자는 무기물 입자들 사이를 연결 및 안정하게 고정시켜주는 바인더 역할을 충실히 수행함으로써, 다공성 코팅층이 도입된 세퍼레이터의 기계적 물성 저하 방지에 기여한다.

- [62] 또한, 상기 바인더 고분자는 이온 전도 능력을 반드시 가질 필요는 없으나, 이온 전도 능력을 갖는 고분자를 사용할 경우 전기화학소자의 성능을 더욱 향상시킬 수 있다. 따라서, 상기 바인더 고분자는 가능한 유전율 상수가 높은 것을 사용할 수 있다. 실제로 전해액에서 염의 해리도는 전해액 용매의 유전율 상수에 의존하기 때문에, 상기 바인더 고분자의 유전율 상수가 높을수록 전해질에서의 염 해리도를 향상시킬 수 있다. 이러한 바인더 고분자의 유전율 상수는 1.0 내지 100 (측정 주파수 = 1 kHz) 범위가 사용 가능하며, 특히 10 이상일 수 있다.
- [63] 전술한 기능 이외에, 상기 바인더 고분자는 액체 전해액 함침시 겔화됨으로써 높은 전해액 팽윤도(degree of swelling)를 나타낼 수 있는 특징을 가질 수 있다. 이에 따라, 상기 바인더 고분자의 용해도 지수, 즉 힐더브랜드 용해도 지수(Hildebrand solubility parameter)는 15 내지 45 $\text{MPa}^{1/2}$ 또는 15 내지 25 $\text{MPa}^{1/2}$ 및 30 내지 45 $\text{MPa}^{1/2}$ 범위이다. 따라서, 폴리올레핀류와 같은 소수성 고분자들보다는 극성기를 많이 갖는 친수성 고분자들이 더 사용될 수 있다. 상기 용해도 지수가 15 $\text{MPa}^{1/2}$ 미만 및 45 $\text{MPa}^{1/2}$ 를 초과할 경우, 통상적인 전지용 액체 전해액에 의해 팽윤(swelling)되기 어려울 수 있기 때문이다.
- [64] 이러한 바인더 고분자의 비제한적인 예로는 폴리비닐리덴 플루오라이드-헥사플루오로프로필렌 (polyvinylidene fluoride-co-hexafluoropropylene), 폴리비닐리덴 플루오라이드-트리클로로에틸렌 (polyvinylidene fluoride-co-trichloroethylene), 폴리메틸메타크릴레이트 (polymethylmethacrylate), 폴리부틸아크릴레이트 (polybutylacrylate), 폴리아크릴로니트릴 (polyacrylonitrile), 폴리비닐피롤리돈 (polyvinylpyrrolidone), 폴리비닐아세테이트 (polyvinylacetate), 에틸렌 비닐 아세테이트 공중합체 (polyethylene-co-vinyl acetate), 폴리에틸렌옥사이드 (polyethylene oxide), 폴리아릴레이트(polyarylate), 셀룰로오스 아세테이트 (cellulose acetate), 셀룰로오스 아세테이트 부틸레이트 (cellulose acetate butyrate), 셀룰로오스 아세테이트 프로피오네이트 (cellulose acetate propionate), 시아노에틸플루란 (cyanoethylpullulan), 시아노에틸폴리비닐알코올 (cyanoethylpolyvinylalcohol), 시아노에틸셀룰로오스 (cyanoethylcellulose), 시아노에틸수크로오스 (cyanoethylsucrose), 플루란 (pullulan) 및 카르복실 메틸 셀룰로오스 (carboxyl methyl cellulose)등을 들 수 있으며, 이에 한정되는 것은 아니다.
- [65] 상기 무기물 입자와 바인더 고분자의 중량비는 예를 들어 50:50 내지 99:1 범위가 바람직하며, 더욱 바람직하게는 70:30 내지 95:5이다. 바인더 고분자에 대한 무기물 입자의 함량비가 상기 범위를 만족하는 경우, 바인더 고분자의

함량이 많아지게 되어 형성되는 코팅층의 기공 크기 및 기공도가 감소되는 문제가 방지될 수 있고, 바인더 고분자 함량이 적기 때문에 형성되는 코팅층의 내필링성이 악화되는 문제도 해소될 수 있다.

- [66] 본 발명의 일 측면에 따른 세퍼레이터는 다공성 코팅층 성분으로 전술한 무기물 입자 및 고분자 이외에, 기타 첨가제를 더 포함할 수 있다.
- [67] 본 발명의 일 실시예에 따른 세퍼레이터는, 알루미나 입자 및 상기 알루미나 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함하는 무기물 입자 및 바인더 고분자를 구비하는, 다공성 코팅층 형성용 조성물을 준비하고, 이러한 조성물을 다공성 기재의 적어도 일면 상에 도포하고, 이를 건조함으로써 제조될 수 있다.
- [68] 먼저, 다공성 코팅층 형성용 조성물은, 바인더 고분자를 용매에 용해시킨 다음 무기물 입자를 첨가하고 이를 분산시켜 제조할 수 있다. 무기물 입자들은 미리 소정의 평균입경을 갖도록 파쇄된 상태에서 첨가할 수 있으며, 또는 바인더 고분자의 용액에 무기물 입자를 첨가한 후 무기물 입자를 불밀법 등을 이용하여 소정의 평균입경을 갖도록 제어하면서 파쇄하여 분산시킬 수도 있다.
- [69] 상기 다공성 코팅층 형성용 조성물을 상기 다공성 기재에 코팅하는 방법은 특별히 한정하지는 않지만, 슬랏 코팅이나 딥 코팅 방법을 사용하는 것이 바람직하다. 슬랏 코팅은 슬랏 다이를 통해 공급된 조성물이 기재의 전면에 도포되는 방식으로 정량 펌프에서 공급되는 유량에 따라 코팅층 두께의 조절이 가능하다. 또한 딥 코팅은 조성물이 들어있는 탱크에 기재를 담그어 코팅하는 방법으로, 조성물의 농도 및 조성물 탱크에서 기재를 꺼내는 속도에 따라 코팅층 두께의 조절이 가능하며 보다 정확한 코팅 두께 제어를 위해 침지 후 메이어바 등을 통해 후계량할 수 있다.
- [70] 이렇게 다공성 코팅층 형성용 조성물이 코팅된 다공성 기재를 오븐에서 건조하여 다공성 기재의 적어도 일면에 코팅층을 형성하게 된다.
- [71] 상기 다공성 코팅층에서는 무기물 입자들은 충전되어 서로 접촉된 상태에서 상기 바인더 고분자에 의해 서로 결합되고, 이로 인해 무기물 입자들 사이에 인터스티셜 볼륨(interstitial volume)이 형성되고, 상기 무기물 입자 사이의 인터스티셜 볼륨(Interstitial Volume)은 빈 공간이 되어 기공을 형성한다.
- [72] 즉, 바인더 고분자는 무기물 입자들이 서로 결합된 상태를 유지할 수 있도록 이들을 서로 부착, 예를 들어, 바인더 고분자가 무기물 입자 사이를 연결 및 고정시키고 있다. 또한, 상기 다공성 코팅층의 기공은 무기물 입자들 간의 인터스티셜 볼륨(interstitial volume)이 빈 공간이 되어 형성된 기공이고, 이는 무기물 입자들에 의한 충전 구조(closed packed or densely packed)에서 실질적으로 면접하는 무기물 입자들에 의해 한정되는 공간이다.
- [73] 본 발명의 일 실시예에 따른 세퍼레이터를 도 1에 도시하였다. 도 1을 참조하면, 상기 세퍼레이터는 다공성 기재(100); 및 상기 다공성 기재(100)의 일 면 상에 형성되고, 알루미나 입자(210) 및 상기 알루미나 입자 보다 평균입경이 작은

수산화알루미늄 입자(220)를 포함하는 다공성 코팅층(200) (바인더 고분자는 미도시)을 구비하고 있다.

- [74] 본 발명의 일 측면에 따른 전기화학소자는 캐소드, 애노드, 상기 캐소드 및 애노드 사이에 개재된 세퍼레이터를 포함하고, 상기 세퍼레이터가 전술한 본 발명의 일 실시예에 따른 세퍼레이터이다.
- [75] 이러한 전기화학소자는 전기 화학 반응을 하는 모든 소자를 포함하며, 구체적인 예를 들면, 모든 종류의 1차, 이차 전지, 연료 전지, 태양 전지 또는 슈퍼 캐패시터 소자와 같은 캐패시터(capacitor) 등이 있다. 특히, 상기 2차 전지 중 리튬 금속 이차 전지, 리튬 이온 이차 전지, 리튬 폴리머 이차 전지 또는 리튬 이온 폴리머 이차 전지 등을 포함하는 리튬 이차전지가 바람직하다.
- [76] 본 발명의 세퍼레이터와 함께 적용될 양극과 음극의 양 전극으로는 특별히 제한되지 않으며, 당업계에 알려진 통상적인 방법에 따라 전극활물질을 전극 전류집전체에 결합된 형태로 제조할 수 있다. 상기 전극활물질 중 양극활물질의 비제한적인 예로는 종래 전기화학소자의 양극에 사용될 수 있는 통상적인 양극활물질이 사용 가능하며, 특히 리튬망간산화물, 리튬코발트산화물, 리튬니켈산화물, 리튬철산화물 또는 이들을 조합한 리튬복합산화물을 사용하는 것이 바람직하다. 음극활물질의 비제한적인 예로는 종래 전기화학소자의 음극에 사용될 수 있는 통상적인 음극활물질이 사용 가능하며, 특히 리튬 금속 또는 리튬 합금, 탄소, 석유코크(petroleum coke), 활성화 탄소(activated carbon), 그래파이트(graphite) 또는 기타 탄소류 등과 같은 리튬 흡착물질 등이 바람직하다. 양극 전류집전체의 비제한적인 예로는 알루미늄, 니켈 또는 이들의 조합에 의하여 제조되는 호일 등이 있으며, 음극 전류집전체의 비제한적인 예로는 구리, 금, 니켈 또는 구리 합금 또는 이들의 조합에 의하여 제조되는 호일 등이 있다.
- [77] 본 발명의 전기화학소자에서 사용될 수 있는 전해액은 A+B와 같은 구조의 염으로서, A⁺는 Li⁺, Na⁺, K⁺와 같은 알칼리 금속 양이온 또는 이들의 조합으로 이루어진 이온을 포함하고 B⁻는 PF₆⁻, BF₄⁻, Cl⁻, Br⁻, I⁻, ClO₄⁻, AsF₆⁻, CH₃CO₂⁻, CF₃SO₃⁻, N(CF₃SO₂)₂⁻, C(CF₂SO₂)₃⁻와 같은 음이온 또는 이들의 조합으로 이루어진 이온을 포함하는 염이 프로필렌 카보네이트(PC), 에틸렌 카보네이트(EC), 디에틸카보네이트(DEC), 디메틸카보네이트(DMC), 디프로필카보네이트(DPC), 디메틸설폭사이드, 아세토니트릴, 디메톡시에탄, 디에톡시에탄, 테트라하이드로퓨란, N-메틸-2-피롤리돈(NMP), 에틸메틸카보네이트(EMC), 감마 부티로락톤 (γ-부티로락톤) 또는 이들의 혼합물로 이루어진 유기 용매에 용해 또는 해리된 것이 있으나, 이에만 한정되는 것은 아니다.
- [78] 상기 전해액 주입은 최종 제품의 제조 공정 및 요구 물성에 따라, 전지 제조 공정 중 적절한 단계에서 행해질 수 있다. 즉, 전지 조립 전 또는 전지 조립 최종 단계 등에서 적용될 수 있다.
- [79] 이하, 본 발명을 구체적으로 설명하기 위해 실시예를 들어 상세하게

설명하기로 한다. 그러나, 본 발명에 따른 실시예는 여러 가지 다른 형태로 변형될 수 있으며, 본 발명의 범위가 아래에서 상술하는 실시예에 한정되는 것으로 해석되어서는 안 된다. 본 발명의 실시예는 당업계에서 평균적인 지식을 가진 자에게 본 발명을 보다 완전하게 설명하기 위해서 제공되는 것이다.

[80]

[81] 실시예 1

[82] PVdF-CTFE (폴리비닐리덴플로라이드-클로로트리플로로에틸렌 공중합체) 10 중량부를 아세톤 95 중량부에 첨가하여 50°C에서 약 12시간 이상 용해시켜 바인더 고분자 용액 10 중량부를 준비하였다. 제조된 바인더 고분자 용액에 알루미나 입자로서 평균입경 0.5 μm 의 알루미나(Al_2O_3)입자와, 수산화알루미늄 입자로서 평균입경 0.2 μm 의 보헤마이트($\gamma\text{-AlO}(\text{OH})$) 입자를 90:10의 중량비이면서, 바인더 고분자/무기물 입자 총합 = 10/90의 중량비가 되도록 첨가하고, 분산하여 슬러리를 제조하였다.

[83] 이와 같이 제조된 슬러리를 딥(dip) 코팅법으로, 두께 9 μm 인 폴리에틸렌 다공성 막(기공도 40%)에 코팅하고, 코팅 두께는 약 5 내지 6 μm 정도로 조절하여, 세퍼레이터를 제조하였다.

[84]

[85] 비교예 1

[86] 무기물 입자로 평균 입경 0.5 μm 의 Al_2O_3 입자만을 사용한 점을 제외하고는 실시예 1과 동일한 방법으로 세퍼레이터를 제조하였다.

[87]

[88] 실시예 2

[89] PVdF-CTFE (폴리비닐리덴플로라이드-클로로트리플로로에틸렌 공중합체) 5 중량부를 아세톤 95 중량부에 첨가하여 50°C에서 약 12시간 이상 용해시켜 바인더 고분자 용액 5 중량부를 준비하였다. 제조된 바인더 고분자 용액에 알루미나 입자로서 평균입경 2 μm 의 알루미나(Al_2O_3)입자와, 수산화알루미늄 입자로서 평균입경 0.2 μm 의 보헤마이트($\gamma\text{-AlO}(\text{OH})$) 입자를 90:10의 중량비이면서, 바인더 고분자/무기물 입자 총합 = 20/80의 중량비가 되도록 첨가하고, 분산하여 슬러리를 제조하였다.

[90] 이와 같이 제조된 슬러리를 딥(dip) 코팅법으로, 두께 20 μm 인 폴리에틸렌 다공성 막(기공도 40%)에 코팅하고, 코팅 두께는 약 10 μm 정도로 조절하여, 세퍼레이터를 제조하였다.

[91]

[92] 비교예 2

[93] 무기물 입자로 평균 입경 500 nm의 Al_2O_3 입자와 평균 입경 20 nm인 Al_2O_3 입자를 90:10의 중량비로 사용한 점을 제외하고는 실시예 2와 동일한 방법으로 세퍼레이터를 제조하였다.

[94]

[95] 비교예 3

[96] 무기물 입자로 평균 입경 500 nm의 보헤마이트(γ -AlO(OH)) 입자와 평균 입경 20 nm인 Al₂O₃ 입자를 90:10의 중량비로 사용한 점을 제외하고는 실시예 2와 동일한 방법으로 세퍼레이터를 제조하였다.

[97]

[98] 실시예 1 및 비교예 1에서 얻어진 세퍼레이터의 두께, 걸리(Gurley) 공기투과도, 및 기계방향(MD)과 횡방향(TD)의 열수축율을 측정하고, 그 결과를 하기 표 1에 기재하였다.

또한, 실시예 1 및 비교예 1에서 얻어진 세퍼레이터의 열수축율 측정시 외관 사진을 도 2 및 도 3에 각각 나타내었다.

[99]

[100] 걸리(Gurley) 공기투과도(통기시간)

[101] 걸리 공기투과도 값은 ASTM D726-94 방법에 의해 측정하였다. 여기서 사용된 걸리는, 공기의 흐름에 대한 저항으로서, 걸리 덴소미터(densometer)에 의해 측정된다. 여기서 설명된 걸리 공기투과도 값은 100 cc의 공기가 12.2 inH₂O의 압력하에서 1 in²의 단면을 통과하는 데 걸리는 시간(초), 즉 통기시간으로 나타낸다.

[102]

[103] 열수축율

[104] 실시예 1 및 비교예 1에서 제조된 세퍼레이터를 50mm * 50mm 크기로 재단하여 A4 용지 사이에 놓고 150℃의 컨벡션 오븐(convection oven)에 30분 동안 넣은 후 기계방향(MD)과 횡방향(TD)의 열수축율을 측정하였다. 이때, 열수축율은 [(최초 길이 - 열처리 후 길이)/(최초 길이)×100]으로 계산하였다.

[105]

[106] [표1]

	실시예 1	비교예 1
두께(μ m)	13.25	14.7
통기시간(Gurley) (s/100cc)	417	572
열수축율(MD/TD) (%) 150℃, 30분	9/9	47/41

[107]

[108] 상기 표 1을 참조하면, 알루미나 입자 및 상기 알루미나 입자보다 입경이 작은 수산화알루미늄 입자를 사용한 다공성 코팅층을 구비한 실시예 1의 세퍼레이터가 알루미나 입자만을 사용한 비교예 1과 비교하여, 통기도와, MD 방향/TD 방향의 열수축율 면에서 우수한 효과를 나타냄을 알 수 있었다.

[109] 특히, 실시예 1의 다공성 코팅층의 두께는 $4.25\mu\text{m}$ 이고, 비교예 1의 다공성 코팅층의 두께는 $5.7\mu\text{m}$ 로서, 실시예 1은 비교예 1에 비하여 더 작은 다공성 코팅층의 두께를 가짐에도 대입경 입자들 사이의 큰 공극 내에, 소입경 입자가 위치할 수 있어, 다공성 코팅층의 치밀성이 개선되어 다공성 기재의 열수축 현상을 더 탁월하게 억제한 것으로 볼 수 있다. 이를 통하여, 밀도가 상대적으로 작은 수산화알루미늄 입자를 사용함으로써, 세퍼레이터 및 전기화학소자의 경량화를 도모하면서도, 전기화학소자가 과열되는 경우에도 캐소드와 애노드 사이의 단락을 억제할 수 있으므로 안정성이 향상된 세퍼레이터와 이를 구비한 전기화학소자를 제공할 수 있을 것이다.

[110] [부호의 설명]

[111] 100: 다공성 기재, 200: 다공성 코팅층

[112] 210: 알루미나 입자, 220: 수산화알루미늄 입자

청구범위

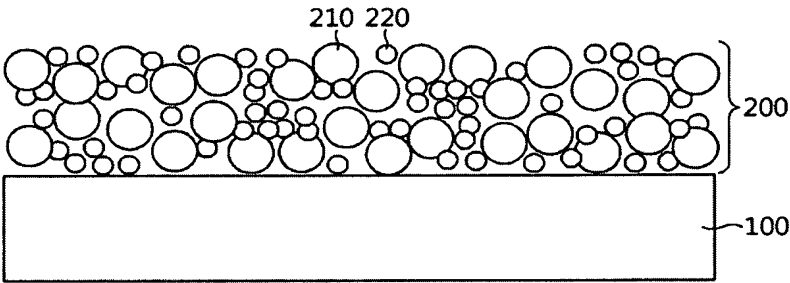
- [청구항 1] 다수의 기공을 갖는 다공성 기재; 및
상기 다공성 기재의 적어도 일면에 또는 상기 다공성 기재의 적어도 일면 및 기공의 일부에 형성되어 있으며, 다수의 무기물 입자 및 상기 무기물 입자의 표면의 일부 또는 전부에 위치하여 상기 무기물 입자 사이를 연결 및 고정시키는 바인더 고분자를 포함하는 다공성 코팅층을 포함하고, 상기 무기물 입자가 알루미나 입자 및 상기 알루미나 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함하는 세퍼레이터.
- [청구항 2] 제1항에 있어서,
상기 수산화알루미늄 입자의 평균입경이 상기 알루미나 입자의 평균입경의 0.01 내지 0.9 배인 세퍼레이터.
- [청구항 3] 제1항에 있어서,
상기 알루미나 입자의 평균입경이 0.5 내지 3 μm 인 세퍼레이터.
- [청구항 4] 제1항에 있어서,
상기 수산화알루미늄 입자의 평균입경이 0.05 내지 0.4 μm 인 세퍼레이터.
- [청구항 5] 제1항에 있어서,
상기 알루미나 입자 대 상기 수산화알루미늄 입자의 함량비가 50:50 내지 97:3인 세퍼레이터.
- [청구항 6] 제1항에 있어서,
상기 수산화알루미늄 입자가 보헤마이트($\gamma\text{-AlO(OH)}$), 유사 보헤마이트($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$), 다이어스포(diaspore, $\alpha\text{-AlO(OH)}$), 바이어리트(bayerlite, $\alpha\text{-AlO(OH)}_3$), 깁사이트(gibbsite, $\gamma\text{-AlO(OH)}_3$), 노드스트란디트(nordstrandite, AlO(OH)_3) 로 이루어진 군으로부터 선택된 1종 이상인 세퍼레이터.
- [청구항 7] 제1항에 있어서,
상기 다공성 기재가 폴리올레핀계 다공성 기재인 세퍼레이터.
- [청구항 8] 제7항에 있어서,
상기 폴리올레핀계 다공성 기재가 폴리에틸렌, 폴리프로필렌, 폴리부틸렌 및 폴리펜텐으로 이루어진 군으로부터 선택된 어느 하나의 고분자 또는 이들의 혼합물로 형성된 세퍼레이터.
- [청구항 9] 제1항에 있어서,
상기 다공성 기재의 두께가 5 내지 50 μm 이고, 기공 크기 및 기공도는 각각 0.01 내지 50 μm 및 10 내지 95%인 세퍼레이터.
- [청구항 10] 제1항에 있어서,
상기 무기물 입자와 바인더 고분자의 중량비가 50:50 내지 99:1인 세퍼레이터.
- [청구항 11] 제1항에 있어서,

상기 바인더 고분자가 폴리비닐리덴
 플루오라이드-헥사플루오로프로필렌 (polyvinylidene
 fluoride-co-hexafluoropropylene), 폴리비닐리덴
 플루오라이드-트리클로로에틸렌 (polyvinylidene
 fluoride-co-trichloroethylene), 폴리메틸메타크릴레이트
 (polymethylmethacrylate), 폴리부틸아크릴레이트 (polybutylacrylate),
 폴리아크릴로니트릴 (polyacrylonitrile), 폴리비닐피롤리돈
 (polyvinylpyrrolidone), 폴리비닐아세테이트 (polyvinylacetate), 에틸렌
 비닐 아세테이트 공중합체 (polyethylene-co-vinyl acetate),
 폴리에틸렌옥사이드 (polyethylene oxide), 폴리아릴레이트(polyarylate),
 셀룰로오스 아세테이트 (cellulose acetate), 셀룰로오스 아세테이트
 부틸레이트 (cellulose acetate butyrate), 셀룰로오스 아세테이트
 프로피오네이트 (cellulose acetate propionate), 시아노에틸플루란
 (cyanoethylpullulan), 시아노에틸폴리비닐알콜
 (cyanoethylpolyvinylalcohol), 시아노에틸셀룰로오스 (cyanoethylcellulose),
 시아노에틸수크로오스 (cyanoethylsucrose), 플루란 (pullulan) 및 카르복실
 메틸 셀룰로오스 (carboxyl methyl cellulose)로 이루어진 군으로부터
 선택된 어느 하나 또는 이들 중 2종 이상의 혼합물인 세퍼레이터.

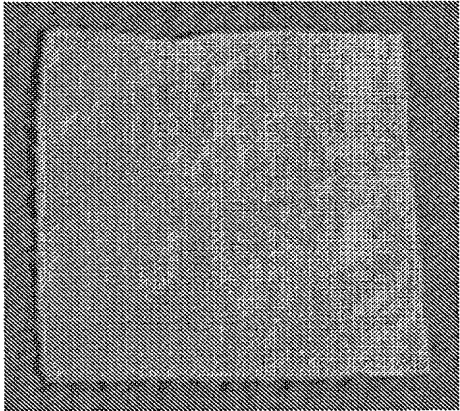
[청구항 12] 캐소드, 애노드, 상기 캐소드 및 애노드 사이에 개재된 세퍼레이터를
 포함하는 전기화학소자에 있어서, 상기 세퍼레이터가 제1항 내지 제11항
 중 어느 한 항의 세퍼레이터인 전기화학소자.

[청구항 13] 제12항에 있어서,
 상기 전기화학소자가 리튬 이차전지인 전기화학소자.

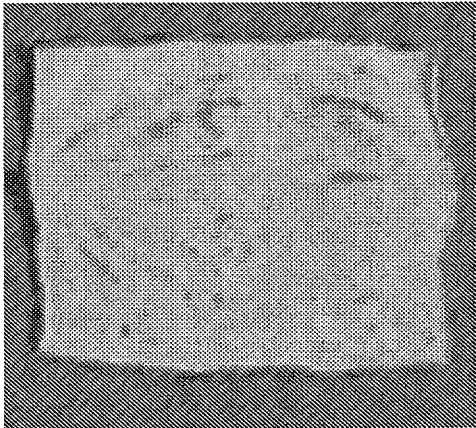
[도1]



[도2]



[도3]



INTERNATIONAL SEARCH REPORT

International application No.
PCT/KR2016/007532

A. CLASSIFICATION OF SUBJECT MATTER																				
<i>H01M 2/16(2006.01)i, H01M 10/052(2010.01)i</i>																				
According to International Patent Classification (IPC) or to both national classification and IPC																				
B. FIELDS SEARCHED																				
Minimum documentation searched (classification system followed by classification symbols) H01M 2/16; H01M 10/05; H01M 10/0525; H05B 3/10; C08J 5/24; B32B 27/36																				
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched Korean Utility models and applications for Utility models: IPC as above Japanese Utility models and applications for Utility models: IPC as above																				
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) eKOMPASS (KIPO internal) & Keywords: separator, alumina, aluminum hydroxide, binder, polyolefin, electrochemical device																				
C. DOCUMENTS CONSIDERED TO BE RELEVANT																				
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.																		
X	KR 10-2012-0079515 A (EIG LTD.) 13 July 2012 See abstract; paragraphs [0005]-[0044]; claims 1-9; and figures 1-3.	1-13																		
A	KR 10-2014-0116415 A (SIHL GMBH.) 02 October 2014 See abstract; paragraphs [0021]-[0052]; and claims 1-11.	1-13																		
A	KR 10-2013-0007740 A (GUMI ELECTRONICS & INFORMATION TECHNOLOGY RESEARCH INSTITUTE et al.) 21 January 2013 See abstract; paragraphs [0024]-[0033]; and claims 1-5.	1-13																		
A	JP 4311208 B2 (TORAY IND. INC.) 12 August 2009 See paragraphs [0052]-[0072]; and claims 1-3.	1-13																		
A	KR 10-2012-0124239 A (LG CHEM. LTD.) 13 November 2012 See abstract; paragraphs [0022]-[0058]; and claims 1-14.	1-13																		
☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.																				
<table><tr><td>* Special categories of cited documents:</td><td>“T”</td><td>later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention</td></tr><tr><td>“A” document defining the general state of the art which is not considered to be of particular relevance</td><td>“X”</td><td>document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone</td></tr><tr><td>“E” earlier application or patent but published on or after the international filing date</td><td>“Y”</td><td>document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art</td></tr><tr><td>“L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)</td><td>“G”</td><td>document member of the same patent family</td></tr><tr><td>“O” document referring to an oral disclosure, use, exhibition or other means</td><td></td><td></td></tr><tr><td>“P” document published prior to the international filing date but later than the priority date claimed</td><td></td><td></td></tr></table>			* Special categories of cited documents:	“T”	later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention	“A” document defining the general state of the art which is not considered to be of particular relevance	“X”	document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone	“E” earlier application or patent but published on or after the international filing date	“Y”	document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art	“L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)	“G”	document member of the same patent family	“O” document referring to an oral disclosure, use, exhibition or other means			“P” document published prior to the international filing date but later than the priority date claimed		
* Special categories of cited documents:	“T”	later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention																		
“A” document defining the general state of the art which is not considered to be of particular relevance	“X”	document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone																		
“E” earlier application or patent but published on or after the international filing date	“Y”	document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art																		
“L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)	“G”	document member of the same patent family																		
“O” document referring to an oral disclosure, use, exhibition or other means																				
“P” document published prior to the international filing date but later than the priority date claimed																				
Date of the actual completion of the international search 21 OCTOBER 2016 (21.10.2016)		Date of mailing of the international search report 25 OCTOBER 2016 (25.10.2016)																		
Name and mailing address of the ISA/KR  Korean Intellectual Property Office Government Complex-Daejeon, 189 Seonsa-ro, Daejeon 302-701, Republic of Korea Facsimile No. 82-42-472-7140		Authorized officer Telephone No.																		

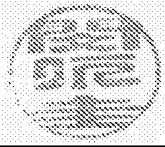
INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/KR2016/007532

Patent document cited in search report	Publication date	Patent family member	Publication date
KR 10-2012-0079515 A	13/07/2012	EP 2662920 A2	13/11/2013
		KR 10-1282067 B1	05/07/2013
		WO 2012-093864 A2	12/07/2012
KR 10-2014-0116415 A	02/10/2014	CN 104115306 A	22/10/2014
		EP 2805366 A1	26/11/2014
		EP 2805366 B1	24/08/2016
		JP 2015-504234 A	05/02/2015
		US 2015-0030933 A1	29/01/2015
		WO 2013-107911 A1	25/07/2013
KR 10-2013-0007740 A	21/01/2013	NONE	
JP 4311208 B2	12/08/2009	CN 1517402 A	04/08/2004
		EP 1441001 A2	28/07/2004
		EP 1441001 B1	28/11/2007
		JP 2004-243760 A	02/09/2004
		KR 10-0996022 B1	22/11/2010
		US 2004-0161620 A1	19/08/2004
		US 7459217 B2	02/12/2008
KR 10-2012-0124239 A	13/11/2012	CN 103733380 A	16/04/2014
		EP 2706591 A2	12/03/2014
		JP 2014-510388 A	24/04/2014
		JP 2016-076492 A	12/05/2016
		JP 5875673 B2	02/03/2016
		KR 10-1358761 B1	07/02/2014
		US 2013-0149587 A1	13/06/2013
		US 2015-0086714 A1	26/03/2015
		US 8927134 B2	06/01/2015
		WO 2012-150838 A2	08/11/2012

국 제 조 사 보 고 서

국제출원번호
PCT/KR2016/007532

A. 발명이 속하는 기술분류(국제특허분류(IPC)) H01M 2/16(2006.01)i, H01M 10/052(2010.01)i		
B. 조사된 분야		
조사된 최소문헌(국제특허분류를 기재) H01M 2/16; H01M 10/05; H01M 10/0525; H05B 3/10; C08J 5/24; B32B 27/36		
조사된 기술분야에 속하는 최소문헌 이외의 문헌 한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC 일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC		
국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우)) eKOMPASS(특허청 내부 검색시스템) & 키워드: 세퍼레이터, 알루미늄, 수산화알루미늄, 바인더, 폴리올레핀, 전기화학소 자		
C. 관련 문헌		
카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
X	KR 10-2012-0079515 A (주식회사 이아이지) 2012.07.13 요약; 단락 [0005]-[0044]; 청구항 1-9; 및 도면 1-3 참조.	1-13
A	KR 10-2014-0116415 A (실 게엠베하) 2014.10.02 요약; 단락 [0021]-[0052]; 및 청구항 1-11 참조.	1-13
A	KR 10-2013-0007740 A (재단법인 구미전자정보기술원 등) 2013.01.21 요약; 단락 [0024]-[0033]; 및 청구항 1-5 참조.	1-13
A	JP 4311208 B2 (TORAY IND INC.) 2009.08.12 단락 [0052]-[0072]; 및 청구항 1-3 참조.	1-13
A	KR 10-2012-0124239 A (주식회사 엘지화학) 2012.11.13 요약; 단락 [0022]-[0058]; 및 청구항 1-14 참조.	1-13
☐ 추가 문헌이 C(계속)에 기재되어 있습니다. ☒ 대응특허에 관한 별지를 참조하십시오.		
* 인용된 문헌의 특별 카테고리: “A” 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌 “E” 국제출원일보다 빠른 출원일 또는 우선일을 가지나 국제출원일 이후에 공개된 선출원 또는 특허 문헌 “L” 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌 “O” 구두 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌 “P” 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌 “T” 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌 “X” 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다. “Y” 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다. “&” 동일한 대응특허문헌에 속하는 문헌		
국제조사의 실제 완료일 2016년 10월 21일 (21.10.2016)	국제조사보고서 발송일 2016년 10월 25일 (25.10.2016)	
ISA/KR의 명칭 및 우편주소 대한민국 특허청 (35208) 대전광역시 서구 청사로 189, 4동 (둔산동, 정부대전청사) 팩스 번호 +82-42-481-8578	심사관 민인규 전화번호 +82-42-481-3326	

국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
KR 10-2012-0079515 A	2012/07/13	EP 2662920 A2 KR 10-1282067 B1 WO 2012-093864 A2	2013/11/13 2013/07/05 2012/07/12
KR 10-2014-0116415 A	2014/10/02	CN 104115306 A EP 2805366 A1 EP 2805366 B1 JP 2015-504234 A US 2015-0030933 A1 WO 2013-107911 A1	2014/10/22 2014/11/26 2016/08/24 2015/02/05 2015/01/29 2013/07/25
KR 10-2013-0007740 A	2013/01/21	없음	
JP 4311208 B2	2009/08/12	CN 1517402 A EP 1441001 A2 EP 1441001 B1 JP 2004-243760 A KR 10-0996022 B1 US 2004-0161620 A1 US 7459217 B2	2004/08/04 2004/07/28 2007/11/28 2004/09/02 2010/11/22 2004/08/19 2008/12/02
KR 10-2012-0124239 A	2012/11/13	CN 103733380 A EP 2706591 A2 JP 2014-510388 A JP 2016-076492 A JP 5875673 B2 KR 10-1358761 B1 US 2013-0149587 A1 US 2015-0086714 A1 US 8927134 B2 WO 2012-150838 A2	2014/04/16 2014/03/12 2014/04/24 2016/05/12 2016/03/02 2014/02/07 2013/06/13 2015/03/26 2015/01/06 2012/11/08

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875						Application or Docket Number 18/631,783								
APPLICATION AS FILED - PART I														
(Column 1)			(Column 2)			SMALL ENTITY		OR		OTHER THAN SMALL ENTITY				
FOR		NUMBER FILED		NUMBER EXTRA		RATE(\$)		FEE(\$)		RATE(\$)		FEE(\$)		
BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A				N/A		320		
SEARCH FEE (37 CFR 1.16(k), (l), or (m))		N/A		N/A		N/A				N/A		700		
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A				N/A		800		
TOTAL CLAIMS (37 CFR 1.16(i))		19 minus 20 =		* 0						OR x =		0		
INDEPENDENT CLAIMS (37 CFR 1.16(h))		3 minus 3 =		*						OR x =		0		
APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).										0		
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))												0		
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL				TOTAL		1820		
APPLICATION AS AMENDED - PART II														
(Column 1)			(Column 2)			(Column 3)			SMALL ENTITY		OR		OTHER THAN SMALL ENTITY	
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)		ADDITIONAL FEE(\$)		RATE(\$)		ADDITIONAL FEE(\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x =				OR x =				
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x =				OR x =				
	Application Size Fee (37 CFR 1.16(s))									OR				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									OR				
						TOTAL ADD'L FEE				OR		TOTAL ADD'L FEE		
(Column 1)			(Column 2)			(Column 3)								
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)		ADDITIONAL FEE(\$)		RATE(\$)		ADDITIONAL FEE(\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x =				OR x =				
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x =				OR x =				
	Application Size Fee (37 CFR 1.16(s))									OR				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									OR				
						TOTAL ADD'L FEE				OR		TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.														